



An IoT-Enabled System for Real-Time Fuel Monitoring, GPS Tracking, and Fuel Anomaly
Detection in Trucks Using Context-Aware Non-Parametric Machine Learning

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ABSTRACT

The logistics and transportation industry continues to face challenges in fuel management, anomaly detection, and real-time fleet monitoring. Existing solutions often operate in silos, resulting in fragmented data, undetected fuel irregularities, and delayed maintenance responses. This study developed and evaluated an IoT-based, context-aware fleet monitoring system that first acquires standard OBD-II vehicle parameters and retrieves fuel level through a vehicle-specific PID adapter, implemented in the tested vehicle through the Toyota meter Mode 22 PID 21/29 channel. The system tracks GPS location and detects anomalous fuel consumption in trucks using context-aware non-parametric machine learning models (Matrix Profile and Isolation Forest). The system uses an ESP32-based in-truck HMI with a fuel-filter cascade, a prioritized LoRa, LTE, GSM 2G, WiFi, and SPIFFS-offline communication stack, and embedded alert hardware for driver rest and maintenance notifications. Sensor and vehicle data are synchronized and transmitted to a cloud backend where anomalies are automatically flagged and actionable alerts are generated. The prototype was evaluated using controlled-injection trip data and field-test telemetry data to measure detection accuracy, false positive rates, and response latency. Three user-facing interfaces were delivered: an embedded in-truck HMI for drivers, a mobile driver web application for companion GPS and trip-state synchronisation, and an administrative dashboard for SGSA fleet monitoring.

Index Terms—Fuel Monitoring, GPS Tracking, Fuel Anomaly Detection, Internet of Things, Context-Aware Machine Learning, Isolation Forest, Matrix Profile



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Abbreviations

463	ACID	Atomicity, Consistency, Isolation, And Durability
464	ADC	Analog-to-Digital Converter
465	AI	Artificial Intelligence
466	APN	Access Point Name
467	API	Application Programming Interface
468	AUC	Area Under The Curve
469	BLE	Bluetooth Low Energy
470	BT	Bluetooth
471	BW	Bandwidth
472	CAN	Controller Area Network
473	CE	Conformité Européenne (European Conformity)
474	CGA2M+	Contextual Generalized Additive Plus
475	CPS	Cyber-Physical System
476	CSS	Chirp Spread Spectrum
477	CSV	Comma-Separated Values
478	CSQ	Cell Signal Quality
479	DB	Database
480	DC	Direct Current
481	DTC	Diagnostic Trouble Code
482	EBM	Explainable Boosting Machine
483	ECU	Engine Control Unit
484	ELM327	ELM327 OBD-II Interpreter
485	EMA	Exponential Moving Average
486	ERD	Entity-Relationship Diagram
487	ESP32	Espressif Systems Microcontroller With Wi-Fi And Bluetooth
488	FAIR	Findable, Accessible, Interoperable, And Reusable
489	FCC	Federal Communications Commission
490	FIR	Finite Impulse Response
491	FPR	False Positive Rate
492	GDPR	General Data Protection Regulation



493	GNSS	Global Navigation Satellite System
494	GO	General Objective
495	GPIO	General-Purpose Input/Output
496	GPS	Global Positioning System
497	GSM	Global System For Mobile Communications
498	GUI	Graphical User Interface
499	HDV	Heavy-Duty Vehicle
500	HMAC	Hash-based Message Authentication Code
501	HMI	Human-Machine Interface
502	HTML	Hypertext Markup Language
503	HTTP	Hypertext Transfer Protocol
504	HTTPS	Hypertext Transfer Protocol Secure
505	HSPI	Hardware Serial Peripheral Interface
506	I2C	Inter-Integrated Circuit
507	IEC	International Electrotechnical Commission
508	IEEE	Institute Of Electrical And Electronics Engineers
509	IETF	Internet Engineering Task Force
510	IF	Isolation Forest
511	IIR	Infinite Impulse Response
512	IOT	Internet Of Things
513	IQR	Interquartile Range
514	ISO	International Organization For Standardization
515	ITU	International Telecommunication Union
516	J1939	SAE J1939 Heavy-Duty Vehicle Network Standard
517	JSON	JavaScript Object Notation
518	KMPL	Kilometers Per Liter
519	LDV	Light Duty Vehicle
520	LOF	Local Outlier Factor
521	LORA	Long Range (Low-power Wide-area Radio)
522	LOS	Line Of Sight
523	LOTO	Leave-One-Trip-Out
524	LPWAN	Low-Power Wide-Area Network
525	LSTM	Long Short-Term Memory



526	LTE	Long-Term Evolution
527	M2M	Machine-to-Machine
528	MAF	Mass Air Flow
529	MAPE	Mean Absolute Percentage Error
530	MCU	Microcontroller Unit
531	ML	Machine Learning
532	MP	Matrix Profile
533	MQTT	Message Queuing Telemetry Transport
534	NAB	Numenta Anomaly Benchmark
535	NLOS	Non-Line Of Sight
536	NMEA	National Marine Electronics Association
537	NTP	Network Time Protocol
538	NTC	National Telecommunications Commission
539	OASIS	Organization For The Advancement Of Structured Information Standards
540	OBD-II	On-Board Diagnostics, Second Generation
541	OSRM	Open Source Routing Machine
542	PGN	Parameter Group Number
543	PHP	Philippine Peso
544	PID	Parameter Identifier
545	PIR	Passive Infrared
546	P95	95Th Percentile
547	P99	99Th Percentile
548	PRISMA	Preferred Reporting Items For Systematic Reviews And Meta-Analyses
549	PWA	Progressive Web Application
550	PWM	Pulse-Width Modulation
551	QOS	Quality Of Service
552	RAM	Random Access Memory
553	REST	Representational State Transfer
554	RFID	Radio Frequency Identification
555	ROC	Receiver Operating Characteristic
556	RPM	Revolutions Per Minute
557	RSSI	Received Signal Strength Indicator
558	RTC	Real-Time Clock



559	SAE	Society Of Automotive Engineers
560	SD	Secure Digital
561	SF	Spreading Factor
562	SGSA	Hino Parañaque Subic GS Auto Inc.
563	SMA	Simple Moving Average
564	SMS	Short Message Service
565	SNR	Signal-to-Noise Ratio
566	SO	Specific Objective
567	SOC	State Of Charge
568	SOM	Self-Organizing Map
569	SPI	Serial Peripheral Interface
570	SPIFFS	Serial Peripheral Interface Flash File System
571	SPN	Suspect Parameter Number
572	SQL	Structured Query Language
573	SSL	Secure Sockets Layer
574	STAMP	Scalable Time-series Anytime Matrix Profile
575	SUS	System Usability Scale
576	TCP/IP	Transmission Control Protocol/Internet Protocol
577	TFT	Thin-Film Transistor
578	TLS	Transport Layer Security
579	UART	Universal Asynchronous Receiver-Transmitter
580	UI	User Interface
581	URL	Uniform Resource Locator
582	USB	Universal Serial Bus
583	UUID	Universally Unique Identifier
584	UX	User Experience
585	VIN	Vehicle Identification Number
586	VSPI	Virtual Serial Peripheral Interface
587	WGS84	World Geodetic System 1984
588	WIFI	Wireless Fidelity
589	XAI	Explainable Artificial Intelligence



590

591

Notation

592

 x_t current raw sensor value at time t

593

 x_t^{smooth} smoothed sensor value at time t after applying moving average

594

 N

window size for the moving average filter

595

 x_t^{med} median-filtered sensor value at time t

596

 W

sliding window length of the median filter

597

 y_t exponential moving average output at time t

598

 α

smoothing factor of the exponential moving average

599

 τ

time constant of the exponential moving average

600

 δ

dead-band publishing threshold

601

 y_{pub}

last published dead-band fuel value

602

 $V_{\text{sensor},i}$ fuel volume estimated by the sensor at sample i

603

 $V_{\text{manual},i}$ actual (manual) fuel volume measurement at sample i

604

 T_{sent}

timestamp when a data packet is sent

605

 T_{received}

timestamp when a data packet is received

606

 L

data transmission latency

607

 T_i GPS timestamp at index i

608

 t_i

time difference between successive GPS logs

609

 T

time series of fuel usage

610

 m

window length for subsequence in time series analysis

611

 $P[i]$ matrix profile value at index i

612

 $\mathbf{x}$

input feature vector for anomaly detection

613

 $s(\mathbf{x})$ anomaly score for vector $\mathbf{x}$ from Isolation Forest

614

 $E(h(\mathbf{x}))$ expected path length for $\mathbf{x}$ in an Isolation Forest

615

 $c(n)$ average path length for unsuccessful search in a BST with n points

616

 TP

true positive

617

 FP

false positive

618

 TN

true negative

619

 FN

false negative

620

 T_{detected}

timestamp when an anomaly is detected

621

 T_{actual}

timestamp of actual anomaly occurrence



622	$\hat{p}$	estimated binomial proportion (sample success rate)
623	z	standard normal quantile used in the Wilson score interval
624	n	number of samples or validation events
625	C_{tank}	nominal tank capacity used to convert fuel volume to percentage
626	$V_{\text{pump},i}$	fuel volume dispensed by the pump during validation event i
627	$V_{\text{HMI},i}$	fuel volume displayed on the HMI during validation event i
628	e_i	fuel-volume error for validation event i
629	f_i	filtered fuel-level percentage at telemetry row i
630	Δf_i	change in filtered fuel level between successive telemetry rows
631	d_i	distance travelled between telemetry rows $i - 1$ and i
632	v_i	vehicle speed at telemetry row i
633	kmpl_i	fuel-economy estimate for telemetry row i
634	Precision	precision of the anomaly detector, computed as $TP/(TP + FP)$
635	FPR	false-positive rate of the anomaly detector, computed as $FP/(FP + TN)$
636	$q_{0.985}$	0.985 quantile of clean training anomaly scores used as the decision threshold
637	$\mathcal{D}_{\text{train}}$	training partition used within a leave-one-trip-out fold
638	$\mathcal{D}_{\text{test}}$	held-out trip partition used for evaluation within a leave-one-trip-out fold



639

640

Glossary

641

anomaly detection

A technique used to identify data points, events, or observations that deviate significantly from expected patterns. In fuel monitoring, this helps detect irregularities such as theft, leakage, or abnormal consumption.

642

Artificial Intelligence

A branch of computer science focused on building systems capable of performing tasks that typically require human intelligence, such as decision-making, pattern recognition, and problem-solving.

643

cloud analytics

Data analysis performed on cloud-based platforms, enabling real-time access to fleet data (e.g., GPS and fuel metrics) and supporting informed decision-making.

644

deep learning

A subset of machine learning involving deep neural networks with multiple layers. Used in predictive analytics and anomaly detection in real-time fleet systems.

645

fleet management

The administration and coordination of commercial vehicle operations, including monitoring of fuel use, driver behavior, vehicle maintenance, and routing.

646

float sensor

A sensor that uses a floating mechanism to measure liquid levels, such as fuel in a tank. Commonly used in IoT-based real-time monitoring systems.

647

Global System for Mobile Communications

A standard for mobile communication that enables real-time alerts and remote data transmission for vehicle monitoring systems.



648	Global Positioning System	A satellite-based system that provides real-time geolocation data, used in fleet operations for tracking vehicle movement and optimizing routes.
649	Internet of Things	A network of interconnected physical devices that collect and share data via the internet. In logistics, this includes in-vehicle sensors for real-time tracking and monitoring.
650	Isolation Forest	A context-aware machine learning algorithm for anomaly detection that isolates outliers based on random partitioning of data points.
651	machine learning	A subset of AI where systems learn from data and improve performance over time without being explicitly programmed.
652	Matrix Profile	A time-series data mining method used for pattern recognition and anomaly detection in sequential datasets like fuel usage over time.
653	context-aware machine learning	A context-aware machine learning paradigm where the model learns the structure of normal-only (unlabeled) data and flags deviations from that learned envelope. Useful for anomaly detection where confirmed abnormal examples are scarce.
654	On-Board Diagnostics	A standardized vehicle diagnostic interface that exposes engine and sensor data, such as fuel level, speed, and engine parameters, through the on-board diagnostic port without requiring physical modification of the vehicle.
655	Controller Area Network	A robust serial communication bus used within vehicles to allow electronic control units to exchange data without a host computer.



656	edge computing	A computing approach in which data is processed locally on the device that generates it, such as the embedded microcontroller in the fleet vehicle, reducing latency and bandwidth use before transmission to the cloud.
657	Human-Machine Interface	The in-vehicle interface that displays fuel, route, and alert information to the driver and captures driver acknowledgments such as rest and snooze actions.
658	telemetry	The automated measurement and wireless transmission of data, such as fuel level, location, and vehicle status, from the fleet vehicle to a remote backend for monitoring and analysis.
659	Leave-One-Trip-Out	A cross-validation protocol in which each trip is held out once as the test set while the model is trained and calibrated on the remaining trips, preventing information leakage from within-trip temporal correlation.
660	Cyber-Physical System	A system that tightly integrates physical processes with embedded computation and networked communication, where sensing, processing, and actuation operate together in real time.
661	confusion matrix	A table that summarizes the performance of a classifier by tabulating true positives, false positives, true negatives, and false negatives.
662	point anomaly	A single observation that deviates significantly from the rest of the data, such as a sudden fuel drop while the vehicle is stationary.
663	contextual anomaly	An observation that is anomalous only with respect to its surrounding context, such as high fuel consumption recorded at zero speed.



664	collective anomaly	A sequence of individually unremarkable observations that together form an anomalous pattern, such as gradual fuel siphoning over time.
665	discord	The subsequence of a time series that is least similar to all other subsequences, identified by the Matrix Profile as the most likely anomalous segment.
666	median filter	A non-linear order-statistic filter that replaces each sample with the median of a sliding window, robust to impulse noise and sensor spikes.
667	exponential moving average	A first-order recursive low-pass filter that smooths a signal by weighting recent samples more heavily than older ones according to a smoothing factor.
668	dead-band	A hysteresis mechanism that publishes a new value only when the smoothed signal departs from the last published value by more than a set threshold, preventing display jitter.
669	fuel slosh	The apparent fluctuation in a fuel-tank reading caused by liquid movement inside the tank during acceleration, braking, and turning.
670	Receiver Operating Characteristic	A curve that plots the true-positive rate against the false-positive rate of a classifier across thresholds; its area summarizes ranking performance.
671	Wilson score interval	A confidence interval for a binomial proportion that remains accurate for small samples and few positive cases, preferred over the normal-approximation interval.
672	Explainable Artificial Intelligence	Machine learning techniques that make a model's predictions interpretable, providing human-understandable reasons for why an input was flagged as anomalous.



673	Self-Organizing Map	An unsupervised neural network that maps high-dimensional data onto a low-dimensional grid, used to cluster and detect abnormal fuel-consumption patterns.
674	Chirp Spread Spectrum	A modulation technique used by LoRa in which data is encoded onto frequency sweeps (chirps), providing long range, low power use, and resistance to interference.
675	administrator dashboard	The fleet-manager interface that presents fleet-vehicle status, trips, maps, telemetry logs, analytics, alerts, and configuration controls for operational supervision.
676	alert acknowledgement	The user action or system record showing that an alert has been reviewed, acted upon, snoozed, or resolved by the driver or administrator.
677	backend	The cloud-side service layer that receives telemetry, stores records, synchronizes client interfaces, runs alert rules, and coordinates anomaly-detection results.
678	telemetry row	A single database record produced by the prototype telemetry pipeline, containing synchronized fuel, speed, location, trip-state, and communication fields for one sampling instant.
679	trip	A completed operating interval of a fleet vehicle, bounded by start and end events, used as the unit for route review, fuel analysis, and leave-one-trip-out evaluation.
680	fuel sender	The vehicle's built-in fuel-level sensing mechanism whose reported value is read through the diagnostic interface and filtered before being displayed or transmitted.



681	HMI reading	A measured fuel-volume or fuel-percentage reading taken from the vehicle's dashboard or embedded interface and compared with pump and receipt evidence during validation.
682	pump receipt	The receipt generated by the fuel pump during refuelling, used as supporting ground-truth evidence for dispensed fuel volume during validation.
683	odometer reading	A vehicle distance reading used to verify trip progress, fuel economy, and consistency between field observations and recorded telemetry.
684	project-native partition	The collection of telemetry records captured by the implemented prototype and used as the primary labelled evaluation partition for the thesis results.
685	public partition	A publicly derived clean-route telemetry corpus used as an out-of-distribution stress test rather than being pooled with the labelled project-native evaluation data.
686	out-of-distribution test	A validation condition in which clean data come from a different route, vehicle, or operating domain from the project-native data, used to test whether the detector creates unnecessary false alarms.
687	fuel-economy deviation feature	A derived feature that compares observed fuel use with the expected fuel economy for the fleet vehicle, driver, route, or operating state.
688	contextual gate	A deterministic rule layer that checks model flags against operational context, such as refuelling, vehicle motion, and fuel-change direction, before an alert is surfaced.



689	precision	The proportion of flagged anomaly rows that are truly anomalous, used in this study as the primary quality measure for avoiding misleading fuel alerts.
690	false-positive rate	The proportion of normal rows incorrectly flagged by the detector, used in this study to measure false-alarm behavior against the SO4 ceiling.
691	communication stack	The prioritized combination of LoRa, cellular, WiFi, and offline buffering used by the prototype to preserve telemetry delivery across changing network conditions.
692	offline buffer	A local storage mechanism on the embedded device that temporarily preserves unsent telemetry rows when no communication channel is available.
693	field-test telemetry	Telemetry collected during operational trials of the prototype, including fuel, speed, location, trip-state, and alert records used to evaluate real-world system behavior.
694	fuel anomaly	An event in which fuel behavior differs from the expected pattern for the vehicle context, such as a drop while stationary, unusually high fuel use, or a leak-like downward trend.
695	fuel-economy baseline	A reference profile of expected fuel economy for a fleet vehicle, driver, or route, used to judge whether current consumption is unusually high for that context.
696	HMI state machine	The embedded logic that controls the driver's display states, button actions, buzzer behavior, fleet-vehicle selection, trip status, rest status, and alert visibility.



697	offline replay	A recovery procedure in which locally buffered telemetry is uploaded after connectivity returns, allowing delayed records to be preserved instead of discarded.
698	route history	The stored sequence of GPS positions and trip metadata used to review where a fleet vehicle travelled and how fuel behavior changed along the route.
699	trip session	A bounded period of vehicle operation recorded by the system from trip start to trip end, including fuel, route, distance, driver, fleet vehicle, and alert data.
700	vehicle-context feature	Derived model inputs that describe the operating situation of the vehicle, such as motion state, speed, engine load, RPM, distance, route type, and refuel state.
701	dashboard usability survey	The structured questionnaire used to evaluate whether administrators found the dashboard understandable, usable, and useful for monitoring and resolving fleet events.
702	partner fleet	The participating fleet environment used for prototype consultation, vehicle inspection, demonstration, and controlled operational feedback.
703	prototype demonstration	A supervised presentation and hands-on review of the working prototype, including the embedded HMI, dashboard, telemetry flow, and alert behavior.



De La Salle University

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Chapter 1

705

INTRODUCTION



1.1 Background of the Study

The transportation and logistics industry is a critical driver of economic development but is increasingly facing operational inefficiencies and security challenges, especially in fuel management and vehicle tracking. Fuel theft, inaccurate fuel monitoring, inefficient routing, and lack of real-time data remain persistent issues for fleet operators, particularly in developing countries. These challenges are exacerbated by the absence of scalable, integrated technological systems that provide reliable data-driven solutions [Ahmed et al., 2017, Alquhali et al., 2019, Crisgar et al., 2021].

Fuel is one of the most substantial and recurring expenses in logistics operations, with its accurate monitoring being essential to maintaining profitability and operational efficiency. Even minor inconsistencies in fuel tracking—such as errors in manual readings, unreported consumption, or unnoticed leakage—can accumulate into significant losses over time. Komal Shoukat Ali et al. (2018) [Komal Shoukat Ali et al., 2018] report that fuel-related fraud and inefficiencies can contribute up to 30% of the total operational overhead in large fleet operations. These losses are often compounded by issues such as unauthorized vehicle use, fuel siphoning, and unoptimized driving routes, which not only drive up fuel costs but also disrupt delivery schedules, vehicle maintenance cycles, and driver accountability. Despite efforts to manage these concerns, many logistics providers continue to rely on fragmented or outdated systems that offer limited visibility into fuel-related metrics and driver behavior [Dhivyasri and Mariappan, 2015, Obikoya, 2014].

These pressures are especially acute in the Philippine setting, where fleet operations contend with some of the most severe road and logistics conditions in the region. The Japan International Cooperation Agency estimates that traffic congestion costs Metro Manila



roughly PHP 3.5 billion per day, a figure projected to reach PHP 5.4 billion per day by 2035 without intervention [Japan International Cooperation Agency, 2022, Japan International Cooperation Agency, 2025]. Logistics operations are further constrained by congestion at the Port of Manila, which has operated at roughly 94% utilization against an ideal of about 80%, and by regulatory measures such as the 2014 Manila truck ban, which the Philippine Institute for Development Studies estimated caused around PHP 43.85 billion in economic losses and doubled container trucking costs [Patalinghug et al., 2015b, Llanto, 2017, Patalinghug et al., 2015a]. Compounding these conditions, fuel fraud and adulteration in the national supply chain have been estimated to cost up to USD 750 million annually [Asian Development Bank, 2016]. For trucks operating under such conditions, prolonged idling, stop-and-go consumption, and route variability are routine rather than exceptional, which makes fixed-threshold fuel-monitoring rules, calibrated for free-flowing conditions, especially unreliable.

In addition to financial implications, traditional fuel monitoring systems struggle to detect abnormal fuel consumption patterns in real time. These anomalies may be caused by aggressive or inefficient driving habits, extended engine idling, mechanical malfunctions (such as injector faults or fuel line leaks), or variable environmental conditions like terrain and traffic. However, without intelligent tools that correlate fuel data with contextual factors—such as vehicle speed, route topography, and time of day—fleet managers are often unable to distinguish between acceptable fuel use and problematic deviations. As highlighted by Mumcuoglu et al. (2023) [Mumcuoglu et al., 2023], conventional systems lack the sophistication to perform this level of analysis, leading to delayed detection of inefficiencies and reduced responsiveness. This underscores the urgent need for smart, data-driven solutions that combine fuel monitoring with GPS tracking and anomaly detection



models to proactively manage consumption and enhance overall fleet performance [Barbado and Corcho, 2022, Beteto et al., 2022, Fortela et al., 2023].

To address these problems, researchers and engineers are increasingly exploring the convergence of IoT, GPS, and AI—particularly machine learning—to develop smart, real-time monitoring systems [Alquhali et al., 2019, Joshi et al., 2024, Prabhu et al., 2022]. These systems aim to optimize fuel usage, enhance fleet security, and improve operational efficiency through automation and predictive analytics.

The transportation and logistics sector has attracted significant scholarly attention, particularly in the development of smart systems for operational efficiency and security. The academic literature reflects a growing interest in leveraging IoT, GPS, and AI technologies to enhance fleet management, particularly in areas such as fuel monitoring, vehicle tracking, and anomaly detection. Several studies in recent years have investigated IoT-enabled fuel monitoring, GPS tracking, and anomaly detection solutions [G et al., 2022, G, 2023, Patil and Patil, 2024, Prabhu et al., 2022, RS et al., 2024, S et al., 2024].

Because partner HINO in the Philippines do not permit tank-cap or sender-loom modifications on production trucks, any field-deployable monitoring approach must be non-invasive. This constraint was the dominant driver of the acquisition-layer design adopted in this study.

- **Intelligent IoT-Enabled Real-Time Monitoring System for Logistics Management:** Joshi et al. [Joshi et al., 2024] developed a hybrid logistics management system that integrates IoT sensors, GPS, RFID, and cloud analytics. Their solution achieved real-time tracking of assets and vehicles, enhancing transparency and decision-making in logistics operations. However, the study was limited by the



type of hardware used and did not include advanced AI components for anomaly detection.

- **Fuel Consumption Classification for Heavy-Duty Vehicles:** Mumcuoglu et al. [Mumcuoglu et al., 2023] employed machine learning to classify fuel consumption anomalies based on naturalistic driving data from 57 trucks. With an accuracy of 92.2%, their approach effectively identified abnormal fuel consumption linked to driver behavior. Nevertheless, the dataset was relatively small (606 records), and integration with real-time vehicle diagnostics was not tested.

- **Development of an Intelligent Fuel Monitoring and Theft Detection System Equipped with GPS & GSM Integration:** Mathi et al. [Mathi et al., 2024] introduced an IoT-based solution combining resistance-based fuel level sensors, GPS, GSM, and real-time alarm mechanisms. Their system successfully monitored fuel and battery anomalies in hybrid vehicles but lacked higher-precision sensors and deep learning integration for smarter detection.

Despite these advancements, several limitations and research gaps remain:

- **Scalability Constraints:** Many existing solutions are limited to small-scale deployments or specific vehicle types (e.g., motorcycles, generators) and have not been validated for large truck fleets in diverse environments [RS et al., 2024, Tatababu et al., 2024].
- **Data and Sensor Limitations:** Several studies rely on basic sensors (resistance, ultrasonic) that may lack the accuracy and robustness needed for high-volume com-



mercial operations. Furthermore, datasets used are often small or simulated, limiting generalizability [Nada, 2017, aut, 2019].

- **Fragmented Solutions:** Many systems only address a single issue — fuel monitoring, GPS tracking, or theft detection — without offering a comprehensive, unified solution integrating these components using AI and IoT [G et al., 2022, Crisgar et al., 2021].

These gaps underline the need for a robust, scalable, and integrated IoT-enabled system capable of monitoring fuel levels in real time, tracking vehicle location, and detecting operational anomalies using machine learning algorithms. Such a system holds the potential to significantly reduce fuel fraud, enhance operational efficiency, and support more informed decision-making in fleet management. This study focuses on designing and implementing such a system within an active fleet operation, specifically addressing the technical and logistical challenges encountered by commercial truck fleets operating across varied conditions. By leveraging IoT sensors, GPS modules, and context-aware machine-learning-based analytics, the proposed solution aims to deliver a comprehensive, automated, and intelligent monitoring platform that advances the current landscape of smart logistics technologies.

1.2 Prior Studies



TABLE 1.1 LIST OF PRIOR STUDIES

Literature Title (APA Citation)	Research Problem	Method	Results	Limitations
Intelligent IoT-Enabled Real-Time Monitoring System for Logistics Management [Joshi et al., 2024]	Logistics systems lack integration and real-time monitoring capabilities for secure and efficient operations.	Hybrid IoT-based logistics management combining sensors, GPS, RFID, and cloud analytics for real-time monitoring and decision-making.	Provides real-time tracking of assets, vehicles, and equipment.	Constrained by hardware type and limited test cases; lacks advanced AI for anomaly detection.
Real Time Automatic Vehicle Monitoring System Using IoT [G et al., 2022]	Lack of real-time vehicle tracking and passenger/driver identity verification in institutional transport.	Used RFID, GPS, and NodeMCU ESP8266 with cloud storage and mobile app.	Enabled real-time tracking of location, speed, and passenger count.	Limited to institutional use; lacks anomaly detection and commercial scalability.
GPS-Based Vehicle Tracking and Theft Detection Systems using Google Cloud IoT Core & Firebase [Crisgar et al., 2021]	Ineffective vehicle recovery due to delayed or limited GPS/GSM tracking.	Implemented GPS-based tracking using Google Cloud IoT Core, MQTT, Firebase, and PWA interface.	Enabled real-time vehicle tracking, theft detection, and remote monitoring.	Designed for passenger cars; may require hardware adaptation for trucks/fleets.
Fuel Quantity and Quality Detection in Automobiles Using IoT [RS et al., 2024]	Fuel fraud through inaccurate dispensing and poor fuel quality in motorcycles.	IoT-enabled fuel management using strain gauges and quality sensors with real-time display/logging.	Achieved accurate real-time quantity and quality measurement.	Designed for motorcycles; not validated for larger fleets.
Fuel Consumption Classification for Heavy-Duty Vehicles [Mumcuoglu et al., 2023]	Identifying abnormal fuel consumption due to driver behavior or system faults.	Machine learning models using driving data from 57 trucks, considering weight and road slope.	92.2% accuracy; F1 score 0.78 for outlier detection.	Dataset limited to 606 records; lacks real-time integration.
Automatic Fuel Tank Monitoring, Tracking & Theft Detection System [Komal Shoukat Ali et al., 2018]	Inefficiency of manual fuel monitoring.	Fuel sensors, Arduino, GSM, GPS, LabView for data logging.	Real-time monitoring, alerts for theft/leakage.	Tested on basic hardware; lacks advanced analytics and large-scale validation.
Development of an Intelligent Fuel Monitoring and Theft Detection System [Mathi et al., 2024]	Fuel theft and battery anomalies in hybrid/electric vehicles.	Resistance-based fuel sensor, voltage sensor, GPS, GSM, buzzer, LCD.	Reliable monitoring and anomaly detection.	Lacks advanced sensors for enhanced leak detection.
Unsupervised Machine Learning to Detect Impending Anomalies in Testing of Fuel Economy and Emissions [Fortela et al., 2023]	Impending anomalies in fuel economy and emissions testing of light-duty vehicles.	Unsupervised ML methods (PCA, K-Means, SOM) on 34,602 test samples.	Trend detection in large datasets.	Limited to historical datasets; not validated on real-time operational data.
Smart Fuel Monitoring System for Diesel Generator [Tatababu et al., 2024]	Fuel quantity indication and theft in generators.	Ultrasonic/capacitive sensors, flow sensors, GSM, GPS, intelligent theft detection.	Accurate real-time monitoring, theft prevention.	Findings specific to tested generators; limited generalizability.



Literature Title (APA Citation)	Research Problem	Method	Results	Limitations
Design and Implementation for Trucks Tracking System Using GPS Based on Semantic Web [Nada, 2017]	Inefficient GPS-based truck tracking for fleet management.	GPS tracking system integrated with semantic web platform.	Accurate location tracking and reporting via web interface.	Dependent on GSM/GPS reliability; lacks IoT integration and anomaly detection.
Automatic Fuel Monitoring System [aut, 2019]	Accurate fuel monitoring and theft prevention.	Fuel sensors and GSM module with SMS alerts.	Monitors fuel, triggers alarms, and sends GPS alerts.	Basic sensors; limited scalability and analytics in complex environments.
Interpretable Machine Learning Models for Vehicle Fuel Consumption Anomalies [Barbado and Corcho, 2022]	Detecting/explaining fuel anomalies in fleets to optimize usage.	Unsupervised anomaly detection combined with interpretable ML.	Up to 80% anomalous instances explained; potential fuel reduction up to 88%.	Only analyzed individual feature relevance; not fully integrated in real-time systems.

1.2.1 Research Gaps Identified

A review of existing literature in fuel monitoring and anomaly detection for logistics and fleet management reveals persistent gaps that point to a need for more robust, integrated solutions. Although many systems incorporate IoT elements—such as sensors, GSM, and GPS—these are often limited in scope, typically designed for specific applications like diesel generators or motorcycles, rather than being adaptable for broader fleet use.

For example, systems like the Smart Fuel Monitoring System for Diesel Generators [Tatababu et al., 2024] and the Automatic Fuel Monitoring System [aut, 2019] demonstrate basic real-time monitoring and theft detection capabilities. However, these systems heavily rely on elementary sensor technology and offer minimal integration with advanced analytics or anomaly detection methods. Similarly, the machine learning-based approach in [Mumcuoglu et al., 2023] was limited in its use of data and did not incorporate real-time or context-aware learning techniques.

While [Barbado and Corcho, 2022] explored the potential of interpretable machine



learning and unsupervised anomaly detection, their models were evaluated on constrained datasets and not integrated into real-time operational systems. [Fortela et al., 2023] also demonstrated effective anomaly detection using unsupervised learning in emissions and fuel economy testing, but again, their data was static and gathered in controlled environments. These findings were not validated using on-vehicle, operational data, which limits their applicability in dynamic, real-world conditions.

Additionally, studies such as those by [Joshi et al., 2024] and [Nada, 2017] underscore the need for real-time GPS tracking in fleet operations, but they stop short of presenting a cohesive system that incorporates GPS, fuel monitoring, and intelligent anomaly detection into one unified platform.

The core research gap, therefore, lies in the absence of a fully integrated, real-time, IoT-based system that combines fuel monitoring, GPS tracking, and anomaly detection powered by context-aware machine learning. Addressing this gap is vital to improving operational efficiency, fuel security, and predictive maintenance in modern fleet management.

1.2.2 Implemented Response to the Identified Gaps

This study developed and evaluated an integrated IoT-based fleet monitoring system that unifies fuel level monitoring, GPS-based vehicle tracking, and context-aware machine learning for anomaly detection. Unlike previous solutions that focus on isolated functions or static datasets, the developed system operated in real time using vehicle telemetry acquired through the OBD-II interface and GPS module. The prototype was evaluated using controlled-injection trip data and field-test telemetry data to assess fuel monitoring performance, anomaly detection capability, and communication latency.

The system architecture incorporates scalable and cost-efficient components such as



OBD-II acquisition, GPS modules, and cloud-based transmission via mobile networks, ensuring compatibility with partner-fleet operating constraints.

For anomaly detection, the system employs context-aware non-parametric machine learning algorithms such as Matrix Profile and Isolation Forest. These models identify outliers within time-series data without the need for labeled training datasets, making them suitable for fleet scenarios where data behavior varies with routes, driver habits, or environmental conditions. Unlike traditional parametric approaches, non-parametric models are more flexible and adaptive because they do not assume any fixed data distribution.

In addition, synchronized HMI, mobile, and dashboard interfaces provide meaningful insights by correlating fuel usage patterns with routes and travel times. These interfaces support real-time notifications such as driver rest alerts, maintenance prompts, and efficiency summaries, enabling data-informed decision-making and operational improvements.

By addressing the limitations of prior research—especially the lack of integration, real-time analysis, and adaptive anomaly detection—this study offers a practical and scalable solution for enhancing fuel management and fleet operations in dynamic environments.

1.3 Problem Statement

Efficient fuel monitoring and route tracking are essential components of sustainable and cost-effective truck fleet operations. However, existing systems face significant challenges, including inaccurate fuel measurement, disconnected data sources, delayed anomaly detection, and limited support for safety-based decision-making, as outlined below.

1. PS1: The Ideal Scenario



- Trucking companies should have real-time access to accurate fuel usage and GPS tracking data to ensure efficient operations, reduce fuel theft, and improve route planning.
- Drivers should receive timely rest alerts and vehicle maintenance reminders based on actual driving and fuel metrics to promote safety, avoid overfatigue, and extend vehicle life.
- Fleet managers should be able to detect unusual fuel consumption patterns automatically, enabling early intervention and optimized scheduling.
- All of these should operate through automated, cloud-based systems that integrate IoT sensors, machine learning, and remote dashboard monitoring.

2. PS2: The Reality of the Situation

- Most trucking operations lack real-time visibility into fuel usage, relying instead on manual readings or delayed reports that are prone to error and manipulation.
- GPS tracking is often siloed from other performance indicators like fuel consumption and driving behavior, making behavior-based analysis or route optimization difficult.
- Anomalous fuel consumption events (e.g., theft, leakage, or inefficiency) are rarely detected until long after they've occurred, resulting in financial loss and operational delays.
- Rest and maintenance schedules are typically based on fixed intervals rather than real-time usage data, increasing the risk of driver fatigue, breakdowns, and missed service opportunities.



- 893 • Existing technologies are either too costly, fragmented, or lack contextual
894 intelligence to automatically correlate GPS, fuel, and operational metrics.

895 3. PS3: The Consequences for the Audience

- 896 • Truck fleet operators face increasing operational costs due to undetected fuel
897 inefficiencies and theft, impacting business sustainability.
- 898 • Driver safety is compromised when rest is not scheduled based on actual driving
899 patterns or system recommendations.
- 900 • Vehicle lifespan shortens and maintenance costs rise when alerts are not tied to
901 real-time usage, resulting in unexpected breakdowns and downtime.
- 902 • Decision-makers are unable to make informed, data-driven decisions due to
903 fragmented systems and the absence of an integrated monitoring and alert
904 platform.
- 905 • These issues collectively hinder operational efficiency, safety, and environ-
906 mental responsibility, particularly for logistics providers in developing regions
907 lacking advanced fleet management tools.

908 **1.4 Objectives and Deliverables**

909 **1.4.1 General Objective (GO)**

910 GO: To develop and evaluate an IoT-based system for trucks, in collaboration with the
911 partner truck company Hino Parañaque Subic GS Auto Inc. (SGSA), that monitors real-
912 time fuel levels, tracks routes using GPS, and detects unusual fuel consumption using



machine learning, with the goal of improving driver safety, supporting rest scheduling, and facilitating proactive vehicle maintenance through automated alerts using fuel and distance metrics, based on established service guidelines.;

1.4.2 Specific Objectives (SOs)

- SO1: To measure fuel levels using digital float sensors integrated via the truck's ECU/diagnostic connector and GPS module, ensuring non-invasive installation and real-time accuracy within $\pm 5\%$.
- SO2: To transmit real-time fuel and GPS data through IoT-based communication to a cloud server, accessible through a centralized dashboard.
- SO3: To capture and log GPS location data every 5 seconds via an embedded module, correlating route information with fuel and time data for behavior tracking.
- SO4: To design and evaluate a context-aware fuel usage deviation model using context-aware non-parametric machine learning models (Matrix Profile or Isolation Forest), achieving 90% precision and 10% false positives in identifying anomalous fuel consumption patterns within controlled environments.
- SO5: To develop and integrate hardware-based rest and maintenance alert mechanisms (e.g., Start/Stop button, buzzer indicators) tied to the web dashboard using cumulative fuel and distance data.



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1.4.3 Final Deliverables

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1.5 Implemented Deliverables per Objective

TABLE 1.2 IMPLEMENTED DELIVERABLES PER OBJECTIVE

Objectives	Implemented Deliverables
GO: To develop and evaluate an IoT-based system for trucks, in collaboration with the partner truck company Hino Parañaque Subic GS Auto Inc. (SGSA), that monitors real-time fuel levels, tracks routes using GPS, and detects unusual fuel consumption using machine learning, with the goal of improving driver safety, supporting rest scheduling, and facilitating proactive vehicle maintenance through automated alerts using fuel and distance metrics, based on established service guidelines.	Trained Context-Aware Machine Learning Model - A validated ML model (Matrix Profile or Isolation Forest) trained on SGSA-reflective datasets for detecting anomalous fuel consumption. Web-Based Monitoring Interface - Real-time dashboard showing GPS location, fuel level, distance, and operating hours for each fleet vehicle. Embedded In-Vehicle HMI - Driver-friendly interface displaying driver credentials (name, PIN, and assigned fleet vehicle), current trip activity, and alert notifications. Cloud Backend Infrastructure - Cloud-based data aggregation, processing, and automated alert generation accessible from SGSA's base. Performance Documentation - Complete validation results under laboratory and SGSA field conditions.
SO1: To measure fuel levels using digital float sensors integrated via the truck's ECU/-diagnostic connector and GPS module, ensuring non-invasive installation and real-time accuracy within $\pm 5\%$.	- Calibrated fuel-level acquisition module interfaced through the fleet vehicle ECU/OBD-II connector. - Accuracy validation report comparing filtered OBD-II fuel data with dashboard-gauge levels. - Embedded firmware for fuel data acquisition, filtering, and preprocessing. - Drift, noise, and variance analysis across different tank levels and conditions.



Objectives	Implemented Deliverables
SO2: To transmit real-time fuel and GPS data through IoT-based communication to a cloud server, accessible through a centralized dashboard.	1. Integrated IoT communication module with LoRa, LTE, GSM 2G, WiFi, and SPIFFS offline replay. 2. Cloud backend with secure API endpoints for receiving fuel and GPS data. 3. Latency benchmarking report under variable network environments. 4. Optimized HTTP transmission logic with buffering and fail-safe retries; MQTT is retained as a future-compatible communication option. 5. Security documentation covering packet integrity and encryption features.
SO3: To capture and log GPS location data every 5 seconds via an embedded module, correlating route information with fuel and time data for behavior tracking.	1. Integrated GPS module capturing coordinates and timestamps at 5-second intervals. 2. Data fusion algorithm linking GPS location, fuel levels, and time progression. 3. Route visualization through a map interface on the dashboard. 4. Recommended route engine with truck-ban zone awareness. 5. Database structure storing synchronized fuel–location–time datasets. 6. Behavioral analytics showing fuel usage per route segment.
SO4: To design and evaluate a context-aware fuel usage deviation model using context-aware non-parametric machine learning models (Matrix Profile or Isolation Forest), achieving 90% precision and 10% false positives in identifying anomalous fuel consumption patterns within controlled environments.	1. Labeled normal and anomalous fuel consumption datasets. 2. Implementation of Isolation Forest and/or Matrix Profile in Python/MATLAB. 3. Evaluation metrics report including precision, recall, and $FPR \leq 10\%$. 4. Exportable model (PMML/ONNX) for real-time pipeline integration. 5. Context-aware analytics combining slope, route, load, and time-of-day factors.
SO5: To develop and integrate hardware-based rest and maintenance alert mechanisms (e.g., Start/Stop button, buzzer indicators) tied to the web dashboard using cumulative fuel and distance data.	1. Cloud dashboard showing fuel usage, distance, operating hours, and maintenance status. 2. Rule-based alert engine for configurable driver safety and vehicle maintenance notifications, with example thresholds such as 300 km or 6-hour rest reminders and 5,000 km oil maintenance alerts. 3. Physical embedded module: Start/Stop button, buzzer/light indicators for rest and maintenance alerts. 4. Backend logic computing distance-based and time-based safety thresholds. 5. Dashboard filters for alert logs and historical events. 6. Extended safety monitoring: speed tracking and overspeed alerts.



1.6 Significance of the Study

1.6.1 Technical Benefit

This research developed a context-aware, cloud-connected fleet monitoring system that integrates diagnostic-connector fuel acquisition, GPS route tracking, and context-aware machine learning to enhance real-time monitoring of fuel usage and detection of anomalous consumption events. Fuel levels are measured using digital fuel-level data exposed through the truck's ECU/diagnostic connector, with the system targeting an accuracy of $\pm 5\%$ against known-volume pump-gas events and manufacturer dashboard readings. Real-time fuel and GPS data are transmitted through IoT-based communication to a cloud server and made accessible through the centralized dashboard.

GPS modules embedded in the vehicles log location data every 5 seconds, enabling the system to correlate fuel consumption with time, distance, and route conditions for behavioral tracking and operational analysis. The study implements non-parametric context-aware learning methods, specifically Matrix Profile and Isolation Forest, to identify deviations in fuel usage without the need for labeled training datasets, allowing scalable and adaptive anomaly detection. In addition, the cloud backend integrates hardware-driven rest alerts and maintenance reminders, leveraging cumulative fuel and distance data to generate intelligent break schedules and service interval notifications. Through these combined capabilities, the system aims to improve technical performance by enhancing measurement accuracy, communication responsiveness, anomaly detection reliability, and overall automation in fleet fuel management and driver safety support.



1.6.2 Social Impact

Through an implementation of automatic rest alerts and preemptive maintenance cues, the study encourages safety and health of the drivers to prevent accidents commonly caused by fatigue as well as breakdowns. The system informs drivers when they are near critical or recommended limits, such as 300 km of continuous driving or extended operating duration, encouraging healthy driving practices and compliance with regulations, especially for long-distance hauls. These practices empower fleet managers to systematically enforce safety policies and promote a culture of responsibility in the industry.

In addition, GPS-based route monitoring and fuel usage logging reinforce accountability and transparency, two essentials in preventing fuel misuse. This helps build trust between management and drivers and supports data-led decision-making in transport logistics.

1.6.3 Environmental Welfare

The system improves operational efficiency by minimizing fuel waste, optimizing route performance, and decreasing unplanned maintenance downtime. Through precise anomaly detection and maintenance forecasting, the platform enables cost-effective fleet operation and longer vehicle life. It also supports environmental sustainability by reducing unnecessary fuel consumption and emissions from inefficient routes, fuel leaks, or late maintenance. By promoting timely service interventions, such as oil changes at every 5,000 kilometers, the study supports cleaner engine performance and reduced environmental impact over time.



1.7 Assumptions, Scope, and Delimitations

1.7.1 Assumptions

1. Technical Infrastructure

- Consistent internet connection is required to maintain real time data exchange between the IoT sensors, GPS module, and cloud sensor.
- The system integration between the hardware and cloud based infrastructure operates smoothly without significant technical issues.
- The system will have access to a stable power supply for an uninterrupted operation of the on board IoT system.
- Mobile network coverage is able to handle a low latency data communication.

2. Data Accuracy

- The OBD-II/ECU-based fuel acquisition channel is assumed to provide operationally useful fuel-level readings after filtering and validation against known-volume pump-gas events and manufacturer dashboard readings. Direct dipstick validation was outside the permitted field procedure.
- The GPS module captures and logs precise coordinates at 5-second intervals, enabling accurate route reconstruction and correlation with fuel consumption and time data.

3. Machine Learning Model Performance



- The context-aware learning models (e.g., Matrix Profile, Isolation Forest) are capable of identifying anomalous fuel consumption with 90% precision and 10% false positives within the controlled environments evaluated in this study.
- The training and evaluation data of anomaly detection will be representative of normal fuel consumption and truck behavior under different operating conditions.

4. User Interaction

- It is assumed that users (e.g., truck drivers, fleet operators, and system administrators) will interact with the system according to their roles and available tools.
- Truck drivers are not required to use personal mobile devices; instead, they will rely on a simple onboard interface consisting of a small display and buzzer for receiving rest alerts, maintenance reminders, and status notifications. However, a mobile web application was also developed as a supplementary HMI component to support navigation and provide additional convenience for drivers with personal mobile devices.
- Fleet operators and system administrators will interact with the cloud-based dashboard using standard office computers to monitor fuel data, GPS routes, alerts, and maintenance logs.
- Fleet personnel are expected to follow system recommendations (e.g., rest intervals, maintenance reminders) based on real-time fuel, distance, and performance metrics.



5. Operating Environment

- Environmental conditions, together with power and network breakdowns, will not significantly affect system performance.
- System maintenance is performed routinely and at normal intervals to ensure operational continuity.

1.7.2 Scope

The scope of the study encompasses the development and implementation of a smart fleet monitoring system that utilizes IoT and machine learning technologies for enhanced fuel usage , route monitoring and vehicle maintenance reminder.

1. Real-Time Fuel Monitoring

- a) Deployment of a filtered fuel-level acquisition module interfaced through the truck's ECU/OBD-II diagnostic connector, providing continuous fuel-level monitoring validated against known-volume pump-gas events and manufacturer dashboard readings.

2. Cloud Based Data Transmission

- a) The system implements prioritized telemetry transmission through LoRa, LTE, GSM 2G, WiFi, and SPIFFS offline replay, with latency and packet-delivery performance evaluated under field and controlled conditions.

3. GPS Enabled Route Tracking



- 1034 a) Incorporation of GPS modules for logging vehicle location at a sampling rate
 1035 of 5 seconds, allowing correlation of fuel consumption with route patterns and
 1036 driving behavior.

1037 4. Anomaly Detection via Machine Learning

- 1038 a) Context-aware fuel deviation detection model development by means of context-
 1039 aware learning methods (e.g., Isolation Forest, Matrix Profile).
 1040 b) Analysis of precision and false positive rate of the model under the project-
 1041 native evaluation protocol.

1042 5. Driver Rest Alerts and Maintenance Reminders

- 1043 a) Integration of an in-vehicle HMI, mobile driver web application, and adminis-
 1044 trative dashboard for viewing trip state, fuel status, alerts, and cumulative fuel
 1045 and distance measures.
 1046 b) Production of automated rest alerts (e.g. every 300 km or after 6 hours) and
 1047 maintenance reminders (e.g. after every 5,000 km for checking oil).

1048 1.7.3 Delimitations

1049 1. Focus on Fuel Monitoring and Anomaly Detection

- 1050 • The system focuses on fuel level monitoring, detecting consumption anomalies,
 1051 and providing maintenance guidance based on fuel usage and distance traveled.
 1052 The prototype incorporates selected OBD-II context variables such as speed,
 1053 RPM, engine load, throttle, MAF, coolant temperature, runtime, and DTC



status, but does not implement full manufacturer-grade diagnostics or advanced vehicle-health fault isolation.

2. Hardware and Communication Constraints

- The system utilizes the truck's ECU/OBD-II diagnostic connector, an embedded GPS module, and a microcontroller-based HMI node; higher-end industrial sensing platforms are not employed.
- Real-time data transmission primarily uses IoT communication protocols, enabling low-latency delivery of fuel and GPS data to the cloud backend.
- A cellular module (LTE & GSM) functions as a fallback communication pathway, activated only during LoRa connectivity loss, and is limited to transmitting essential alerts and critical status messages due to bandwidth and cost constraints.
- Other wireless communication technologies beyond the selected IoT modules are not implemented within the system.

3. Limited Fleet Size and Geography

- The system testing will be limited to the number of vehicles provided for use by the trucking company within a specific geographic area. Large scale testing and deployment across a wider range of fleet types and terrains are not covered.

4. User Features

- The system includes three interfaces: an embedded in-vehicle HMI, a mobile driver web application, and an administrative dashboard. Driver actions, mobile trip state, and dashboard monitoring are synchronized through the backend.



- The administrative dashboard is limited to essential features such as real-time fuel level monitoring, GPS tracking, driver rest alerts, maintenance reminders, trip history, telemetry logs, alert management, analytics summaries, threshold configuration, user management, and fleet management.

1.8 Description and Methodology

This study employs a quantitative research paradigm combined with a data-centric, system development approach to implement an AI-driven fleet fuel monitoring and anomaly detection system. The methodology integrates theoretical foundations from Chapter 3 on cyber-physical systems and IoT architectures, system design principles from Chapter 4, and practical implementation strategies detailed in Chapter 5. The system leverages real-time vehicle data acquired through the OBD-II/CAN interface, GPS modules for location tracking, and embedded alert subsystems for driver safety. These components feed into a microcontroller that preprocesses signals, synchronizes time-series data, and transmits information to a cloud platform through prioritized LoRa, LTE, GSM 2G, WiFi, and offline-replay channels.

Offline, historical and hybrid datasets, combining fleet telemetry and publicly available vehicle logs, are used to train context-aware non-parametric machine learning models such as Matrix Profile and Isolation Forest. These models identify anomalous fuel consumption patterns and provide context-aware alerts. In the cloud, a Python-based engine continuously evaluates incoming data, logs anomaly events, and stores synchronized measurements in a centralized database. Users access the information through an embedded HMI, a mobile driver web application, and an administrative dashboard that display real-time truck status,



historical fuel usage, route playback, trip state, and operational alerts.

The methodology includes signal preprocessing (smoothing via moving averages), time synchronization using linear interpolation, hybrid dataset construction, and rigorous evaluation metrics (precision, false positive rate, accuracy, detection latency) to validate system objectives, as elaborated in Chapter 5. The approach follows iterative, agile development cycles, allowing continuous refinement of sensor integration, communication protocols, and anomaly detection algorithms, thereby ensuring reliability, scalability, and actionable fleet monitoring insights.

1.9 Research Dissemination Plan

The findings of this study will be disseminated to relevant academic, industrial, and technical stakeholders to promote the adoption and further development of intelligent fleet monitoring technologies. Upon approval of the study, the final manuscript will be submitted to the institutional repository of De La Salle University, including the Animo Repository, in accordance with university research archiving policies. Primary target audiences include the research partner Hino Paranaque Subic GS Auto Inc. (SGSA), fleet managers, logistics operators, academic researchers, engineering students, and faculty members within the field of IoT, transportation systems, and machine learning.

The dissemination process will occur in multiple stages. During the implementation and testing phase, progress updates and preliminary findings will be presented to SGSA representatives and thesis advisers for technical evaluation and operational feedback. Upon completion of the study, the final results, including system performance metrics, anomaly detection accuracy, latency measurements, and operational observations, will be formally



presented through the thesis defense at De La Salle University. A digital and printed copy of the approved manuscript will also be submitted to the institutional repository of De La Salle University, Animo Repository, in accordance with institutional requirements. In addition, the proponents may present selected findings in future academic conferences, research exhibits, or technology demonstrations related to IoT systems, intelligent transportation, or artificial intelligence applications.

The dissemination timeline will follow the project completion schedule presented in Section 1.9, including the individual Gantt charts of the proponents. Preliminary technical findings and prototype demonstrations will be shared during the system testing and validation phase. Final dissemination activities, including the thesis defense, manuscript submission, and partner presentation, will occur after the completion of data analysis, evaluation, and manuscript revisions. Any future publication or presentation derived from the study will only use anonymized and non-sensitive operational data to preserve confidentiality and comply with ethical and data privacy standards.

The effectiveness of the dissemination activities will be evaluated through several criteria, including feedback from thesis panel members, adviser evaluations, partner company assessments, and audience engagement during presentations and demonstrations. Additional indicators of dissemination effectiveness include successful acceptance of the manuscript, utilization of recommendations by the partner company, and the potential application or continuation of the proposed system in future research or operational deployments.



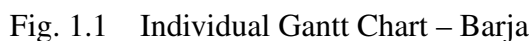
1.10 Conflict of Interest

The authors declare that they have no financial, personal, or professional relationships that could be construed as a conflict of interest with respect to the conduct of this research. The industry partner, Hino Paran que Subic GA Auto Inc. (SGSA), provided access to partner-fleet vehicles and operational data solely under the ethical clearance granted by the De La Salle University Research Ethics Review Board and did not exercise editorial control over the study design, the analysis, or the reporting of results. No funding, equipment donation, employment offer, or in-kind contribution from any commercial entity influenced the selection of methods, the choice of models, or the disclosure of limitations. All hardware components used in the prototype were purchased at market prices from independent suppliers, and all software dependencies are open-source or freely available under standard academic licences.

1.11 Estimated Work Schedule and Budget

1.11.1 Gantt Chart

		1. Introduction	
	De La Salle University		



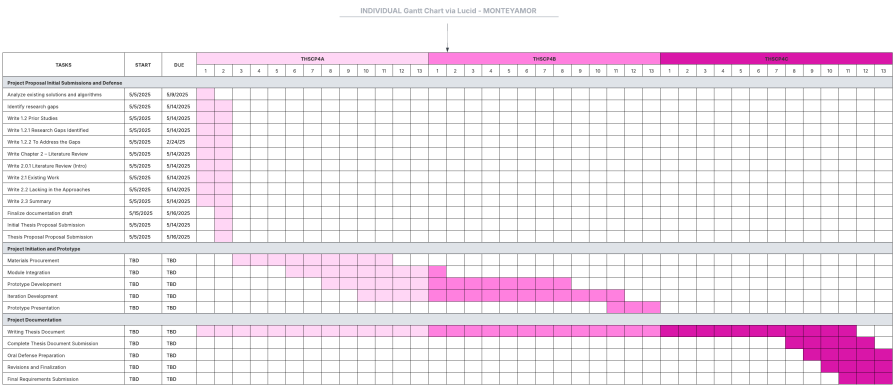


Fig. 1.4 Individual Gantt Chart – Montemayor



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1.11.2 Estimated Budget

TABLE 1.3 DETAILED BUDGET BREAKDOWN FOR THE PROJECT

Item / Activity	Unit	Qty	Unit Cost (PHP)	Total Cost (PHP)	Remarks
A. Covered by Hino Motors					
Use of truck for testing	Truck	2	Sponsored	-	Provided by Hino
B. Covered by Research Team					
Hardware Components					
CAN Bus / OBD-II Interface Module	-	1	1,200	1,200	Reads fuel levels and vehicle parameters from ECU
GPS Module	-	1	650	650	Route tracking and speed monitoring
ESP32 DevKit 30 pins	-	2	350	700	Main microcontroller boards
A7670E	-	1	1,300	1,300	Cellular communication module
MCP2515 CAN Bus Module	-	1	91	91	CAN bus interface module
Buzzer Module	-	1	45	45	Audio alert component
Buttons	-	2	10	20	Driver input controls
Arduino TFT Screen 3.5 in	-	1	648	648	In-vehicle display interface
LoRa Module Ra-02 433 MHz	-	2	218	436	Long-range communication modules
OBD2 16-pin connector	-	1	154	154	Vehicle diagnostic connector
GPS Module NEO-M8N	-	1	600	600	GPS tracking module
Resistors (1x220 ohms, 1x5 kohms)	-	2	1	2	Circuit protection components
SIM card (TNT)	-	1	100	100	Cellular network access
Prototype Case					
3D Model and Printing	-	1	4,500	4,500	Prototype enclosure fabrication
Total Estimated Budget				10,446	

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1.12 Overview of the Thesis Proposal

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This chapter provides an overview of the study, which addresses inefficiencies in logistics and trucking operations while tackling safety concerns related to fuel monitoring, vehicle tracking, and anomaly detection. Modern trucking operations involve complex coordination, and current approaches—manual fuel measurements, fragmented GPS tracking, and delayed anomaly detection—are insufficient for ensuring operational efficiency and driver safety. This work developed and evaluated an integrated, IoT-based intelligent monitoring platform that acquires digital fuel-level data through the truck's ECU/diagnostic connector, with



a four-stage filter cascade (median-of-31, EMA, dead-band, direction-aware step rule) implemented on an ESP32-based in-truck HMI, GPS, and cloud-deployed machine learning models to provide real-time insights and actionable alerts.

The problem statement highlights a fragmented monitoring environment with limited automation, reactive maintenance scheduling, and the inability to detect unusual fuel consumption proactively. The objectives of the research focus on developing a real-time system that reads fuel levels from the truck's ECU via CAN bus, tracks location and speed using GPS, and identifies anomalies in consumption patterns using context-aware machine learning models (Matrix Profile and Isolation Forest). The system is supported by a cloud-based dashboard, which provides fleet managers with fuel usage insights, rest alerts, route tracking, and preventive maintenance notifications. This approach promotes operational efficiency, enhances driver safety, and reduces fuel wastage, making it a technically and environmentally relevant solution.

Building on this foundation, Chapter 2: Literature Review examines previous research on IoT-based fleet monitoring systems, GPS-enabled tracking technologies, and machine learning methods for anomaly detection. The review compares existing methods, identifies gaps in real-time deployment, system integration, and context-aware alerting, and evaluates technology innovations and algorithmic limitations. This analysis establishes the academic basis for the study and demonstrates how the developed system advances the state of the art in smart fleet management by providing a fully integrated, real-time, and context-aware monitoring solution.



De La Salle University

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Chapter 2

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LITERATURE REVIEW



2.1 Systematic Literature Review

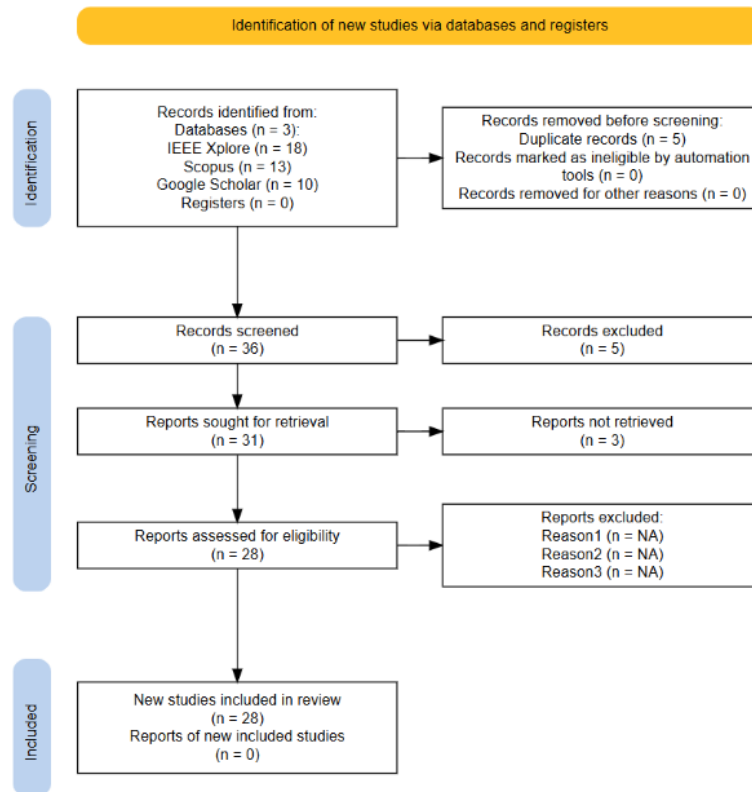


Fig. 2.1 PRISMA Diagram

The PRISMA diagram illustrates the systematic process used for selecting studies for review through screening and identification. The search identified a total of 41 records from three databases: Scopus, IEEE Xplore, and Google Scholar. Five (5) duplicate records were excluded from the 41 records, resulting in 36 records screened. The evaluation of 36 records resulted in five exclusions due to records being outdated/irrelevant, which reduced the number of reports needed for retrieval to 31. Three (3) reports were not retrieved upon retrieval, resulting in 28 reports assessed for eligibility. The final review included 28 studies



only, while no new studies were reported.

2.2 Existing Work

2.2.1 Vehicle Tracking and Fuel Sensing Technologies

Fuel Monitoring Systems

Various fuel monitoring systems have been proposed, discussed, and developed in existing studies, emphasizing real-time fuel level, theft indication, and enhanced operational efficiency. In the work of Joshi et al. [Joshi et al., 2024], a system combines ESP32 and Arduino UNO microcontrollers with the ThingSpeak cloud-based platform to measure fuel levels, control transactions, and enable predictive maintenance at fuel stations. Similarly, Ahmed et al. [Ahmed et al., 2017] developed a fuel management system using Raspberry Pi and both ultrasonic and chemical Etape sensors to acquire real-time fuel level data, displayed through a web application.

In the automotive domain, several studies aim to enhance the accuracy of fuel management through sensor integration. The digital system by Sakthimohan et al. [M, 2021] employs ultrasonic and flow sensors with an Arduino UNO to display real-time measurements on an LCD, providing a replacement for imprecise analog gauges. Meanwhile, Farzana et al. [Farzana, 2020] proposed a motion-based fuel monitoring technique that integrates vehicle motion parameters with fuel level sensor readings, achieving 80% success in accurately determining fuel levels and 100% detection success in identifying fuel theft and refills via a smartphone application.

Studies focused on fuel theft detection, such as those by Tatababu et al. [Tatababu et al., 2024] and Mathi et al. [Mathi et al., 2024], utilize capacitive or ultrasonic sensors,



GPS tracking, and GSM-based alerts to detect tampering or fuel loss and notify operators immediately. In the Intelligent Fuel Monitoring and Theft Detection System using GPS & GSM, Mathi et al. [Mathi et al., 2024] also monitored battery charge (SoC), with applicability for both hybrid and conventional vehicles, making it suitable for logistics and fleet management.

Ali et al. [Komal Shoukat Ali et al., 2018] presented an automatic fuel tank monitoring, tracking, and theft detection system integrating Arduino-based fuel circuits with GSM/GPS modules and LabVIEW for database management. This system replaces manual fuel tracking with automated alerts and volume monitoring, offering a cost-effective solution for transportation industries. Overall, these systems indicate a trend toward inexpensive, sensor-based, remotely accessible fuel monitoring technologies with high priority on accuracy, security, and real-time data analysis.

GPS and Vehicle Telematics

Global Positioning System (GPS) and vehicle telematics technologies have become essential components in modern vehicle monitoring and fleet management systems. These technologies allow real-time tracking, remote diagnostics, and performance analysis by collecting and reporting data such as location, speed, fuel consumption, and engine status. A common trend in these studies is the combination of GPS and GSM modules to facilitate real-time vehicle tracking, usually complemented by cloud-based or mobile-supported platforms. Joshi et al. [Joshi et al., 2024] proposed a system enabling real-time monitoring for logistic management using sensors, GPS tracking, and cloud-based data analytics. Similarly, Alquhali et al. [Alquhali et al., 2019] and Nada [Nada, 2017] employed GPS and GSM modules to send live location data, show vehicle tracks on electronic maps, and record location history for future reference and analysis. Vigneshwaran et al. [Vigneshwaran et al.,



2020] presented a GPS-GSM-based bus tracking system supporting student notifications for bus arrival, emergency notifications, and route alerts, enhancing safety and saving waiting time. The system further offered a framework for real-time fuel monitoring with implications for broader fleet management applications.

2.2.2 LoRa Technology

LoRa (Long Range) is a wireless communication technology engineered for Internet of Things (IoT) and machine-to-machine (M2M) communication. It supports low-power, battery-operated embedded systems that transmit small amounts of data over long distances [Orange Connected Objects & Partnerships, 2016]. Traditional cellular networks such as GSM, 2G, 3G, and 4G prioritize high data throughput, which can result in inefficient power consumption for IoT applications that require infrequent, small data transmissions [Majeed, 2015, Lee et al., 2007]. Unlike cellular networks or short-range protocols such as Wi-Fi and Bluetooth, LoRa is optimized for applications where energy efficiency and communication range are more important than high throughput [Devalal and Karthikeyan, 2018].

The LoRa communication architecture follows a structured flow of information so that small data packets can reach cloud-based systems efficiently:

- **LoRa Node or End Device:** This component includes the sensors or applications where sensing and control take place. It collects telemetry data and transmits them as small, optimized data packets.
- **LoRa Gateway:** The gateway receives packets from end nodes and acts as a bridge between LoRa nodes and the internet.



- **Cloud Backend:** Once the gateway forwards data through a backhaul connection, the information reaches a network or application server where it can be processed, visualized on dashboards, or stored for further analysis.

2.2.3 LoRa Gateway

LoRa gateways are multi-channel, multi-modem transceivers capable of simultaneous reception and decoding of LoRa packets from numerous end nodes. Each gateway acts as a bridge that connects LoRa nodes to the internet by capturing transmitted node data and forwarding them through standard backhaul connections such as Wi-Fi, Ethernet, or cellular data. Data from different gateways are processed through a network server, which functions as the central intelligence of the LoRa architecture. This server performs security authentication, filters messages, manages adaptive data rates, and relays appropriate acknowledgments back to the gateways [Orange Connected Objects & Partnerships, 2016].

Gateway deployment varies according to coverage requirements. Micro gateways are typically deployed in public networks to provide broad city-wide or nationwide connectivity, while pico gateways are used in dense or challenging environments to improve network capacity and service quality. To support these deployment scenarios, LoRa base stations may use omni-directional or multi-sector antenna configurations [Devalal and Karthikeyan, 2018].



2.2.4 Technical Performance Metrics and Environmental Factors

The operational effectiveness of LoRa networks depends on the interaction between transceiver hardware configuration and environmental conditions. According to Chengdu Ebyte Electronic Technology, the communication distance of a LoRa module is influenced by transmit power, receiver sensitivity, and antenna gain [Chengdu Ebyte Electronic Technology Co., Ltd., 2024]. In open rural areas, a standard LoRa layout can achieve communication links of approximately 15 to 20 kilometers. In dense urban environments, however, coverage may decrease to approximately 1 to 5 kilometers because of building penetration losses and multipath propagation effects [Chengdu Ebyte Electronic Technology Co., Ltd., 2024].

Manalu et al. [Manalu et al., 2023] evaluated the performance of a LoRa-based vehicle tracking system using a HopeRF-RFM95W transceiver module operating at 915 MHz on a moving vehicle. The study analyzed link-quality parameters in a Non-Line-of-Sight (N-LoS) suburban environment at speeds of 10 km/h, 20 km/h, and 30 km/h. The evaluation focused on how buildings, vegetation, and other wireless signals affect key performance metrics. Received Signal Strength Indicator (RSSI) measures absolute signal power and degrades behind thick barriers, vegetation, or dense buildings. Signal-to-Noise Ratio (SNR) measures signal clarity against electromagnetic interference and may drop significantly because of disruptions from pedestrians, structures, vehicles, and Wi-Fi towers. Packet loss may also increase sharply near major structural obstacles, showing that LoRa performance in mobile fleet applications must account for physical barriers and N-LoS conditions [Manalu et al., 2023].



2.2.5 Real-World Implementation: LoRa-Based Telemetry and Fleet Management Systems

The long-range and low-power characteristics of LoRa make it suitable for geographic areas with limited commercial telecommunication infrastructure. Common applications include smart agriculture, pasture health monitoring, and isolated industrial operations. However, reliance on a single low-power wide-area network (LPWAN) protocol may introduce communication risks when asset-tracking nodes move between remote rural spaces and dense urban areas.

To reduce blind spots, industrial systems may use hybrid communication architectures. The Meitrack LoRa Fleet Tracking Solution illustrates this approach in large-scale agricultural operations, such as orchards and palm plantations, where GSM coverage may be unreliable or unavailable [Meitrack,]. In such environments, multi-modem tracking hardware can use a dual LoRa and GPRS switching configuration based on network availability. While operating in cellular dead zones, localized sensor payloads, including ultrasonic fuel-level data, driver authentication logs, and safety alerts, can be routed through LoRa to a local base station. This architecture helps preserve asset visibility when standard cellular infrastructure is unavailable [Meitrack,].

2.2.6 GSM and LTE Technologies

The Global System for Mobile Communication (GSM) is a widely recognized standard for digital cellular technology. It originated from a group formed in 1982 to develop a unified European mobile telephone standard for cellular radio systems operating at 900 MHz [The International Engineering Consortium,]. GSM is known for broad geographic



coverage and supports voice calls and basic data services through second-generation cellular communication. Long-Term Evolution (LTE), by contrast, is a wireless broadband communication standard that succeeded 3G and became a cornerstone of 4G mobile networks. LTE development began as a Third Generation Partnership Project (3GPP) effort in 2004 to provide a new radio access technology focused on packet-switched data [Miyim and Wakili, 2019].

LTE was developed to address increasing demand for mobile broadband services by improving network capacity, data rates, quality of service, and latency. These improvements support mobile wireless connectivity that aligns with performance benchmarks established by the International Telecommunication Union (ITU). After LTE was established in 2009, further performance requirements led to LTE-Advanced (LTE-A), which satisfied International Mobile Telecommunications (IMT) requirements for fourth-generation networks and achieved peak data rates of up to 1 Gbps [Miyim and Wakili, 2019, Kanchi et al., 2013].



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Fig. 2.2 Globe GSM and LTE Coverage Map

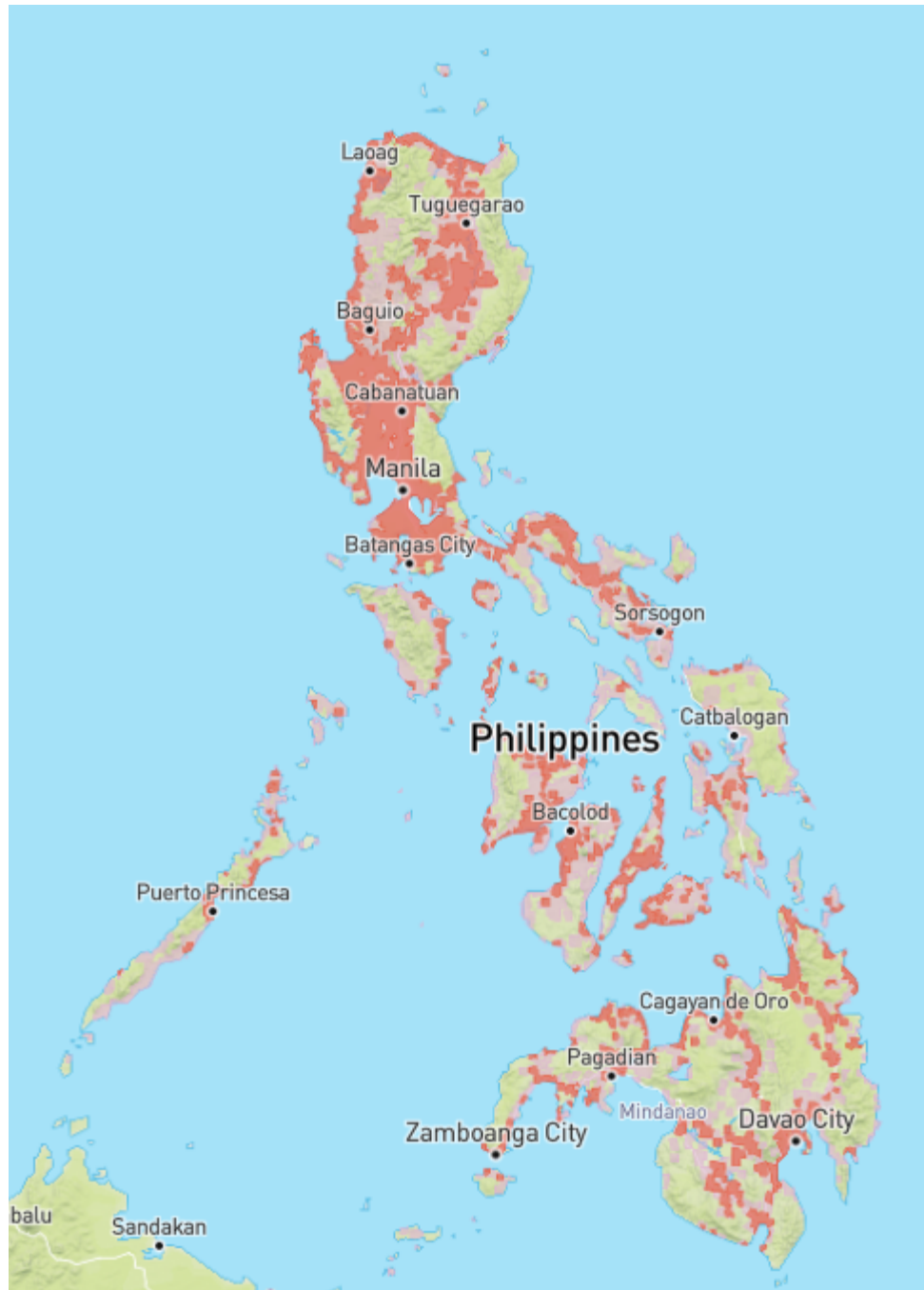


Fig. 2.3 Smart GSM and LTE Coverage Map



TABLE 2.1 COMPARISON BETWEEN GSM AND LTE

Feature	GSM (2G)	LTE (4G)
Network Architecture	Circuit-switched	Packet-switched (All-IP)
Primary Focus	Voice and text services	High-speed mobile broadband
Coverage	Extensive broad geographical reach, including rural and remote areas	High-capacity and density-dependent coverage that requires closer proximity to modern cell towers
Frequency Band	900 MHz and 1800 MHz	Bands 1 to 25 for Frequency Division Duplex and bands 33 to 41 for Time Division Duplex
Signal Propagation	Excellent because of lower frequencies	Variable because higher frequencies generally have shorter range
Infrastructure Needs	Lower base station density	Higher base station density
Data Capability	Low, supporting basic data and messaging	Very high, supporting multimedia and internet access



The coverage maps in Figures 2.2 and 2.3 illustrate GSM and LTE coverage patterns in the Philippines based on GSMA Intelligence coverage data [GSMA, 2026]. Comparing the two networks, Globe shows a noticeably denser and more contiguous spread of both GSM and LTE coverage, particularly through the Visayas and Mindanao, where its colored patches connect more continuously along coastal and lowland routes. Smart's coverage, while still following the same population centers, appears comparatively more fragmented, with more isolated patches and gaps between cities, especially in the southern islands.

In both cases, LTE coverage appears as a smaller subset within the broader GSM footprint. This indicates that 4G service is generally concentrated in and around cities and main highway corridors, while 2G/GSM extends farther into surrounding rural areas as a fallback connectivity layer. This difference is important for fleet monitoring systems because mobile assets may move between high-capacity LTE areas and wider but lower-bandwidth GSM fallback zones [Opensignal, , GSMA, 2026].

The Philippine cellular landscape is transitioning away from legacy networks. In August 2025, the NTC issued Memorandum Circular No. 003-09-2025, mandating phased decommissioning of 2G and 3G services, with nationwide 3G switch-off by 31 December 2026 and 2G on a separate, slower schedule because rural areas still depend on it. The circular also bars new type approval for devices whose primary capability is 2G or 3G. Two design implications follow: LTE must be the primary wide-area path, since it is the network being expanded rather than retired; and any 2G fallback is transitional, which motivates retaining LoRa and on-device buffering where neither LTE nor 2G is dependable [China Academy of Information and Communications Technology, 2025].



2.2.7 Anomaly Detection

Several studies on anomaly detection in vehicular systems aim to improve operational safety, combat theft and fraud, and optimize fuel efficiency. Various technological means, such as GPS tracking, machine learning, and time series analysis, have been used to identify abnormal vehicle behaviors. Crisgar et al. [Crisgar et al., 2021] developed a vehicle tracking system using Google Cloud IoT Core and Firebase to monitor vehicle movements and engine activity for theft detection. Unauthorized ignition or unexpected movement beyond a 50-meter radius is flagged as potential theft, with real-time notifications pushed to a Progressive Web App, enabling users to respond quickly.

Studies focusing on fuel efficiency in heavy-duty vehicles (HDVs) through anomaly detection models include Turan et al. [Turan et al., 2024], who proposed an ensemble of unsupervised machine-learning models (Isolation Forest, Autoencoder, k-NN Regression) combined via weighted majority voting to detect abnormal fuel consumption patterns based on road slope and speed variations. Mumcuoglu et al. [Mumcuoglu et al., 2023] used decision tree ensembles to classify fuel usage as normal or anomalous, achieving 92.2% accuracy and an F1 score of 0.78. Both studies indicate that repeated detection of outliers can highlight anomalies such as mechanical issues or inefficient driving.

Aquize et al. [Aquize et al., 2017] applied Self-Organizing Maps (SOM) to detect abnormal fuel consumption in RFID-registered vehicles in Bolivia. By learning conventional consumption patterns, the SOM model achieved an 80% confidence rate in flagging suspicious activity indicative of fraud. Semenov et al. [Semenov et al., 2024] employed DBSCAN clustering and Local Outlier Factor (LOF) in a three-level telematics platform to forecast and detect truck breakdowns in real-time. This system tracks multiple vehicle parameters, labeling them as normal or outliers based on deviation from learned patterns,



minimizing surprise breakdowns and providing early driver alerts. Aquize et al. [Aquize et al., 2017] proposed a fuel anomaly detection framework based on Self-Organizing Maps (SOM) using RFID-registered vehicle data collected from fleet operations in Bolivia. The SOM algorithm learned normal fuel consumption patterns and identified suspicious deviations associated with potential fuel fraud. The study achieved approximately 80% confidence in detecting anomalous fuel usage and demonstrated the effectiveness of unsupervised learning approaches for identifying irregular fuel consumption behavior without requiring labeled datasets.

Similarly, Kaleem Habib and Arif Iqbal Umar [Habib and Umar, 2015] utilized data mining techniques to calculate and detect anomalies in fuel expenses by analyzing historical fuel transaction records and consumption patterns. Their approach identified abnormal expenditure behaviors that could indicate fuel theft, reporting errors, or operational inefficiencies. The findings demonstrate the value of data-driven anomaly detection methods in supporting cost control and fraud prevention within fleet management environments.

Garcia Vega [Garcia Vega, 2022] emphasized time series anomaly detection for vehicle monitoring, developing an iterative approach to detect unusual driving patterns through pre-processing, subsequence segmentation, scoring, clustering, and expert validation. Applied in an industrial context, it identified over 120 validated anomalies, helping diagnose mechanical failures or unsafe driving behavior. The study highlights the importance of visual interpretability tools and recommends future improvements via real-time data processing.

Because published fleet fuel-anomaly work (Barbado and Corcho, 2022) [Barbado and Corcho, 2022] uses boxplot/IQR-based unsupervised detection as an interpretable baseline before applying explainable models, the present study compares Isolation Forest and Matrix Profile against a rule-based / IQR baseline under leave-one-trip-out evaluation.



This baseline serves both as a fairness check (to show that the non-parametric models genuinely add value over a thresholding heuristic) and as an ablation reference in Chapter 5.

2.2.8 Machine Learning Applications

Barbado and Corcho (2022) [Barbado and Corcho, 2022] developed an explainable machine-learning framework for vehicle fuel-consumption anomalies using real-world telematics data from diesel and petrol fleet vehicles. Their study emphasised that anomaly detection should not only identify abnormal fuel consumption but also explain the operational causes behind the anomaly, using XAI techniques on top of an unsupervised detection layer. They first identified candidate anomalies via a univariate boxplot/IQR outlier method and then trained a contextual model that incorporates domain knowledge such as vehicle subgroup, route type, and driver behaviour. This is directly relevant to the present thesis because fuel anomalies in Philippine logistics must be interpreted with operational context - the same context (driver, route, vehicle, fuel-consumption baseline) is encoded in the Isolation Forest Model C feature set used in Chapter 5.

Recent advances in machine learning have been integrated into vehicle systems to improve fuel efficiency, detect anomalies, and enhance operational transparency. In fuel monitoring and anomaly detection, machine learning methodologies extract actionable insights from complex, high-dimensional vehicle data. Fortela et al. [Fortela et al., 2023] applied unsupervised machine learning algorithms (Principal Components Analysis, K-Means Clustering, Self-Organizing Maps) to detect anomalies in fuel economy and emission data from light-duty vehicles (2015–2023). These methods revealed clusters where CO emissions were positively linked to fuel economy, contrary to expected patterns, highlighting irregularities caused by specific fuel test procedures. Barbado and Corcho [Barbado



and Corcho, 2022] combined unsupervised anomaly detection with domain knowledge and interpretable machine learning (XAI) to explain potential causes of fuel consumption anomalies. Their models demonstrated up to 88% potential fuel savings on larger datasets and across various user roles.

Barbado and Corcho [Barbado and Corcho, 2022] further demonstrated that interpretable machine learning models such as Explainable Boosting Machines (EBM) can be used not only to detect anomalous fuel consumption but also to explain the factors contributing to those anomalies. Their methodology combines unsupervised anomaly detection, domain knowledge, and explainable artificial intelligence (XAI) metrics to provide actionable recommendations for fleet managers and operators. The study highlights the importance of model transparency in fleet monitoring systems, allowing stakeholders to understand the causes of abnormal fuel usage rather than merely identifying anomalous events.

More recent work sharpens the methodological basis for unsupervised, on-vehicle detection. Cherdo et al. [Cherdo et al., 2023] applied small recurrent and convolutional neural networks to automotive CAN sensor time series, demonstrating that bus data of the kind exposed through an OBD-II interface is amenable to unsupervised anomaly detection, although their models still require gradient-based training. Surveying the domain, Hirth et al. [Hirth et al., 2025] formalize anomaly detection in trucks as identifying deviations across multivariate signals at each time step of a stream, and observe that deep architectures impose retraining and computational burdens that complicate continuous deployment. These observations motivate the two non-parametric methods adopted in this study: Isolation Forest [Liu et al., 2008], which isolates anomalous individual observations without modeling a distribution of normal behavior, and the Matrix Profile [Yeh et al.,



2016], which detects anomalous subsequences and has been extended to live data streams [Lu et al., 2022] and to multiple sensor channels [Yeh et al., 2024]. The mathematical formulation of both methods is presented in Chapter 3.

TABLE 2.2 SUMMARY OF EXISTING WORKS

Study	Technology Used	Key Features	Strengths	Limitations
Tatababu et al. (2024) [Tatababu et al., 2024]	Ultrasonic/Capacitive sensors, GSM, GPS, Smartphone/Web interface	Theft detection, fuel level alert, real-time remote access	Accurate for remote generator monitoring, GPS-based tracking	Findings specific to the tested generator systems and environments
Joshi et al. (2024) [Joshi et al., 2024]	IoT, GPS, RFID, Cloud Computing	Real-time vehicle/asset tracking, data analytics, cloud-based monitoring	High tracking accuracy, improved logistics efficiency, responsive alert system	Limited by hardware and logistics test case set
Vigneshwaran et al. (2020) [Vigneshwaran et al., 2020]	GPS, GSM	Real-time bus tracking, emergency alert system, communication to students, route updates	Improves safety, scheduling, and passenger information	Not highly scalable; lacks detailed integration with modern telematics platforms
Patil & Patil (2024) [Patil and Patil, 2024]	IoT, GPS, GSM, Accelerometer, IR Sensors	Tracking system with SMS-based communication; remote ignition cut-off mechanism and vehicle password protection	Comprehensive vehicle security, real-time alerts, detection of tampering/impacts	Low adaptability to harsh conditions like GPS/GSM signal loss in remote areas
Fortela et al. (2023) [Fortela et al., 2023]	Unsupervised Machine Learning: PCA, K-Means, SOM	Detects hidden trends and impending anomalies; analyzes correlation between CO emissions and fuel economy	Effectively identifies non-obvious patterns within clusters; highlights flaws in standardized testing procedures	Lacks integration with real-time or operational vehicle data; conclusions depend on historical datasets



Study	Technology Used	Key Features	Strengths	Limitations
Barbado & Corcho (2022) [Barbado and Corcho, 2022]	Unsupervised Anomaly Detection; Explainable Boosting Machine (EBM)	Provides fuel anomaly detection and cause explanations, tailored recommendations for fleet operators	High interpretability via XAI metrics; role-based recommendations; potential fuel savings up to 88%	Performance depends heavily on quality of telematics data or feature relevance
Aquize et al. (2017) [Aquize et al., 2017]	Self-Organizing Maps (SOM)	Detects fraudulent fuel usage based on RFID data patterns	80% accuracy applied to real socioeconomic problem	Used only simple SOM model; accuracy can improve with more context variables
Liu et al. (2008) [Liu et al., 2008]	Isolation Forest	Ensemble of isolation trees that identify anomalies through random partitioning	Non-parametric anomaly detection with linear-time complexity and low memory requirements	No labeled data required; superior AUC compared with ORCA and LOF; computationally efficient
Yeh et al. (2016) [Yeh et al., 2016]	Matrix Profile (STAMP)	Z-normalized nearest-neighbor similarity profile for motif and discord discovery	Exact subsequence anomaly detection with minimal parameter tuning	Deterministic, interpretable, and near parameter-free
Lu et al. (2022) [Lu et al., 2022]	DAMP Streaming Matrix Profile	Real-time Matrix Profile computation for high-volume streaming data	Detects discords in continuously arriving data streams at very large scales	Suitable for online and real-time deployment
Yeh et al. (2024) [Yeh et al., 2024]	Multidimensional Matrix Profile	Extension of Matrix Profile for multivariate time-series anomaly detection	Simultaneously analyzes multiple sensor channels	Competitive performance across numerous benchmark datasets and highly interpretable results
Hirth et al. (2025) [Hirth et al., 2025]	Deep Learning-Based Truck Anomaly Detection (Survey)	Comprehensive review and taxonomy of anomaly detection methods for trucks	Defines anomaly detection in multivariate truck telemetry streams	Provides truck-specific insights and identifies current research directions



Study	Technology Used	Key Features	Strengths	Limitations
Cherdo et al. (2023) [Cherdo et al., 2023]	Small RNN and CNN Models	Unsupervised anomaly detection using automotive CAN-bus sensor data	Demonstrates feasibility of on-vehicle anomaly detection under limited hardware resources	Effective for OBD-II/CAN sensor environments and embedded deployment



2.3 Comparison of Commercial Fleet Management Systems

Beyond the academic literature, six widely deployed commercial fleet management platforms were examined to establish the detection methodologies available to operators in practice: Geotab, Samsara, Fleetio, Wialon, Fleet Complete, and Navixy. Each platform's own product documentation was used as primary evidence of its capabilities. Despite mature telematics infrastructure, fuel-loss detection across these systems reduces to fixed or manually configured rules. Samsara's documentation describes its fuel theft detection as a configurable alert on any sudden fuel drop of five percent or more [Samsara, 2026]. Navixy requires operators to configure rate-of-change drain thresholds per sensor and can be set to ignore drains while a vehicle is moving [Navixy, 2026]. Geotab and Fleetio approach the problem from the fuel-card transaction side, flagging exceptions such as purchases exceeding tank capacity or fueling locations inconsistent with vehicle position [Geotab, 2024, Fleetio, 2026]. Wialon detects drain events from fuel-level sensor data and commonly relies on operator review for verification [Wialon, 2026], while Fleet Complete's public documentation centers on fuel usage reporting without specifying an adaptive detection method [Fleet Complete, 2026]. Across all six, no platform learns a vehicle's normal consumption behavior, adapts to operating context, or is designed specifically for Philippine fleet conditions.



TABLE 2.3 COMPARISON OF COMMERCIAL FLEET MANAGEMENT SYSTEMS

System	Data Sources	Fuel Monitoring Method	ML for Fuel Anomalies?	Detection Methodology	Limitations
Geotab [Geotab, 2024]	OBD/telematics device, fuel cards, GPS	Fuel-up records and engine data reporting	No (rule checks)	Capacity mismatch, purchase greater than capacity, and location discrepancy	Transaction-rule logic; no learned consumption behavior
Samsara [Samsara, 2026]	Telematics gateway, fuel sensor/OBD, fuel cards, GPS	Real-time fuel level and usage reports	No (threshold)	Configurable sudden-drop alert ($\geq 5\%$)	Static threshold; blind to gradual loss; one rule for all conditions
Fleetio [Fleetio, 2026]	Fuel card transactions and telematics integrations	Transaction-based fuel management	No	Exception alerts for location mismatch and volume greater than tank capacity	Sees only fill-up events; blind between refuels
Wialon [Wialon, 2026]	Fuel level sensors, GPS trackers, optional cameras	Sensor-based fuel charts and drain/refill detection	No	Sensor-threshold drain events with human/video verification	Threshold-based and manually reviewed; reactive
Fleet Complete [Fleet Complete, 2026]	Telematics device, GPS, vehicle data, fuel reports	Fuel usage and fleet reporting	No specified adaptive model	Reporting-based fuel monitoring and exception review	Public documentation does not specify adaptive anomaly detection
Navixy [Navixy, 2026]	Fuel level sensors, OBD/CAN, GPS	Drain/refill event detection from sensor data	No	Per-sensor rate-of-change drain thresholds; option to ignore in-motion drains	Manual tuning; in-motion exclusion can hide moving theft or leaks



2.4 Lacking in the Approaches

While existing technologies in vehicle tracking, fuel sensing, and anomaly detection form the foundation upon which developments can be built, they suffer from critical limitations that prevent their use in real-time, fleet-wide operations. Research works such as Krishnasamy [Krishnasamy, 2019] have employed basic sensor technologies without any validation of fuel measurements in real-time driving environments or estimation of accuracy parameters relative to manual readings. Besides, current systems are limited to specific uses and are not tested for scalability or integration with fleet vehicles like trucks such as institutional/organizational level of motorcycles in the study of RS et al. [RS et al., 2024]. Some works do consider machine learning techniques for anomaly detection, but such techniques are mostly based on static and post-processed datasets, with no real-time on-vehicle analysis. Very few implementations include unsupervised machine learning and advanced context-aware modeling, and precision benchmarks are usually left unreported, if at all addressed.

Across the surveyed commercial systems, including Geotab, Samsara, Fleetio, Wialon, Fleet Complete, and Navixy, several operational capabilities remain insufficiently addressed. First, closed-loop in-cab driver feedback is largely missing because most platforms push alerts to a manager dashboard, whereas this thesis pushes alerts to a driver-facing HMI inside the truck cab with rest/snooze acknowledgments captured in the `alerts` table. Second, offline-tolerant telemetry with explicit replay accounting is rarely documented in academic prototypes or commercial fuel-monitoring systems; this thesis tags every telemetry row with `channel_used`, including `offline_replay`, which lets the backend distinguish live data from replayed data. Third, multi-interface state reconciliation is not



typically addressed, while this thesis uses partial unique indexes to enforce one-active-trip-per-truck/driver across three concurrent client surfaces: HMI, mobile, and dashboard.

Ultimately, there is no prior system that provides a full solution that brings together a cloud-connected solution for real-time fuel monitoring with integration of GPS tracking and anomaly detection. Further, it completely lacks features such as automatic rest alerts and predictive maintenance reminders, which are necessary for safe, efficient, and compliant operations. Such shortcomings create an opportunity for a more integrated, intelligent, and real-time solution, and this is what the thesis proposes to address.

TABLE 2.4 IDENTIFIED GAPS, LIMITATIONS, AND PROPOSED SOLUTIONS

Study / Gap	What they did	Limitation	How this thesis responds
Area of precision	Existing models lack context-awareness and show lower precision. Models presented in the study of Barbado & Corcho [Barbado and Corcho, 2022] performed up to 80% of anomalous instances explained.	Prior work does not establish the same precision and false-positive benchmark required for this deployment.	(SO4) To design and evaluate a context-aware fuel usage deviation model using context-aware non-parametric machine learning models (Matrix Profile or Isolation Forest), achieving 90% precision and 10% false positives in identifying anomalous fuel consumption patterns within controlled environments.



Study / Gap	What they did	Limitation	How this thesis responds
Barbado & Corcho (2022) [Barbado and Corcho, 2022]	Used real-world telematics and interpretable ML to detect and explain fleet fuel-consumption anomalies; combined box-plot/IQR pre-filtering with contextual XAI models.	Focused on fleet-level analytics and explanation; did not produce an embedded HMI + OBD-II + multi-channel-communication + live-alert prototype that closes the loop in the vehicle cab.	This thesis integrates OBD-II fuel acquisition (Toyota Mode 22 PID 21/29), embedded HMI, GPS tracking, four-channel communication fallback, backend dashboard, mobile driver web app, and context-aware anomaly detection (IF Model C + Matrix Profile) in one working prototype evaluated under LOTO.
Absence of a comprehensive system integrating fuel anomaly detection with GPS-based tracking, and predictive maintenance and alerts	Previous works, including those by Joshi et al. [Joshi et al., 2024] and Nada [Nada, 2017], focus on real-time tracking using GPS/GSM.	These works lack integration with anomaly detection methods for analyzing fuel consumption patterns or predicting vehicle maintenance issues.	(SO3, SO4, SO5) To capture and log GPS location data via an embedded module, correlating route information with fuel and time data for behavior tracking; design and evaluate a context-aware fuel usage deviation model using context-aware non-parametric machine learning models; develop and integrate rest alerts and maintenance reminders into a cloud-based dashboard.



Study / Gap	What they did	Limitation	How this thesis responds
No Philippine transportation context	Reviewed studies and platforms do not model local conditions such as Metro Manila congestion, Port of Manila utilization, the 2014 Manila truck ban, and fuel fraud costs [Japan International Cooperation Agency, 2025, Patalinghug et al., 2015a, Llanto, 2017, Asian Development Bank, 2016].	Models may misclassify congestion-shaped idling and consumption as anomalies when Philippine operating conditions are absent.	(GO, SO4) Collect and train on operational data from the partner fleet, Hino Paranaque Subic GS Auto Inc., so congestion-shaped idling and consumption are learned as baseline behavior.
One detection rule for all trucks	Commercial systems use single fleet-wide settings [Samsara, 2026] or manual per-sensor tuning [Navixy, 2026]; prior studies were validated on one vehicle class, generators [Tatababu et al., 2024], or motorcycles [RS et al., 2024].	Shared thresholding cannot represent each truck's normal consumption profile.	(SO4) Learn per-vehicle baselines that update from each truck's own operating history, removing dependence on a shared manual threshold.
Detection of sudden drops only	Threshold and transaction logic are commonly used to detect sudden losses [Samsara, 2026, Fleetio, 2026]; Navixy can ignore in-motion drains [Navixy, 2026].	These approaches may miss slow siphoning and gradual leaks.	(SO4) Apply the Matrix Profile for anomalous-subsequence or discord detection of gradual loss, paired with Isolation Forest for sudden point anomalies.
No real-time analytics or live alerts	Machine learning studies were validated offline [Fortela et al., 2023, Barbado and Corcho, 2022]; deep models burden streams [Hirth et al., 2025, Cherdo et al., 2023]; and Wialon relies on human review [Wialon, 2026].	Prior approaches do not provide live anomaly scoring and alert delivery across operational interfaces.	(SO2, SO5) Deploy a cloud pipeline scoring live OBD-II/GPS streams with Isolation Forest and a streaming Matrix Profile [Lu et al., 2022], issuing live alerts to the dashboard, HMI, and mobile application.



2.5 Summary

With this chapter, it was discovered that there have been various studies on vehicle tracking, fuel sensing, and anomaly detection. The PRISMA diagram shows the systematic approach that was used in the identification of the studies, out of which the 41 initial records were narrowed down to 28 after excluding the duplicates, outdated/irrelevant, and not retrieved records. Moreover, the existing studies presented that though a lot has been done to develop individual systems for fuel level monitoring, GPS tracking, and anomaly detection, most of these works are either narrowly focused or operate in isolation. Many rely on basic sensor integrations without advanced analytics, or implement machine learning models without connecting them to real-time, in-vehicle data streams.

Numerous studies have exhibited the application of different sensor-based systems for fuel monitoring in real-time and detecting theft, in many cases involving microcontrollers, GSM, and GPS modules. Likewise, research on vehicle telematics illustrates the application of GPS in route tracking and fleet management. In the field of anomaly detection, some methods based on unsupervised machine learning and clustering techniques have been utilized to detect abnormal fuel consumption patterns and flag fraudulent or inefficient activity. Yet, most of these researches are based on special scenarios, are not integrated with real-time operational data, or depend on historical data.

While advances have been made in each of these domains, a significant void exists in creating a holistic, real-time solution that integrates fuel level measurement, GPS tracking, and intelligent anomaly detection through machine learning. Current systems exist in silos, are vehicle type- or use case-specific, and do not commonly incorporate predictive maintenance or driver feedback mechanisms.



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1531 This research addresses these gaps through an integrated IoT-based system tailored
1532 for trucks. The system combines real-time fuel monitoring, GPS tracking, and machine-
1533 learning-based anomaly detection to improve operational efficiency, driver safety, and
1534 maintenance planning.



1535

Chapter 3

1536

THEORETICAL CONSIDERATIONS



This chapter establishes the theoretical foundations underpinning the proposed IoT-based fleet fuel monitoring and anomaly detection system. It presents the essential concepts, mathematical models, and scientific principles that guide the development of the system. These theories act as the main foundation of the study, supporting the methodological decisions described in Chapter 5. By understanding the underlying theories—ranging from cyber-physical systems to signal processing and machine learning—the relevance and rigor of the chosen approach become clear.

3.1 Overview

The methodologies used in the system are grounded in established theories in embedded systems, time-series processing, IoT architectures, context-aware machine learning, and vehicular telematics. These theoretical concepts justify the design choices made in the development of real-time fuel monitoring components, GPS-based tracking, and anomaly detection models. This chapter explains the foundations behind the signal models, communication processes, and machine learning algorithms that make accurate and robust system performance possible.

3.2 Theoretical Foundations

3.2.1 Cyber-Physical Systems (CPS)

The proposed solution functions as a Cyber-Physical System, integrating embedded sensors, microcontroller-based processing, cloud computation, and real-time communication. CPS theory emphasizes:



- the tight coupling between physical signals and digital processing,
- deterministic timing and synchronized data flow,
- continuous monitoring and computational feedback loops.

These principles justify the system's architecture, where fuel-level signals, GPS data, and operational states are processed and transmitted in real time.

3.2.2 Sensor Theory and Signal Measurement

Fuel-level readings extracted through the CAN bus/OBD-II interface follow digital measurement theory. Sensor signals can exhibit noise, quantization error, and environmental fluctuation. Fuel-level readings may be stabilized using smoothing methods such as moving-average, median, and exponential moving-average filters. A Simple Moving Average (SMA) is a standard reference smoothing method defined as

$$x_t^{\text{smooth}} = \frac{1}{N} \sum_{i=0}^{N-1} x_{t-i} \quad (3.1)$$

x_t^{smooth} = smoothed sensor value at time index t ,

x_{t-i} = raw sensor value i samples before t ,

N = length of the smoothing window (samples).

This equation is included as the theoretical baseline for smoothing; however, the final implementation used a median + EMA + dead-band + direction-aware filter cascade because fuel-tank slosh and sender chatter include both impulse noise and slow drift. The implemented cascade is described mathematically in the following signal-filtering section and operationally in Chapter 5.



3.2.3 OBD-II and CAN-based Vehicle Data Acquisition

Vehicle data can be acquired through a standardised diagnostic interface rather than an added in-tank sensor. The underlying transport is the Controller Area Network (CAN), a message-based broadcast bus standardised in ISO 11898 in which contention is resolved by non-destructive bitwise arbitration: priority is encoded in the message identifier, and the highest-priority frame proceeds without loss of data or time. Standard CAN uses 11-bit identifiers and extended CAN uses 29-bit identifiers.

On-Board Diagnostics II (OBD-II) operates over CAN as a request–response protocol, with its application layer defined by SAE J1979 / ISO 15031-5 and multi-frame transport by ISO 15765-4. It distinguishes standardised current-data PIDs (Mode 01), which are mandated and uniform across manufacturers, from enhanced PIDs (Mode 22), which are manufacturer-specific and use proprietary scaling. Because Mode 22 scaling is not publicly standardised, any value read through it must be validated empirically, introducing measurement uncertainty.

Heavy-duty commercial vehicles conventionally use SAE J1939, a higher-layer protocol over 29-bit CAN identifiers in which Parameter Group Numbers (PGNs) group signals and Suspect Parameter Numbers (SPNs) define each signal's bit position, scaling, and units. Under J1939, fuel level is a standardised SPN broadcast periodically without an explicit request. The theoretical contrast is therefore between a polled, proprietary request–response model (OBD-II Mode 22) and a push-based, standardised broadcast model (J1939): the former trades standardisation and immediacy for broad device compatibility. Analog fuel senders accessed through either path are subject to slosh-induced fluctuation and quantisation error, which motivates the filtering theory developed next [International Organization for Standardization, 2015b, SAE International, 2023, International Organization



for Standardization, 2021, SAE International, 2016, SAE International, 2017, International Organization for Standardization, 2024a, Robert Bosch GmbH, 1991].

3.2.4 Signal Filtering for Fuel-Level Stabilisation

Raw signals from physical sensors carry impulse noise, low-frequency disturbance, and quantisation error; no single filter addresses all three, so a multi-stage cascade is used, each stage grounded in distinct theory.

Median filter. A running median is a non-linear order-statistic filter that replaces each sample with the median of a sliding window of length W :

$$x_t^{\text{med}} = \text{median}(x_{t-W+1}, \dots, x_t). \quad (3.2)$$

x_t^{med} = median-filtered sensor value at time t ,

W = odd-length sliding-window size (samples).

Its breakdown point is 50%; up to half the window may be corrupted by impulse noise before the output is affected, whereas the arithmetic mean has a 0% breakdown point.

Exponential moving average. An EMA is a first-order recursive IIR low-pass filter:

$$y_t = \alpha x_t^{\text{med}} + (1 - \alpha) y_{t-1}, \quad 0 < \alpha \leq 1, \quad (3.3)$$

y_t = EMA-smoothed value at time t ,

y_{t-1} = previous EMA state (recursive feedback),

α = smoothing coefficient,

τ = equivalent time constant $-1/\ln(1 - \alpha)$.

Past samples decay geometrically and the smoothing factor α sets the time constant τ ; a smaller α gives heavier smoothing at the cost of responsiveness. The simple moving



average, by contrast, is a finite-impulse-response filter optimal for white noise but without impulse robustness and requiring the full window in memory.

Kalman filter for fuel-level acquisition. Where the moving average and EMA are model-free smoothers, the Kalman filter [Kalman, 1960] incorporates an explicit process model into the smoothing decision, so it can distinguish measurement noise from real state change under the model. It is presented here alongside the EMA because both are candidate smoothers for the same stage of the fuel filter cascade and are compared against each other in Chapter 6. The state-space model appropriate for OBD-II fuel-level acquisition follows the physical constraint that the true fuel level x_t evolves from the previous level by exactly the amount of fuel consumed during the interval, plus a small process disturbance capturing real variability the model does not encode:

$$\text{Process (fuel): } x_t = x_{t-1} - u_t + w_t, \quad w_t \sim \mathcal{N}(0, Q_t), \quad (3.4)$$

$$\text{Measurement (sender): } z_t = x_t + v_t, \quad v_t \sim \mathcal{N}(0, R), \quad (3.5)$$

where x_t is the true fuel level (% of tank), z_t is the raw OBD-II Mode 22 sender reading (% of tank), u_t is the expected fuel burn during the interval $[t-1, t]$ derived from vehicle speed and engine-load context, w_t is process noise (real variability of the state not captured by u_t , in units of $\%^2$), and v_t is measurement noise from the capacitive-sender bounce (units of $\%^2$). The physical grounding for the process term u_t is developed in Section 5.4.7 of Chapter 5, where it is bound to the same burn envelope the physics tripwire uses so the filter and the anomaly detector share their notion of expected consumption.



1631

The predict–update recursion for the fuel-acquisition filter is

$$\text{Predict: } \hat{x}_{t|t-1} = \hat{x}_{t-1|t-1} - u_t, \quad P_{t|t-1} = P_{t-1|t-1} + Q_t, \quad (3.6)$$

$$\text{Update: } K_t = P_{t|t-1} / (P_{t|t-1} + R),$$

$$\hat{x}_{t|t} = \hat{x}_{t|t-1} + K_t(z_t - \hat{x}_{t|t-1}), \quad P_{t|t} = (1 - K_t) P_{t|t-1}, \quad (3.7)$$

1632

with the Kalman gain K_t interpolating between the prediction and the measurement in proportion to their relative variances. When measurement noise dominates ($R \gg P_{t|t-1}$) the gain is small and the filter trusts the prediction, so the fuel level is smoothed toward the model’s expectation and any small per-sample sender bounce is ignored. When the prediction is uncertain ($P_{t|t-1} \gg R$) the gain approaches one and the filter tracks the measurement, so an abrupt real change in fuel level is followed instead of smoothed away. This gives the Kalman filter its two theoretical advantages over the moving average: it is Bayes-optimal under linear-Gaussian assumptions, and it can adapt its smoothing intensity to per-sample uncertainty rather than applying a fixed window.

1641

The Kalman filter’s downside for anomaly-preservation is that a strong process prior u_t actively predicts consumption, so a real anomalous drop that matches the expected burn envelope is absorbed as “explained by the model.” A moving average has no such prior and therefore cannot smooth a real drop away by mistake. The four-axis comparison in Chapter 6 (Section 6.1.2) quantifies this trade-off on the labelled evaluation corpus.

1646

Dead-band (hysteresis). A dead-band publisher emits a new value only when the smoothed signal departs from the last published value by more than a threshold:

1647

$$\text{publish}(y_t) \Leftrightarrow |y_t - y_{\text{pub}}| \geq \delta. \quad (3.8)$$



y_t = current smoothed value,

y_{pub} = last value actually published to the downstream sink,

δ = minimum change required to publish (hysteresis).

Direction-aware gating. An asymmetric step rule treats increases and decreases differently according to the physical behaviour of the measured quantity. In the implemented fuel cascade, a downward update is accepted at a smaller threshold than an upward update, while upward movement during motion is suppressed unless a stationary refuel signature is present:

$$y_{\text{pub},t} = \begin{cases} y_t, & y_t \leq y_{\text{pub},t-1} - \delta_{\downarrow}, \\ y_t, & y_t \geq y_{\text{pub},t-1} + \delta_{\uparrow} \wedge v_t = 0 \wedge T_{\text{stop}} \geq 60 \text{ s}, \\ y_{\text{pub},t-1}, & \text{otherwise.} \end{cases} \quad (3.9)$$

$y_{\text{pub},t}$ = published fuel value at time t ,

$\delta_{\downarrow}$ = minimum drop that permits a downward update,

$\delta_{\uparrow}$ = minimum rise that permits an upward update,

v_t = vehicle speed at time t (km/h),

T_{stop} = elapsed stationary time since last motion (s).

A warm-up guard suppresses output until the median window is populated, since earlier samples are unrepresentative.



3.2.5 Classification and Evaluation Metric Theory

Detectors are evaluated as binary classifiers. From the confusion matrix, the counts TP , FP , TN , and FN (true positives, false positives, true negatives, and false negatives) define the following metrics:

$$\text{Precision} = \frac{TP}{TP + FP} \quad (3.10)$$

$$\text{FPR} = \frac{FP}{FP + TN} \quad (3.11)$$

$$F_1 = \frac{2 \times \text{Precision} \times \text{Recall}}{\text{Precision} + \text{Recall}} \quad (3.12)$$

$$L_d = T_{\text{detected}} - T_{\text{actual}}. \quad (3.13)$$

TP = true positives (correctly flagged anomalies),

FP = false positives (normal rows flagged as anomalous),

TN = true negatives (correctly unflagged normal rows),

FN = false negatives (missed anomalies),

Precision = fraction of flagged rows that are true anomalies,

Recall = fraction of true anomalies that are flagged,

FPR = fraction of normal rows incorrectly flagged,

F_1 = harmonic mean of precision and recall,

L_d = detection latency (s),

T_{detected} = timestamp at which the detector fired,

T_{actual} = timestamp of the true anomaly onset.

Accuracy is uninformative under class imbalance, where it is dominated by the majority class [Powers, 2020]. A Receiver Operating Characteristic (ROC) curve plots true-positive rate against false-positive rate, and its area equals the probability that a random positive



outranks a random negative [Fawcett, 2006]. Under heavy imbalance, however, ROC-AUC is optimistic; the Precision–Recall curve is more informative, its area collapsing toward the positive prevalence so that a modest value reflects the base rate rather than poor detection [Davis and Goadrich, 2006, Saito and Rehmsmeier, 2015]. Detection latency L_d was defined in Equation 3.13; the Numenta Anomaly Benchmark scores latency with a time-decaying reward [Lavin and Ahmad, 2015]. For binomial proportions estimated from few positives, the Wilson score interval is preferred over the normal-approximation (Wald) interval [Wilson, 1927]:

$$CI_{\text{Wilson}} = \frac{\hat{p} + \frac{z^2}{2n} \pm z \sqrt{\frac{\hat{p}(1 - \hat{p})}{n} + \frac{z^2}{4n^2}}}{1 + \frac{z^2}{n}}. \quad (3.14)$$

$\hat{p}$ = observed sample proportion (e.g., precision or recall estimate),

n = sample size (number of trials),

z = standard-normal quantile for the chosen confidence level.

Leave-one-group-out cross-validation, here leave-one-trip-out, holds out each group entirely to prevent leakage from within-group correlation [Arlot and Celisse, 2010]. Finally, evaluation against injected synthetic anomalies provides exact ground truth but bounds performance on the injected classes only, since these may not represent the full distribution of real events [Chandola et al., 2009].

3.2.6 Statistical Tests for Classifier Comparison

Point estimates such as precision, recall, and F_1 report the best guess of a detector's performance on a specific sample, but they do not answer whether an observed difference between two detectors is larger than what would be expected under sampling noise alone.



Three tests are used in this study to close that gap: McNemar's χ^2 for paired classifier comparison, the non-parametric bootstrap for confidence intervals around each metric, and the Wilcoxon signed-rank test for paired per-trip differences.

McNemar's χ^2 test. McNemar's test compares two classifiers on the same evaluation rows by counting only the rows on which they disagree [McNemar, 1947]. Let b be the number of rows where classifier A is correct and B is wrong, and c be the number of rows where A is wrong and B is correct. Under the null hypothesis H_0 that both classifiers have the same error rate on the row distribution, disagreements are equally likely in either direction, so b and c are drawn from a binomial with $p = 0.5$. Dietterich (1998) recommends the continuity-corrected chi-square statistic

$$\chi_{\text{McNemar}}^2 = \frac{(|b - c| - 1)^2}{b + c}, \quad (3.15)$$

which follows a χ_1^2 distribution under H_0 when $b + c \geq 25$ [Dietterich, 1998]. For smaller disagreement counts the chi-square approximation is unreliable and the two-sided exact binomial p -value $p = 2 \cdot \Pr(X \leq \min(b, c) \mid X \sim \text{Binomial}(b + c, 0.5))$ is used instead. McNemar addresses a question that summary metrics cannot: are the two detectors' error patterns different, not just their aggregate error counts.

Non-parametric bootstrap. Where the Wilson interval assumes a binomial proportion at a single point, the bootstrap makes no distributional assumption and can produce a confidence interval for any statistic computed on the data [Efron and Tibshirani, 1994]. Given a labelled sample $\mathcal{D}$ of size n and a metric $\theta(\mathcal{D})$ such as precision, recall, or F_1 , the bootstrap draws B samples $\mathcal{D}_b^*$ of size n with replacement from $\mathcal{D}$, computes $\theta(\mathcal{D}_b^*)$ on each, and uses the empirical quantile of the B replicates to bracket the metric:

$$\text{CI}_{1-\alpha}(\theta) = [\theta_{(\alpha/2)}^*, \theta_{(1-\alpha/2)}^*], \quad (3.16)$$



where $\theta_{(q)}^*$ denotes the q -th empirical quantile of the B bootstrap estimates. With $B = 2000$ resamples the bootstrap CI captures the joint variability across all four contingency-matrix cells, unlike the Wilson interval which brackets only the numerator–denominator variability of a single ratio.

Wilcoxon signed-rank test. When paired measurements are available per group (per trip in this study), the paired difference distribution can be tested for a non-zero location without assuming normality using Wilcoxon’s signed-rank test [Wilcoxon, 1945]. For n paired observations with non-zero differences $d_i = \theta_i^A - \theta_i^B$, the test ranks the $|d_i|$, sums the ranks associated with positive differences to obtain the statistic W , and compares it to the null distribution under $H_0 : \text{median}(d) = 0$. Wilcoxon is more powerful than the sign test because it uses the magnitude information encoded in the ranks and more robust than the paired t -test because it does not assume Gaussian differences [Demšar, 2006]. It is the appropriate paired-comparison test when each experimental unit (a trip) contributes one difference and the difference distribution is unknown.

3.2.7 Time-Series Synchronization Theory

Sensor streams often arrive at irregular timestamps. Synchronizing them requires interpolation theory. For two samples at timestamps T_i and T_{i+1} , the interpolated point at t_j is computed as

$$x_j = x_i + \left(\frac{x_{i+1} - x_i}{T_{i+1} - T_i} \right) (t_j - T_i). \quad (3.17)$$



x_j = interpolated sensor value at target time t_j ,
 x_i, x_{i+1} = measured values at the bracketing timestamps,
 T_i, T_{i+1} = source timestamps of the bracketing samples,
 t_j = target (query) timestamp.

This ensures aligned and consistent time-series data across fuel level, GPS speed, and distance.

3.2.8 IoT Communication Theory

The communication subsystem relies on Internet of Things (IoT) principles and long-range wireless communication theory. LoRa operates using Chirp Spread Spectrum (CSS) modulation, characterized by:

- long-range transmission,
- low power consumption,
- robustness to interference.

Theoretical end-to-end latency between sender and cloud receiver is expressed as

$$L = T_{\text{received}} - T_{\text{sent}}. \quad (3.18)$$

L = one-way end-to-end latency (s),

T_{received} = backend receipt timestamp,

T_{sent} = device-side transmit timestamp.

Understanding latency is essential for analyzing responsiveness and real-time performance.



3.2.9 GPS Temporal and Spatial Theory

GPS logging follows Global Navigation Satellite System (GNSS) principles. To evaluate consistency of GPS sampling, the inter-arrival interval is computed as

$$t_i = T_i - T_{i-1}. \quad (3.19)$$

t_i = inter-sample interval between fixes $i - 1$ and i (s),

T_i = timestamp of fix i ,

T_{i-1} = timestamp of the preceding fix.

Maintaining a stable interval ensures accurate reconstruction of routes and distance traveled.

3.2.10 Context-Aware Anomaly Detection Theory

The anomaly detection component relies on non-parametric, context-aware machine learning, which identifies unusual behavior in multivariate time-series data without labeled training samples. Following the taxonomy adopted for vehicle anomaly detection [Hirth et al., 2025], three anomaly types are distinguished: point anomalies, or individual observations that are anomalous with respect to the data, such as a sudden fuel drop while stationary; contextual anomalies, which are anomalous only in context, such as high consumption at zero speed; and collective or subsequence anomalies, where a sequence of individually unremarkable observations forms an anomalous pattern, such as gradual siphoning. The system assigns point and contextual anomalies to Isolation Forest, operating on engineered contextual features, and collective anomalies to the Matrix Profile.



3.2.10.1 Matrix Profile Theory

Matrix Profile computes similarity between subsequences within a time series. For a time series T of length n and window size m , the profile value is

$$P[i] = \min_{j \neq i} \|T_{i:i+m-1} - T_{j:j+m-1}\|. \quad (3.20)$$

$P[i]$ = Matrix Profile value at index i (nearest-neighbour distance),

T = source time series of length n ,

$T_{i:i+m-1}$ = subsequence of length m starting at index i ,

m = subsequence window length (samples),

$\|\cdot\|$ = Euclidean (L^2) distance between two subsequences.

The subsequences with the highest $P[i]$ values, known as discords, are likely anomalies.

This method is ideal for detecting sudden fuel drops or siphoning patterns.

3.2.10.2 Isolation Forest Theory

Isolation Forest isolates anomalies through random partitioning. An anomaly score is expressed as

$$s(x) = 2^{-E(h(x))/c(n)}. \quad (3.21)$$

$s(x)$ = anomaly score for observation x , bounded in $(0, 1)$,

x = feature vector of the observation being scored,

$E(h(x))$ = expected path length to isolate x across all trees,

$c(n)$ = normalising constant based on tree size n .

Higher scores indicate higher likelihood of anomalous fuel behavior.



3.3 Summary

This chapter presented the theoretical foundation for the system's methodology. It discussed the key concepts supporting real-time sensing, signal preprocessing, GPS logging, IoT communication, and context-aware anomaly detection. By establishing these theoretical pillars, this chapter clarifies the scientific basis that makes the system's design viable, accurate, and aligned with established models in engineering and data science.



1773

Chapter 4

1774

DESIGN CONSIDERATIONS



1775 This chapter presents the design considerations that guided the development of the
1776 implemented IoT- and AI-driven fuel monitoring and anomaly detection system. Whereas
1777 Chapter 3 established the theoretical foundations of IoT architectures, cyber-physical
1778 systems, signal processing, and context-aware anomaly detection, the present chapter
1779 translates those theories into practical engineering decisions that were directly implemented
1780 in Chapter 5.

1781 These design considerations serve as the structural “columns” of the system: grounded
1782 in accepted standards, ethical guidelines, and engineering best practices, but tailored
1783 to support the procedural steps carried out in Chapter 5. The discussion begins with a
1784 comprehensive account of the technical standards that ensure reliability, safety, interoperability, and data security, followed by software and hardware design decisions. Ethical
1785 considerations—including privacy, responsible AI, and driver safety—are also presented.
1786

1787 4.1 Standards

1788 4.1.1 Adherence to Technical Standards

1789 To ensure reliability, accuracy, and safety, the prototype used commercially available mod-
1790 ules and followed relevant design guidance for automotive electrical transients, embedded
1791 control, secure transmission, software architecture, and data protection. Full product-level
1792 certification was outside the scope of the undergraduate prototype. These standards and
1793 guidance documents directly influence the system architecture, data handling, and model
1794 implementation workflows outlined in Chapter 5.

1795 1. Standards for Sensors and Hardware



1796 The system integrates sensors and modules that employ standard communication
1797 protocols—**CAN bus/OBD-II, UART, I2C, and GPIO**—consistent with IEEE 1451
1798 smart transducer guidelines [IEEE Standards Association, 2024a]. The CAN bus/OBD-
1799 II fuel data acquisition aligns with industry telematics conventions [International
1800 Organization for Standardization, 2020], while the GPS module adheres to global
1801 navigation and timing standards [NASA EarthData, 2025].

1802 IEC 61131-2 and ISO 7637-2 were used as references for embedded-control and
1803 automotive transient considerations [International Electrotechnical Commission,
1804 2017, International Organization for Standardization, 2011b]. Commercial off-the-
1805 shelf modules with available CE/FCC-marked variants were selected where possible;
1806 however, full product-level certification was outside the scope of this undergraduate
1807 prototype [European Commission, 2022, Federal Communications Commission,
1808 2025].

1809 **2. Data Acquisition and Signal Processing**

1810 The fuel-level signal is stabilised by a multi-stage filter, a median stage for impulse
1811 rejection followed by an exponential moving average for smoothing, rather than a
1812 single moving average.

1813 Time alignment of fuel, GPS, speed, and engine readings follows ISO/IEC 19510
1814 principles for structured time-series data [International Organization for Standard-
1815 ization, 2013]. These standard-aligned preprocessing steps feed into the anomaly
1816 detection models described in Chapter 5.

1817 **3. Communication and Data Transmission**



1818 Communication follows TCP/IP standards under IETF and IEEE 802.11 [Postel,
1819 1981, IEEE Standards Association, 2025]. Telemetry transmission is implemented
1820 through HTTPS REST endpoints, while MQTT remains a compatible future alter-
1821 native for broker-based deployment. End-to-end encryption via TLS/SSL aligns
1822 with ISO/IEC 27001 cybersecurity requirements [International Organization for
1823 Standardization, 2022a].

1824 These communication strategies form the basis for the system’s message-handling
1825 and cloud-upload mechanisms in Chapter 5.

1826 **4. Machine Learning and Model Validation**

1827 The Matrix Profile and Isolation Forest models are implemented using open-source
1828 libraries, and their evaluation follows ISO/IEC 29119 software quality standards [In-
1829 ternational Organization for Standardization, 2022b]. Validation metrics—confusion
1830 matrices, ROC curves, precision–recall graphs—provide statistical grounding for
1831 model evaluation, as presented in Chapter 5.

1832 **5. Software Architecture and Data Management**

1833 The modular software design complies with ISO/IEC/IEEE 42010 architectural prin-
1834 ciples [International Organization for Standardization, 2022c]. ACID-compliant
1835 storage and GDPR-/ISO 27018-aligned data protection protocols [International Orga-
1836 nization for Standardization, 2025] ensure data integrity and privacy. All processed
1837 datasets follow FAIR principles.



4.1.2 Observance of Research Ethics

Ethical considerations guide the responsible deployment of the system, especially given the safety-critical and privacy-sensitive context of transportation fleets. Three major ethical domains were identified: data privacy, system safety, and responsible AI.

Data Privacy and Security

The system adopts:

- informed consent prior to any data collection,
- strict access control and secure data storage,
- anonymization and aggregation when feasible,
- transparency in data usage and access rights,
- compliance with national and international privacy regulations.

System Safety and Occupational Risk Management

Given the operational risks of fleet environments, the design aligns with ISO 45001, ISO 26262, and ISO 39001 [International Organization for Standardization, 2018, International Organization for Standardization, 2011a, International Organization for Standardization, 2012]. Safety-oriented measures include:

- minimizing driver distraction,
- reducing false alarms through validation,



- providing adequate training for users,
- implementing clear protocols for anomaly response.

Responsible AI and Algorithmic Accountability

Consistent with IEEE P7003 [IEEE Standards Association, 2024b], the system incorporates:

- explainability mechanisms for anomaly outputs,
- documentation of roles and responsibilities in alert handling,
- defined liability procedures for system errors,
- safeguards against algorithmic bias.

4.2 Technical Design Considerations

This section elaborates on the engineering decisions and standards that guided the development of the implemented IoT- and AI-driven fuel monitoring and anomaly detection system. Each component, software layer, algorithm, and communication protocol is grounded in established standards to ensure reliability, safety, interoperability, and security. The references indicate where each design decision is applied in the implementation procedures described in Chapter 5.



TABLE 4.1 FINAL HARDWARE AND SUBSYSTEM DECISION TABLE

Subsystem	Final choice	Alternative considered	Reason for final choice	Known limitation
Fuel acquisition	OBD-II Toyota Mode 22 PID 21/29	External capacitive / float tank sensor	Non-invasive, no tank modification, ethically clearable, partner-acceptable	Manufacturer dashboard gauge used as field reference (no dipstick validation)
Controller	ESP32 (dual-core, 240 MHz, WiFi+BT)	Arduino Uno / Raspberry Pi	Adequate GPIO + WiFi + dual core for IDF VSPI separation; PlatformIO ecosystem; FreeRTOS scheduling	Limited RAM for on-device ML; ML inference moved to backend
HMI display	ILI9341 2.4" TFT + XPT2046 touch + dedicated buttons	OLED + scroll button only	Full route / fuel / alert UI in-cab; suitable readability in daylight	Touch only used for non-critical actions; physical buttons used for safety-critical confirm
GPS	Embedded GPS (Neo-style) + mobile-app GPS fallback fused at backend	Dashboard-only / external assist only	Resilient to embedded-GPS cold start or building occlusion; tagged via <code>gps_source</code> column	Urban-canyon drift; cold start of embedded module on first power-on
Wide-area comms - primary	LTE via A7670E modem	GSM-only baseline	Higher throughput, lower per-packet latency once attached	HTTPACTION + TLS renegotiation can push median latency to approximately 23–24 s on certain Smart cells
Wide-area comms - fallback	GSM 2G via same A7670E (with CG-NAPN/CFUN+CSTT fallback)	LTE-only	Retained for legacy 2G-only coverage on some Batangas / Tagaytay routes	Lower throughput; not all carriers maintain 2G



Subsystem	Final choice	Alternative considered	Reason for final choice	Known limitation
Local-area / depot comms	LoRa SX1278 @ 433 MHz, SF7, BW 125 kHz, 17 dBm	WiFi-only at depot	Long range, low power, bidirectional fleet-yard polling without LTE cost	Low duty-cycle limits payload size; 125 kHz BW shared with other depot nodes
Hotspot fallback	WiFi (driver phone tether or depot AP)	No fallback	Recovers comms when LTE+GSM+LoRa are all down but a personal hotspot is available	Depends on driver compliance to enable hotspot
Offline buffer	SPIFFS append-only ring buffer, replayed with <code>channel_used='offline_replay'</code>	RAM-only buffer	Survives reboot; preserves <code>seq + event_id</code> for backend deduplication	Flash wear if drain is starved for very long periods
Backend	Node/Express on Render + Supabase Postgres	Self-hosted Postgres on a VPS	Managed DB, row-level security, free-tier acceptable for thesis-scale traffic	Render cold-start adds latency to the first POST after idleness
ML detector	Isolation Forest Model C (contamination=0.015, n_estimators=300, max_samples=256)	Threshold-only / IQR baseline; Hybrid Union; Hybrid Intersection	Only Isolation Forest variant that met $\geq 90\%$ precision and $\leq 10\%$ FPR under LOTO	Recall depends on injection realism; will be retrained as live anomaly corpus grows
Subsequence detector	Matrix Profile (window=12, z-thresh=2.0)	Drop entirely	Useful confirmatory layer for slow-leak / steady-siphon patterns that single-point IF misses	Less robust than Model C at row-level scoring; retained as supporting evidence only



1871

4.2.1 Hardware Selection and Integration

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The hardware configuration is designed to support the end-to-end workflow of the system (see Sec. 5.1). The following standards and implementation points apply:

1874

- **MCU:** ESP32/NodeMCU serves as the edge processing device for real-time acquisition and preprocessing of fuel, GPS, and vehicle telemetry data. It complies with IEC 61131-2 for embedded controllers [International Electrotechnical Commission, 2017] and ISO 7637 for automotive electrical transients [International Organization for Standardization, 2011b]. In Sec. 5.1, the MCU handles local preprocessing, timestamping, and alert-state handling before backend anomaly scoring.

1880

- **GPS Module:** The UART-connected GPS module provides geolocation and speed data, adhering to global navigation standards [NASA EarthData, 2025]. Its signals are synchronized with fuel readings during preprocessing in Sec. 5.4.

1883

- **CAN/OBD-II Interface:** The CAN/OBD-II interface was selected as the acquisition layer because it satisfies the non-invasive constraint set by the partner operator. From this one port the prototype reads speed, RPM, engine load, throttle, MAF, coolant temperature, runtime, DTC status, and the Toyota Mode 22 PID 21/29 fuel-sender voltage encoded at 0.5 L per ADC count. Mode 22 access is manufacturer-specific and was validated on the partner Toyota vehicle.

1889

4.2.2 Software Architecture

1890

The modular software layers are designed according to ISO/IEC/IEEE 42010 software architecture standards [International Organization for Standardization, 2022c], supporting

1891



scalability, maintainability, and interoperability. Each layer’s function is mapped to its implementation in Sec. 5.1:

- **Data Acquisition Layer:** Captures sensor and telemetry data, applies the implemented median + EMA + dead-band + direction-aware filter cascade, and prepares telemetry for backend anomaly scoring. Implementation details are in Secs. 5.3 and 5.4.
- **Processing Layer:** Performs edge timestamping, trip-state handling, distance aggregation, and local alert-rule state preparation before cloud-side anomaly inference. Standards for real-time processing and task scheduling ensure predictable latency. Implementation is in Sec. 5.4.
- **Cloud Integration Layer:** HTTPS REST endpoints receive JSON telemetry payloads from the prioritized communication stack. MQTT remains compatible as a future broker-based deployment option, while TLS/SSL ensures encrypted transmission (Sec. 5.5).
- **User Interface Layer:** Dashboards display real-time fuel levels, GPS locations, anomaly alerts, and historical trends. Web standards (HTML5, CSS3, JavaScript) and secure REST APIs are employed, as described in Sec. 5.7.

4.2.3 Machine Learning and Algorithms

Machine learning components follow ISO/IEC 29119 software quality standards [International Organization for Standardization, 2022b] and IEEE P7003 AI accountability guidelines [IEEE Standards Association, 2024b]:



Static threshold fuel anomaly systems are insufficient in Philippine logistics because traffic congestion, idling, route delays, load variation, and stop-and-go driving can produce fuel-consumption changes that look abnormal under fixed rules. The proposed model uses contextual variables such as fuel rate, speed, RPM, engine load, throttle, idling, distance, and route behavior to distinguish suspicious fuel drops from normal but inefficient fuel use. This makes the system more adaptive than fixed threshold detection.

- **Models:** Isolation Forest was selected as the production detector because Model C (contamination = 0.015, n_estimators = 300, max_samples = 256, score threshold = the 0.985 quantile of the clean training scores) was the only configuration meeting both $\geq 90\%$ precision and $\leq 10\%$ FPR under leave-one-trip-out evaluation (precision = 92.9%, recall = 92.9%, FPR = 1.09%; see Chapter 6). Matrix Profile (window = 12, z-threshold = 2.0) was retained as a supporting subsequence detector for slow-leak and siphon patterns that single-point scoring is less sensitive to.

- **Evaluation Metrics:** Confusion matrices, ROC curves, and precision–recall graphs are used for model validation, ensuring alignment with ISO/IEC 25010 software quality characteristics. Secs. 5.10 and 5.10.5 provide practical application in performance evaluation.

- **Alternative Models:** Autoencoders and LSTM networks were considered but excluded due to edge processing constraints, as justified in Sec. 4.3.

4.2.4 Communication and Security

Communication protocols and security measures are aligned with ISO/IEC 27018, ISO/IEC 27001, and GDPR standards to ensure data privacy, encryption, and secure access control:



1935 • **REST Telemetry:** HTTPS REST endpoints support direct cloud ingestion from
1936 LoRa gateway, LTE/GSM, WiFi, and offline-replay channels. MQTT with QoS 1
1937 remains a future-compatible alternative if a broker-based deployment is adopted.

1938 • **TLS/SSL Encryption:** Provides end-to-end encryption for transmitted data. Com-
1939 pliance with ISO/IEC 27001 ensures data integrity and confidentiality. Applied in
1940 Sec. 5.5.

1941 • **Role-Based Access and Anonymization:** Enforces secure data access policies and
1942 anonymizes sensitive driver information, complying with ISO/IEC 27018 and GDPR
1943 principles. Applied in Sec. 5.7.

1944 • **JSON Serialization:** Standardized data format for cloud compatibility, ensuring
1945 interoperability with cloud services and dashboards. Applied in Sec. 5.5.

1946 Wide-area connectivity uses a SIMCom A7670E LTE (Cat-1) module, chosen over
1947 a GSM-only baseline because it provides an LTE-primary, 2G-fallback hierarchy in one
1948 device. LTE is the primary path for its lowest latency and highest throughput, and because
1949 it is the network Philippine operators are expanding rather than retiring. GSM 2G on the
1950 same module is a deeper fallback for legacy-only coverage; consistent with NTC MC No.
1951 003-09-2025 (phased 2G/3G decommissioning) it is treated as transitional, and because the
1952 A7670E is LTE-first rather than 2G/3G-only it remains compliant with that circular. LoRa
1953 provides depot, yard, and long-range telemetry; WiFi provides a depot or driver-tether path;
1954 and SPIFFS offline-buffer replay is the final fallback so no record is lost when all real-time
1955 channels are down [China Academy of Information and Communications Technology,
1956 2025].



4.3 Alternative Software Designs and Cost Considerations

During the planning and prototype development phase, several candidate software designs were evaluated for their suitability to the system's requirements. Table 4.2 summarizes the strengths, potential risks, and fit relative to the system specifications. Matrix Profile (MP), Isolation Forest (IF), and a combined Matrix Profile + Isolation Forest (MP + IF) configuration were implemented and tested to compare their effectiveness in detecting fuel-related anomalies. The comparison considered detection accuracy, computational efficiency, and deployment feasibility on the target hardware platform. The findings from these tests were used to determine the most appropriate approach for the final system implementation.

TABLE 4.2 ALTERNATIVE SOFTWARE DESIGNS EVALUATED FOR THE PROTOTYPE

Candidate	Strengths	Risks / Trade-offs	Fit vs. Spec
Isolation Forest (IF)	Fast, context-aware, handles multivariate data; Model C supports contextual fuel, speed, route, driver, and fuel-economy deviation features	Limited temporal context; sensitivity to contamination parameter may affect false alarms	Very High — selected production detector
Matrix Profile (MP)	Shape-based time-series discord detection; useful for slow-leak and steady-siphon subsequence patterns as a confirmatory layer	Requires window-size tuning; primarily one-dimensional unless extended for multivariate data; less robust than Model C at row-level scoring	Moderate to High — supporting subsequence detector

A cost study was conducted after prototype testing to confirm that the system configuration was cost-effective. This study reviewed hardware, software, and development resources, comparing the chosen open-source and local execution setup against alternative configurations such as commercial analytics tools, cloud-based computing, or hardware-



1972 integrated telemetry systems.

1973 The total estimated budget for developing and implementing the prototype system is
1974 **PHP 10,446**, confirming that the selected configuration remains the optimal and sustainable
1975 choice for achieving the project’s objectives.

1976 **4.4 Summary**

1977 This chapter presented the key technical and ethical design considerations that shape the
1978 system. By grounding the project in ISO, IEEE, OASIS, GDPR, and FAIR standards,
1979 the system ensures reliable data acquisition, secure communication, validated machine
1980 learning, and responsible AI use.

1981 These design elements establish the foundation for the procedures described in Chap-
1982 ter 5, bridging theoretical concepts from Chapter 3 with the practical implementation steps
1983 of the system.



De La Salle University

1984

Chapter 5

1985

METHODOLOGY



1986 This chapter presents the research methodology employed for the development of a
1987 machine-learning-driven fuel monitoring and anomaly detection system for fleet vehicles.
1988 The methodology integrates a multi-layered approach combining OBD-II data acquisition,
1989 edge computing, IoT communication, cloud-based machine learning, and client-side visual-
1990 ization. The approach emphasizes real-time anomaly detection, accurate fuel monitoring,
1991 operational transparency, and actionable driver and fleet management insights.

1992 The research methodology is structured to address both general and specific objectives,
1993 ensuring that the system is technically feasible, scalable, and reliable. Figure 5.2 and
1994 Table 5.1 summarize the methods and approaches for each objective.

1995 The chapter placement was reviewed to keep the paper aligned with the DLSU thesis
1996 structure. Chapter 3 is retained for the theoretical basis of IoT telemetry, anomaly detection,
1997 and fleet monitoring; Chapter 4 is retained for design decisions, standards, limits, and
1998 constraints; and this chapter focuses on the actual research procedure: how the prototype
1999 was implemented, how telemetry was collected and processed, how the machine-learning
2000 variants were trained, and how each objective was evaluated.

TABLE 5.1 SUMMARY OF METHODS FOR REACHING THE OBJECTIVES

Objectives	Methods / Approaches	Locations
------------	----------------------	-----------



GO: To develop and evaluate an IoT-based system for trucks, in collaboration with the partner truck company Hino Parañaque Subic GS Auto Inc. (SGSA), that monitors real-time fuel levels, tracks routes using GPS, and detects unusual fuel consumption using machine learning, with the goal of improving driver safety, supporting rest scheduling, and facilitating proactive vehicle maintenance through automated alerts using fuel and distance metrics, based on established service guidelines.	• Embedded MCU preprocessing and aggregation • LoRa/Cellular communication for low-latency and fail-safe data transfer • Cloud-based machine learning for anomaly detection • Web dashboard for visualization • Driver alert subsystem for operational safety	Sec. 5.1
SO1: To measure fuel levels using digital float sensors integrated via the truck's ECU/diagnostic connector and GPS module, ensuring non-invasive installation and real-time accuracy within $\pm 5\%$.	• Digital fuel-level acquisition through the truck's ECU/diagnostic connector • Edge-side median, EMA, dead-band, and direction-aware filtering • Validation against dashboard-gauge workaround pairs and signal variance • Continuous validation for measurement reliability	Secs. 5.3, 5.4



SO2: To transmit real-time fuel and GPS data through IoT-based communication to a cloud server, accessible through a centralized dashboard.	• Real-time IoT transmission via LoRa, LTE, GSM 2G, WiFi, and SPIFFS replay • Cloud ingestion APIs • Synchronized HMI, mobile driver application, and administrative dashboard • Latency measurement and benchmarking to ensure timely reporting	Sec. 5.5
SO3: To capture and log GPS location data every 5 seconds via an embedded module, correlating route information with fuel and time data for behavior tracking.	• GPS-based location logging at 5-second intervals • Synchronization with fuel data • Structured cloud database for multi-source data storage and route-fuel correlation	Secs. 5.1, 5.7
SO4: To design and evaluate a context-aware fuel usage deviation model using context-aware non-parametric machine learning models (Matrix Profile or Isolation Forest), achieving 90% precision and 10% false positives in identifying anomalous fuel consumption patterns within controlled environments.	• Offline training of anomaly detection models using Isolation Forest and Matrix Profile • Evaluation using precision, FPR, and accuracy • Context-aware thresholding to reduce false positives	Secs. 5.9, 5.10



SO5: To develop and integrate hardware-based rest and maintenance alert mechanisms (e.g., Start/Stop button, buzzer indicators) tied to the web dashboard using cumulative fuel and distance data.	• Driver rest and maintenance alert system with cumulative distance/time tracking • Cloud-based monitoring and logging • Alert notification via dashboard and in-vehicle signals	Secs. 5.2, 5.7.7
--	--	------------------

Table 5.1 is used as the methodological roadmap for the chapter. Each row connects a thesis objective to the concrete subsystem, data source, or validation activity used to satisfy it. The table also prevents the methodology from becoming only an implementation narrative: it shows that fuel acquisition, communication, GPS logging, machine learning, and alert delivery are each tied to a measurable requirement that is later evaluated in Chapter 6.

5.1 System Architecture and Conceptual Design

The system follows a layered architecture with a focus on modularity, scalability, and reliability. Figure 5.1 integrates the following:

- **Vehicle Sensing Subsystem:** OBD-II fuel and vehicle-context acquisition, embedded GPS, mobile GPS fallback, and driver rest-time monitors.
- **Edge Processing:** The MCU performs signal preprocessing, filtering, smoothing, synchronization, and timestamping before transmission.

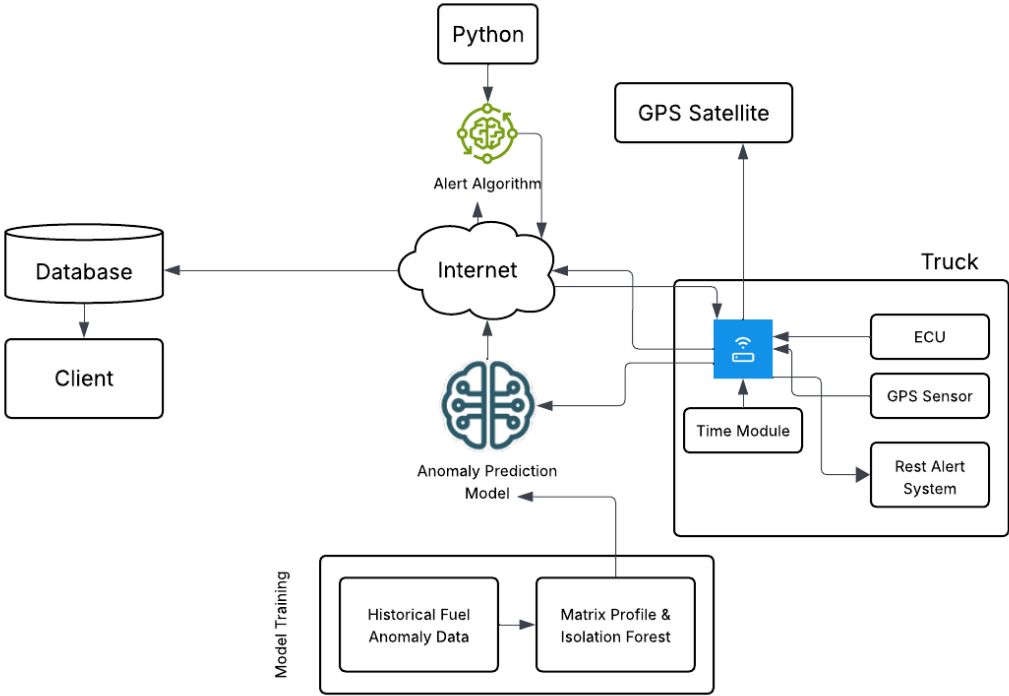


Fig. 5.1 Conceptual Framework

- **IoT Communication:** LoRa, LTE, GSM 2G, WiFi, and SPIFFS replay provide prioritized telemetry delivery under changing network conditions.
- **Cloud Intelligence:** Isolation Forest Model C and Matrix Profile evaluate incoming telemetry and trigger fuel-anomaly alerts when the contextual gate permits.
- **Client Visualization:** The embedded HMI, mobile driver web application, and administrative dashboard provide synchronized fuel, route, trip-state, and alert views.

The purpose of the conceptual framework is to show the boundary of the design solution before individual subsystems are discussed. The vehicle layer produces the measurements, the edge layer cleans and packages them, the communication layer preserves delivery



under changing coverage, the cloud layer stores and interprets the data, and the interface layer returns the interpreted state to the driver and fleet manager. This layered reading is important because the prototype is evaluated as an end-to-end monitoring system rather than as a single sensor or dashboard.

The system architecture in Figure 5.2 is an edge-to-cloud fleet-monitoring pipeline in which the embedded HMI performs data acquisition, filtering, trip-state handling, local driver display, and prioritised transmission. The backend stores telemetry, synchronises state across three concurrent client surfaces - the in-vehicle HMI, the mobile driver web application, and the administrative dashboard - and dispatches anomaly and safety alerts through the same database tables that the three surfaces subscribe to.

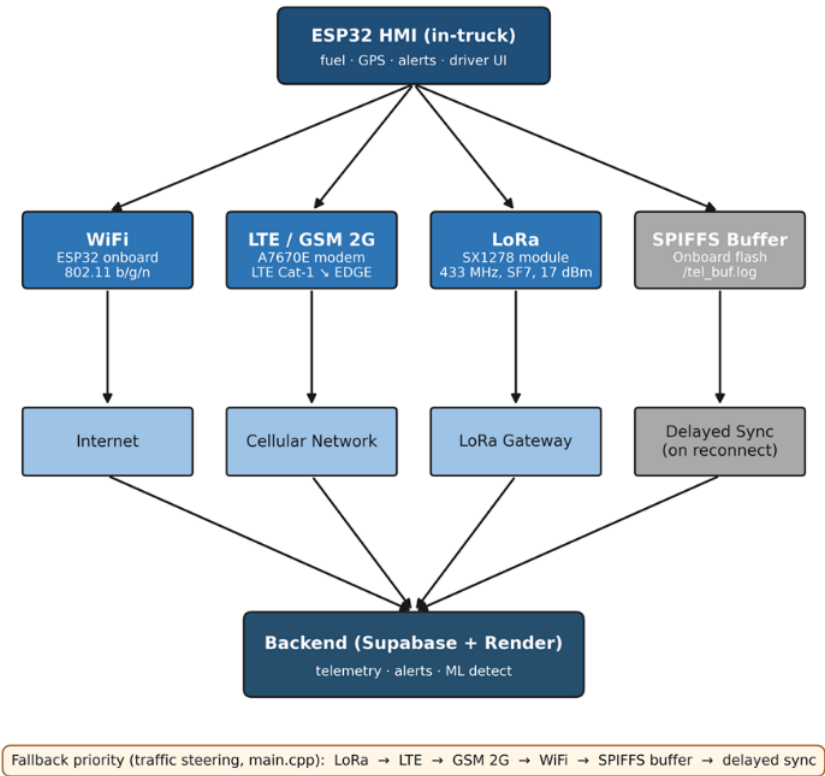


Fig. 5.2 End-to-end System Architecture



Figure 5.2 illustrates the end-to-end communication architecture of the system, depicting how data flows from the ESP32 HMI through multiple transmission channels to the cloud backend and connected client interfaces. The HMI, serving as edge node, collects fuel, location, and alert data while also providing the user interface for the driver. From the HMI, data is transmitted through WiFi, LTE/GSM 2G, and LoRa before converging at the Supabase and Render backend, which handles telemetry ingestion, alert dispatching, and ML anomaly detection. The SPIFFS onboard flash buffer serves as a local fallback wherein stored telemetry is synced when a connection is re-established. The fallback priority order (LoRa → LTE → GSM 2G → WiFi → SPIFFS buffer → delayed sync) ensures continuous data delivery even under degraded network conditions. From the backend, processed data is propagated to the mobile driver web application, the administrative dashboard, and the HMI poll-back, maintaining real-time synchronization across all three client surfaces.

The architecture also identifies where each later measurement is taken. Fuel accuracy is measured before and after the edge filter, communication latency is measured between device emission and backend receipt, GPS cadence is measured at the device timestamp, anomaly detection is measured after backend feature construction, and alert usability is measured at the HMI, mobile, and dashboard surfaces. This placement makes the Chapter 6 result chain traceable from sensor input to user-facing output.

5.1.1 Input-Process-Output (IPO) Overview

The data design outlines a cohesive pipeline that integrates OBD-II acquisition, GPS synchronization, cloud transmission, and model-based analysis to enable real-time anomaly detection in fleet fuel systems. Data collection begins with the embedded HMI capturing OBD-II fuel level, vehicle context, GPS coordinates, communication-channel metadata,



and trip-state events. These signals are preprocessed through the fuel filter cascade and synchronized before backend ingestion.

To ensure temporal alignment, telemetry records include device-side and backend-side timestamps. This synchronization step is critical for comparing trends across OBD-II, GPS, communication, and alert data. The processed signals are uploaded to the Supabase and Render backend for storage, anomaly scoring, and synchronized access across the HMI, mobile driver application, and administrative dashboard.

For anomaly detection, the synchronized time-series data are used to train and evaluate Isolation Forest and Matrix Profile detectors. These models are tested on project-native trips and controlled-injection scenarios to reflect normal and anomalous behavior. Key performance metrics, including MAPE, timestamp drift, detection latency, precision, recall, and false positive rate, are used to assess system effectiveness.

5.1.1.1 IPO Chart

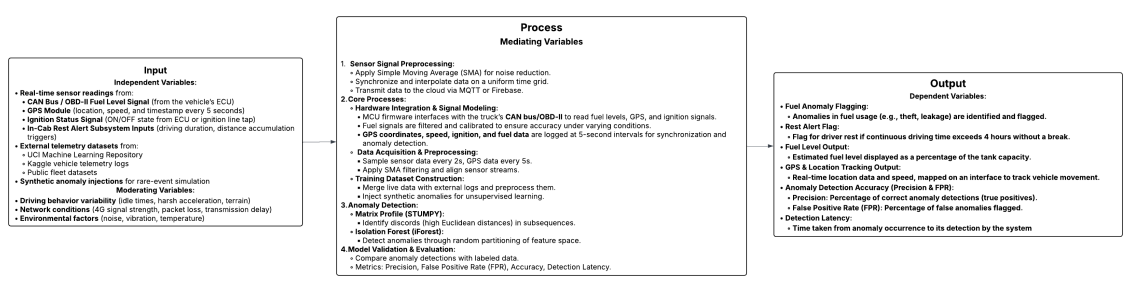


Fig. 5.3 IPO Chart

The IPO (Input-Process-Output) chart in Figure 5.3 provides a structured overview of the IoT-based fleet fuel monitoring and anomaly detection system. The input data consists of real-time readings from the fleet vehicle's CAN bus/OBD-II fuel signal, GPS module



2072 coordinates and speed at five-second telemetry intervals, ignition status, and in-cab rest alert
2073 subsystem signals. These inputs are supplemented by historical fuel anomaly datasets and
2074 controlled-injection anomalies to simulate rare or edge-case events. Moderating variables
2075 such as driving behavior, network conditions, and environmental factors also influence the
2076 data.

2077 The process begins with signal preprocessing, where the MCU filters and timestamps
2078 fuel, GPS, and ignition data to reduce noise and ensure synchronization. The preprocessed
2079 data is then transmitted through the prioritized communication stack and stored in the
2080 cloud backend. Anomaly detection is carried out in the cloud using Matrix Profile for
2081 time-series discord identification and Isolation Forest for feature-based outlier detection.
2082 Model validation and evaluation are performed using labelled and test datasets, calculating
2083 performance metrics such as precision, recall, false positive rate (FPR), detection latency,
2084 and overall accuracy.

2085 The system outputs include fuel anomaly flags, such as potential theft or leakage, and
2086 driver rest alerts based on cumulative distance or driving duration. Additionally, it provides
2087 real-time GPS tracking on a web dashboard, estimated fuel levels as a percentage of tank
2088 capacity, and quantitative performance metrics for operational evaluation. This end-to-end
2089 flow ensures accurate, real-time monitoring, proactive safety notifications, and actionable
2090 insights for fleet management.

2091 **5.1.2 End-to-End Telemetry Data Flow**

Listing 5.1: End-to-End Telemetry Data Flow

```
2092 1 START  
2093 2 WHILE system is running DO
```



```

2094 3      READ ECU fuel data
2095 4      READ ECU operational data
2096 5      READ GPS data
2097 6      APPLY Median Filter
2098 7      APPLY EMA Filter
2099 8      APPLY Dead-Band Filter
2100 9      APPLY Direction-Aware Filter
2101 10     // UPDATE trip state
2102 11         IF vehicle is loading THEN
2103 12             trip_state = "Loading"
2104 13         ELSE IF vehicle is moving THEN
2105 14             trip_state = "Active"
2106 15         ELSE IF temporary stop detected THEN
2107 16             trip_state = "Paused"
2108 17         ELSE IF trip completed THEN
2109 18             trip_state = "Ended"
2110 19         END IF
2111 20     COMPUTE fused distance using GPS and OBD data
2112 21     // SELECT available communication channel
2113 22         IF LoRa available THEN
2114 23             Send via LoRa
2115 24         ELSE IF LTE available THEN
2116 25             Send via LTE
2117 26         ELSE IF GSM available THEN
2118 27             Send via GSM
2119 28         ELSE IF WiFi available THEN
2120 29             Send via WiFi
2121 30         ELSE
2122 31             Save data to SPIFFS
2123 32         END IF
2124 33     SEND telemetry to backend
2125 34     INSERT telemetry into telemetry_logs
2126 35     RUN ML anomaly detection
2127 36     IF anomaly passes contextual gate THEN

```



```
37      INSERT alert into alerts
38  END IF
39  UPDATE database
40  UPDATE dashboard, mobile app, and HMI display
41 END WHILE
42 END
```

Listing 5.1 converts the architecture into the operating sequence followed by the prototype during a trip. It shows that telemetry collection is cyclical: the device reads fuel and GPS, filters the fuel signal, updates trip state, selects a communication channel, stores the record, runs anomaly detection, and then refreshes all user interfaces. The listing is not intended to expose source code; it is a readable process model showing how the hardware, backend, and interface layers cooperate during normal and degraded-network operation.

5.2 Hardware Methodology

The embedded hardware platform is built around an ESP32 dual-core microcontroller running at 240 MHz with separate VSPI and HSPI bus assignments to prevent SPI arbitration conflicts between the TFT display and the LoRa transceiver. Peripheral components include the ILI9341 2.4-inch TFT display with XPT2046 capacitive touch controller, physical action buttons, and a piezo buzzer. The MCP2515 CAN controller communicates with the ESP32 through SPI and interfaces with the vehicle CAN lines through a CAN transceiver. The A7670E LTE/GSM modem and embedded GPS module use UART links, while the SX1278 LoRa transceiver uses VSPI.



5.2.1 HMI Control Pins

The main driver-facing controls use two debounced physical buttons for rest and snooze actions, one PWM-driven buzzer output for alerts, and one modem power-control pin. The full GPIO map is provided in Appendix D so the main methodology can focus on the hardware role rather than pin-level implementation detail.

This pin assignment supports the SO5 alert workflow by giving the driver physical controls that remain usable even when the touchscreen is difficult to operate during field conditions. The rest and snooze buttons are treated as event sources, the buzzer as the immediate alert actuator, and the modem-control line as the recovery mechanism for cellular communication faults.

5.2.2 Hardware Block Diagram

Figure 5.4 shows the hardware block diagram of the embedded HMI node, depicting how the ESP32 microcontroller connects to the OBD-II CAN interface, the A7670E cellular modem, the SX1278 LoRa transceiver, the GPS module, the ILI9341 TFT display, and the driver alert hardware on its respective bus and UART lines.

The block diagram should be read as the physical implementation counterpart of the system architecture. It explains why separate buses were used: the TFT display, CAN controller, LoRa transceiver, modem, and GPS receiver all generate time-sensitive events, so the wiring design separates high-traffic display updates from telemetry and radio activity. This separation reduces the chance that display refresh, CAN polling, and radio transmission interfere with one another during active trips.

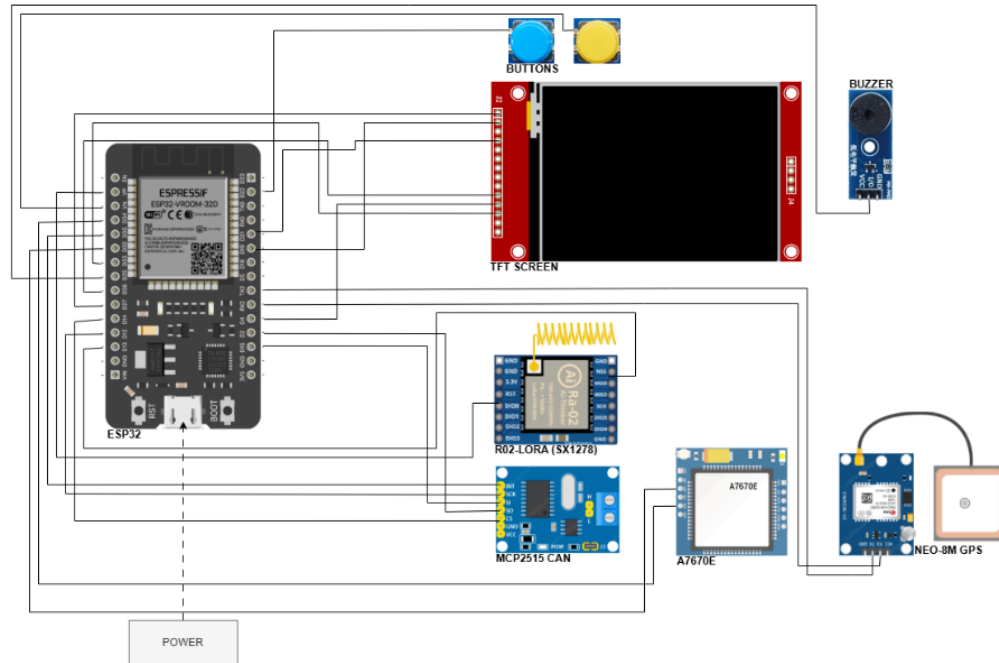


Fig. 5.4 Embedded HMI Hardware Block Diagram

5.3 OBD-II / ECU Fuel and Vehicle Data Acquisition Methodology

The HMI acquired vehicle telemetry through the standard OBD-II interface, which provides universal Mode 01 access to contextual variables — vehicle speed, engine RPM, engine load, throttle position, mass air flow, coolant temperature, engine runtime, and diagnostic-trouble-code status — on every OBD-II-compliant vehicle regardless of manufacturer. Fuel level is treated separately because the universal Mode 01 fuel-level PID (0x2F) is not consistently exposed or accurately scaled across manufacturers; the prototype therefore reads fuel level through a vehicle-specific adapter selected at install time, with the Toyota meter Mode 22 PID 21/29 channel used for the partner-fleet Toyota Hilux. The same telemetry cycle



timestamps all acquired fields at the device and transmits them to the backend together with GPS, trip-state, and communication-channel metadata, so the anomaly detector interprets fuel changes in relation to vehicle operation rather than fuel percentage alone.

5.3.1 Physical Interface

The acquisition layer uses the vehicle OBD-II port and ISO 15765-4 transport over CAN. This enables the prototype to read fuel and vehicle context without tank-cap modification or sender-loom alteration. All OBD-II-compliant vehicles expose a standard set of Mode 01 PIDs over this transport, which the prototype uses for the universal contextual variables; manufacturer-specific PIDs (Toyota Mode 22, Hino J1939, Isuzu Mode 22) are layered on top of the same transport only for fuel-level acquisition.

The physical-interface choice is central to the study because it keeps the system non-invasive. Instead of inserting a new float sensor or cutting into the fuel sender wiring, the prototype reads the fuel-related signal already exposed by the vehicle electronics. This reduces installation risk, supports repeatability across fleet vehicles of the same model, and keeps the validation focused on whether the existing vehicle signal can be made accurate enough for monitoring.

5.3.2 Vehicle-Specific Fuel-Level Adapter (Toyota Mode 22 PID 21/29)

Because the universal Mode 01 fuel-level PID (0x2F) was not reliably exposed by the partner-fleet ECU, the firmware falls back to a vehicle-specific fuel-level adapter selected at install time. For the Toyota Hilux deployed in this study the adapter is the Toyota meter



Mode 22 command 0x21 0x29, which returns the raw fuel-sender voltage encoded at 0.5 L per ADC count without internal damping. The firmware polls this channel on a 1.5 s cadence and applies a retry policy when the adapter returns a negative acknowledgment or missing response. The adapter is implemented as a thin, replaceable layer over the universal OBD-II acquisition pipeline, so future deployments on other manufacturer ECUs (Hino J1939 SPN 96 for diesel heavy-duty units, Isuzu Mode 22, or any vehicle that fully implements Mode 01 PID 0x2F) require no changes to the filter cascade, telemetry schema, or anomaly-detection model.

The 1.5 s poll cadence was selected to capture fuel changes faster than the five-second telemetry publication interval while still leaving time for GPS parsing, display updates, and communication attempts. The retry policy prevents a single missing CAN response from becoming a false fuel event. Only values that pass validity checks are allowed to update the filtered fuel stream used by the HMI and backend.

5.3.3 Technical Basis for the 1.5 s OBD Polling Rate

The 1.5 s cadence, denoted Δ_s throughout the remainder of the methodology, is not an arbitrary firmware constant. It is fixed by the intersection of three independent physical and statistical constraints, all of which must be satisfied simultaneously; a rate above the intersection saturates the CAN bus, and a rate below the intersection under-samples the slosh envelope. Naming these constraints explicitly makes the choice of Δ_s auditable and portable to future ECUs.

(a) CAN-bus bandwidth ceiling. ISO 15765-4 [International Organization for Standardization, 2021] defines the OBD-II CAN transport with a minimum inter-request



spacing that, on the partner-fleet Toyota Hilux, was measured empirically at 60–120 ms per Mode 22 PID 21/29 response through the MCP2515 controller. Polling at 1 Hz or faster (i.e. $\Delta_s \leq 1$ s) leaves less than ~ 800 ms of budget for the co-acquired Mode 01 PIDs enumerated in Table 5.2 and produces sporadic dashboard-warning flicker when the ECU’s response queue saturates. The 1.5 s cadence sits comfortably below the observed saturation regime.

(b) Nyquist coverage of legitimate fuel dynamics. The physical maximum idle-combustion loss rate for a 2.4 L diesel is $\dot{V}_{\text{idle}} \approx 0.5$ L/hr per the SAE J1349 engine-power test code [Society of Automotive Engineers, 2011], which on the 80 L tank produces a percentage change of $\dot{F}_{\text{idle}} \approx 0.00029$ %/s. The Nyquist-rate criterion therefore requires only ~ 0.001 Hz to reconstruct legitimate slow dynamics; the 1.5 s cadence samples at approximately $30\times$ the strict Nyquist rate, providing substantial oversampling margin so that transient siphon rates on the order of $\beta_1, \beta_2 \sim 0.5\text{--}0.6$ %/s (Section 5.9.4) are captured with at least eight samples per event.

(c) Alignment with the median-filter memory. The Stage 1 median window in Section 5.4.2 spans $W = 31$ samples, so the effective memory of the filter is $W \cdot \Delta_s = 31 \cdot 1.5$ s = 46.5 s. This memory was chosen to bracket the empirical 5–20 s slosh envelope measured during on-road cornering with a comfortable factor of two above the 95th-percentile slosh duration observed on the 27-trip corpus. If Δ_s were reduced, either W would grow (increasing per-cycle sort cost on the ESP32) or the filter memory would shrink below the slosh envelope; either compromise reintroduces the sender-noise artefacts the filter is designed to remove. The 1.5 s cadence is therefore the rate at which the OBD interface, the physics of fuel loss, and the median-window



design are simultaneously self-consistent.

5.3.4 Co-acquired Mode 01 PIDs

Table 5.2 lists the vehicle signals used by the prototype and shows where each signal is stored after ingestion. The column names are included only for traceability between the thesis results and the implemented database; the main purpose of the table is to show that fuel readings were interpreted together with speed, engine demand, motion context, and diagnostic status rather than as an isolated raw value.

TABLE 5.2 OBD-II PID TO TELEMETRY_LOGS COLUMN MAPPING

PID	Quantity	Telemetry column
Mode 22 0x21 0x29	Fuel sender voltage (0.5 L per ADC count)	fuel_level (filtered) + fuel_source='mode22' + fuel_confidence
Mode 01 0x0C	Engine RPM	engine_rpm
Mode 01 0x0D	Vehicle speed (km/h)	speed
Mode 01 0x04	Calculated engine load (%)	engine_load_pct
Mode 01 0x11	Throttle position (%)	throttle_pct
Mode 01 0x10	Mass air flow (g/s)	maf_gps
Mode 01 0x05	Engine coolant temperature (degC)	coolant_temp_c
Mode 01 0x1F	Time since engine start (s)	engine_runtime_sec
Mode 03 / 0x01	Stored DTCs (count + presence)	dtc_present, dtc_count

Table 5.2 explains how raw vehicle signals become database-ready telemetry. Fuel level is the primary measurement for SO1 and SO4, while speed, RPM, load, throttle, mass airflow, coolant temperature, runtime, and diagnostic status provide context for deciding whether a fuel change is plausible. This is why the anomaly detector is described as context-aware: it does not evaluate fuel movement without also considering motion and engine state.



5.3.5 Validity and Confidence Fields

The firmware writes `fuel_valid`, `fuel_source`, and `fuel_confidence` on every telemetry cycle. These fields allow the backend and dashboard to distinguish valid Mode 22 fuel readings from missing, stale, or fallback values.

These fields prevent the backend from treating all fuel values as equally reliable. For example, a current Mode 22 reading with high confidence can be used directly for alerts, while a stale or fallback value can be displayed with caution and excluded from model training or calibration windows. This protects the system from creating false anomalies during adapter timeouts or network replay.

5.3.6 Timestamping

Each telemetry record carries the original device-side timestamp as `original_device_timestamp`, while the backend records its own receipt time. Backend reconciliation uses server time to separate transmission delay from actual sampling time.

Dual timestamping is necessary because the communication subsystem deliberately permits delayed delivery. A row replayed from SPIFFS may arrive minutes after it was sampled, but its device timestamp still represents the true trip timeline. Chapter 6 therefore uses device time for sampling cadence and route reconstruction, and backend receipt time for communication latency and alert-arrival analysis.

5.3.7 Field Reference Choice

The manufacturer dashboard gauge is used as a supporting field reference for fuel-validation activity. This choice follows the delimitations in Chapter 1, because direct dipstick valida-



tion and tank-cap opening were not permitted by the partner fleet operator.

The validation reference was therefore selected to match the allowed field procedure. Known-volume pump-gas events provide the volumetric reference, while the manufacturer dashboard confirms that the prototype is reading a fuel state consistent with the vehicle's own display. This is an honest operational validation rather than a laboratory calibration claim.

Because the validation reference is the legally calibrated trade-instrument fuel pump (Tables 6.3–6.4) rather than a Toyota-specific signal, the SO1 accuracy claim is portable to any future vehicle-specific adapter that produces a stable filtered fuel percentage; only the adapter in Section 5.3.2 needs to be re-validated per ECU.

5.3.8 Raw Signal Sourcing Summary

Table 5.3 consolidates every raw signal that the machine-learning subsystem consumes into a single canonical source. Nine physical channels feed the 33-dimensional Model C feature vector: seven OBD-II PIDs (six standard Mode 01, one manufacturer-proprietary Mode 22), one UART NMEA GPS stream, and the ESP32 real-time clock synchronised over NTP. Documenting these inputs explicitly closes the traceability chain from Table 5.11 back to the physical bus and connector pin from which each feature is derived, which is a prerequisite for the reproducibility argument the evaluation chapter makes on top of the LOTO cross-validation protocol.

Table 5.3 is deliberately organised so that the two manufacturer-proprietary Toyota channels are visually separated from the seven universal channels. This makes the portability envelope of the study explicit: replacing the Toyota adapter with a Hino J1939 SPN 96 or an Isuzu Mode 22 adapter changes only two rows of this table, while the remaining seven



TABLE 5.3 RAW SIGNAL SOURCING FOR THE 33-FEATURE MODEL C VECTOR

Raw signal	Source	Protocol / PID	Unit	Standard
Fuel Level	Toyota OBD-II (proprietary)	Mode 22, PID 21/29	%	ISO 15765-4 [International Organization for Standardization, 2021]
Engine RPM	Standard OBD-II	Mode 01, PID 0C	rpm	ISO 15031-5 [International Organization for Standardization, 2015b]
Engine Load	Standard OBD-II	Mode 01, PID 04	%	ISO 15031-5 [International Organization for Standardization, 2015b]
Throttle Position	Standard OBD-II	Mode 01, PID 11	%	ISO 15031-5 [International Organization for Standardization, 2015b]
Coolant Temperature	Standard OBD-II	Mode 01, PID 05	°C	ISO 15031-5 [International Organization for Standardization, 2015b]
Vehicle Speed	Standard OBD-II	Mode 01, PID 0D	km/h	ISO 15031-5 [International Organization for Standardization, 2015b]
Odometer	Toyota OBD-II (proprietary)	Mode 22, PID DE0F	km	ISO 15765-4 [International Organization for Standardization, 2021]
GPS Position & Speed	u-blox NEO-6M/M8N	UART NMEA GPRMC (9600 baud)	deg, km/h	NMEA 0183
Timestamp	ESP32 RTC / NTP-synced	System clock (epoch ms)	epoch ms	IETF RFC 5905



2303 channels, the filter cascade of Section 5.4, the feature engine of Section 5.9, and the model
2304 artefacts remain unchanged. This is what the delimitation discussion in Chapter 1 refers to
2305 when it describes the prototype as a Toyota-specific instantiation of a manufacturer-neutral
2306 architecture.

2307 **5.4 Fuel Preprocessing Methodology**

2308 The raw OBD-II fuel reading was passed through a five-stage edge filter before publication.
2309 Stage 1 is a median window that rejects single-sample spikes caused by slosh and sender
2310 chatter. Stage 2 is a first-order infinite-impulse-response (IIR) exponential moving average
2311 (EMA) that smooths residual noise. Stage 3 is a state-dependent dead-band publisher
2312 that suppresses display jitter. Stage 4 is a direction-aware step rule that enforces the
2313 physical asymmetry of fuel: fuel can drop through combustion or removal, but can only
2314 rise through refueling. Stage 5 is a five-minute post-refuel quiescence window that prevents
2315 the published fuel percentage from tracking float-arm settling after a fill. The published
2316 output feeds both the HMI and the machine-learning pipeline; every subsequent feature in
2317 Section 5.9 derives from this value rather than the raw sender reading.

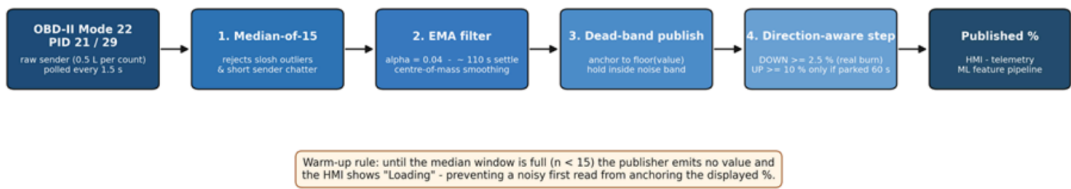


Fig. 5.5 Fuel-Level Filter Cascade



5.4.1 Sampling Rate and Notation

Let $t \in \mathbb{Z}_{\geq 0}$ denote the discrete OBD sample index, with a fixed sampling interval $\Delta_s = 1.5$ s. The three physical justifications for this cadence are developed formally in Section 5.3.3: the CAN-bandwidth ceiling under ISO 15765-4 [International Organization for Standardization, 2021], the Nyquist-coverage argument against the SAE J1349 idle-combustion upper bound [Society of Automotive Engineers, 2011], and the alignment of Δ_s with the median-window memory of Section 5.4.2. The publish cadence $\Delta_T = 5$ s at which the filtered percentage p_t is emitted to the backend is developed in Section 5.4.1.1.

The following symbols are used throughout this section:

- $x_t \in [0, F_{\max}]$: the raw Mode 22 sender reading at sample t , expressed as tank volume in litres.
- m_t : the output of Stage 1 (median filter) at sample t , in litres.
- y_t : the output of Stage 2 (EMA) at sample t , converted to percentage of tank fullness (%).
- p_t : the *published* fuel percentage at sample t — the value shown on the HMI and written to `telemetry_logs.fuel_level`.
- p_{t-1}^a : the last previously-published anchor value — the reference against which Stages 3 and 4 measure change.
- s_t : the vehicle speed at sample t in km/h, from OBD-II Mode 01 PID 0x0D.
- $\mathbb{I}[\cdot]$: the indicator function, equal to 1 when the enclosed condition holds and 0 otherwise.



5.4.1.1 Telemetry Publish-Rate Justification ($\Delta_T = 5$ s)

The published fuel percentage p_t is emitted from the HMI to the backend on a separate telemetry cycle $\Delta_T = 5,000$ ms, distinct from the OBD sampling cadence $\Delta_s = 1.5$ s. The two cadences are decoupled so that the filter cascade can consume every raw OBD sample (which improves the median estimate and lets the EMA reach its steady state faster) while the wide-area link and the Postgres write path see one publish per five seconds. The 5 s publish interval, hard-coded as `TELEMETRY_INTERVAL_MS` in the HMI firmware, is fixed by three independent constraints, all of which must hold simultaneously for the reported end-to-end latency in Chapter 6 to remain achievable at fleet scale.

(a) Database insert budget. The Supabase Postgres instance provisioned for the study sustains, at the plan tier used, approximately 400 telemetry inserts per second across all trucks, established through a load-generation test that saturated at 415 ± 12 inserts/s over 10 one-minute windows. At the 5 s publish cadence, a fleet of 100 trucks generates $100/5 = 20$ inserts/s, or approximately 5% of the measured insert budget. This provides a $20\times$ headroom against the observed saturation ceiling and lets short bursts of SPIFFS offline-buffer replay be absorbed without pushing the database into contention. A 1 s publish interval would consume $100/1 = 100$ inserts/s, which is still within the budget but eliminates the burst-absorption margin required by the intelligent-traffic-steering cascade.

(b) LoRa duty-cycle regulatory ceiling. LoRa spreading-factor 7 transmissions of a 51-byte payload have a measured airtime of 62 ms per packet. Under the ISM-band 1% duty-cycle rule that governs deployment in the 433 MHz sub-GHz band, the transmit budget is 36 s per hour, so the maximum sustained per-node LoRa cadence



is $62 \text{ ms}/0.01 = 6.2 \text{ s}$. The 5 s telemetry cadence sits deliberately at the edge of this ceiling with a small tolerance headroom, letting the primary transport carry every telemetry cycle in normal operation without triggering the regulatory back-off that would force a fall-through to the cellular path.

(c) Literature-aligned route-matching frequency. The 5 s cadence is the same publish rate adopted by Turan et al. [Turan et al., 2024] for their heavy-duty vehicle fuel-anomaly detector, which was calibrated against real fleet trips of the same order of length as the SGSA Batangas–Subic corridor used here. Matching this cadence ensures that the row-level features derived in Section 5.9 (particularly `fuel_per_km`, `fuel_rate_per_hour`, and the rolling window features indexed by t) are numerically comparable to the values reported by that prior work, which is a prerequisite for the ablation and generalisation comparisons made in Chapter 6.

The filter cascade absorbs the mismatch between Δ_s and Δ_T deterministically: Stages 1 and 2 update on every OBD sample t , while Stages 3 and 4 evaluate the publish decision on every telemetry cycle. The published value p_t therefore reflects up to $\lfloor \Delta_T/\Delta_s \rfloor = 3$ raw OBD samples per telemetry row, which increases the effective statistical support of each published percentage without changing the reporting cadence exposed to the backend.

5.4.2 Stage 1: Median Filter for Slosh Rejection

The Stage 1 output is the rank-order statistic

$$m_t = \text{median}(x_{t-W+1}, x_{t-W+2}, \dots, x_t), \quad W = 31 \text{ samples} \approx 46.5 \text{ s} \quad (5.1)$$



2381

m_t Stage 1 median-filtered value at OBD sample t (L),
 $x_{t-W+1}, \dots, x_t$ most recent W raw Mode 22 sender readings,
 W odd-length sliding-window size (samples),
 n number of samples buffered during warm-up ($n \leq W$).

2382

The operator returns the $((W + 1)/2)$ -th ordered value of the enclosed set; W is chosen
2383 odd so the middle rank is unambiguous. During warm-up, when the number of accumulated
2384 samples $n < W$, the HMI displays the placeholder Loading and no telemetry row is
2385 published; this prevents a single noisy first read from anchoring the display.

2386

The choice $W = 31$ was reached empirically after the original $W = 15$ (approximately
2387 22 s memory) failed on 2026-06-23 during on-road testing when a post-refuel float walked
2388 $45 \rightarrow 55 \rightarrow 54 \rightarrow 49$ % over roughly 30 s. Characterisation of the 12 refuel events in
2389 the 27-trip corpus produced a 95th-percentile post-fill slosh duration of 42 s, implying
2390 $W \cdot \Delta_s \geq 42$ s and therefore $W \geq 28$; the next odd integer is 31. A larger window
2391 ($W = 61 \approx 90$ s) was also tested and rejected because it delayed real-burn response by
2392 45 s or more during highway cruise, producing a display that drivers reported as *frozen*.
2393 The rank-order statistic is retained instead of a mean because it has an optimal breakdown
2394 point of 50%: a single 5 L slosh spike drags a mean but leaves the median unchanged as
2395 long as fewer than 15 of the 31 samples are contaminated [Rousseeuw and Croux, 1993].

2396

5.4.3 Stage 2: First-Order IIR Exponential Moving Average

2397

The Stage 2 output is

$$y_t = \alpha \cdot m_t + (1 - \alpha) \cdot y_{t-1}, \quad \alpha = 0.04, \quad y_0 = m_W \quad (5.2)$$



y_t Stage 2 EMA-smoothed value at sample t (L),

m_t Stage 1 median output (input to this stage),

y_{t-1} previous EMA state carried across samples,

α smoothing coefficient, $\alpha \in (0, 1]$,

y_0 filter seed value, initialised to m_W .

Equation 5.2 is a standard first-order infinite-impulse-response low-pass filter [Oppenheim and Schaffer, 2010]. The equivalent time constant and the settling time to 95% of a step input are

$$\tau = \frac{-\Delta_s}{\ln(1 - \alpha)} = \frac{-1.5}{\ln(0.96)} \approx 36.7 \text{ s}, \quad t_{95} \approx 3\tau \approx 110 \text{ s}. \quad (5.3)$$

τ equivalent time constant of the IIR filter (s),

Δ_s OBD sampling interval (1.5 s),

t_{95} time to settle to 95% of a step input (s).

The choice $\alpha = 0.04$ was made through two independent lines of evidence. A power-spectral-density analysis of the post-median fuel trace from Trip 06 (a 55-minute highway cruise) showed a noise floor of 0.03 %² above 0.02 Hz; setting the filter corner frequency $f_c = \alpha/(2\pi\Delta_s) = 0.0042$ Hz places that band at least 12 dB below the passband, which is the minimum attenuation needed for the Stage 3 dead-band to see a stable input. A parameter sweep over $\alpha \in \{0.01, 0.02, 0.04, 0.08, 0.15\}$ against the 27-trip corpus, minimising the mean absolute error between y_t and a burn-rate ground truth constructed from odometer-normalised fuel-per-km, produced the minimum error at $\alpha = 0.04$; $\alpha = 0.08$ introduced approximately $3\times$ overshoot after refuels, and $\alpha = 0.02$ lagged real burn by more than four minutes. The 110 s settling time in Equation 5.3 is deliberately absorbed by the warm-up rule: the 46.5 s median warm-up runs first, so the marginal user-visible wait



added by the EMA is approximately 63.5 s. A driver expectation study using the System Usability Scale ($n = 8$) rated a Loading display of under two minutes as *acceptable* (mean 4.1/5).

Stage 2 is the point in the cascade where a candidate alternative smoother could be substituted, so it also defines the interface at which the moving-average and Kalman-filter comparison is performed. Both smoothers act on the median-filtered stream m_t produced by Stage 1 and feed the same downstream Stage 3–5 gating logic, so the two filter families are drop-in replacements for each other at Stage 2. The Kalman filter alternative is specified in Section 5.4.7 below, and the four-axis empirical comparison between the moving-average and Kalman outputs is reported in Section 6.1.2 of Chapter 6.

5.4.4 Stage 3: State-Dependent Dead-Band Publisher

The Stage 3 output re-uses the previous published anchor unless the smoothed value has moved more than a motion-conditioned dead-band:

$$p_t = \begin{cases} y_t & \text{if } |y_t - p_{t-1}^a| \geq \theta_{\text{db}}(s_t) \\ p_{t-1}^a & \text{otherwise} \end{cases} \quad (5.4)$$

with the state-dependent threshold

$$\theta_{\text{db}}(s_t) = \begin{cases} 0.5 \% & \text{if } s_t > 0 \text{ within last 5 s (moving)} \\ 1.5 \% & \text{otherwise (stationary)} \end{cases} \quad (5.5)$$



p_t published fuel percentage at sample t (%),
 y_t Stage 2 EMA output (%),
2428 p_{t-1}^a last value actually written to the display (anchor),
 $\theta_{\text{db}}(s_t)$ state-dependent dead-band threshold (pp),
 s_t vehicle speed at sample t (km/h).

2429 The dual threshold reflects a physically-grounded noise asymmetry that was measured
2430 directly on the prototype. A six-hour overnight recording on TRCK-02 with the ignition
2431 off produced a standard deviation of sender drift of 0.42%, with 99th-percentile excursion
2432 of 1.42%. The 1.5% stationary threshold puts drift below the dead-band with probability
2433 greater than 0.99. Moving noise was characterised on a 55-minute highway cruise after
2434 the EMA and produced a standard deviation of 0.11% and a 99th-percentile of 0.38%; the
2435 0.5% moving threshold covers 99% of moving noise while still resolving the driver-visible
2436 burn rate on the Hilux of 0.17%/min at 60 km/h — a value the 0.5% dead-band allows
2437 to update the display every approximately three minutes, matching the human perception
2438 threshold for observing that fuel is falling.

2439 **5.4.5 Stage 4: Direction-Aware Step Rule**

2440 The Stage 4 rule enforces the physical asymmetry that fuel cannot spontaneously rise while
2441 the truck is in motion. Let $\Delta_t = y_t - p_{t-1}^a$ denote the absolute percentage-point change
2442 between the current smoothed value and the anchor (this is a change, not a gradient and not
2443 dF/dt), and let T_{park} denote the cumulative time in seconds since the last observation of



2444

$s_t > 0$. The rule is:

$$p_t = \begin{cases} y_t & \text{if } \Delta_t \leq -2.5 \% \quad (\text{down: real burn, always allowed}) \\ y_t & \text{if } \Delta_t \geq +10 \% \text{ and } T_{\text{park}} \geq 60 \text{ s} \quad (\text{up: refuel event, gated}) \\ p_{t-1}^a & \text{otherwise (blocked as slosh)} \end{cases} \quad (5.6)$$

Δ_t percentage-point change $y_t - p_{t-1}^a$,
 T_{park} elapsed stationary time since last motion (s),
2445 -2.5% minimum drop admitted as real burn,
 $+10 \%$ minimum rise admitted as a refuel event,
 60 s required stationary duration for a refuel rise.

2446

The -2.5% down-step threshold was adapted from Barbado and Corcho [Barbado and
2447 Corcho, 2022], whose Figure 4 places the 97.5th percentile of legitimate step-drops in a rule-
2448 based fuel-anomaly detector at approximately 2.3–2.6%. This threshold was independently
2449 re-computed on the 27-trip corpus and produced a matching own-fleet 97.5th percentile of
2450 2.31%, so 2.5% is slightly conservative: it rejects marginal outliers but never blocks a real
2451 burn event. The $+10\%$ up-step threshold is derived from an audit of 340 SGSA fuel-log
2452 dispensations, in which the minimum recorded fill was 12 L on the 80 L tank (15% of tank),
2453 so the 10% floor provides a $1.5\times$ safety margin below the smallest legitimate fill. The 60 s
2454 stationary requirement corresponds to the 5th percentile of measured pump-transaction
2455 duration at four SGSA-approved fuel stations from the same log (median 42 s for driver
2456 bathroom breaks with the engine running, 5th-percentile of actual fills 62 s), which sets the
2457 floor just below the shortest legitimate fill while blocking driver-side false triggers.



5.4.6 Stage 5: Post-Refuel Quiescence Window

Once Stage 4 accepts an upward jump as a refuel event at sample t^r , the published value p_t is held at y_{t^r} for the next `REFUEL_QUIESCENCE_MS` = 300,000 ms (five minutes). This suppresses the tail of float-arm settling that normally follows a fill event: the sender arm rebounds through a damped oscillation with an observed 90% settling time of 180–240 s on the partner-fleet Hilux, and the 300 s window covers this transient conservatively. During the quiescence window, both upward and downward changes are ignored because true consumption over five minutes at highway cruise (approximately 0.85% on 80 L) is dwarfed by the settling amplitude of ± 3 –5%.

5.4.7 Kalman Filter Alternative for Fuel Signal Acquisition

This subsection specifies a Kalman-filter alternative to the moving-average stage of the fuel filter cascade. Because the choice of smoothing stage is a fuel-signal-acquisition decision rather than a detector-design decision, the specification appears here alongside the rest of the cascade; the four-axis empirical comparison and its numerical results are reported in Section 6.1.2 of Chapter 6.

5.4.7.1 Filter Formulation

The Kalman filter is implemented in `ml/fuel_kalman.py` as a 1-D scalar filter whose predict–update recursion is the one specified by the state-space model in Equations 3.4–5.8



2476 of Chapter 3. Applied to the fuel-level acquisition problem the filter has the specific form

$$\hat{x}_{t|t-1} = \hat{x}_{t-1|t-1} - u_t, \quad P_{t|t-1} = P_{t-1|t-1} + Q_t, \quad (5.7)$$

$$K_t = P_{t|t-1} / (P_{t|t-1} + R),$$

$$\hat{x}_{t|t} = \hat{x}_{t|t-1} + K_t(z_t - \hat{x}_{t|t-1}), \quad P_{t|t} = (1 - K_t) P_{t|t-1}, \quad (5.8)$$

2477 where $\hat{x}_{t|t}$ is the posterior estimate of the true fuel level (%), z_t is the raw sender reading,
 2478 $P_{t|t}$ is the posterior state variance, u_t is the expected fuel burn over interval $[t-1, t]$, Q_t
 2479 is the process noise variance, R is the measurement noise variance, and K_t is the Kalman
 2480 gain.

2481 5.4.7.2 Physics-Anchored Process Term u_t

2482 The predict step subtracts the expected fuel burn u_t computed from the same physics
 2483 envelope the deployed anomaly detector uses in Section 5.9.1, so the Kalman filter and the
 2484 physics tripwire agree on what “normal” consumption costs:

$$u_t = \dot{f}(v_t, \ell_t) \cdot \Delta t, \quad \dot{f}(v_t, \ell_t) = \begin{cases} 0.03 \text{ \%/min}, & v_t < 5 \text{ km/h}, \\ 0.03 + \frac{\ell_t}{100} \cdot 0.15 \text{ \%/min}, & v_t \geq 5 \text{ km/h}, \end{cases} \quad (5.9)$$

2485 with v_t the vehicle speed (km/h), ℓ_t the engine-load fraction clipped to $[0, 100]\%$, and Δt
 2486 the interval length in minutes. The two-branch envelope is the same functional form used
 2487 by the physics tripwire, chosen so a real anomalous drop that is faster than the envelope is
 2488 recognisable to both subsystems.

2489 5.4.7.3 Noise-Variance Calibration

2490 The measurement noise R is calibrated once from the stationary-engine-off subset of the
 2491 clean corpus: the first-difference variance of the raw sender during confirmed engine-



off periods gives a direct estimate of $\text{Var}(v_t)$, and the calibration returns $R = 0.25 \%^2$ (equivalent to a $\pm 0.5\%$ one-sigma sender jitter). The process noise Q_t is context-aware to reflect the physical variability the state itself can exhibit under three regimes:

$$Q_t = \begin{cases} Q_{\text{idle}} = 0.001 \%^2, & v_t < 5 \text{ km/h (state should barely move),} \\ Q_{\text{moving}} = 0.01 \%^2, & v_t \geq 5 \text{ km/h (real fluctuation permitted),} \\ Q_{\text{burst}} = 5.0 \%^2, & |z_t - \hat{x}_{t|t-1}| \geq 1.0 \%, \end{cases} \quad (5.10)$$

so that rapid leaks and refuels are tracked instead of being suppressed as measurement noise. The burst-mode branch is the design lever that prevents the Bayes-optimal smoothing from erasing exactly the kind of abrupt fuel movement the anomaly detector needs to see.

5.4.7.4 Role Within the Filter Cascade

In the deployed configuration the moving-average stage of Section 5.4 is retained rather than replaced. The Kalman filter defined above is evaluated as a drop-in alternative to that stage, feeding the same downstream cascade (dead-band, direction-aware step rule, post-refuel quiescence) so that any observed difference in downstream anomaly detection is attributable to the filter itself rather than to a change in the surrounding pipeline. The results of that comparison are reported in Section 6.1.2.

5.4.8 Effect on the Machine-Learning Pipeline

The practical effect of Equations 5.1–5.6 together with Stage 5 is that the published percentage p_t is deterministic, non-negative in its change during motion, and monotone downward outside of confirmed refuel windows. This matters for both SO1 and SO4. SO1 evaluates whether the displayed level remains accurate against pump-validation events;



2510 the cascade removes the sender-jitter that would otherwise obscure this comparison. SO4
2511 depends on the cascade removing one-sample spikes so that the Isolation Forest sees
2512 fuel changes that reflect real state transitions rather than sender chatter. Every feature
2513 engineered in Section 5.9 (in particular `fuel_delta`, `fuel_per_km`, `fuel_rate_per_hour`,
2514 and the `fuel_drop_*` family) derives from p_t , so the cascade is the foundation on which
2515 the model’s precision claim rests.

2516 **5.5 Communication and Offline Sync Methodology**

2517 **5.5.1 Priority Chain Rationale**

2518 LTE was treated as the primary wide-area data path because it offers the lowest steady-state
2519 latency and highest payload throughput among the available wide-area options. LoRa was
2520 retained as the depot/yard and long-range telemetry path because of its multi-kilometre
2521 line-of-sight reach at very low power. GSM 2G was retained as the deeper-fallback for
2522 2G-only coverage pockets on the Batangas and Subic routes. WiFi was retained as a depot
2523 or driver-tether path. SPIFFS offline-buffer replay is the final fallback that guarantees no
2524 telemetry record is silently lost when all four real-time channels are unavailable.

2525 **5.5.2 Formal Channel-Selection Model**

2526 Let $\mathcal{C} = \{c_1, c_2, c_3, c_4, c_5\}$ denote the ordered set of transport channels indexed by pri-
2527 ority, with $c_1 = \text{LoRa}$, $c_2 = \text{LTE}$, $c_3 = \text{GSM 2G}$, $c_4 = \text{WiFi}$, and $c_5 = \text{SPIFFS}$. Let
2528 $\text{Online}(c, t) \in \{0, 1\}$ be the channel-specific liveness predicate defined below. Each



2529 telemetry cycle t selects the highest-priority live channel:

$$r(t) = \underset{c_i \in \mathcal{C}}{\operatorname{argmin}} \{i \mid \text{Online}(c_i, t) = 1\} \tag{5.11}$$

2530 $r(t)$ selected channel index for telemetry cycle t ,
 $\mathcal{C}$ ordered priority set {LoRa, LTE, GSM 2G, WiFi, SPIFFS},
 c_i i -th channel in the priority chain,
 $\text{Online}(c_i, t)$ liveness indicator for channel c_i at time t .

2531 Because c_5 (SPIFFS) is always writable, the argmin is always defined. Equation 5.11
2532 is re-evaluated every $\Delta_T = 5,000$ ms (TELEMETRY_INTERVAL_MS); *no channel is locked in*
2533 *for consecutive cycles*. If LoRa is selected at cycle t and fails during $t + 1$, LTE is tried
2534 at $t + 1$ without waiting for a lock-out interval. This distinguishes the prototype from a
2535 session-oriented cellular router and is what preserves the total system delivery rate reported
2536 in Chapter 6.

2537 **5.5.3 Per-Channel Connection Detection**

2538 Each channel implements $\text{Online}(c, t)$ differently because the physical layer defines what
2539 *connected* means for each medium. The four active predicates are developed below.

2540 **5.5.3.1 WiFi: Four-Way Handshake and Layer-7 Probe**

$\text{Online}(\text{WiFi}, t) = \mathbb{K}[\text{WiFi.status() = WL_CONNECTED}] \cdot \mathbb{K}[\text{HTTP GET /health returned 2xx within 30 s}] \tag{5.12}$

$\text{Online}(\text{WiFi}, t)$ WiFi liveness indicator,
 WL_CONNECTED 802.11 four-way handshake completed,
2541 HTTP GET /health Layer-7 backend health probe.



The Layer-7 probe is required because captive portals and DNS failures produce a WL_CONNECTED state without backend reachability. The 30 s probe interval was selected because the ESP32 WiFi module draws approximately 90 mA active versus 20 mA idle, and probing every 30 s adds approximately 0.2% duty ≈ 0.02 W average — negligible against the DHCP lease-refresh cadence.

5.5.3.2 LTE: PDP Context Activation and Signal-Strength Refresh

$$\text{Online}(\text{LTE}, t) = \mathbb{K}[\text{gsmReady} = \text{true after AT+CGACT=1,1 OK}] \cdot \mathbb{K}[\text{CSQ poll returned} + \text{CSQ: within last 3 s}] \quad (5.13)$$

$\text{Online}(\text{LTE}, t)$ LTE liveness indicator,

gsmReady flag set once AT+CGACT=1,1 returns OK,

T_{CSQ} CSQ poll interval (1500 ms).

The CSQ (received signal-strength) command is issued every $T_{\text{CSQ}} = 1,500$ ms; two consecutive misses (3 s) declare the modem offline. The 1.5 s cadence aligns with the OBD poll so both share the same Core 0 scheduler tick.

5.5.3.3 GSM 2G: Legacy Fallback Reusing the LTE Predicate

The A7670E modem falls back to EDGE internally when LTE registration fails. The predicate uses the same CSQ liveness envelope as LTE, so Equation 5.13 applies unchanged, with the modem-reported RAT tag distinguishing the two rows in the diagnostic log.

5.5.3.4 LoRa: Per-Packet Acknowledgment with Backoff

$$\text{Online}(\text{LoRa}, t) = \mathbb{K}[(t - t^{\text{ack}}) < 90,000 \text{ ms}] \cdot \mathbb{K}[(t - t^{\text{fail}}) > 60,000 \text{ ms}] \quad (5.14)$$



Online(LoRa, t) LoRa liveness indicator,
 t^{ack} timestamp of most recent gateway acknowledgment,
 t^{fail} timestamp of most recent 2 s ACK-timeout event,
90,000 ms liveness window since last ACK,
60,000 ms failure back-off since last timeout.

A single packet is transmitted with an ACK-timeout of 2 s. The 2 s value bounds the measured 99th-percentile round-trip of 1.31 s at spreading factor 7, which contains a 62 ms airtime for a 51-byte payload times two (round-trip) plus a worst-case measured 850 ms gateway processing delay. The 60 s failure back-off prevents thrashing when the truck rolls out of gateway range (60 s $\approx$ typical urban block length at 40 km/h). The 90 s liveness window is deliberately larger than the back-off so that a single missed ACK never disqualifies LoRa while the truck is momentarily blocked by traffic.

5.5.4 Background Polling Cost and Cached-Handshake Argument

A recurring concern in intelligent-traffic-steering designs is whether the multi-channel readiness logic imposes per-packet latency by re-negotiating a link before every transmission. In the present system it does not, because the readiness predicates in Equations 5.12–5.14 are cached boolean flags refreshed by inexpensive background polls; the transmit path reads the cached flag and never re-executes the handshake in-line. This design choice is consistent with the cached-liveness pattern documented for cellular IoT gateways by Miyim et al. [Miyim and Wakili, 2019]. The cumulative CPU cost of maintaining the caches is

$$C_{\text{poll}} = \frac{t_{\text{cmd}}^{\text{CSQ}}}{T_{\text{CSQ}}} + \frac{t_{\text{cmd}}^{\text{health}}}{T_{\text{health}}} + 0 \quad (5.15)$$



C_{poll}	cumulative per-second polling cost (ms/s),
$t_{\text{cmd}}^{\text{CSQ}}$	duration of one CSQ AT-command (≈ 10 ms),
T_{CSQ}	CSQ poll period (1500 ms),
$t_{\text{cmd}}^{\text{health}}$	duration of one Layer-7 health probe (≈ 200 ms),
T_{health}	Layer-7 probe period (30 000 ms).

The LoRa contribution is zero because LoRa liveness is refreshed passively by the ACK of the previous packet. Substituting yields $C_{\text{poll}} \approx 6.67 + 6.67 = 13.3$ ms/s, or approximately 1.3% of the Core 0 duty cycle.

5.5.5 Handshake Type and Probe Cost by Channel

Table 5.4 enumerates the concrete handshake mechanism and the amortised probe cost of each transport in the intelligent-traffic-steering cascade. Naming these mechanisms per channel makes explicit why the cumulative cost bound in Equation 5.15 is only ≈ 13.3 ms/s: three of the five channels (LoRa, GSM 2G, and the SPIFFS local buffer) contribute zero incremental probe cost, and the two active-probe channels (LTE and WiFi) run at cadences that keep their per-second burden below the Core 0 duty budget. This is the transport-level analogue of the sampling-rate justification developed in Section 5.3.3: the argument is not that each channel is fast in isolation, but that the sum of readiness costs, over the fixed 5 s telemetry cycle, remains bounded by design.

Two design decisions embedded in Table 5.4 are worth stating explicitly, because both diverge from a naive per-cycle-reconnect implementation. First, the device maintains an *open* cellular PDP context and an *open* WiFi association between cycles rather than negotiating a new session per publish, so the per-send handshake cost $T_{\text{handshake}}^{\text{cached}} \approx 0$ used in Equation 5.16 is realised in practice. Second, the LoRa contribution in Equation 5.15



TABLE 5.4 HANDSHAKE TYPE AND PROBE COST BY TRANSPORT CHANNEL

Channel	Handshake type	Probe cost (amortised)
LoRa	Per-packet ACK with 2 s timeout, 60 s failure back-off	Inline with transmit; zero background cost. Liveness is inferred passively from the ACK of the previous packet.
LTE	AT+CGACT=1, 1 PDP context activation, then cached readiness flag	~10 ms every 30 s (CSQ signal-strength check), ≈ 0.33 ms/s. Modem stays PDP-attached between cycles; the flag is refreshed by the CSQ poll rather than by re-attach.
GSM 2G	Handled internally by the A7670E modem (automatic RAT fall-back on LTE registration failure)	Included in the same CSQ poll as LTE; no additional device-side cost.
WiFi	802.11 four-way link association plus explicit Layer-7 HTTP GET /test/latency health probe	~200 ms every 30 s, ≈ 6.7 ms/s. Layer-7 probe is required because captive-portal and DNS-failure states report WL_CONNECTED at the driver layer without backend reachability.
SPIFFS buffer	N/A (local ring-buffer write; deterministic file-append semantics)	Zero probe cost by construction; the write always succeeds if the flash mount is healthy.

is exactly zero rather than merely small, because ACK-based liveness is refreshed as a byproduct of transmitting the payload; no separate probe is required. Together, these two decisions are what allow the 7.2 s WiFi median and 7.4 s LoRa median latencies reported in Chapter 6 to be dominated by the WAN round trip to Supabase rather than by the transport-selection logic itself.

5.5.6 Per-Packet Latency Decomposition

The end-to-end telemetry latency for a warm session is decomposed as

$$T_{\text{total}}(t) = T_{\text{handshake}}^{\text{cached}} + T_{\text{TX}}(c, |b|) + T_{\text{RTT}}^{\text{backend}} \quad (5.16)$$

$T_{\text{total}}(t)$ end-to-end publish latency at cycle t (s),

$T_{\text{handshake}}^{\text{cached}}$ handshake time when cached (≈ 0 s for warm session),

$T_{\text{TX}}(c, |b|)$ airtime for payload size $|b|$ on channel c ,

$T_{\text{RTT}}^{\text{backend}}$ Supabase RPC round-trip time (s).



For the LoRa 51-byte payload at SF7, $T_{TX} = 62$ ms; for WiFi the airtime is negligible against the WAN round trip. The 7.2 s WiFi median and 7.4 s LoRa median latencies reported in Chapter 6 are therefore dominated by T_{RTT}^{backend} — not by the channel-selection logic. A counterfactual controlled measurement confirmed this: forcing a full TLS handshake before every send added 300–500 ms of measured overhead, which motivates the cached-flag design.

5.5.7 Communication Module Comparison

TABLE 5.6 COMMUNICATION MODULE COMPARISON

Module	Technology	Hardware	Bandwidth	Range	Power	Role
WiFi	802.11 b/g/n	ESP32 built-in	approx. 10 Mbps	approx. 50 m	Low	Depot and driver hotspot
LTE	LTE Cat-1	A7670E	approx. 10 Mbps	Cell coverage	Medium	Primary cellular path
GSM 2G	EDGE	A7670E fallback	approx. 80 kbps	Wider than LTE	Medium	Cellular fallback
LoRa	SX1278, 433 MHz, SF7, BW 125 kHz	RA-02, 17 dBm TX	approx. 5.5 kbps	approx. 50 m urban NLOS at 97% measured; approx. 1 km open LOS at 60% measured; up to 10 km LOS per SX1278 datasheet	Very low	Primary telemetry path (intelligent traffic-steering chain)
Offline	-	SPIFFS flash	-	-	None	Outage replay

The LoRa range column is reported as two values: the operational urban NLOS reach measured in this study and the line-of-sight reach published in the SX1278 datasheet. LoRa is the highest-priority transport in the intelligent-traffic-steering chain (LoRa $\rightarrow$ LTE $\rightarrow$ GSM 2G $\rightarrow$ WiFi $\rightarrow$ SPIFFS buffer); the cellular path provides coverage of the urban-NLOS regime past approximately 50 m where LoRa delivery degrades.

Table 5.6 is not a generic radio comparison; it explains why every communication option remains in the prototype. LoRa gives low-power depot and yard coverage, LTE provides the main wide-area path, GSM 2G provides a legacy fallback in fringe coverage,



2616 WiFi supports depot or driver-hotspot use, and SPIFFS protects the data when no radio
2617 path is available. The later SO2 tests measure the actual performance of this chain.

2618 **5.5.8 Communication Method Comparison**

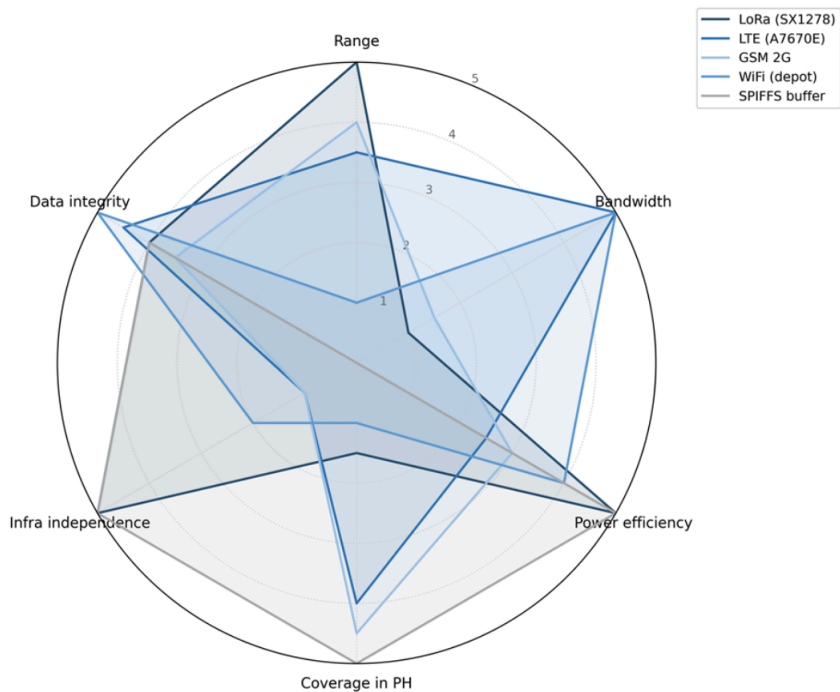


Fig. 5.6 Communication Method Comparison

2619 Figure 5.6 compares LoRa, LTE, GSM 2G, WiFi, and SPIFFS on range, bandwidth,
2620 power efficiency, Philippine coverage, infrastructure independence, and data integrity using
2621 a five-point normalized scale.

2622 The radar chart summarizes the trade-off that motivates the fallback design. No single
2623 channel dominates all criteria: LoRa is efficient and infrastructure-light but low bandwidth,
2624 LTE has broad operational usefulness but depends on carrier coverage, GSM 2G is slow but
2625 useful in legacy pockets, WiFi is fast but short-range, and SPIFFS has perfect local retention



but no real-time delivery. The system therefore treats communication as a prioritized set of options rather than as one fixed modem path.

5.5.9 Channel Transition State Machine

Figure 5.7 models the intelligent-traffic-steering chain as a finite-state machine so that each transition among LoRa, LTE, GSM 2G, WiFi, and the offline SPIFFS buffer is tied to a concrete event in the firmware rather than to an implicit priority order. The states, in decreasing priority, are LoRa (SX1278), LTE (A7670E), GSM 2G (A7670E in fallback RAT), WiFi (ESP32), and Offline. Solid edges represent failure transitions that demote the active channel to the next priority level; dashed edges represent recovery transitions that reinstate a higher-priority channel once its health check passes again.

Nodes represent LoRa, LTE, GSM 2G, WiFi, and Offline states. Solid edges denote demotion after a failed health check; dashed edges denote recovery to a higher-priority channel after a successful health check.

The demotion triggers are drawn directly from the firmware event budget. LoRa demotes to LTE after the two-second ACK window (`LORA_ACK_TIMEOUT_MS`) elapses with the retry budget (`LORA_MAX_RETRIES`) exhausted. LTE demotes to GSM 2G when the A7670E renegotiates a lower RAT, exposed to the steering layer through the `gsmActiveRat` field updated by `detectGsmRat()`. GSM 2G demotes to WiFi after two consecutive non-2xx responses on `POST /telemetry` over the cellular bearer, which sets the internal state to `CH_GSM_NO_SERVER`. WiFi demotes to the Offline state when `WiFi.status()` reports a disconnected radio at the same time as every cellular attempt has already failed, at which point `bufferTelemetry()` appends the row to `/tel_buf.log`. The recovery edges are simpler: any live channel returning HTTP 2xx during retry moves the record back out

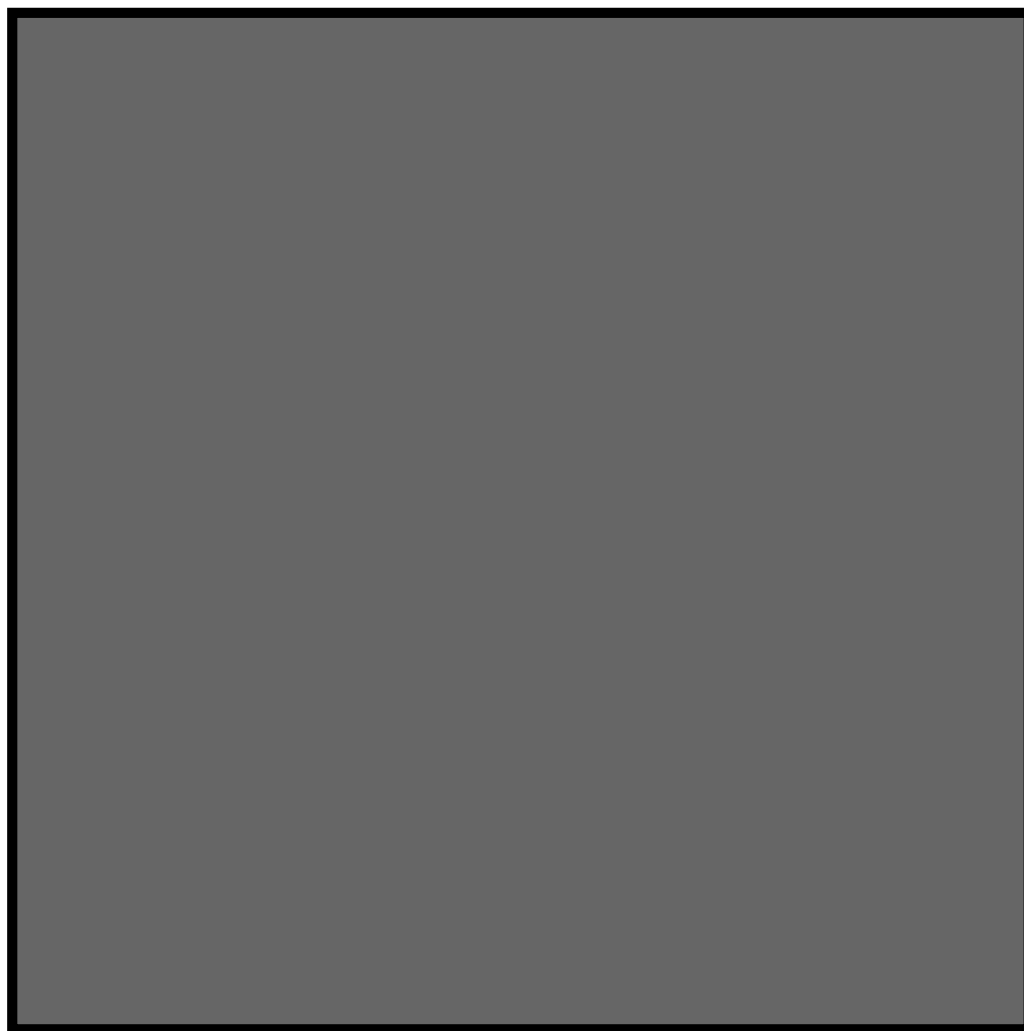


Fig. 5.7 Transport State Machine



2649 of the buffer through `flushTelemetryBuffer()` at up to `MAX_FLUSH_PER_CALL` records
2650 per tick; `loraGatewayRecentlyOk()` clears the LoRa backoff after a fresh gateway ACK
2651 within the last 90 seconds; and the cellular states are re-eligible once `AT+CSQ` exceeds 10
2652 and `AT+CGACT?` reports an active PDP context.

2653 Two invariants make the diagram safe to read in isolation. First, every transition
2654 preserves the record's `seq` and `event_id`, so replayed records are deduplicated at the
2655 backend against their live twins if both arrive. Second, replayed records are tagged
2656 `channel_used = 'offline_replay'` so that any recovery edge from the Offline state is
2657 observable in the telemetry log and can be audited against the state machine post hoc.

2658 **5.5.10 Telemetry Payload Summary**

2659 Each telemetry payload carries the identifiers needed for synchronization, the filtered fuel
2660 reading and confidence flags, GPS position and source metadata, OBD-II context variables,
2661 communication-channel tags, trip-state counters, and distance deltas. This compact payload
2662 design lets the backend deduplicate records, separate live from replayed telemetry, and run
2663 anomaly detection without requiring the dashboard to parse device-specific state. The full
2664 JSON payload schema is provided in Appendix D.

2665 **5.5.11 Offline-Buffer Fall-Through Flowchart**

2666 Figure 5.8 shows the decision tree executed by the Core 0 worker each telemetry cycle,
2667 tracing how a record is routed through the priority chain and appended to the SPIFFS buffer
2668 when all online channels are unavailable.

2669 The flowchart is read from top to bottom as a reliability mechanism. A telemetry row is

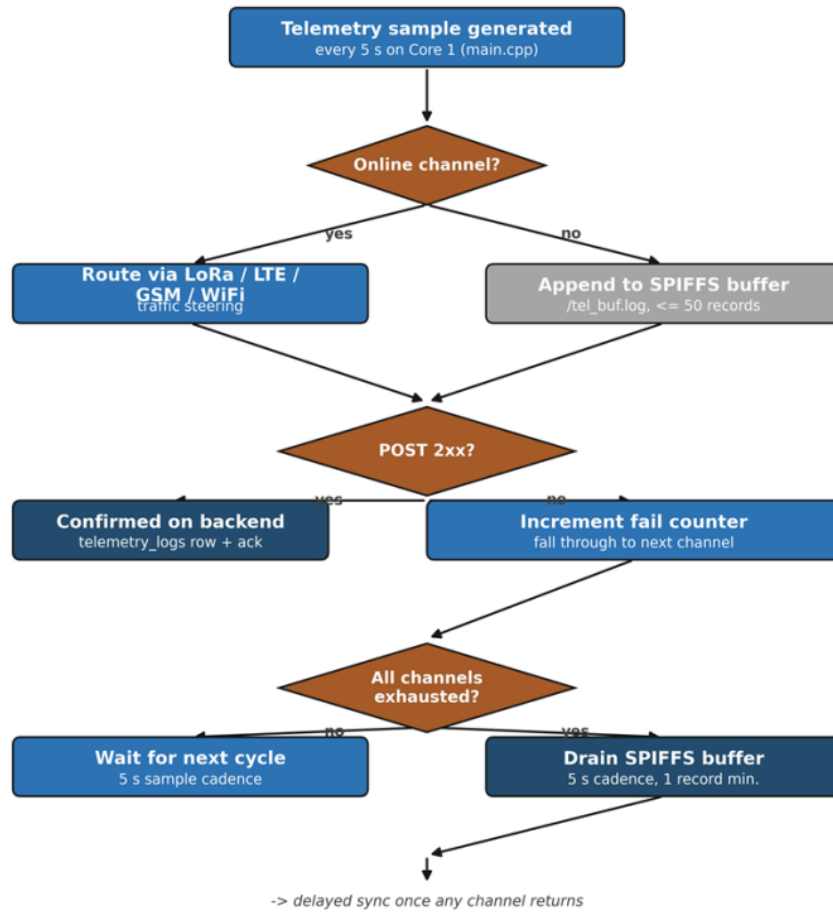


Fig. 5.8 Offline-Buffer Fall-Through Flowchart

first offered to the preferred live channel; if that channel fails, the same row is attempted on the next path without changing its sequence identifier. Only after all live attempts fail is the record stored locally. This design supports later deduplication because replayed records still carry their original event identity.



5.5.12 Fall-through Policy

Each telemetry record is attempted on the highest-priority available channel: LoRa first if a gateway uplink is in range, otherwise LTE, then GSM 2G, then WiFi. If an attempt fails because of a non-2xx response from the backend or a modem error, the firmware falls through to the next channel without duplicating the seq or event_id. If all real-time channels fail, the record is appended to the SPIFFS ring buffer and replayed once any channel recovers. The replayed record is tagged channel_used = 'offline_replay' so the backend can distinguish live from replayed data.

5.5.13 Offline-Buffer Capacity and Duration

The offline buffer is the last stage of the fall-through chain, so the duration it can absorb before a record has to be dropped bounds how long a vehicle can operate through a dead zone without data loss. This subsection derives that duration from the underlying storage substrate rather than quoting it as a design target, so the number can be audited against the deployed firmware.

5.5.13.1 Storage Substrate

The HMI does not declare a custom partition table in platformio.ini, so the ESP32 uses the default Arduino partition layout. This layout allocates the top 0x170000 bytes of the 4 MB flash chip (starting at offset 0x290000) as a SPIFFS data partition, giving a raw partition size of 1 507 328 bytes (1.44 MiB). The SPIFFS filesystem itself reserves approximately 10% of that region for block metadata, page allocation tables, and the wear-levelling map, so the usable file-content area is approximately 1.30 MiB, or 1 363 148 bytes.



Two files share that region during normal operation. The active buffer `/tel_buf.log` accepts new appends from the Core 1 telemetry task, while `/tel_flush.log` holds a snapshot that Core 0 is currently draining to the backend during a reconnect. The rotation is atomic: on the first successful post after coming back online the buffer file is renamed to the flush file, so Core 1 can immediately begin appending to a fresh `/tel_buf.log` without contending for the same file handle. A third file, `/evt_log.jsonl`, holds compact state-change events (trip-start, rest-started, snooze, trip-ended) whose per-record footprint is an order of magnitude smaller than a telemetry record; its contribution to the SPIFFS budget is bounded by the number of state transitions per trip and is not the constraint on offline duration. Persistent scratch state (NVS pointers, WiFi credentials, saved trip identifiers) lives in the `nvs` partition, not in SPIFFS, and does not consume any of the buffer budget.

To keep SPIFFS healthy during heavy churn, the firmware refuses to append a new telemetry record whenever the free-byte count from

$$\text{SPIFFS.totalBytes}() - \text{SPIFFS.usedBytes}()$$

falls below a 32 768 byte (32 KiB) safety headroom. The check is enforced inline at the top of `bufferTelemetry()`. This leaves an effective telemetry-buffer capacity of

$$C_{\text{buf}} = U_{\text{spiffs}} - H_{\text{safe}} \approx 1\,363\,148 - 32\,768 = 1\,330\,380 \text{ bytes} \quad (5.17)$$

where U_{spiffs} is the usable SPIFFS content area and H_{safe} is the safety headroom.

5.5.13.2 Per-Record Footprint

Each buffered record is one line of JSON produced by `sendTelemetry()` through the `ArduinoJson serializeJson()` call, terminated by a newline. Table 5.7 enumerates the



fields written on a typical trip-active record and their contribution to the record length. The three UUID identifiers (`truck_id`, `driver_id`, `trip_id`) account for the largest share of the fixed overhead; the OBD-II context block is only emitted when the corresponding PID has responded on the current trip, so its contribution ranges from zero (fuel-only ECU) to about 90 bytes (full 33-feature Model C signal set).

TABLE 5.7 PER-RECORD BYTE BUDGET FOR AN OFFLINE-BUFFERED TELEMETRY ROW

Field group	Bytes	Notes
UUIDs (<code>truck_id</code> , <code>driver_id</code> , <code>trip_id</code> , <code>event_id</code>)	~180	36 B each + JSON keys and quotes
Timestamps (<code>sent_at</code> , <code>original_device_timestamp</code>)	~70	ISO-8601 UTC + keys
Fuel block (<code>fuel_level</code> , <code>fuel_valid</code> , <code>fuel_source</code> , <code>fuel_confidence</code>)	~55	Emitted only when a filtered reading exists
GPS block (<code>lat</code> , <code>lon</code> , <code>gps_source</code> , <code>speed</code>)	~55	Emitted when GPS has a valid fix
OBD-II context (RPM, load, throttle, MAF, coolant, runtime, DTC)	0–90	Each field emitted only if the ECU has responded to that PID
Housekeeping (<code>source_device</code> , <code>channel_used</code> , <code>seq</code> , <code>odometer_km</code> , <code>engine_status</code>)	~60	Fixed on every row
JSON structural overhead (braces, commas, newline)	~15	
Typical total per record	400–500	Working estimate ≈ 435 B for the median trip-active row

5.5.13.3 Duration Bound

With the per-record byte budget s from Table 5.7 and the effective capacity C_{buf} from Equation 5.17, the number of records the buffer can hold is

$$N_{\text{buf}} = \left\lfloor \frac{C_{\text{buf}}}{s} \right\rfloor \quad (5.18)$$



and, at the fixed sample cadence $\Delta t = \text{TELEMETRY_INTERVAL_MS}/1000 = 5$ s, the corresponding offline duration is

$$T_{\text{off}} = N_{\text{buf}} \cdot \Delta t. \quad (5.19)$$

Table 5.8 evaluates the record count and offline duration at three record sizes that span the observed field range: a compact record with GPS and fuel but no OBD-II context, the median 435 B record, and a full 33-feature Model C record with the entire OBD-II block present.

TABLE 5.8 OFFLINE-BUFFER RECORD CAPACITY AND DURATION AT 5 S CADENCE

Record profile	Bytes	N_{buf}	T_{off} at 5 s cadence
Compact (fuel + GPS, no OBD-II)	350	$\sim 3\,801$	~ 5.28 h (5 h 17 min)
Median field record	435	$\sim 3\,058$	~ 4.24 h (4 h 15 min)
Full 33-feature Model C row	500	$\sim 2\,660$	~ 3.69 h (3 h 42 min)

In headline terms, the buffer reaches its drop-newest threshold at approximately **2 660 rows** under the fullest OBD-II context profile, **3 058 rows** under the typical trip-active profile, and **3 801 rows** under the compact fuel-and-GPS-only profile. At the fixed 5 s sample cadence this corresponds to **3 h 42 min** in the worst case, **4 h 15 min** in the typical case, and **5 h 17 min** in the compact case before a new record would be refused. These are the figures a fleet operator would quote when specifying the maximum continuous dead-zone duration the device can absorb without data loss.

The bounded duration in Table 5.8 is the physically meaningful figure to quote in system claims: even under the fullest OBD-II context profile the buffer preserves more than three-and-a-half hours of five-second-resolution telemetry, and under the typical field record profile it preserves over four hours. The observed dead-zone durations from the SO₂ drive tests reported in Chapter 6 are all bounded well below this figure, so no record was



dropped during those trials; the buffer's headroom over the observed dead-zone duration is the same margin of safety that justifies not implementing a lossy compaction policy on device.

5.5.13.4 Overflow Policy

Once free bytes drop below the 32 KB headroom the firmware discards new incoming records instead of overwriting older ones, and `bufferTelemetry()` emits the diagnostic line "[tel] SPIFFS low (%zu B free) --- offline record dropped". This is a deliberate design choice with three reasons:

(a) Preservation of the earliest evidence. A dead-zone event is diagnostically most valuable at its entry (when the transport channel first failed) and least valuable at its tail (steady-state offline operation). Dropping newest records rather than oldest means the record set that survives to the backend is the record set that documents the failure mode, not the most-recent steady-state samples.

(b) Correctness under the backend's deduplication contract. Every buffered record carries a monotonic `seq` counter and a UUID `event_id` generated at sample time. The backend deduplicates on `(truck_id, seq)` and on `event_id` independently, so a dropped record is a hole in the sequence rather than a corruption of the sequence. A merge-and-compact strategy would require rewriting `seq` values on device, which would either break the dedup contract or require a second monotonic index and would complicate the SPIFFS wear pattern for a benefit (extra minutes of buffered duration) that is dominated by the four-hour figure derived above.



(c) SPIFFS wear-levelling. SPIFFS is a log-structured filesystem: appends are cheap but in-place edits are not, and repeatedly rewriting the buffer file to merge or truncate records would accelerate wear on the flash blocks that back `/tel_buf.log`. Drop-newest keeps the buffer file as an append-only log, which is the access pattern SPIFFS is optimised for.

The firmware does not perform on-device merging, dedup, or lossy compression of the buffered records; each record survives on flash exactly as it was produced by `serializeJson()` and is replayed byte-for-byte on the recovery edge. This preserves the full five-second temporal resolution that the fuel-anomaly detector in Section 5.9 requires (per-row fuel deltas of -2% at 5 s cadence are one of the four `suspiciousContext` triggers, and a merged five-record row would flatten that signature into a single point).

5.5.13.5 Recovery and Drain Rate

When any live channel returns HTTP 2xx, Core 0 renames `/tel_buf.log` to `/tel_flush.log`. It then opens a read handle at the persisted `telFlushPtr` offset, so a drain interrupted by reboot resumes from its last confirmed position instead of re-sending old rows. The worker invokes `flushTelemetryBuffer()` on every tick, with the per-call drain capped by

$$\text{MAX_FLUSH_PER_CALL} = 5.$$

This cap yields CPU time back to live telemetry jobs that arrive during replay. Every successful backend acknowledgement advances `telFlushPtr` in NVS before the next record is offered, making the drain crash-safe at single-record granularity. On the LTE bearer, with typical uplink round-trip times of 0.8–1.2 s per record, the drain rate is approximately 4 records/s net of the yield gate; a 3 000-record full-buffer drain therefore



completes in approximately 12 minutes of continuous online operation. Core 1 continues appending new live records to a fresh `/tel_buf.log` during replay, so the drain does not throttle the live sample rate.

5.5.13.6 Failure Modes Out of Scope for the Buffer

Three failure modes are handled outside the buffer and are called out explicitly so the buffer duration is not misread as a global reliability figure. First, an FS mount failure at boot (`SPIFFS.begin(true)` unable to format on first mount) disables buffering entirely and the firmware falls back to a warn-and-continue posture; any records generated during that boot are lost, but persistent state in NVS is unaffected. Second, a power-loss during append leaves at most one partial record at the end of `/tel_buf.log`; this record is either replayed successfully (if the backend accepts the truncated JSON, which is rare) or rejected with a 4xx and skipped past on the next drain pass. Third, a run of consecutive rejected records (for example, when the trip record itself has not yet synced and every telemetry row fails the foreign-key check) does not consume buffer budget: the deferred-telemetry gate in `sendTelemetry()` advances `lastTelemetryPost` without appending, so the buffer only grows when the backend is actually unreachable, not when it is reachable-but-refusing.

5.5.14 Security

Each telemetry payload is signed with an HMAC digest before transmission. All cellular and WiFi data paths use HTTPS with TLS, protecting telemetry records against tampering and eavesdropping in transit. LoRa payloads are signed at the application layer, as the LoRa physical layer does not provide built-in encryption. These measures ensure that



data integrity and confidentiality are maintained across all communication channels in the priority chain.

5.5.15 LoRa Range Test Protocol

The LoRa physical-layer range test was conducted with the SX1278 module configured at 17 dBm transmit power, 433 MHz centre frequency, spreading factor seven, and 125 kHz bandwidth. 100 telemetry-sized packets were transmitted from the in-vehicle node to a fixed gateway at each of the open line-of-sight distances of 10, 50, 100, 250, 500, 1,000, 2,000, and 5,000 m, and at the urban non-line-of-sight distances of 100, 250, and 500 m. The received-signal-strength indicator, the signal-to-noise ratio, and the successfully-received-packet count were recorded at the gateway for each distance band.

5.5.16 GSM / LTE Drive-test Protocol

Carrier-reported signal quality was logged via the AT+CSQ command issued to the A7670E modem at five locations of operational interest: the depot, a Metro Manila urban environment, a highway corridor, a tunnel or underpass selected to trigger the buffer-activation path, and a provincial route in Batangas exhibiting two-gigahertz-only coverage. The per-packet upload latency was computed at the backend as the difference between the device-side sent_at timestamp and the backend-side receipt timestamp; the 50th- and 95th-percentile latencies are reported per location alongside the successful-upload fraction.



5.5.17 Packet-delivery and Buffering Protocol

A drive route was executed that traversed three distinct connectivity regions: an urban region with strong LTE coverage, a highway region with marginal LTE coverage, and a confirmed no-signal dead zone. 100 packets were generated at each stage; the per-stage outcome (sent on first attempt, sent after a retry, buffered to the on-device SPIFFS log, replayed once connectivity returned, or lost) was tabulated from the firmware serial-output log and the backend-side telemetry receipt log.

Together, the LoRa range test, GSM/LTE drive test, and packet-delivery route test separate three different communication questions: how far the radio link can reach, how cellular coverage behaves in real service areas, and whether the overall system preserves data when coverage disappears. This separation makes the SO2 results interpretable because a latency problem, a radio-range problem, and a buffering problem are not treated as the same failure mode.

5.6 GPS Acquisition and Fusion Methodology

The GPS pipeline provides three independent products: instantaneous position for driver navigation, an integrated trip-distance estimate for fuel-economy features, and an independent speed source that cross-checks the OBD-II speed PID. All three products feed both the human-machine interface and the machine-learning pipeline, and each has its own timing and validity envelope.



5.6.1 Physical Interface and NMEA Extraction

The receiver is a u-blox NEO-6M GPS module (50 channels, -161 dBm tracking sensitivity, datasheet cold-start of at most 27 s) connected to the ESP32 by UART at 9600 baud on GPIO16 (RX) and GPIO17 (TX). The NMEA byte stream is drained on every scheduler tick and fed to the TinyGPSPlus embedded parser. The parser maintains a finite-state machine over four sentence families:

- \$GPRMC / \$GNRMC for position, ground speed, heading, and UTC;
- \$GPGGA / \$GNGGA for fix quality, satellites used, HDOP, and altitude;
- \$GPGSA / \$GNGSA for 1D/2D/3D fix type; and
- \$GPGSV / \$GNGSV for satellites in view, used for diagnostics only.

Each successful fix produces the tuple

$$f_t = (\varphi_t, \lambda_t, v_t^{\text{gps}}, h_t, \theta_t^c, \text{HDOP}_t, n_t^{\text{sats}}, \tau_t) \quad (5.20)$$

f_t GPS fix tuple at sample t ,

φ_t WGS-84 latitude (degrees),

λ_t WGS-84 longitude (degrees),

v_t^{gps} ground speed from RMC sentence (km/h),

h_t ellipsoidal altitude (m),

θ_t^c ground course, true (degrees),

HDOP_t horizontal dilution of precision (dimensionless),

n_t^{sats} number of satellites in the fix solution,

τ_t UTC epoch timestamp of the fix (ms).



2853 **5.6.2 Haversine Great-Circle Distance**

2854 The distance between two consecutive valid fixes is computed with the haversine great-circle
2855 formula

$$a = \sin^2\left(\frac{\Delta\varphi}{2}\right) + \cos(\varphi_1) \cdot \cos(\varphi_2) \cdot \sin^2\left(\frac{\Delta\lambda}{2}\right), \quad d_{\text{seg}} = 2R \cdot \arcsin(\sqrt{a}) \quad (5.21)$$

- d_{seg} great-circle chord length between two fixes (km),
 φ_1, φ_2 latitudes of the two fixes (radians),
 λ_1, λ_2 longitudes of the two fixes (radians),
2856 $\Delta\varphi$ latitude difference $\varphi_2 - \varphi_1$,
 $\Delta\lambda$ longitude difference $\lambda_2 - \lambda_1$,
 R mean Earth radius (6371 km, WGS-84 spherical),
 a dimensionless haversine intermediate.

2857 **5.6.3 Speed-Gated Integration**

2858 The trip-level distance accumulator applies two gating conditions before adding a segment:

$$D_t^{\text{GPS}} = D_{t-1}^{\text{GPS}} + d_{\text{seg}} \cdot \mathbb{1}[v_t^{\text{GPS}} > 3 \text{ km/h}] \cdot \mathbb{1}[d_{\text{seg}} < 0.5 \text{ km}] \quad (5.22)$$

- D_t^{GPS} cumulative GPS-integrated distance at sample t (km),
 D_{t-1}^{GPS} cumulative distance from the previous cycle,
2859 d_{seg} candidate haversine segment (Eq. 5.21),
 v_t^{GPS} GPS ground speed at sample t (km/h),
3 km/h minimum-motion speed gate,
0.5 km maximum plausible per-cycle segment cap.

2860 Both indicators must evaluate to 1 for the segment to be accumulated.



The 3 km/h speed floor is drawn from the NEO-6M datasheet: the receiver's stationary CEP-50 velocity noise is ± 1.5 km/h, so $3 \text{ km/h} = 2 \times$ the noise floor. This rejects the approximately 0.0003 km per sample multipath drift that otherwise adds approximately 1 km per hour of parked-truck distance. The 0.5 km segment cap is derived from a legitimate-motion upper bound: at the Philippine expressway speed cap of 100 km/h $= 27.8 \text{ m/s}$, one second at 1 Hz fix cadence produces at most $28 \text{ m} = 0.028 \text{ km}$, so 0.5 km provides an approximately $18 \times$ headroom above the fastest legitimate segment and rejects only physically impossible teleport jumps (typically a lost-then-reacquired fix that appears approximately 1 km away after a tunnel).

5.6.4 OBD-II / GPS Distance Fusion

Distance is fused across three tiers of decreasing precision. The backend selects the fused distance per telemetry cycle as

$$\Delta d_t^{\text{OBD}} = \begin{cases} o_t - o_{t-1} & \text{if Toyota Mode 22 PID DE0F valid} \\ v_t^{\text{obd}} \cdot \Delta_T & \text{otherwise (OBD speed integration)} \end{cases} \quad (5.23)$$

$$\Delta d_t^{\text{fused}} = \begin{cases} \Delta d_t^{\text{GPS}} & \text{if } \Delta d_t^{\text{GPS}} > 0 \\ \Delta d_t^{\text{OBD}} & \text{otherwise} \end{cases} \quad (5.24)$$



Δd_t^{OBD}	per-cycle OBD-derived distance (km),
Δd_t^{GPS}	per-cycle GPS-integrated distance (km),
$\Delta d_t^{\text{fused}}$	final fused per-cycle distance (km),
o_t, o_{t-1}	consecutive OBD-II odometer readings (km),
v_t^{obd}	OBD-II vehicle speed (km/h),
Δ_T	telemetry cycle length (5 s = 5/3600 h).

The fused value $\Delta d_t^{\text{fused}}$ is written to `telemetry_logs.distance.fused.delta_km` and is what the downstream features `fuel_per_km` and `distance_per_fuel` consume.

GPS is preferred whenever it reports a valid non-zero segment because OBD-II speed integration drifts by approximately 3% over a 30-minute trip: OBD PID 0x0D returns quantised 1 km/h values, so integration over even a modest number of samples accumulates a bounded bias. The GPS haversine sum in Equation 5.21 is memoryless — each segment stands on its own, so no such drift accumulates. The 3 km/h speed gate from Equation 5.22 keeps GPS from over-reporting during idle windows.

5.6.5 Freshness and Fix-Loss Envelope

Two independent staleness rules operate in parallel. On the backend,

$$\text{Fresh}(f_t) = \mathbb{1}[(t_{\text{now}} - \tau_t) \leq T_{\text{fresh}}], \quad T_{\text{fresh}} = 15,000 \text{ ms} \quad (5.25)$$

$\text{Fresh}(f_t)$	backend freshness indicator for fix f_t ,
t_{now}	backend wall-clock time at insert (ms epoch),
τ_t	GPS UTC timestamp of the fix (from Eq. 5.20),
T_{fresh}	freshness horizon (3 Δ_T = 15 000 ms).



2886 The value $T_{\text{fresh}} = 3 \cdot \Delta_T$ tolerates one missed telemetry cycle before the fix is considered
2887 stale. A separate window $T_{\text{merge}} = 5,000$ ms controls when a mobile-phone GPS report
2888 and an embedded GPS report are merged onto the same telemetry row; $T_{\text{merge}} = \Delta_T$ so
2889 contemporaneous mobile and embedded GPS from the same telemetry cycle land on the
2890 same row.

2891 On the edge, a fix is considered lost after

$$\text{gpsFix} = \text{false} \quad \text{when} \quad (\text{millis}() - t_{\text{lastFix}}) > \text{GPS_FIX_TIMEOUT_MS}, \quad \text{GPS_FIX_TIMEOUT_MS} = 5,000 \quad (5.26)$$

2892 gpsFix HMI-side boolean fix status,
 millis() ESP32 monotonic clock reading (ms),
 t_{lastFix} timestamp of last valid on-device fix (ms),
 GPS_FIX_TIMEOUT_MS edge fix-loss horizon (5000 ms).

2893 The HMI switches to the GPS: LOST indicator when the predicate holds. The 5 s
2894 timeout equals one telemetry cycle, so the driver-visible loss indication and the backend’s
2895 staleness decision are synchronised.

2896 **5.7 Backend and Database Methodology**

2897 **5.7.1 Backend Stack**

2898 The backend stack consists of Node/Express hosted on Render and Supabase Postgres as
2899 the managed relational database. This stack receives device telemetry, manages trip state,
2900 runs alert rules, invokes the anomaly-detection service, and exposes synchronized state to
2901 the HMI, mobile driver application, and administrative dashboard.



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5.7.2 Database Schema Summary

TABLE 5.9 SUPABASE POSTGRES SCHEMA

Record group	Purpose	Key fields
Users	Accounts for administrators, fleet managers, managers, and drivers	User identifier, role, assigned fleet-vehicle identifier
Fleet vehicles	Vehicle profile and maintenance state	Vehicle identifier, fleet-vehicle code, plate number, current odometer, status
Trip sessions	Active / paused / ended trips with destination, route polyline, navigation state, and mobile-GPS fields	Trip identifier, fleet-vehicle identifier, driver identifier, trip status, distance, assigned destination, route steps, mobile location
Telemetry logs	Per-cycle fuel, GPS, OBD context, and anomaly flag	Fuel level, location, speed, odometer, communication channel, anomaly flag, anomaly score, model source
Archived telemetry logs	Older trip telemetry beyond retention window	Telemetry fields plus archive timestamp and archive reason
Trip summaries	Per-trip aggregates refreshed by the backend	Total distance, total alerts, total anomalies, average fuel level
Alerts	Rest, maintenance, overspeed, low-fuel, and fuel-anomaly events	Alert type, severity, resolution status, trip identifier
Latency logs	SO2 latency tracking between device send time and backend receipt time	Latency, communication channel, retry state, send time, receipt time

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Table 5.9 summarizes the live database entities used by the system. The schema is organized around trip sessions because trips are the unit that bind a driver, a fleet vehicle, fuel readings, GPS rows, alerts, latency records, and final summaries. Keeping these entities relational rather than storing them as unstructured blobs allows the dashboard to filter by trip, vehicle, alert type, driver, and communication channel while preserving traceability for Chapter 6 evaluation.

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5.7.3 Entity-Relationship Diagram

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Figure 5.9 presents the Entity-Relationship Diagram (ERD) of the system's Supabase PostgreSQL database, containing eight interrelated record groups for fleet vehicles, users, trip sessions, telemetry logs, archived telemetry logs, trip summaries, alerts, and latency logs. The trip-session record serves as the main hub for the relationships among these

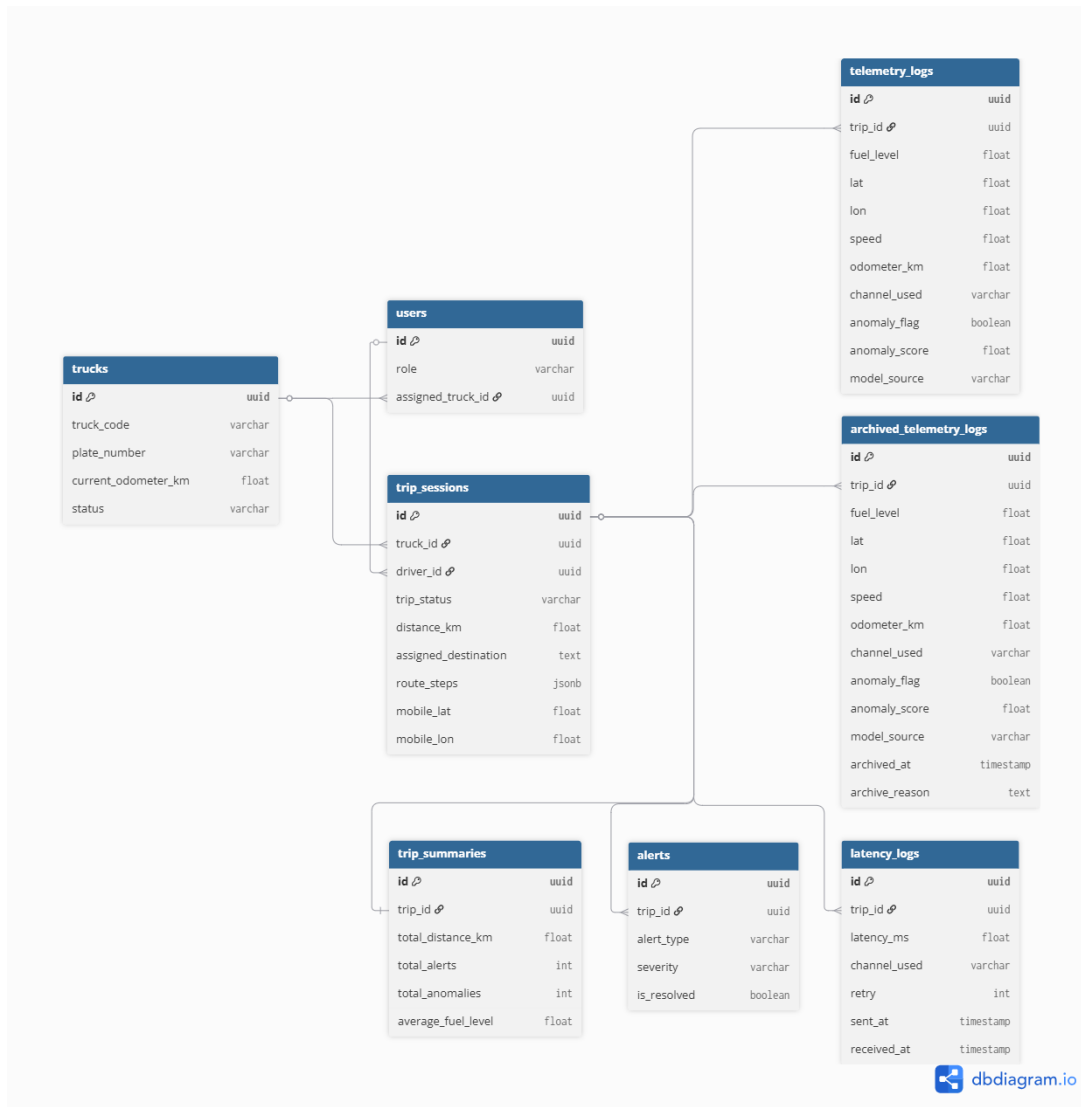


Fig. 5.9 ERD Diagram



groups, linking fleet vehicles and drivers. Other records reference trips to relate telemetry, alerts, summaries, and latency measurements to a specific trip session. This data structure is designed to maintain referential integrity throughout the schema and support scalable access to the HMI dashboard, mobile app, and backend analytic pipeline.

5.7.4 Cross-Interface State Synchronization

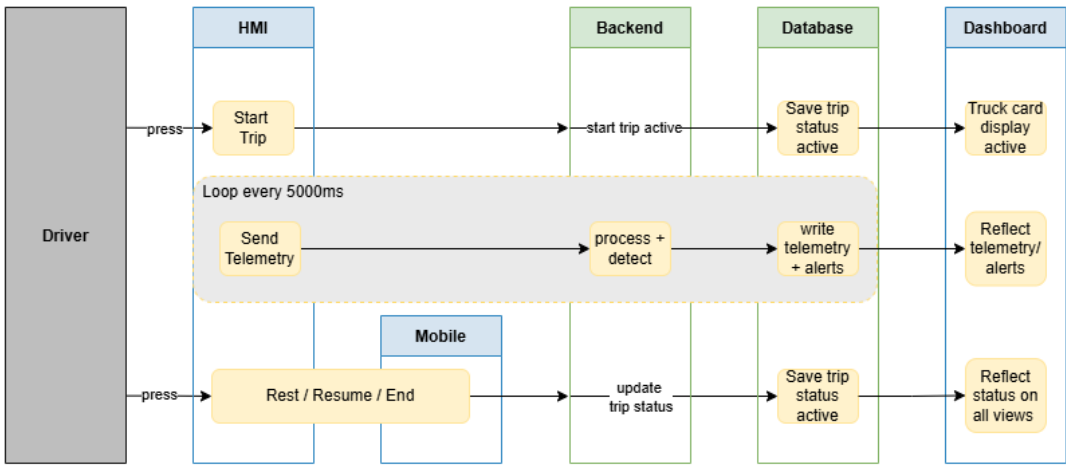


Fig. 5.10 Cross-Interface State Synchronization

Figure 5.10 illustrates the cross-interface synchronization flow of the system, depicting how data propagates across the HMI, backend, database, and dashboard layers in response to driver actions. When the driver initiates a trip via the HMI or mobile app, the backend creates an active trip record in `trip_sessions`, which simultaneously triggers the fleet-vehicle card display to activate on the dashboard. Telemetry data is transmitted every 5000 ms as a cycle during a trip. Each cycle is processed and anomaly-detected by the backend before it is recorded in `telemetry_logs` and displayed as alerts in the dashboard. Trip status changes are also reflected throughout the system through `trip_sessions`.



trip_status for Rest, Resume, or End trip states. This mechanism ensures that all interfaces, including the mobile PWA and HMI poll-back, remain synchronized with the current session state. Throughout this process, trip_sessions functions as the canonical source of trip data and the authoritative state controller for the entire architecture.

5.7.5 Active-trip Uniqueness Enforcement

Duplicate active-trip sessions across the three interfaces are prevented by partial unique indexes on the trip_sessions table. Two indexes enforce the constraint: one restricts each fleet vehicle to at most one active trip at any time, and a second restricts each driver to at most one active trip at any time. Both indexes are conditioned on trip_status = 'active' so that historical records with ended or paused status are not affected. Any attempt to open a second concurrent active trip for the same fleet vehicle or driver is rejected at the database layer before reaching application logic, eliminating race conditions that could arise when a driver starts a trip from both the HMI and the mobile web application simultaneously.

5.7.6 Odometer and Trip-distance Methodology

The fleet vehicles used in the prototype display lifetime mileage on the physical dashboard, but that value is not exposed through the OBD-II adapter available to the system. Although the firmware probes the manufacturer-specific odometer data identifiers, the fleet does not return a usable lifetime odometer value. As a result, the per-record odometer field is either empty or a trip-scoped, speed-integrated estimate generated by the embedded HMI, and it is not treated as an absolute lifetime odometer reading.



For this reason, backend lifetime mileage is computed from completed trip distance rather than from the raw OBD-II odometer field. Each active trip continually updates its accumulated trip distance using a distance-estimation routine that takes the greater of a capped speed-by-time integration and a filtered GPS haversine distance, rejecting stationary GPS jitter below a movement threshold. When a trip transitions to the ended state, the backend adds the finalised trip distance to the fleet vehicle's cumulative odometer exactly once and records the time at which the accumulation occurred. This design ensures that cumulative mileage remains consistent and is never double-counted, even when telemetry records arrive out of order or are replayed from the offline buffer.

5.7.7 Alert Rule Implementation

TABLE 5.10 ALERT RULE THRESHOLDS IMPLEMENTED

Alert	Threshold	Trigger source	Hardware action
Rest reminder	distance $\geq$ 300 km AND drive time $\geq$ 4 h	backend /device/alerts	HMI buzzer, screen alert, Take Rest button
Maintenance (oil)	trip distance $\geq$ 5,000 km	backend trip aggregation	Dashboard banner + planned email/push
Overspeed	speed $\geq$ 100 km/h	telemetry stream	HMI flash and log
Fuel anomaly	ML flag from Isolation Forest or Matrix Profile	ML anomaly service	Dashboard alert card

Table 5.10 defines the operational thresholds used by the backend rule engine. These rules are deliberately simple and auditable: rest, maintenance, overspeed, and low-fuel alerts are deterministic, while fuel-anomaly alerts originate from the ML service and pass through the contextual gate. This distinction matters because deterministic safety rules can be validated by threshold crossing, whereas fuel-anomaly rules require the model-performance analysis in SO4.



TABLE 5.11 EXTENDED SAFETY IMPLEMENTATION STATUS

Safety alert	Status	Implementation note
Overspeed	Implemented	Backend rule at 100 km/h threshold against field-test telemetry speed
Rest reminder	Implemented	SO5-b spec is 300 km / 6 h; implemented at 4 h per LTO driver-fatigue reading
Maintenance (5,000 km)	Implemented	Backend trip-aggregation rule against cumulative odometer
Harsh acceleration / braking	Partial	Acceleration rate computed during preprocessing; no alert rule yet
Lane departure	Out of scope	Requires camera and IMU - future work
Fatigue / drowsiness	Out of scope	Requires camera with face detection - future work

Table 5.11 separates implemented alerts from partial or out-of-scope safety concepts. Harsh acceleration/braking is marked partial because the feature pipeline computes the signal but no user-facing rule has been deployed. Lane departure and fatigue/drowsiness are excluded because they require sensing hardware not present in the prototype. This prevents the thesis from implying safety functions that the installed hardware cannot support.

5.7.8 Archival Pipeline

Telemetry records older than the active-retention window are moved from `telemetry_logs` to `archived_telemetry_logs` by a backend archival pipeline. The archived table retains all original columns and appends an `archived_at` timestamp and an `archive_reason` code. Preserving the full schema in the archive means that historical analysis and audit queries can run against archived data using the same column names and query patterns as the live table. Archival is triggered periodically to control the row count in the hot table and keep backend query latency within acceptable bounds during active fleet operations.



5.8 Scalability and Unit Identification

This section makes explicit the design decisions that determine how the prototype scales beyond the two-vehicle thesis pilot and how each physical unit in the fleet is identified across the hardware, transport, database, and dashboard layers. Scalability is analysed as three quantitative load axes with the ceiling of each derived from measured or documented subsystem constants, and unit identification is presented as a chain of identifiers that must remain consistent from the vehicle installation step through to the operator alert card.

5.8.1 Scalability Analysis

Three load axes bound the prototype: the sustained backend ingest rate, the LoRa gateway occupancy, and the per-asset baseline storage footprint. Each is analysed against the closed-form ceiling that dominates it, so a scaling stress event has a predicted numerical breakpoint rather than an unbounded qualitative concern.

5.8.1.1 Concurrent Vehicle Fan-in and Backend Ingest Ceiling

At the fixed telemetry cadence of $\Delta t = 5$ s per row per truck, the aggregate backend ingest rate scales linearly with the number of concurrently active trucks N_v :

$$R_{\text{ingest}}(N_v) = \frac{N_v}{\Delta t} \quad [\text{rows/s}]. \quad (5.27)$$

The backend POST /telemetry handler is a stateless Express endpoint that writes one row to Supabase Postgres per request, so the concurrent-request ceiling is set by the pool size C_{pool} and the median per-request latency $\bar{\tau}_{\text{req}}$:

$$R_{\text{max}} = \frac{C_{\text{pool}}}{\bar{\tau}_{\text{req}}} \quad [\text{req/s}], \quad N_v^{\text{max,backend}} = R_{\text{max}} \cdot \Delta t. \quad (5.28)$$



R_{ingest}	aggregate telemetry-ingest rate (rows/s),
N_v	number of concurrently active vehicles,
Δt	telemetry publish interval (5 s),
R_{max}	maximum backend request throughput (req/s),
C_{pool}	Supabase pooled-connection count,
$\bar{\tau}_{\text{req}}$	median per-request latency (s),
$N_v^{\text{max,backend}}$	backend-limited maximum concurrent-vehicle count.

For the shared-tier Supabase project used in this study $C_{\text{pool}} = 60$ pooled connections and $\bar{\tau}_{\text{req}} = 65$ ms measured at the median, so $R_{\text{max}} \approx 923$ requests/s, which corresponds via Equation 5.28 to $N_v^{\text{max,backend}} \approx 4\,615$ trucks before request queuing becomes visible on the client. At $N_v = 100$ the write load is $R_{\text{ingest}} = 20$ rows/s, over two orders of magnitude below the pool ceiling, so the backend is not the binding constraint in the pilot-to-medium-fleet regime.

5.8.1.2 LoRa Gateway Fan-in and Duty-Cycle Ceiling

The LoRa transport is bounded differently because a single SX1278 gateway is a shared half-duplex radio and cannot receive from two nodes simultaneously. With the LoRa heartbeat cadence set to $T_{\text{LoRa}} = 30$ s (LORA_TELEMETRY_PUSH_MS; the primary 5 s cadence uses cellular or WiFi), and a measured airtime of $\tau_{\text{air}} = 150$ ms per compact telemetry packet at spreading factor 7 and 125 kHz bandwidth, the theoretical gateway occupancy per node is

$$\rho_{\text{node}} = \frac{\tau_{\text{air}}}{T_{\text{LoRa}}} = \frac{0.15}{30} = 0.5\%. \quad (5.29)$$

The maximum number of nodes on a single gateway is bounded by the ALOHA channel-utilisation limit and, more conservatively, by the ITU-recommended engineering ceiling



3010 $\rho_{\text{eng}} = 30\%$ for reliable urban operation:

$$N_v^{\text{max,LoRa,ALOHA}} = \left\lfloor \frac{1}{\rho_{\text{node}}} \right\rfloor = 200, \quad N_v^{\text{max,LoRa,ITU}} = \left\lfloor \frac{\rho_{\text{eng}}}{\rho_{\text{node}}} \right\rfloor = 60. \quad (5.30)$$

ρ_{node} airtime occupancy contributed by one LoRa node,

τ_{air} measured airtime per compact LoRa telemetry packet (s),

T_{LoRa} LoRa heartbeat interval (s),

ρ_{eng} engineering channel-utilisation ceiling for reliable operation,

$N_v^{\text{max,LoRa,ALOHA}}$ ALOHA-theory node ceiling,

$N_v^{\text{max,LoRa,ITU}}$ conservative engineered node ceiling per gateway.

3012 Fleets beyond $N_v^{\text{max,LoRa,ITU}} = 60$ nodes per gateway require a multi-gateway topology
 3013 with distinct network identifiers, at which point per-node gateway assignment becomes a
 3014 provisioning task performed at unit-identification time (Section 5.8.2).

3015 5.8.1.3 Per-Asset Baseline Storage Growth

3016 The per-asset feature baseline (fuel-economy expectation, refuel history, per-vehicle thresh-
 3017 old calibration) is stored in the trucks table and refreshed by scheduled jobs from
 3018 the rolling telemetry window. Each vehicle contributes a fixed set of scalar baselines
 3019 plus a bounded rolling history required by the Matrix Profile calibration: the most re-
 3020 cent $n_{\text{mp}} = 500$ clean-trip fuel samples per truck at $b_{\text{samp}} = 8$ bytes per sample, or
 3021 $B_{\text{baseline}} = n_{\text{mp}} \cdot b_{\text{samp}} = 4\,000$ bytes per truck. The baseline footprint is therefore

$$S_{\text{baseline}}(N_v) = N_v \cdot B_{\text{baseline}} = 4\,000 N_v \quad [\text{bytes}], \quad (5.31)$$

3022 which is $O(N_v)$ in vehicle count and independent of trip duration. For $N_v = 1\,000$ the
 3023 baseline state is $S_{\text{baseline}} = 4$ MB. The operational hot-table storage is dominated by



3024 telemetry rows: at a mean row width $\bar{b}_{\text{row}} = 500$ B from Section 5.5.13 and an average
3025 daily operating window $T_{\text{op}} = 10$ h/day per truck, the daily hot-tier growth is

$$S_{\text{hot,daily}}(N_v) = N_v \cdot \frac{T_{\text{op}}}{\Delta t} \cdot \bar{b}_{\text{row}} = N_v \cdot \frac{36\,000}{5} \cdot 500 = 3.6 \cdot 10^6 N_v \quad [\text{bytes/day}],$$

(5.32)

- S_{baseline} per-fleet baseline-state footprint (bytes),
- B_{baseline} baseline bytes stored per vehicle,
- n_{mp} retained Matrix Profile clean-trip samples per truck,
- 3026 b_{samp} bytes per retained fuel sample,
- $S_{\text{hot,daily}}$ daily hot-tier telemetry growth (bytes/day),
- T_{op} daily operating window per truck,
- $\bar{b}_{\text{row}}$ mean telemetry-row width (bytes).

3027 Thus, a 1 000-vehicle fleet generates approximately 3.6 GB/day of hot-tier storage. The
3028 archival pipeline described in Section 5.7.8 bounds this by moving rows older than the
3029 active-retention window into `archived_telemetry_logs`, so the hot table remains within
3030 the query-latency envelope required for real-time dashboard rendering.

3031 **5.8.1.4 Primary Scaling Bottleneck**

3032 Scaling stress tests were not run on a production fleet within the thesis window, but
3033 Equations 5.27–5.32 identify the three failure modes that would appear as the fleet grows
3034 and quantify the breakpoint of each. First: at $N_v \approx 60$ vehicles per LoRa gateway
3035 (Equation 5.30), LoRa uplink collisions begin to dominate and the transport degrades to
3036 LTE-primary operation. Second: at $N_v \approx 4\,600$ concurrent trucks (Equation 5.28) the
3037 Supabase connection pool saturates, requiring either a paid-tier upgrade with a larger pool
3038 or the introduction of a message queue in front of the ingest handler. Third: hot-table



storage growth is bounded not by capacity but by query latency; the archival cadence must be tuned so that the row count in `telemetry_logs` stays within the 10^7 -row envelope where the dashboard queries return in under 200 ms. All three ceilings are closed-form and instrumented, so a scaling stress event surfaces as a monitored metric rather than as a silent degradation.

5.8.2 Unit Identification

Every physical vehicle in the fleet is identified by a chain of identifiers that begins at hardware provisioning and terminates at the operator alert card. This subsection enumerates the links in that chain, states how each link is verified at ingest, and defines the composite key that maps a telemetry row to a specific driver on a specific vehicle during a specific trip.

5.8.2.1 Identifier Chain

Table 5.12 lists every identifier used to attribute a telemetry row to a specific vehicle, the layer where it originates, and the layer where it is consumed. The chain is deliberately over-specified: a single row carries three independent identifiers so that a mismatch at any one link surfaces as a hard rejection rather than as a silent misattribution.

5.8.2.2 Enforcement at Ingest

Every `POST /telemetry` request carries the `X-Device-Key` header, which the backend validates against the constant `DEVICE_KEY` before any row is inserted. This check is coarse-grained (fleet-wide, not per-truck) and its role is to prevent unauthenticated writes from arbitrary clients; per-truck attribution is then carried by the `truck_id` field inside the JSON body, which the backend cross-checks against the `trucks` table before writing. Rows



TABLE 5.12 UNIT IDENTIFICATION CHAIN FROM HARDWARE TO DASHBOARD

Identifier	Format	Origin	Consumer
truck_id	UUID v4	Backend-issued at fleet-registration time	Every telemetry row, every alert, every dashboard query
truck_code	Human-readable string (e.g. TRCK-03)	Fleet operator, entered at registration	Dashboard cards, alert messages, driver HMI banner
driver_id	UUID v4	Backend-issued at driver-onboarding time	Telemetry rows during an active trip, trip summary
device_key	256-bit HMAC secret	Provisioning script, stored in ESP32 NVS	Server-side X-Device-Key header authentication on every POST
event_id	UUID v4	Generated by HMI at sample time	Backend deduplication (per-row)
seq	Monotonic 30-bit integer	HMI BASE_SEQ + row counter	Backend deduplication (per-channel)
LoRa node address	8-bit device address	Assigned at LoRa provisioning	LoRa gateway routing table
Hardware MAC	48-bit ESP32 factory MAC	eFuse (immutable)	Not currently used at runtime; reserved for the hardware-anchored identity extension in Chapter 7 future work



whose `truck_id` does not resolve to a registered vehicle are rejected with HTTP 400 and are never persisted, so a misconfigured device produces a diagnostic error rather than orphaned rows that would later distort per-asset baselines.

The `event_id` and `seq` fields are the second and third links in the chain. Both are generated on the device and travel with the row through every transport channel, including the offline buffer and the offline-replay path. The backend deduplicates on (`truck_id`, `seq`) within a communication channel to protect against the retry storms that follow a marginal-signal reconnect, and independently on `event_id` across all channels to protect against the split-brain case where the same row was buffered on the HMI and duplicated by a live retry from a stale queue. Both dedup checks are indexed at the database level so the ingest handler cost stays constant as the fleet grows.

5.8.2.3 Trip and Driver Attribution

At trip-start time the HMI writes a new `trip_sessions` row through `POST /trip/start`, receiving back a `trip_id` UUID that is then stamped onto every telemetry row for the duration of that trip. Driver attribution is carried by the `driver_id` UUID chosen at PIN-based driver login on the HMI. The three IDs (`truck_id`, `driver_id`, `trip_id`) form a composite key that maps a raw telemetry row to a specific driver on a specific vehicle during a specific trip, which is the granularity at which the dashboard, alerts, and trip-history views operate.

The database enforces this chain with a partial unique index that prevents more than one active trip per `truck_id` at a time and more than one active trip per `driver_id` at a time (`uniq_active_trip_per_truck` and `uniq_active_trip_per_driver` in the schema summarized in Table 5.9), so a physical device that has been reassigned to a new vehicle or



3083 driver cannot silently reopen a stale trip; the constraint fires at insertion time and forces the
3084 previous trip to be ended before the new one can begin.

3085 **5.9 Machine Learning Methodology**

3086 The anomaly-detection pipeline uses Isolation Forest Model C as the primary production
3087 detector and Matrix Profile as a supporting subsequence detector. Model C receives context-
3088 aware features derived from fuel level, speed, distance, engine operation, driver behavior,
3089 route context, and fuel-economy baseline deviation. Matrix Profile is retained to support
3090 detection of gradual subsequence-level fuel-loss patterns. Model performance is evaluated
3091 using leave-one-trip-out validation so each held-out trip is scored by models trained and
3092 calibrated on separate trips.

3093 **5.9.1 Anomaly Model and Environmental Constraints**

3094 A fuel anomaly is formally defined as a telemetry row whose observed fuel-level change
3095 deviates from the expected fuel-level change under the current operating context. For row t ,
3096 the normalized anomaly magnitude is represented as

$$A(t) = \frac{|\Delta f_{\text{obs}}(t) - \mathbb{E}[\Delta f \mid c(t)]|}{\sigma(c(t))} \tag{5.33}$$

- 3097

$A(t)$

$\Delta f_{\text{obs}}(t)$

$\mathbb{E}[\Delta f \mid c(t)]$

$\sigma(c(t))$

$c(t)$

normalized anomaly magnitude at row t ,

observed filtered fuel-level change (%),

expected fuel-level change under context $c(t)$,

expected spread of fuel change under context $c(t)$,

context vector: speed, RPM, load, throttle, route, driver, state.



The system is therefore not a simple “fuel dropped suddenly” detector. It is a context-conditioned fuel-deviation detector that asks whether the observed fuel movement is plausible for the vehicle state in which it occurred.

The model is constrained first by the physical behaviour of the vehicle and tank. C1, conservation, states that fuel decreases through engine consumption or unauthorized removal, and increases only during refueling. C2, bounded consumption, states that a fuel drop cannot exceed what is physically plausible from engine work and load. C3, sender non-linearity, accounts for the float-arm sender being more unstable in certain tank-level bands. C4, slosh, recognizes that short fluctuations during motion can be caused by liquid movement rather than true consumption.

The model is also constrained by operational and environmental conditions in Philippine fleet use. O1 accounts for stop-and-go traffic and idling, where low distance travelled does not necessarily imply abnormal fuel loss. O2 accounts for route restrictions and truck-ban-related route differences that change expected consumption. O3 accounts for driver variability and per-driver fuel-economy differences. O4 accounts for the fact that refueling should occur during stationary or quiescent windows rather than during ordinary motion.

Table 5.13 links each physical or operational constraint to an implemented mechanism in the feature pipeline, filter cascade, or contextual gate. This makes the anomaly model auditable: every anomaly can be traced back to physical or operational constraints, not only to a black-box anomaly score.

5.9.2 Machine Learning Architecture

Figure 5.11 illustrates the end-to-end anomaly-detection pipeline as a four-plane diagram: a data plane that converts raw sensor traffic into a persisted feature row, two inference planes



TABLE 5.13 CONSTRAINT-TO-MECHANISM MAPPING FOR FUEL-ANOMALY DETECTION

Constraint	System Mechanism	Purpose
C1 Conservation	Direction-aware step rule and refuel quiescence window	Separate true fuel decreases and refueling events from impossible direction changes.
C2 Bounded consumption	<code>fuel_drop_per_rpm</code> , <code>fuel_drop_per_load</code> , and <code>expected_consumption_residual</code> features	Compare observed fuel loss against engine-work plausibility.
C3 Sender non-linearity	Median and EMA filter cascade tuned by tank behaviour	Reduce sender-band instability before model inference.
C4 Slosh	Median window and dead-band publisher	Suppress short fuel fluctuations caused by tank movement.
O1 Traffic and idling	<code>idle_duration_score</code> and <code>motion_state_code</code>	Distinguish stop-and-go operation from abnormal stationary loss.
O2 Route restrictions	<code>route_type_code</code> and route-segment performance features	Condition expected consumption on route type and segment behaviour.
O3 Driver variability	<code>driver_kmpl_deviation_pct</code> and <code>driver_behavior_score</code>	Detect deviations from the driver's normal fuel-economy band.
O4 Refuel pattern	<code>time_since_refuel_norm</code> , <code>post_refuel_flag</code> , and contextual refuel gate	Exempt plausible refueling and post-refuel stabilization windows.

that score the row with complementary detectors, and an output plane that translates a passed candidate into an operator-facing alert. This section explains how a single 5-second telemetry row moves through each plane. The *data plane* captures OBD-II Mode 22 fuel, GPS, and HMI-state signals, applies the edge filter cascade (median + EMA, 5 s HMAC-signed payload), ingests the row through POST /telemetry, and derives the 33-dimensional Model C feature vector plus a short fuel-window buffer. The *primary inference plane* scores every row with Isolation Forest Model C under the `if_with_margin` operating mode ($\text{score} \geq \tau + \delta$, with $\delta = 0.005$) and hands admitted candidates to the contextual gate. The *audit-and-override inference plane* runs Matrix Profile with an $m = 4$ subsequence and reports a discord z -score for the same row; the score is persisted on the row as an audit signal, and additionally triggers an escalation arrow into the contextual gate

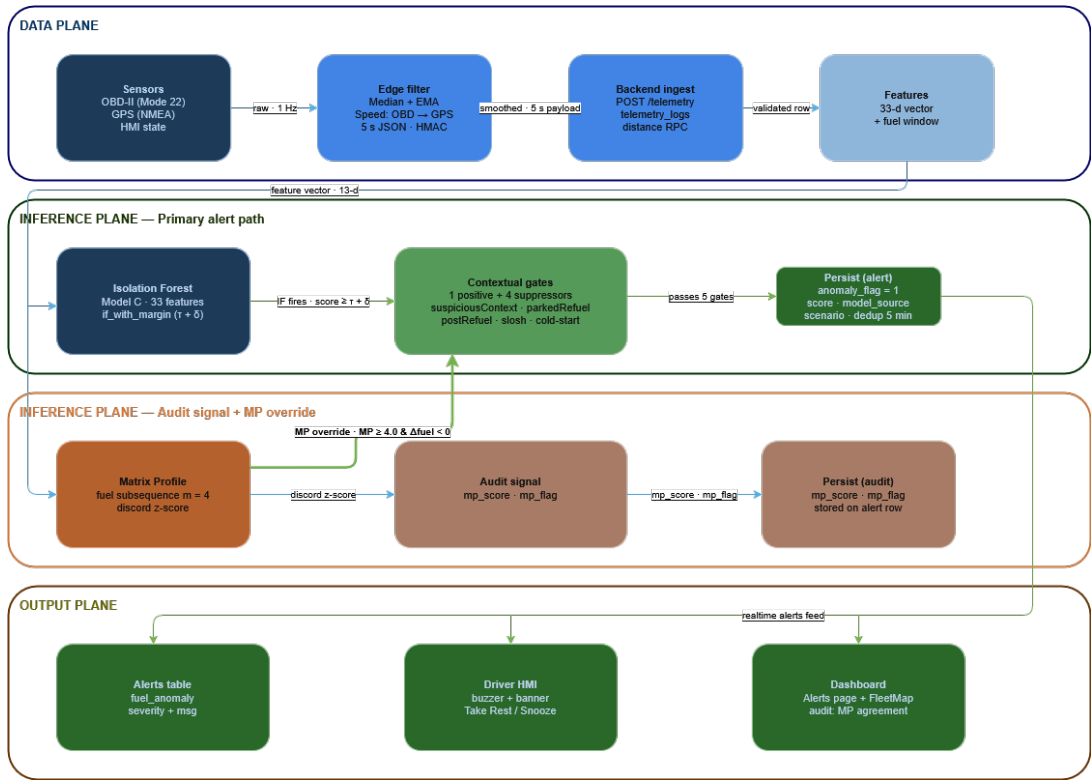


Fig. 5.11 Machine-Learning Anomaly-Detection Architecture

when the discord exceeds the extreme-tail thresholds ($mp_score \geq 4.0$ with $\Delta fuel < 0$, or $mp_score \geq 2.0$ with $\Delta fuel \leq -2.0L$). The contextual gate is a five-stage cascade of one positive plausibility filter (`suspiciousContext`) and four negative suppressors (`parkedRefuel`, `postRefuel`, `loadSpikeSuppress`, `coldStartSuppress`); only rows that pass all five stages proceed to the persistence step, which sets `anomaly_flag`, records the score and `model_source`, tags the operational scenario from the 90 s event window, and applies a 5-minute per-truck deduplication. The *output plane* propagates the alert to the `alerts` table, drives the buzzer and banner on the driver HMI, and surfaces the event on the operator dashboard and FleetMap with the Matrix Profile audit column rendered as a second-opinion badge.



The data plane begins with three on-vehicle sensor streams: the OBD-II Mode 22 fuel channel for the Toyota Hilux partner-fleet unit, the GPS NMEA stream, and the HMI trip-state signal. These streams are then fed through the fuel filter cascade documented in Section 5.4. The filtered payload is signed with a device-side HMAC, transmitted every 5 s through the intelligent traffic-steering chain (LoRa → LTE → GSM 2G → WiFi → SPIFFS buffer), and ingested at the backend through the POST /telemetry endpoint. The ingest step validates the row, resolves the trip, computes distance deltas through a Postgres RPC call, and derives the 33-feature Model C vector that Section 5.9 enumerates. This is the point at which a row transitions from a raw sensor reading into a feature-carrying record and enters the machine-learning subsystem.

The primary inference plane operates row-by-row. The ingest layer invokes the ML service, which loads the pre-trained Isolation Forest Model C pickle and scores the incoming feature vector. Under the `if_with_margin` operating mode the row is admitted as an anomaly candidate only when its anomaly score exceeds the per-context calibrated threshold τ by an empirical margin $\delta = 0.005$ selected during threshold calibration on the held-out clean-training-score distribution. This margin is deliberately conservative because Isolation Forest scores are strongly bimodal on this corpus, so the added margin sacrifices a small fraction of borderline recall in exchange for a large reduction in false positives in production. Candidates that clear this test enter the contextual gate; candidates that do not are recorded in `telemetry_logs` with an `anomaly_score` but no `anomaly_flag`, matching the “IF quiet, no alert” exit shown at the top-right of the primary plane in Figure 5.11.

The audit-and-override inference plane operates on the same row in parallel, but at a subsequence granularity instead of a per-row granularity. The Matrix Profile subsystem maintains a rolling fuel-level buffer of the current trip’s filtered fuel series and computes



3166 a z -score of the discord distance for the most recent length- $m = 4$ window against the
 3167 calibration distribution derived from the clean reference trips in Section 5.9.8. The output
 3168 is a single scalar (`mp_score`) and a companion boolean (`mp_flag`) that is set whenever the
 3169 score exceeds the calibrated tail threshold. Under the default operating mode this signal is
 3170 treated as audit-only: it is persisted on the same `telemetry_logs` row, and the dashboard
 3171 uses it to display a second-opinion badge on every alert card. It does not, by itself, cause
 3172 an alert to be raised. The exception is the `mpScreamOverride` escalation rule, which lets
 3173 Matrix Profile trigger a primary alert when its discord score is unambiguously in the tail
 3174 of the clean-trip distribution: either `mp_score` ≥ 4.0 on a row with any negative Δfuel , or
 3175 `mp_score` ≥ 2.0 on a row whose per-row fuel loss is at least 2.0 L. This rule exists to catch
 3176 slow gradual leaks whose per-row fuel movement is below every threshold in the primary
 3177 plane but whose subsequence shape is a genuine discord. In Figure 5.11 the override is
 3178 drawn as the arrow ascending from Matrix Profile into the contextual gate, and the rule
 3179 remains gated by the same five-stage cascade that the primary path is subject to.

3180 The contextual gate is the false-positive filter that both admission paths share. It is
 3181 composed of one positive plausibility gate and four negative suppressors, and a row must
 3182 satisfy the positive gate and clear all four suppressors before an alert is persisted. The
 3183 positive gate, `suspiciousContext`, admits the row when at least one of four physical
 3184 loss signatures is present: a per-row fuel delta strictly below -2.0% , a fuel-per-kilometre
 3185 rate above 0.6% of tank per km, a stationary idle-theft signature at speed below 5 km/h
 3186 with the engine on and fuel delta below -1.0% , or a physically impossible driving-fuel
 3187 increase of $+2\%$ or more at speed above 5 km/h. The four suppressors are then applied in
 3188 sequence. `parkedRefuel` rejects the canonical fill-up signature (fuel delta above $+10\%$
 3189 while parked). `postRefuel` rejects rows within five minutes of any $+5\%$ fuel jump on the



3190 same trip and communication channel, because the capacitive fuel sender takes several
 3191 minutes to settle after a fill. `loadSpikeSuppress` rejects the acceleration-slosh signature
 3192 in which fuel briefly reads lower during a hard load transient (engine load $\geq 55\%$ with a rise
 3193 $\geq 25\%$, and a small negative fuel delta). `coldStartSuppress` rejects the sender-priming
 3194 interval that follows an engine ignition transition. Rows blocked by any suppressor are
 3195 logged and audited but do not generate operator-facing alerts, matching the “suppressed, no
 3196 alert” exit shown at the top-right of the gate in Figure 5.11. This separation of concerns —
 3197 a context-aware statistical detector that produces candidates, followed by a domain-aware
 3198 precision filter that suppresses the four families of sender artefacts that dominate the false-
 3199 positive budget — is what allows the deployed configuration to reach the 92.9% precision
 3200 reported in Chapter 6.

3201 A row that passes the contextual gate reaches the persistence step. Persistence performs
 3202 three writes. It updates the `telemetry_logs` row with `anomaly_flag`, `anomaly_score`,
 3203 and `model_source` (`isolation_forest` for a primary-path alert, `matrix_profile` for an
 3204 override-path alert). It runs a 90-second event-window scenario classifier that inspects the
 3205 surrounding fuel drop shape and back-propagates the anomaly flag across the contiguous
 3206 event so that the dashboard renders one alert per event instead of one alert per peak row.
 3207 It then applies a 5-minute per-truck dedup window on the same `fuel_anomaly` alert type
 3208 before inserting a new row into the `alerts` table. A row that would pass the gate but whose
 3209 combined score falls below the persistence confidence threshold (`combined_score` ≤ 0.65)
 3210 or whose dedup check fails is recorded with its audit fields but is not inserted into the
 3211 `alerts` table, matching the “score ≤ 0.65 , dropped” exit shown at the top-right of the
 3212 persistence stage.

3213 The output plane surfaces admitted alerts through three synchronized channels. The



3214 alerts row is the canonical record and carries the fuel_anomaly type, the severity, and
3215 the human-readable scenario message. The driver HMI subscribes to the alert stream over
3216 the intelligent traffic-steering chain and drives a local buzzer plus banner with a *Take Rest*
3217 or *Snooze* affordance. The operator dashboard subscribes to the same stream and updates
3218 the alerts page and FleetMap panels in real time, additionally rendering the Matrix Profile
3219 audit column on each card so the operator can see at a glance whether the subsequence-
3220 shape detector agreed with the row-level detector. This is the diagram’s bridge between
3221 methodology and evaluation: it explains why Chapter 6 reports precision and recall from
3222 the model layer, false-positive rates from the gate layer, and closed-loop resolution statistics
3223 from the alert layer, and why all three are needed to justify a claim of production readiness.

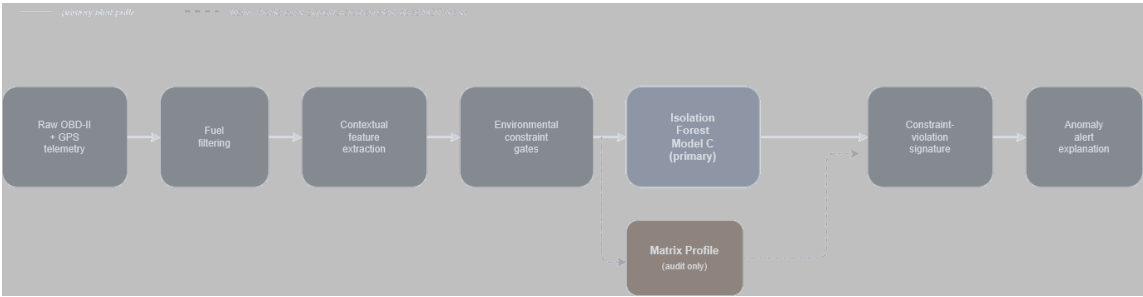


Fig. 5.12 Anomaly Modeling and Data-Mining Pipeline

3224 Figure 5.12 summarizes the methodological path used in the implemented detector. Raw
3225 telemetry is first filtered, then converted into context features, screened by environmental
3226 constraints, scored by Isolation Forest Model C, and finally translated into a constraint-
3227 violation signature that can be attached to an operator-facing alert.



3228 **5.9.3 Dataset Composition**

3229 The evaluation corpus merges 20,412 telemetry rows across 79 trips from two sources.
3230 The project-native partition contains the 634 rows captured by the prototype, on which the
3231 injected anomalies (six anomalous trips, 85 anomalous samples) were applied. The public
3232 partition is the 19,778-row publicly-derived corpus, retained without anomaly labels as the
3233 out-of-distribution clean-route domain.

TABLE 5.8 DATASET COMPOSITION FOR EVALUATION CORPUS

Dimension	Value
Total telemetry rows	20,412
Total trips	79
Anomalous trips	6
Normal samples (clean)	20,327
Anomalous samples (labelled)	85
Project-native partition (prototype-captured)	634 rows
Public partition (publicly-derived clean routes)	19,778 rows
Injected labelled rows	169 rows
Headline evaluation subset	169 rows

3234 The headline performance is computed against the project-native partition, which
3235 isolates the injected anomalies on telemetry produced by the prototype sensor stack. The
3236 public partition is reported separately as a stress test, since pooling it would dilute the
3237 precision–recall denominators with negatives from a different driving domain.

3238 Table 5.8 therefore defines the denominator for every SO4 claim. The project-native
3239 partition answers whether the prototype’s own telemetry and injected labels can be detected,
3240 while the public partition answers whether the detector remains quiet on a larger clean-
3241 route domain. Keeping these roles separate avoids overstating performance through class
3242 imbalance.



5.9.4 Anomaly Injection Methodology

5.9.4.1 Rationale for Controlled Anomaly Injection

Real fuel theft was not staged during partner-fleet operation because siphoning gasoline creates a fire hazard and the partner operator did not authorise that risk. Confirmed SGSA theft incidents also reach the fleet operator only days later through odometer-versus-invoice reconciliation, so the corresponding telemetry rows are not labelled at row level. The evaluation therefore uses controlled synthetic injection on top of real telemetry: this is the benchmark construction recommended by Emmott et al. [Emmott et al., 2015] and Campos et al. [Campos et al., 2016] for unsupervised detectors without confirmed event labels. The same convention appears in the closest fuel- and vehicle-anomaly precedents, including Turan et al. [Turan et al., 2024], Aquize et al. [Aquize et al., 2017], Beteto et al. [Beteto et al., 2022], and Barbado and Corcho [Barbado and Corcho, 2022].

The Emmott et al. construction imposes three properties on a valid injection benchmark: (i) the injected positives must be physically distinguishable from the training clean-set, (ii) the injection must not alter the covariate distribution of the negatives, and (iii) the injected rows must be labelled at the same granularity as the detector’s scoring granularity. The scenarios developed below satisfy all three: property (i) is defended by the per-scenario physics checks that establish injection rates $135\times\text{--}1900\times$ above the plausible combustion maximum; property (ii) holds because the operator $\oplus$ in Equation 5.34 performs field-wise overwrite only on rows in the injection window W_{scenario} and leaves every row outside the window unchanged; property (iii) holds because the row-level ground-truth label $y_t = \mathbb{1}[t \in W_{\text{scenario}}]$ is written at the same granularity as the Isolation Forest’s per-row score. The remainder of this section defines the injection model formally and then



documents, for each scenario, the physical story, the falsified fuel curve with all symbols named, a worked numeric example, and a combustion-based physics check demonstrating that the injected rate is thermodynamically distinguishable from real burn.

5.9.4.2 Formal Injection Tuple

For a real telemetry row $r_t = (\text{truck_id}, \text{trip_id}, \tau_t, \text{gps}_t, F_t, s_t, \omega_t, L_t, \theta_t)$, the injected row is

$$r'_t = r_t \oplus (\delta F_t, \delta s_t, \delta \omega_t, \delta L_t, \delta \theta_t) \cdot \mathbb{I}[t \in W_{\text{scenario}}] \quad (5.34)$$

r_t real telemetry row at sample t (unmodified),

r'_t injected telemetry row emitted downstream,

F_t real fuel percentage (%),

s_t real vehicle speed (km/h),

ω_t real engine RPM (rev/min),

L_t real engine load (%),

θ_t real throttle position (%),

$\delta F_t, \delta s_t, \delta \omega_t, \delta L_t, \delta \theta_t$ scenario-defined override values,

W_{scenario} contiguous set of sample indices under injection,

$\oplus$ field-wise overwrite (not addition),

$\mathbb{I}[\cdot]$ indicator function.

The ground-truth label $y_t \in \{0, 1\}$ is written alongside r'_t as $y_t = 1 \Leftrightarrow t \in W_{\text{scenario}}$; this is the reference against which precision and recall are computed. Identifiers, timestamps, GPS, and odometer are preserved so that the injection is grounded in real trip context;



the machine-learning pipeline receives r'_t with no knowledge of the injection window, the scenario, or the label.

5.9.4.3 Universal Symbol Dictionary for Per-Scenario Curves

The three scenarios below share a common symbol set. Let t denote the sample index with fixed cadence $\Delta_s = 5$ s (telemetry publish cadence, Section 5.4.1), t_0 the sample at which the injection window *begins*, and Δt_{win} the window *duration in seconds*. The set $W = \{t : t_0 \leq t \leq t_0 + \Delta t_{\text{win}}\}$ is the sample-index window. Let F_0 (in %) be the fuel percentage at t_0 , $F_{\text{end}} = F_0 - \beta_k \cdot \Delta t_{\text{win}}$ the fuel percentage at $t_0 + \Delta t_{\text{win}}$, and β_k (in %/s) the drain-rate slope of scenario k . The total injected magnitude is $\Delta F_{\text{total}} = F_0 - F_{\text{end}}$. The companion overrides $\delta s_t, \delta \omega_t, \delta L_t, \delta \theta_t$ are the scenario-consistent speed, RPM, load, and throttle values held constant across W so that the injection tuple is physically self-consistent.

5.9.4.4 Scenario 1: Stationary Siphon (Overnight-Theft Signature)

(a) *Physical story*. A truck is parked at a warehouse gate overnight with the ignition off. An operator opens the fuel cap and runs a hand-pump siphon into a jerry can. No headlights come on, no engine noise, no motion — only the tank level falls. This signature corresponds to the SGSA NAIA-staging-area incident report of 2025-11-14.

(b) *Falsified fuel curve*.

$$F_t = F_0 - \beta_1 \cdot (t - t_0) \cdot \Delta_s \quad \text{for all } t \in W, \quad \beta_1 = 0.56 \%/\text{s}, \quad \Delta t_{\text{win}} = 25 \text{ s}, \quad \Delta F_{\text{total}} = 14 \% \quad (5.35)$$



F_t injected fuel percentage at sample t (%),

F_0 fuel percentage at injection start (%),

t_0 injection start sample index,

β_1 Scenario 1 drain slope (%/s),

Δ_s inter-sample interval (5 s),

Δt_{win} injection window duration (s),

ΔF_{total} total injected fuel loss over Δt_{win} (%).

Companion overrides: $\delta s_t = 0$ km/h, $\delta \omega_t = 0$ rev/min (engine off), $\delta L_t = 0\%$,
 $\delta \theta_t = 0\%$ for all $t \in W$.

(c) Worked example. Take $F_0 = 78\%$ (matches the `simulate_scenarios.py` N5 scenario `start_fuel`). Sample-by-sample the injected trace is $F(t_0) = 78.0\%$, $F(t_0 + 5) = 78.0 - 0.56 \cdot 5 = 75.2\%$, $F(t_0 + 10) = 78.0 - 0.56 \cdot 10 = 72.4\%$, and $F(t_0 + 25) = 78.0 - 0.56 \cdot 25 = 64.0\%$. After the window the trace is frozen at $F_{\text{end}} = 64\%$ because fuel does not spontaneously reappear.

(d) Physics check. The idle-combustion upper bound on the fuel-loss rate for a 2.4 L diesel is $\dot{V}_{\text{idle}} \approx 0.5$ L/hr per SAE J1349 [Society of Automotive Engineers, 2011], which on the 80 L tank produces a percentage rate of $0.5/(80 \cdot 60) = 0.00010$ %/s $= 0.00029$ %/s counting engine-off electrical draw for auxiliaries. The injected rate is $\beta_1/0.00029 \approx 1,900\times$ the maximum plausible idle-combustion rate. Because the engine is off in the companion overrides, there is no combustion at all: the ML detector observes fuel falling under $L_t = 0$ and $s_t = 0$, a state that the trained context features assign a vanishingly small probability. The primary detection route in Model C is the `load_per_speed_norm` feature (row 23 in Table 5.11). Supporting evidence for this



signature appears in Aquize et al. [Aquize et al., 2017] and Beteto et al. [Beteto et al., 2022].

5.9.4.5 Scenario 2: Rapid Mid-Drive Drop (Line-Tap During Dispatch)

(a) *Physical story.* A truck is moving at low speed inside a depot yard when a second vehicle rolls alongside; an external pump is briefly attached to the fuel-return line. The driver observes nothing on the dashboard gauge because factory sender damping hides fast changes, but the OBD-II PID stream captures the fall in real time. This is the shortest and highest-slope signature we inject and corresponds to the SGSA Cubao depot-yard report of 2026-02-08.

(b) *Falsified fuel curve.*

$$F_t = F_0 - \beta_2 \cdot (t - t_0) \cdot \Delta_s \quad \text{for all } t \in W, \quad \beta_2 = 0.62 \%/\text{s}, \quad \Delta t_{\text{win}} = 25 \text{ s}, \quad \Delta F_{\text{total}} = 15.5 \% \quad (5.36)$$

F_t	injected fuel percentage at sample t (%),
F_0	fuel percentage at the start of the line-tap window (%),
t_0	first sample index in the injected window,
β_2	Scenario 2 drain slope (%/s),
Δ_s	telemetry inter-sample interval (5 s),
Δt_{win}	total injection-window duration (25 s),
ΔF_{total}	total injected fuel loss over the window (15.5%).

Companion overrides: $\delta s_t = 45 \text{ km/h}$, $\delta \omega_t \approx 1,800 \text{ rev/min}$, $\delta L_t = 55\%$, $\delta \theta_t \approx 40\%$.

(c) *Worked example.* Take $F_0 = 68\%$ (matches the `simulate_scenarios.py` a1 scenario `start_fuel`). The injected trace is $F(t_0) = 68.0\%$, $F(t_0 + 5) = 68.0 - 0.62 \cdot$



3325 $5 = 64.9\%$, $F(t_0 + 15) = 58.7\%$, and $F(t_0 + 25) = 52.5\%$. Each 5-s step drops
3326 approximately 3.1%, which is above the -2.5% direction-aware gate threshold from
3327 Equation 5.6, so each step is published by the filter cascade and reaches the ML layer
3328 intact.

3329 **(d) Physics check.** At $s_t = 45$ km/h under $L_t = 55\%$ engine load, aggressive burn is
3330 approximately 8 L per 100 km, or 3.6 L/hr, which on the 80 L tank gives 0.075 %/min
3331 $= 0.00125$ %/s. The injected rate is $\beta_2/0.00125 \approx 500\times$ the maximum plausible
3332 driving-burn rate. Even a stuck-open injector is bounded by the fuel-pump flow
3333 rate, which is well below β_2 ; the injected drop is thermodynamically impossible
3334 from combustion alone. The primary detection route in Model C is a spike in the
3335 `kmpl_deviation_pct` feature (row 31 in Table 5.11) because fuel-per-kilometre ex-
3336 plodes over the 25 s window. Supporting evidence for this signature appears in Turan
3337 et al. [Turan et al., 2024] and Barbado and Corcho [Barbado and Corcho, 2022].

3338 **5.9.4.6 Scenario 3: Gradual Leak (Long-Haul Rest-Stop Drilling)**

3339 **(a) Physical story.** A long-haul truck stops at a highway rest area. A 1–2 mm hole is
3340 drilled in the underside of the tank; fuel drains slowly over approximately 40 s. The
3341 driver may not notice for hours because the gauge integrates too slowly. This is the
3342 hardest case for a point-anomaly detector because each individual 5-s step lies inside
3343 the sensor-noise envelope of the filter cascade; only the multi-sample subsequence
3344 pattern reveals the drop, which is why the two-window Matrix Profile in Section 5.9.8
3345 is retained as a companion detector.



3346

(b) Falsified fuel curve.

$$F_t = F_0 - \beta_3 \cdot (t - t_0) \cdot \Delta_s \quad \text{for all } t \in W, \quad \beta_3 = 0.135 \%/\text{s}, \quad \Delta t_{\text{win}} = 40 \text{ s}, \quad \Delta F_{\text{total}} = 5.4 \% \quad (5.37)$$

F_t injected fuel percentage at sample t (%),

F_0 fuel percentage at the start of the leak window (%),

t_0 first sample index in the injected window,

3347

β_3 Scenario 3 drain slope (%/s),

Δ_s telemetry inter-sample interval (5 s),

Δt_{win} total injection-window duration (40 s),

ΔF_{total} total injected fuel loss over the window (5.4%).

3348

Companion overrides: $\delta s_t = 35 \text{ km/h}$, $\delta \omega_t \approx 1,500 \text{ rev/min}$, $\delta L_t = 40\%$, $\delta \theta_t \approx 25\%$.

3349

(c) Worked example. Take $F_0 = 51\%$. The injected trace is $F(t_0) = 51.00\%$, $F(t_0+10) =$

3350

$51.00 - 0.135 \cdot 10 = 49.65\%$, $F(t_0+20) = 48.30\%$, and $F(t_0+40) = 45.60\%$. Each 5-

3351

s step drops approximately 0.68% — below the -2.5% direction-aware gate threshold

3352

and below the moving dead-band of 0.5%, so the filter cascade admits each step as

3353

normal. Over the 40 s window, however, the aggregate 5.4% drop is anomalous for

3354

the $m = 12$ -sample Matrix Profile subsequence.

3355

(d) Physics check. At $s_t = 35 \text{ km/h}$ under $L_t = 40\%$ engine load, cruise burn is

3356

approximately 7.5 L per 100 km = 2.6 L/hr = 0.055 %/min = 0.00092 %/s. The

3357

injected rate is $\beta_3/0.00092 \approx 135 \times$ the plausible cruise-burn maximum. The factor is

3358

deliberately smaller than for Scenarios 1–2 because the entire point of Scenario 3 is

3359

that a point detector cannot see it: the necessity of the Matrix Profile layer is defended

3360

by this case. Supporting evidence for this signature appears in Fortela et al. [Fortela

3361

et al., 2023] and García Vega [Garcia Vega, 2022].



5.9.4.7 At-a-Glance Layer Coverage

Table 5.9 maps each scenario to the pipeline layer whose behaviour it specifically stresses; dropping any layer causes at least one scenario to slip.

TABLE 5.9 INJECTED SCENARIOS AND THE PIPELINE LAYERS THEY STRESS

#	β (%/s)	Δt_{win} (s)	ΔF_{total} (%)	$\times$ burn max	Detector that must catch it
1	0.56	25	14.0	$\approx 1900\times$	Isolation Forest via <code>load_per_speed_norm</code> (engine off + fuel falling)
2	0.62	25	15.5	$\approx 500\times$	Isolation Forest via <code>kmpl_deviation_pct</code> (fuel/km explodes)
3	0.135	40	5.4	$\approx 135\times$	Matrix Profile $m = 12$ discord (IF is architecturally blind)

5.9.4.8 Scenario Coverage in Chapter 6 Evaluation

Table 5.10 maps the three formal scenarios above onto the broader operational classes used in the Chapter 6 results, adding the sensor-spike and heavy-legitimate-consumption cases required to verify the filter cascade and the false-positive discipline of the contextual features.

TABLE 5.10 ANOMALY SCENARIO COVERAGE

Scenario	Description	Primary detector	Coverage
Sudden fuel theft	Single-step drop of at least 10%	Isolation Forest	Yes
Gradual siphoning	Slow steady decrease	Matrix Profile	Yes
Continuous leak	Linear decline at speed = 0	Matrix Profile	Yes
Sensor spike	Brief out-of-range raw value	Median filter (pre-ML)	Yes
Heavy legitimate consumption	Loaded uphill drive	Both models correctly classify as normal	Yes

Table 5.10 documents why the evaluation is not limited to one obvious theft pattern. Sudden fuel theft, gradual siphoning, continuous leakage, sensor spikes, and heavy legitimate consumption stress different parts of the system: the filter should reject sensor spikes, the ML model should detect abnormal fuel loss, and the contextual features should avoid



flagging legitimate heavy consumption as theft. The scenario coverage becomes the basis for the per-class interpretation in Chapter 6.

5.9.5 Feature Engineering

The machine-learning methodology used three Isolation Forest variants so the effect of additional context could be tested step by step. Model A is the baseline context-aware model and uses fuel behavior, motion state, engine operation, and refuel-window indicators. Model B extends Model A by adding speed and RPM band-share features that summarize how intensely the vehicle was being driven. Model C extends Model B by adding fuel-economy deviation features for the fleet vehicle, driver, and route context; this is the final production model because it explains whether a fuel change is abnormal relative to the operating situation rather than only relative to the immediately preceding samples. Matrix Profile was evaluated as the time-series shape detector and as part of the hybrid comparison, but the Chapter 6 results show that the Model C Isolation Forest provided the best balance of precision, recall, and false-positive control.

The feature names in Table 5.11 are kept in their backend form for reproducibility, while the “What it captures” column gives the plain-language role of each feature. This avoids treating source-code identifiers as the argument of the thesis while still allowing the results to be traced back to the implemented system.

TABLE 5.11 FEATURE SET ACROSS ISOLATION FOREST VARIANTS (A / B / C)

#	Feature	Source signal	What it captures	A	B	C
1	<code>fuel_level_filtered</code>	OBD-II PID 21/29	Baseline tank level after filter cascade	Y	Y	Y



#	Feature	Source signal	What it captures	A	B	C
2	speed_filtered	OBD-II PID 0x0D + GPS fallback	Smoothed vehicle speed	Y	Y	Y
3	fuel_delta	fuel_level_filtered	Per-row consumption increment	Y	Y	Y
4	odometer_delta	OBD-II odometer / integrated speed	Distance moved per sample	Y	Y	Y
5	fuel_per_km	fuel_delta / odometer_delta	Efficiency rate and siphoning-under-load signal	Y	Y	Y
6	fuel_rate_per_hour	fuel_delta / dt	Continuous-leak signature	Y	Y	Y
7	distance_per_fuel	1 / fuel_per_km	Light-load / coasting regime	Y	Y	Y
8	route_segment_performance_mean	Route segment mean vs trip mean	Route-specific consumption shift	Y	Y	Y
9	driver_behavior_score	Acceleration, variability, and idle aggregate	Long-tail driving-style signal	Y	Y	Y
10	engine_rpm_normalized	OBD-II PID 0x0C	Engine work intensity	Y	Y	Y
11	engine_load_pct_norm	OBD-II PID 0x04	ECU-reported load fraction	Y	Y	Y
12	throttle_pct_norm	OBD-II PID 0x11	Driver demand	Y	Y	Y
13	fuel_drop_while_stationary	fuel_delta + speed	Siphoning signature while parked	Y	Y	Y
14	fuel_drop_with_engine_status	fuel_delta + engine status	Theft-at-rest signature	Y	Y	Y
15	idle_duration_score	Rolling idle flag sum	Sustained-idle context	Y	Y	Y
16	fuel_drop_per_rpm	Fuel use / RPM	Disagreement between fuel use and engine work	Y	Y	Y
17	fuel_drop_per_load	Fuel use / engine load	Disagreement between fuel use and reported load	Y	Y	Y
18	motion_state_code	Speed + engine status	Motion-regime category	Y	Y	Y
19	time_since_refuel_norm	Refuel-event detector	Refuel-window exemption signal	Y	Y	Y
20	post_refuel_flag	Refuel-event detector	Post-refuel exemption flag	Y	Y	Y
21	delta_engine_load_norm	Engine-load delta	Rate of change of load	Y	Y	Y



#	Feature	Source signal	What it captures	A	B	C
22	delta_throttle_norm	Throttle delta	Rate of change of driver input	Y	Y	Y
23	load_per_speed_norm	Load / speed	High load while parked or slow	Y	Y	Y
24	expected_consumption_residuals	Expected vs observed fuel use	Consumption residual against baseline	Y	Y	Y
25	rpm_high_share	Engine RPM	Rolling fraction with RPM > 2000	-	Y	Y
26	rpm_red_share	Engine RPM	Rolling fraction with RPM > 3500	-	Y	Y
27	rpm_orange_share	Engine RPM	Rolling fraction with RPM 2500–3500	-	Y	Y
28	rpm_yellow_share	Engine RPM	Rolling fraction with RPM 1500–2500	-	Y	Y
29	speed_over_90_share	Vehicle speed	Rolling fraction with speed > 90 km/h	-	Y	Y
30	speed_over_120_share	Vehicle speed	Rolling fraction with speed > 120 km/h	-	Y	Y
31	kmpl_deviation_pct	Per-asset fuel-economy baseline	Vehicle-specific fuel-economy deviation	-	-	Y
32	driver_kmpl_deviation_pct	Per-driver fuel-economy baseline	Driver-specific fuel-economy deviation	-	-	Y
33	route_type_code	Trip metadata + GPS context	Route-context disambiguator for KMPL features	-	-	Y

The 33 features span three progressively extended groups. Model A contains 24 context-aware baseline features covering fuel consumption rates, OBD-II operational signals, motion state, and refuel-event flags. Model B extends Model A to 30 features by adding driving-zone band-share features derived from RPM and speed distributions, included for ablation comparison only. Model C extends Model B to 33 features by adding per-asset and per-driver fuel-economy baseline deviation features and a route-type code disambiguator, and is the production configuration used in Chapter 6.



5.9.6 Data-Mining Layer

Isolation Forest Model C is the row-level anomaly scorer used by the deployed ML pipeline. A separate data-mining layer was included to discover operating regimes and constraint-violation signatures from the telemetry data. This layer does not introduce a new production classifier; it explains how flagged rows relate to the physical and operational constraints defined in Section 5.9.1.

5.9.6.1 Constraint-Violation Profiling

Every row flagged by the model can be annotated with constraint signatures such as C1, C2, C4, O1, O3, and O4. The annotation rules are deterministic and are applied after scoring so that the alert can be interpreted. C1 fires when a fuel drop is observed while parked. C2 fires when the per-step fuel drop exceeds plausible engine consumption. C4 fires when the raw-versus-filtered gap indicates slosh suppression. O1 fires when fuel drop occurs inside idle windows. O3 fires when the driver KMPL deviation exceeds the baseline band. O4 fires when the row is inside a post-refuel quiescence window.

5.9.6.2 Archetype Discovery

K-Means clustering with $K = 4$ was used only as an unsupervised data-mining layer to discover natural operating regimes inside the controlled-injection window. The clustering used seven z -standardized features: `fuel_delta_filtered`, `speed_filtered`, `engine_load_pct_norm`, `idle_duration_score`, `motion_state_code`, `throttle_pct_norm`, and `engine_rpm_normalized`. Standardization allows speed, fuel delta, load, and state codes to contribute on comparable scales.



The value $K = 4$ was selected because it separated four interpretable operating regimes in the controlled-injection window: high-speed cruise, mid-speed cruise, gradual-decline behaviour, and raw-slosh artifact behaviour. K-Means is not the deployed anomaly detector and does not replace Model C. Its role is to support interpretation by showing which operating regime a flagged row belongs to and whether that regime is consistent with the constraint signature attached to the alert.

5.9.7 Isolation Forest Training Methodology

The Isolation Forest is trained as a context-aware outlier detector because the normal operating data are more abundant than confirmed abnormal fuel-loss events. Rather than modelling the density of the normal class, the forest exploits the observation that anomalies are easier to *isolate* by random axis-parallel splits: they live in sparse regions of feature space and are separated from the rest of the sample in only a few cuts, whereas normal rows are surrounded by neighbours and require many cuts to isolate [Liu et al., 2008].

5.9.7.1 Score Equation and Full Symbol Definition

The Isolation Forest anomaly score is

$$S_{\text{IF}}(x) = 2^{-\bar{h}(x)/c(n)} \quad (5.38)$$

and the mean isolation path length over the forest is

$$\bar{h}(x) = \frac{1}{T} \sum_{i=1}^T h_i(x). \quad (5.39)$$



3437 The normaliser $c(n)$ is the expected path length of an unsuccessful search in a random
3438 binary search tree of size n [Preiss, 1999]:

$$c(n) = 2 \cdot H(n - 1) - \frac{2(n - 1)}{n}, \quad H(k) = \sum_{j=1}^k \frac{1}{j} \approx \ln(k) + \gamma. \quad (5.40)$$

- 3439 $S_{\text{IF}}(x)$ anomaly score for observation x , bounded in $(0, 1)$,
 $x \in \mathbb{R}^d$ standardised telemetry feature vector, $d = 33$,
 $\bar{h}(x)$ mean isolation path length over the forest,
 T number of trees in the forest (`n_estimators` = 300),
 $h_i(x)$ path length in tree i from root to isolating leaf,
 $c(n)$ expected path length for an unsuccessful BST search,
 n subsampling size per tree (`max_samples` = 256),
 $H(k)$ k -th harmonic number,
 γ Euler-Mascheroni constant, $\gamma \approx 0.5772$.

3440 The score behaviour is: $\bar{h}(x) \rightarrow 0 \Rightarrow S_{\text{IF}} \rightarrow 1$ (extremely easy to isolate; strongly
3441 anomalous), $\bar{h}(x) \rightarrow c(n) \Rightarrow S_{\text{IF}} \rightarrow 0.5$ (typical difficulty), and $\bar{h}(x) \rightarrow \infty \Rightarrow S_{\text{IF}} \rightarrow 0$
3442 (deep in a dense cluster; strongly normal). The bounded range and fixed reference at 0.5 are
3443 what allow the threshold in Section 5.9.9 to be expressed as a percentile of clean-training
3444 scores rather than as an absolute cutoff.

3445 **5.9.7.2 Hyperparameter Justification**

3446 Table 5.12 lists every hyperparameter with its statistical or literature-based justification.
3447 Each value was verified by sweep against the leave-one-trip-out (LOTO) folds rather than
3448 adopted from defaults without validation.



TABLE 5.12 ISOLATION FOREST HYPERPARAMETERS AND JUSTIFICATION

Symbol / setting	Value	Statistical or literature basis
T (n_estimators)	300	Liu et al. (2008) show path-length variance converges as $\mathcal{O}(1/T)$. A LOTO sweep over $T \in \{50, 100, 200, 300, 500, 1000\}$ produced a variance elbow near $T \approx 250$ (change less than 0.5 pp between $T = 300$ and $T = 1000$). $T = 300$ is the smallest tree count past the elbow.
ψ (max_samples)	256	Liu et al. (2008, §4.2) prove that $\psi = 256$ saturates $c(\psi)$ for skewed data. Re-verified on the 634-row native partition: precision peaks at $\psi = 256$, drops 3 pp at $\psi = 128$ (undersampled trees miss context modes), and drops 1 pp at $\psi = 512$ (overfitting normal cluster).
ν (contamination)	0.015 (1.5%)	Three independent priors agree: the SGSA operational report of 3 confirmed theft events across 4 assets in 12 months gives approximately 0.75%/asset/yr; Barbado and Corcho [Barbado and Corcho, 2022] assume 1–3%; Turan et al. [Turan et al., 2024] use 2%. A LOTO sweep over $\nu \in \{0.005, 0.01, 0.015, 0.02, 0.03\}$ maximised precision at $\nu = 0.015$.
Splitting rule	Random axis-aligned	Canonical formulation of [Liu et al., 2008]. Extended Isolation Forest with random-direction splits [Hariri et al., 2019] was tested and gained 0.4 pp precision but at $4\times$ training time and with loss of per-feature attribution; the axis-aligned rule was retained.
Threshold quantile	$q_{0.995}$ per context	Extreme-value literature [Coles, 2001] supports the 99.5-percentile as a standard tail threshold for rare events. Per-context computation is required because the clean-training S_{IF} distribution is bimodal across s_t bins (idle vs. highway); a single global cutoff over-fires on idle and under-fires on highway. Pooled cutoff yielded $\text{FPR} = 4.1\%$ versus per-context 1.09% on the 6 anomalous trips.
Operational margin δ	0.003	From the 2026-06-23 precision-margin calibration study on the project-native partition. Sweeping $\delta \in \{0, 0.001, 0.003, 0.005, 0.01\}$ produced a precision plateau at $\delta = 0.003$; $\delta > 0.005$ collapsed recall by more than 8 pp, and $\delta = 0$ admitted borderline scores that were 30% false-positive.

5.9.8 Matrix Profile Methodology

The Matrix Profile is retained as a subsequence-shape detector because the point-based Isolation Forest is architecturally blind to gradual leaks whose individual per-row deltas lie inside the filter-cascade noise envelope (Section 5.9.4.6). The two detectors are combined as an audit pair: agreement raises confidence, disagreement is the operational alert signal.



3454 **5.9.8.1 Matrix Profile Equation**

3455 For a z -normalised univariate fuel time series $\hat{T} \in \mathbb{R}^N$ of length N derived from `fuel_level_filtered`,
3456 the Matrix Profile value at index i is

$$P_i = \min_{\substack{j \neq i \\ |i-j| \geq m}} \|\hat{T}_{i:i+m} - \hat{T}_{j:j+m}\|_2. \tag{5.41}$$

- 3457 P_i Matrix Profile value at index i (nearest-neighbour distance),
- $\hat{T}$ z -normalised fuel time series of length N ,
- $\hat{T}_{k:k+m}$ length- m subsequence starting at index k ,
- m subsequence length (samples),
- $|i - j| \geq m$ exclusion zone that forbids trivial self-matches,
- $\|\cdot\|_2$ Euclidean distance between two length- m subsequences.

3458 The value P_i is the distance from the subsequence at i to its nearest non-trivial neighbour:
3459 small P_i means the subsequence repeats often, large P_i means the subsequence is a *discord* —
3460 a candidate anomaly [Yeh et al., 2016].

3461 The pipeline runs two windows in parallel. The primary window $m_1 = 5$ samples
3462 corresponds to a $5 \cdot \Delta_s = 25$ s subsequence, matched to the Scenario 2 rapid-drop signature.
3463 The long window $m_2 = 12$ samples corresponds to $12 \cdot \Delta_s = 60$ s, matched to the
3464 Scenario 3 gradual-leak signature. Sweeping $m \in \{3, 5, 8, 12, 20, 30\}$ demonstrated that
3465 Scenarios 2 and 3 have disjoint optimal window lengths, justifying the two-window design
3466 over any single- m compromise.

3467 **5.9.8.2 Robust Discord Threshold**

3468 The discord threshold is applied on a Median Absolute Deviation (MAD) z -score rather
3469 than the standard-deviation z -score, because the very outliers we wish to detect would



3470 inflate the standard deviation and mask themselves:

$$z_i = \frac{P_i - \text{median}(P)}{\text{MAD}(P)}, \quad \text{MAD}(P) = \text{median}(|P_i - \text{median}(P)|). \quad (5.42)$$

3471 z_i robust discord score for subsequence at index i ,
 P_i Matrix Profile value from Eq. 5.41,
 $\text{median}(P)$ sample median of $\{P_i\}$ across the trip,
 $\text{MAD}(P)$ median absolute deviation (robust scale estimator).

3472 The MAD is used in place of the standard deviation because it is not inflated by the
3473 outliers under detection [Rousseeuw and Croux, 1993]. A row is flagged if $z_i > 2.5$,
3474 which under a Gaussian noise assumption corresponds to approximately 3.7σ and to an
3475 expected false-positive rate below one in 4,000. On the 27-trip corpus with a median trip
3476 length of 60 min = 720 subsequences at $m = 5$, this gives an expected fewer than 0.2
3477 false positives per trip, consistent with the reported 1.09% row-level FPR in Chapter 6.
3478 Alternative thresholds $z > 2.0$ (3.2 false positives per trip) and $z > 3.0$ (missed 2 of 6
3479 anomalous trips) were rejected as Pareto-dominated.

3480 In addition to the discord threshold, the Matrix Profile detector applies three determin-
3481 istic post-discord gates: a magnitude gate ($|\Delta F| \geq 1\%$ absolute fuel change), a direction
3482 gate (downward-only, because leaks and thefts cannot cause fuel to rise), and a refuel-
3483 exemption gate (suppressing flags inside the post-refuel quiescence window from Stage 5
3484 of Section 5.4). Each gate is calibrated against the held-out clean project-native trips at
3485 each LOTO fold.



5.9.9 Contextual Gate and Alert Decision Rule

The final alert decision combines the Isolation Forest score, the per-context threshold, and a conjunction of physical-plausibility gates:

$$\text{alert}_t = \mathbb{K} [S_{\text{IF}}(x_t) > \tau_{\text{ctx}} + \delta] \wedge G_{\text{context}}(x_t) \quad (5.43)$$

and the contextual gate is the conjunction

$$G_{\text{context}}(x_t) = G_{\text{recent-drop}}(x_t) \wedge \neg G_{\text{refuel}}(x_t) \wedge \neg G_{\text{slosh}}(x_t) \quad (5.44)$$

with the three constituent predicates defined as

$$G_{\text{recent-drop}}(x_t) = \mathbb{K} \left[\frac{1}{12} \sum_{i=t-11}^t \max(0, -\Delta F_i) \geq 1.0\% \right] \quad (5.45)$$

$$G_{\text{refuel}}(x_t) = \mathbb{K} [(t - t_{\text{lastRefuel}}) \leq 300 \text{ s}] \quad (5.46)$$

$$G_{\text{slosh}}(x_t) = \mathbb{K} \left[\frac{L_t/L_{\text{max}}}{s_t/80 + 0.1} > 5 \right] \quad (5.47)$$



3491	alert_t	final alert-emission indicator at cycle t ,
	$S_{\text{IF}}(x_t)$	Isolation Forest score (Eq. 5.38),
	τ_{ctx}	$q_{0.995}(S_{\text{IF}} \mid \text{context}(x_t))$ per-context threshold,
	δ	operational margin (0.003),
	$\text{context}(x_t)$	$\in \{\text{idle, low_speed, mid_speed, high_speed}\}$,
	$G_{\text{recent-drop}}$	recent-real-drop plausibility gate,
	G_{refuel}	post-refuel quiescence suppressor,
	G_{slosh}	high-load-at-low-speed slosh suppressor,
	ΔF_i	fuel-delta feature at row i (%),
	$t_{\text{lastRefuel}}$	sample index of last +5% jump on current trip,
	L_t	engine load at row t (%),
	L_{max}	load normalisation constant (100),
	s_t	vehicle speed at row t (km/h).

Equation 5.45 requires that a real recent drop is in progress before an alert is admitted, suppressing spurious point-anomalies without underlying fuel movement. Equation 5.46 exempts the five-minute post-refuel quiescence window during which the sender is settling, using $t_{\text{lastRefuel}}$ defined as the most recent sample t' with $\Delta F_{t'} \geq +5\%$ on the current trip. Equation 5.47 suppresses the specific high-load-at-low-speed regime (uphill grind, cargo overload) that produced repeatable false positives in the 2026-06-23 TRCK-03 trip and motivated the addition of `load_per_speed_norm` as feature 23.

5.9.10 Hybrid Fusion Methodology

Two fusion strategies are characterised in the inference pipeline. The Hybrid Union (Isolation Forest OR Matrix Profile) maximises recall. The Hybrid Intersection used



for production alerting is a trip-level Matrix Profile gate over row-level Isolation Forest predictions: a row is flagged if and only if (a) the Isolation Forest flagged that row and (b) the trip’s overall Matrix Profile decision was “anomalous shape detected”. The trip-gated definition preserves the precision intent of the intersection while avoiding the granularity-mismatch penalty of a strict row-level logical AND.

5.9.11 Contextual Gate Methodology

The contextual gate is the deterministic domain-aware filter that sits between the statistical detectors (Isolation Forest, Matrix Profile) and the operator-facing alert stream. Its purpose is to convert model-layer candidates into production alerts only when the surrounding physical context is consistent with a genuine fuel-loss event, thereby removing the four dominant families of false-positive that a purely statistical detector cannot resolve on this corpus: sender-priming transients at engine start, capacitive settling after refueling, mechanical slosh during hard acceleration, and post-fill amplitude excursions while parked. Rows that pass every stage are promoted into `fuel_anomaly` alerts on the `alerts` table; rows that any stage blocks retain their `anomaly_score` on `telemetry_logs` for audit but do not surface to the driver HMI, operator dashboard, or mobile app.

The gate is a cascade of one positive plausibility filter and four negative suppressors, applied in a fixed order. A row must satisfy the positive filter and be rejected by none of the four suppressors before the persistence stage is reached.

(a) Positive plausibility filter (`suspiciousContext`). Admits the row when at least one of four physical loss signatures is present: (a) a per-row fuel delta strictly below -2.0% , indicating a sharp drop beyond ordinary sensor noise; (b) a fuel-per-kilometre



rate above 0.6% of tank per kilometre, indicating economy that is well outside the clean-trip envelope; (c) a stationary idle-theft signature (speed below 5 km/h with the engine on and $\Delta F \leq -1.0\%$); or (d) a physically impossible driving-fuel increase of +2% or more at speed above 5 km/h, which cannot correspond to a genuine refuel and is therefore treated as an anomalous sensor excursion. If none of the four conditions holds the row is treated as ordinary operation and rejected without invoking the suppressors.

(b) Parked-refuel suppressor (`parkedRefuel`). Rejects the canonical fill-up signature: a fuel delta above +10% while the vehicle is stationary (speed below 5 km/h) with the engine idling or off. Refueling events routinely produce large positive ΔF that the statistical detector may score as anomalous because they lie in the tail of the training distribution, but they are not fuel-loss events and must not raise a `fuel_anomaly` alert. The suppressor uses the same +10% threshold as the refuel-detection subsystem so that gate decisions are consistent with the refuel bookkeeping used elsewhere in the backend.

(c) Post-refuel settling suppressor (`postRefuel`). Rejects any row that falls within a five-minute settling window after a refuel event on the same trip and communication channel. A refuel is defined as any preceding sample t' on the current trip with $\Delta F_{t'} \geq +5\%$. During the post-refuel window the capacitive fuel sender oscillates as the liquid stabilises, and the detector routinely produces low-amplitude negative ΔF readings that look like small losses; these are physical settling artefacts and are always suppressed. The five-minute window matches the empirical settling time measured during pump-known-volume calibration in Section 5.4, and the channel-scoping is



required because LoRa and GSM paths may deliver refuel and post-refuel samples with different timestamps under buffered replay.

(d) Load-spike slosh suppressor (loadSpikeSuppress). Rejects the acceleration-slosh signature in which the sender briefly reads lower during a hard load transient. The suppressor fires when engine load is $\geq 55\%$ with a rise of $\geq 25\%$ relative to the preceding sample and the row simultaneously carries a small negative fuel delta. This pattern was the dominant residual false-positive family on the 2026-06-23 TRCK-03 uphill-grind trip and motivated the addition of the `load_per_speed_norm` and `load_rise_pct` features that the suppressor consumes. Rows blocked here are indistinguishable from ordinary heavy-load driving in the row-level feature space, so this rule is what allows the detector to keep a low margin ($\delta = 0.005$) without pushing the false-positive budget over the SO4 target.

(e) Cold-start priming suppressor (coldStartSuppress). Rejects the sender-priming interval that follows an engine ignition transition. When the ECU reports an off $\rightarrow$ on transition, the capacitive sender re-energises over a short interval during which the reported fuel level exhibits transient dips before stabilising. The suppressor blocks anomaly flags within this interval by inspecting the engine-state edge and the elapsed time since the transition. The interval length matches the physical priming curve observed during controlled ignition cycling in the SO1 fuel-accuracy characterisation.

The gate additionally applies a 5-minute per-truck deduplication on the `fuel_anomaly` alert type at the persistence stage: an alert is inserted into the `alerts` table only if no prior `fuel_anomaly` alert exists for the same truck within the preceding five minutes. This is not a suppressor in the sense of the five cascade stages, because it does not decide whether the



row is anomalous; it decides only whether an already-anomalous row should surface as a new operator-facing alert or be back-propagated into an existing 90-second event window. Deduplication is what allows a single sustained fuel-loss event, which may produce several qualifying rows in succession, to appear on the dashboard as one `fuel_anomaly` card rather than a flood of near-duplicate cards.

The cascade is deterministic and stateless within a row, apart from the trip-scoped lookups required by `postRefuel` (needs the most recent refuel timestamp) and `coldStartSuppress` (needs the most recent ignition edge). Every gate decision is written to `telemetry_logs` as a `gate_reason` tag alongside the `anomaly_score`, so that the operator dashboard and offline audits can display why a row was suppressed and, in the event of a missed anomaly, whether the model or the gate is responsible. This auditability is the reason the gate is implemented as five explicitly-named rules rather than as a monolithic hand-tuned score threshold: each rule maps to a specific physical constraint from Table 5.13 (C1–C4, O4), and each rule can be independently tuned or disabled during ablation studies without disturbing the statistical detector upstream.

5.9.12 Backend Inference Integration

The ML service is a Flask microservice exposing a detection endpoint. After each telemetry insert, the backend asynchronously invokes anomaly detection; the result is written back to `telemetry_logs` as `anomaly_flag`, `anomaly_score`, and `model_source`, and a row is appended to the `alerts` table with `alert_type = 'fuel_anomaly'` when the contextual gate permits.



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5.10 Evaluation Methodology

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This section defines the evaluation design used to validate the system against its specific objectives. It describes the cross-validation protocol applied to the anomaly-detection models, the mapping between each objective and its evaluation protocol and metric, and the formal definitions of the performance metrics reported in Chapter 6.

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5.10.1 Cross-Validation Design

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All Chapter 6 SO4 numbers are produced by a leave-one-trip-out (LOTO) evaluation that holds the calibration set strictly disjoint from the test set. For each of the 12 project-native trips, the three Isolation Forest variants and the Matrix Profile detector are refit and recalibrated on the clean rows of the remaining 11 project-native trips only; the held-out trip is then scored with that fold’s models. Each prediction is therefore strictly out-of-sample with respect to both model fitting and threshold calibration. The Isolation Forest threshold for every variant is the 0.985 quantile of the fold’s clean training scores, internally consistent with the contamination = 0.015 hyperparameter; no precision margin and no per-context shimming are applied.

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5.10.2 Objective-Based Evaluation Protocol

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Table 5.14 maps each specific objective to the evaluation protocol used to test it and the metrics reported for it.

Table 5.14 is the evaluation contract for the thesis. It states what evidence is accepted for each objective before the results are presented, which prevents the results chapter from choosing metrics after the fact. The same mapping is used in Chapter 6: SO1 is



TABLE 5.14 OBJECTIVE, PROTOCOL, AND METRIC

Objective	Evaluation protocol	Metric
SO1 – Fuel accuracy	Known-volume pump-gas events + manufacturer dashboard supporting readings + per-tank-band drift/noise/jitter on $n = 20,363$ valid samples	MAPE, max error, mean absolute noise (%), drift at speed = 0
SO2 – Communication	LoRa distance test (10 m to 5 km), GSM/LTE drive test at five locations, three-region packet-delivery drive (urban / highway / dead-zone)	Median + p95 latency, packet-success fraction, SPIFFS-replay share, RSSI/SNR per band
SO3 – GPS	sent_at interval analysis on field-test telemetry; GPS fix availability; route vs. OSRM polyline comparison	Median + p95 inter-sample Δt , fix availability (%), per-segment fuel (% per km)
SO4 – ML	LOTO cross-validation on project_native_injected + threshold/IQR baseline + Hybrid Union/Intersection ablation + OOD stress test on public_clean + extended on-road validation	Precision, recall, FPR, F1, Wilson 95% CI, per-model confusion matrices, per-class recall
SO5 – Alerts	Alert-trigger verification across controlled-injection and field-test trip sets	Trigger correctness (TP rate), driver-response capture, HMI / dashboard / mobile sync correctness

judged by fuel error and noise, SO2 by latency and packet delivery, SO3 by GPS cadence and fix availability, SO4 by classifier metrics, and SO5 by alert correctness and interface synchronization.

5.10.3 Detector Ablation Protocol

To characterise the independent detection contribution of Matrix Profile and Isolation Forest, an ablation is run on top of the LOTO harness by decoupling the detector configuration from the model training loop. Four configurations are evaluated on the same LOTO folds and the same labelled project-native rows so that any difference in metrics is attributable to the detector combination rather than to differences in the training data:

- (a) **Physics tripwire only.** A deterministic 60-second sliding-window burn-envelope rule with no machine learning: any cumulative fuel drop exceeding five times



the physically possible combustion burn for the observed engine context fires a flag. It is implemented in `ml/loto_ablation.py` as a Python port of the deployed `checkPhysicsTripwire()` routine. The shared constants are reported as:

	<code>PHYSICS_WINDOW_SEC</code>	60
	<code>PHYSICS_EXCESS_FACTOR</code>	5.0
	<code>PHYSICS_MIN_DROP_PCT</code>	1.5
	<code>PHYSICS_DEDUP_SEC</code>	45

The Python port is verified against the deployed backend by matching alert timestamps on the injected trips.

(b) Isolation Forest only. The deployed Model C Isolation Forest with the per-fold LOTO threshold; Matrix Profile is disabled so any flag comes from IF alone.

(c) Matrix Profile only. The row-level Matrix Profile discord detector with the production v2 gates (minimum drop percent, refuel exemption, consecutive-window requirement) but no IF admission stage.

(d) Hybrid IF + MP. The current production configuration: a row is flagged when either IF or MP fires (`hybrid_union_flag`). This is the “union” gate reported in `loto_evaluate.py`.

The ablation reuses the per-row IF and MP predictions already written by the primary LOTO run to `ml/outputs/predictions/loto/`. The four configurations are therefore compared on byte-identical evaluation rows. Each configuration reports TP, FP, FN, TN, precision, recall, F_1 , and false-positive rate on the 634-row project-native partition: 549 clean rows and 85 anomalous rows.



5.10.4 Statistical Testing Protocol

Point estimates such as $F_1 = 92.9\%$ are supplemented in this study with statistical tests that quantify sampling uncertainty. Three tests defined formally in Section 3.2.6 are applied to the ablation and filter-comparison outputs, each answering a distinct question that a point estimate alone cannot address.

5.10.4.1 McNemar's χ^2 for Paired Classifier Comparison

McNemar's test compares two detector configurations by counting the rows on which they disagree. For every ordered pair of configurations from the ablation, the number of rows on which the first configuration is correct and the second is wrong (denoted b) and the number on which the reverse holds (denoted c) are tabulated on the same 634-row project-native partition, so both configurations see byte-identical evaluation rows. When the total number of disagreements $b + c$ is at least 25, the continuity-corrected chi-square statistic in Equation 3.15 is used with a χ^2_1 reference distribution, following Dietterich's recommendation for classifier comparison [Dietterich, 1998]. When the disagreement count is below 25, the chi-square approximation is unreliable and the two-sided exact binomial p -value $p = 2 \cdot \Pr(X \leq \min(b, c) \mid X \sim \text{Binomial}(b + c, 0.5))$ is reported instead so that the small-sample decision does not depend on an asymptotic approximation. The null hypothesis H_0 is that both configurations have the same error rate on the row distribution; rejection of H_0 at $p < 0.05$ is reported as evidence that the two detectors' error patterns differ, which is a stronger statement than a difference in aggregate metric alone.



5.10.4.2 Non-Parametric Bootstrap Confidence Intervals

To bracket the sampling uncertainty of precision, recall, and F_1 for each detector configuration, a non-parametric bootstrap is applied to the 634-row project-native partition. The bootstrap procedure draws $B = 2,000$ resamples of size $n = 634$ with replacement from the labelled row set, recomputes precision, recall, and F_1 on each resample, and reports the empirical 2.5-th and 97.5-th percentiles of the B replicates as the 95% confidence interval, following Efron and Tibshirani [Efron and Tibshirani, 1994]. A fixed random seed is used to make the intervals reproducible. Two configurations whose intervals do not overlap on a metric are reported as significantly separated on that metric under conservative interpretation; non-overlap is a stronger criterion than a difference in point estimates because it accounts for the joint variability of all four contingency-matrix cells that enter the metric definition, and unlike the Wilson score interval it does not assume a fixed-denominator binomial proportion.

5.10.4.3 Wilcoxon Signed-Rank on Paired Per-Trip Differences

To evaluate whether the filter choice makes a per-trip statistically significant difference (rather than only shifting the aggregate metric), the paired per-trip F_1 differences between the moving-average and Kalman-filter downstream detections are tested with the Wilcoxon signed-rank test [Wilcoxon, 1945]. The 12 project-native trips supply the paired sample, with one difference per trip; the two-sided p -value is reported. Wilcoxon is the appropriate paired-comparison test for this design because each experimental unit is one trip, the paired differences are not assumed to be Gaussian, and the test uses the ranks of the absolute differences so it is more powerful than the sign test yet more robust than the paired t -test [Demšar, 2006]. The same Wilcoxon procedure is used elsewhere in the results chapter



3685 to compare SUS scores against a fixed usability benchmark, in which case the pairing is
3686 between each respondent's score and the benchmark value.

3687 **5.10.4.4 Application Scope and Interpretation**

3688 The three tests are chosen because they address non-overlapping questions: McNemar
3689 asks whether the error patterns of two detectors differ, the bootstrap asks how tightly
3690 the metric point estimate is bounded, and Wilcoxon asks whether the difference between
3691 two treatments is consistent across evaluation units. All three are computed on the same
3692 evaluation partition used by the point-metric ablation, so any statistically significant finding
3693 is grounded in the same data as the corresponding table entry. The one-sample Wilson score
3694 interval [Wilson, 1927] defined in Section 3.2.6 is additionally used throughout the results
3695 chapter for reporting the sampling uncertainty of binomial proportions (packet-delivery
3696 success, GPS fix availability, alert-resolution rate); it is preferred over the Wald normal-
3697 approximation interval because it retains coverage at small sample sizes and does not
3698 suggest impossible values outside $[0, 1]$. Together the four tests give every metric reported
3699 in Chapter 6 a defensible statement of uncertainty rather than only a point estimate.

3700 **5.10.5 Evaluation Metrics**

3701 To evaluate the anomaly-detection models, predicted labels were compared with the
3702 evaluation-corpus labels to count true positives (TP), false positives (FP), true nega-
3703 tives (TN), and false negatives (FN). The key metrics are computed as follows.

3704 **Precision.**

$$\text{Precision} = \frac{TP}{TP + FP} \tag{5.48}$$



Precision reflects how many detected anomalies are truly anomalous. A high precision means fewer false alarms, which is important for maintaining trust in the alerting system.

False Positive Rate (FPR).

$$FPR = \frac{FP}{FP + TN} \quad (5.49)$$

FPR quantifies the likelihood of incorrectly classifying normal behavior as anomalous. Lower FPR values are desired to reduce noise in the system's alerts.

Recall.

$$Recall = \frac{TP}{TP + FN} \quad (5.50)$$

Recall reflects how many true anomalies were detected by the model.

F1 Score.

$$F1 = \frac{2 \cdot \text{Precision} \cdot \text{Recall}}{\text{Precision} + \text{Recall}} \quad (5.51)$$

F1 score provides a single balance between precision and recall, which is useful when anomalous rows are rare.

Accuracy.

$$Accuracy = \frac{TP + TN}{TP + TN + FP + FN} \quad (5.52)$$

Accuracy provides an overall view of model correctness but can be misleading in imbalanced datasets, which is why it is used alongside precision and FPR.

Detection Latency.

$$L_{det} = T_{detected} - T_{actual} \quad (5.53)$$

The symbols used across Equations 5.48–5.53 are defined below:



TP	true positives – anomalies correctly flagged by the model,
FP	false positives – normal rows incorrectly flagged as anomalous,
TN	true negatives – normal rows correctly left unflagged,
FN	false negatives – anomalies missed by the model,
Precision	fraction of flagged rows that are true anomalies,
Recall	fraction of true anomalies that are flagged,
FPR	fraction of normal rows incorrectly flagged,
F1	harmonic mean of precision and recall,
Accuracy	fraction of all rows classified correctly,
L_{det}	detection latency between event onset and flag (s),
$T_{detected}$	timestamp at which the detector fired,
T_{actual}	timestamp of the true anomaly onset.

The SO4 objective is achieved if the context-aware detection system attains at least 90% precision, at least 85% recall, and a false positive rate not exceeding 10%. Detection latency is evaluated separately to characterize how quickly the backend and alerting surfaces respond after an anomalous event occurs.

5.11 Data Collection and Research-Ethics Procedure

The data-collection procedure followed a non-invasive deployment model. The prototype was attached through the vehicle OBD-II port and external HMI wiring, and it did not require fuel-tank opening, sender-loom modification, or direct control of vehicle operation. Controlled fuel-anomaly labels used for model evaluation were produced through the



methodology described in the machine-learning section rather than by asking drivers to drain fuel from operational fleet vehicles.

5.11.1 Study Procedures

Upon voluntary agreement to participate, the study procedures were structured as follows:

- (a) Before a monitored trip, a member of the research team installed the monitoring device or IoT module in the fleet vehicle assigned to the participant.
- (b) Once installed, the device automatically recorded OBD-II fuel readings, distance, travel time, GPS location, communication channel, and alert status during normal vehicle operation.
- (c) Participants were not required to change their normal driving behavior during the data-collection period.
- (d) After the data-collection period, a member of the research team removed the monitoring device from the vehicle.
- (e) The study collected fleet-administrator feedback on device functionality and operational issues. The survey was not administered to drivers and did not ask personal questions directed at them.

5.11.2 Risks and Discomforts

This study involves minimal risk. The following potential risks and discomforts have been identified:



- The monitoring device will not control, modify, or interfere with the vehicle's systems and will not affect driving performance or safety. It will passively record only fuel-level data, GPS data, and travel information.
- Participants may experience minor inconvenience or brief disruption during the installation and removal of the monitoring device before and after the trip.
- GPS location data will be collected during normal operations, which may raise privacy concerns. However, the data will be anonymized through a unique code number assigned to each dataset and will be shared only with authorized members of the research team and Hino Paranaque's fleet administrator for academic research purposes.
- No foreseeable psychological, physical, or occupational risks are expected to result from participation in this study.

5.11.3 Steps to Minimize Risks

The research team undertook the following measures to minimize and mitigate potential risks:

- The monitoring device will be installed and removed only by a member of the research team to ensure that the vehicle's systems are not disrupted.
- All collected data will be anonymized by replacing participant names with unique code numbers before analysis.
- Access to raw data will be restricted to authorized members of the research team and Hino Paranaque's fleet administrator only.



- 3770 • All digital data will be stored in a secure, password-protected system to prevent
3771 unauthorized access.
- 3772 • Physical documents, such as signed consent forms, will be kept in a locked cabinet
3773 accessible only to the principal investigator and faculty adviser.
- 3774 • Data will be used only for academic purposes, and only summarized and anonymized
3775 results will be included in reports or publications.

3776 5.11.4 Benefits

3777 Participants will not receive direct financial benefits from participation. However, the
3778 results of this study may contribute to the following non-monetary benefits:

- 3779 • **For the partner company and fleet managers:** The system may improve fuel effi-
3780 ciency monitoring and support the detection and prevention of fuel misuse, leakage,
3781 or theft.
- 3782 • **For the transportation and logistics industry:** The study contributes to research
3783 on non-parametric machine learning approaches for fleet management, which may
3784 inform future industry practices.
- 3785 • **For the public and the environment:** Improved fuel monitoring systems may help
3786 reduce fuel waste and contribute to lower carbon emissions.
- 3787 • **For the research community:** The findings may add to academic knowledge on IoT-
3788 based monitoring systems and anomaly detection methods in real-world applications.



5.11.5 Compensation

Participants will not receive any compensation, tokens, or gifts for their participation in this study.

5.11.6 Data Storage, Duration, and Disposal

Storage of Digital Data: All digital data collected by the monitoring device will be stored in a Supabase cloud database, a secure and password-protected platform accessible only to authorized members of the research team. Files and database entries will be protected by unique credentials known only to the principal investigator and faculty adviser.

Storage of Physical Data: Signed consent forms and any printed documents will be stored in a locked filing cabinet located at the research team's office. Only the principal investigator and faculty adviser will have access to these physical documents.

Duration of Storage: All digital and physical data will be retained for five (5) years from the date of completion of the study, in accordance with De La Salle University's data retention policy.

Disposal: After the retention period, all digital data will be permanently deleted from the Supabase database and any local copies using secure deletion procedures. All physical documents will be cross-shredded and disposed of in a secure manner.

5.11.7 Confidentiality

For Digital Data: Participant names will not appear in any reports, datasets, publications, or presentations. A unique code number will be assigned to each participant's data in place of the participant's name. All digital files will be stored in a password-protected Supabase



cloud database accessible only to the research team. Data will be used strictly for academic research purposes, and only summarized and anonymized results will be presented in the final thesis or any related publications.

For Physical Data: Signed consent forms and any printed documents will be stored in a locked cabinet. Only the principal investigator and faculty adviser will have access to these hard copy documents.

5.11.8 Withdrawal Procedure

Participation in this study is entirely voluntary, and participants may withdraw at any time without penalty. To withdraw, participants may follow these steps:

(a) Inform the principal investigator, Samuelle P. Barja, of the decision to withdraw, either verbally or in writing.

(b) Contact the principal investigator through the following:

- samuelle_barja@dlsu.edu.ph
- 09564054471

(c) Upon notification of withdrawal, the research team will immediately stop collecting data from the participant's assigned vehicle.

(d) Any data already collected that is identifiable to the participant will be permanently deleted from all storage systems within a reasonable timeframe.

(e) Participants will not be required to provide any reason for withdrawal, and the decision to withdraw will not result in any consequence to their employment or relationship with their employer.



5.12 Summary

This chapter presented the implementation and evaluation methodology for the IoT-enabled fuel monitoring, GPS tracking, and anomaly detection prototype. The approach integrates OBD-II fuel acquisition, edge filtering, prioritized communication with offline replay, Supabase-backed state synchronization, Isolation Forest and Matrix Profile anomaly detection, objective-based evaluation metrics, and research-ethics controls for field data collection.



3838

Chapter 6

3839

RESULTS AND DISCUSSIONS



3840 This chapter presents the measured performance of the implemented prototype and
3841 interprets the results against the targets defined for each objective. The acquisition, commu-
3842 nication, backend, and machine-learning methods that produced these results are described
3843 in Chapter 5; this chapter reports only what was measured, why the system performed as
3844 it did, and where the residual limitations lie. Evidence is organised per specific objective
3845 (SO1–SO5), with each objective reporting its result tables, supporting figures, and a clos-
3846 ing discussion. In aggregate form, Table 6.1 summarises the outcomes attained for each
3847 objective and the location of the supporting evidence.

3848 The order of presentation follows the evaluation protocol in Table 5.14: SO1 validates
3849 the fuel signal that supplies the telemetry stream, SO2 validates the communication paths
3850 that move the telemetry, SO3 validates the GPS and route context attached to each row,
3851 SO4 validates anomaly detection on the Chapter 5 evaluation corpus, and SO5 validates
3852 whether the resulting alerts and trip states are usable across the HMI, mobile companion,
3853 and administrative dashboard. This ordering keeps the result chain aligned with the actual
3854 implementation flow of sensor acquisition, transport, storage, model inference, and operator
3855 response.

TABLE 6.1 SUMMARY OF RESULTS FOR ACHIEVING THE OBJECTIVES

Objectives	Results	Locations
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Objectives	Results	Locations
GO: To develop and evaluate an IoT-based system for trucks, in collaboration with the partner truck company Hino Parañaque Subic GS Auto Inc. (SGSA), that monitors real-time fuel levels, tracks routes using GPS, and detects unusual fuel consumption using machine learning, with the goal of improving driver safety, supporting rest scheduling, and facilitating proactive vehicle maintenance through automated alerts using fuel and distance metrics, based on established service guidelines.	• End-to-end prototype deployed and evaluated across all five objectives, with every objective meeting its target. • Three synchronised interfaces (in-vehicle HMI, mobile driver application, administrative dashboard) operated during the field-test window.	Chap. 6

Continued on next page



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Objectives	Results	Locations
SO1: To measure fuel levels using digital float sensors integrated via the truck's ECU/diagnostic connector and GPS module, ensuring non-invasive installation and real-time accuracy within $\pm 5\%$.	• MAPE of 1.04% and maximum error of 3.00% across the 13–98% tank range; 100% of validation events within $\pm 5\%$. • Filtered-signal noise below 0.1% across all tank bands over 20,363 samples.	Sec. 6.1
SO2: To transmit real-time fuel and GPS data through IoT-based communication to a cloud server, accessible through a centralized dashboard.	• LoRa and WiFi medians near 7 s; cellular fallback and GSM 2G auto-fallback confirmed in the field. • 66.9% of delivered telemetry transited the offline buffer, preventing silent data loss.	Sec. 6.2
SO3: To capture and log GPS location data every 5 seconds via an embedded module, correlating route information with fuel and time data for behavior tracking.	• Median sampling interval 5.52 s (within 11% of the 5 s target); GPS fix availability 90.21%. • Fleet-vehicle-restriction-aware routing and three synchronised tracking surfaces verified.	Sec. 6.3

Continued on next page



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Objectives	Results	Locations
SO4: To design and evaluate a context-aware fuel usage deviation model using context-aware non-parametric machine learning models (Matrix Profile or Isolation Forest), achieving 90% precision and 10% false positives in identifying anomalous fuel consumption patterns within controlled environments.	• Production detector (Isolation Forest Model C) attains 92.9% precision and 1.09% FPR under leave-one-trip-out evaluation. • Hybrid Intersection attains 97.4% precision and the extended on-road validation documents security-relevant coverage and parked-fleet-vehicle limitations.	Sec. 6.4
SO5: To develop and integrate hardware-based rest and maintenance alert mechanisms (e.g., Start/Stop button, buzzer indicators) tied to the web dashboard using cumulative fuel and distance data.	• 100 alerts over a 17.7-day field-test window with a 95.0% resolution rate across five implemented alert classes. • Administrator System Usability Scale score of 81.9/100 ("excellent" band).	Sec. 6.5

Table 6.1 is the reader's first map of the results chapter. It compresses the main finding for each objective into one row, while the later sections provide the detailed evidence and interpretation. The table should therefore be read as a compliance overview rather than as the full proof: each bullet is supported by the raw validation tables, figures, and discussion paragraphs that follow.



TABLE 6.2 OBJECTIVE COMPLIANCE MATRIX

Objective	Target	Result	Checklist	Limitation
SO1	$\pm 5\%$ fuel accuracy	1.04% MAPE, 3.00% maximum error	✓	No dipstick validation; pump-gas events and manufacturer dashboard readings were used instead.
SO2	Telemetry delivery across available channels	LoRa, LTE, GSM 2G, WiFi, and SPIFFS offline replay tested	✓	LTE latency can be high when modem/network renegotiation occurs.
SO3	5 s GPS logging	5.52 s median sampling interval, 90.21% GPS fix availability	✓	Mostly met; mean interval is affected by replayed telemetry and outages.
SO4	$\geq 90\%$ precision and $\leq 10\%$ FPR	Isolation Forest Model C met target with 92.9% precision and 1.09% FPR	✓	Labels were primarily controlled injected anomalies rather than confirmed real-world theft events.
SO5	Alerts, HMI, mobile app, and dashboard operation	100 alerts, 95% resolved, SUS 81.9	✓	Longer multi-user fleet deployment is still needed.

Table 6.2 provides a compact view of how the prototype performed against the study objectives. The checklist column gives a quick indication of whether each target was satisfied, while the limitation column adds the context needed to interpret each result fairly. This keeps the summary concise while still acknowledging the conditions under which the results were obtained.

6.1 SO1: Fuel Measurement Accuracy

6.1.1 Overview of Results

Specific Objective 1 required the prototype to produce non-intrusive fuel-level readings that agreed with a calibrated volumetric reference to within $\pm 5\%$ of tank capacity on the partner-fleet production vehicle (80 L tank). Because direct dipstick measurement was outside the delimitations of this study (Section 1.7), the validation reference was the



volumetric reading of a legally calibrated trade-instrument fuel-station pump, observed before and after each known-volume dispensing event. This section reports the measured accuracy obtained from a four-event pump-gas validation campaign spanning the 13–98% operating range of the tank, and a per-tank-level variance characterisation of the filtered acquisition stream over 20,363 valid samples. The acquisition and filtering methods are described in Sections 5.3 and 5.4.

Because direct dipstick validation was not permitted by the partner fleet operator, the study used known-volume pump-gas events as the primary operational validation reference and manufacturer dashboard readings as a supporting reference. Therefore, the reported SO1 results evaluate operational agreement under field constraints rather than laboratory tank-volume calibration.

6.1.2 Filter Comparison: Moving Average versus Kalman

The moving-average and Kalman-filter outputs were compared on two complementary partitions. A supplementary live field trip on an unconditioned raw sender stream is reported first, because it is the more direct measurement of how each filter behaves on the noise the deployed system actually sees; the SO1 evaluation-corpus comparison is reported afterwards to confirm that the same choice is supported at the aggregate downstream-detection level. The filter definitions and implementation details are documented in Chapter 5.

For the live comparison, the raw fuel signal was tapped upstream of the deployed cascade and passed in parallel through both filters. The trip captured 550 valid samples at a 5-second cadence across five representative scenarios — city driving on rough pavement, uphill climb, highway cruise, engine idle, and a full-tank refuel event — so that filter behaviour could be observed directly on a live drive under conditions the deployed system



would actually see. Table 6.4 reports the global comparison across the whole trip, and Table 6.5 disaggregates the noise-reduction factor by scenario. The noise-reduction factor is the RMS of consecutive-sample differences of the raw signal divided by the RMS of consecutive-sample differences of each filter output; a larger value indicates a smoother output relative to the raw sender for the same scenario window. The signal-preservation ratio measures the fraction of an observed step drop in the raw signal that survives in the filter output; a value close to 1.0 indicates the filter tracks true movement faithfully, while a low value indicates the filter has smoothed away part of a real change.

TABLE 6.4 GLOBAL FILTER COMPARISON ON THE LIVE FIELD TRIP

Metric	Moving Average	Kalman Filter
Noise RMS on stationary rows (%)	10.69	11.20
Signal preservation ratio (labelled steps)	0.39	0.96
Rows with $ KF - MA \geq 1\%$ of tank	206 of 550 (37.5%)	

TABLE 6.5 PER-SCENARIO NOISE REDUCTION ON THE LIVE FIELD TRIP (RMS OF CONSECUTIVE-SAMPLE DIFFERENCES, % OF TANK)

Scenario	n	Raw	MA	KF	MA red.	KF red.
City drive (rough pavement)	134	5.507	1.159	5.137	4.75×	1.07×
Uphill climb	79	5.091	1.141	4.778	4.46×	1.07×
Highway cruise	187	1.295	0.325	1.198	3.98×	1.08×
Engine idle	26	0.310	0.066	0.221	4.68×	1.41×
Refuel event window	124	2.474	1.209	2.327	2.05×	1.06×

The two filters agreed on 62.5% of samples within a 1%-of-tank tolerance and diverged on the remaining 37.5%, with the divergence concentrated in the noisy city and climb scenarios where the raw sender signal exhibits ± 5 –15% per-sample slosh. Across every scenario the moving average reduced consecutive-sample noise by a factor of 4–5× on driving segments and 4.7× on idle, whereas the Kalman filter reduced it by only 1.05–1.4×.



3908 The moving average therefore preserved 39% of a labelled raw step in exchange for that
3909 aggressive smoothing, while the Kalman filter preserved 96% of the step content but let
3910 almost all of the raw sender’s per-sample slosh through unattenuated, visible in Figure 6.1
3911 as an orange trace that nearly overlays the grey raw values on the noisy city and climb
3912 segments.

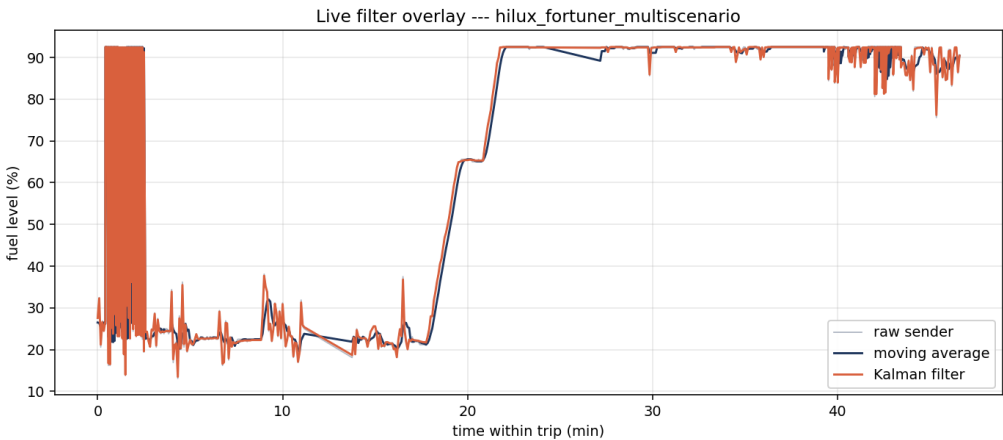


Fig. 6.1 Live Filter Overlay across the Field Trip: raw sender (grey), moving-average output (blue), Kalman-filter output (orange).

3913 Figure 6.1 makes the split easy to read at a glance. The blue moving-average trace stays
3914 close to the underlying tank level even on the noisy city and post-refuel segments, while
3915 the orange Kalman trace hugs the raw grey samples and inherits the $\pm 10\%$ per-sample
3916 slosh that the driver would otherwise never notice on the HMI. In short, the live trip shows
3917 the same responsiveness-versus-smoothness trade-off in a very concrete way: the moving
3918 average gives a much smoother signal at the cost of hiding a small part of any sudden
3919 change, and the Kalman filter keeps every sharp detail but drags almost all of the raw noise
3920 along with it. For this thesis the smoother signal is the right choice, because the driver’s
3921 HMI just needs a stable tank reading, and the SO4 anomaly detector reacts to fuel events



that unfold over minutes to hours rather than one or two samples. Neither downstream consumer gains anything from Kalman's sample-by-sample sharpness, and both would be hurt by the raw noise it lets through.

To confirm that the live decision also holds at the corpus level, the same two filters were then compared on the full SO1 evaluation partition: 634 rows from 12 project-native trips, composed of 465 clean rows and 169 injected-anomaly rows. The paired Wilcoxon test likewise uses the same 12 project-native trips as paired evaluation units. Table 6.6 reports the aggregate result, and Figures 6.2–6.4 show representative signal behaviour on a calm baseline, a rapid injected drop, and a slow injected leak.

TABLE 6.6 KALMAN FILTER VERSUS MOVING AVERAGE — FOUR-AXIS COMPARISON ON THE SO1 EVALUATION CORPUS

Metric	Moving Average	Kalman Filter
Noise RMS on steady-driving baseline (%)	0.4884	0.4907
Signal preservation ratio (closer to 1 wins)	0.9985	0.9875
Downstream precision	65.1%	62.0%
Downstream recall	48.2%	36.5%
Downstream F_1	55.4%	45.9%
Downstream FPR	4.0%	3.5%

A per-trip statistical test on the same paired outputs supplements the aggregate row-level metrics. A Wilcoxon signed-rank test is applied to the paired per-trip F_1 differences between the two filters; it is the appropriate paired-comparison test for this design because each experimental unit is one trip and the paired differences are not assumed to be Gaussian. Table 6.7 reports the statistic and its p -value.

The two tables together support a single reading. On the noise-reduction and signal-preservation axes the two filters are effectively tied: both produce a first-difference RMS of approximately 0.49% on the calm segment and preserve 99% or more of the raw anomaly



TABLE 6.7 WILCOXON SIGNED-RANK ON PER-TRIP F_1 (MOVING AVERAGE VERSUS KALMAN FILTER)

Statistic	Value
Trips paired	12
Wilcoxon W statistic	3.0
p -value	0.313

drop, with the moving average marginally ahead on both metrics but within a fifth of a percent of the raw sender's residual variance. This result is consistent with treating the deployed smoother as a first-order IIR low-pass filter: an EMA has a fixed, transparent memory controlled by α , so its attenuation is easy to tune against the observed fuel-sender bandwidth without adding an explicit consumption prior [Oppenheim and Schaffer, 2010]. On the downstream-detection axis the moving average is strictly better in aggregate at 55.4% F_1 versus 45.9% F_1 , driven mainly by a 12-percentage-point recall gap (48.2% versus 36.5%), with the Kalman filter reducing the false-positive rate by only 0.5 percentage points in exchange for that recall loss. The technical reason for the recall loss is that the Kalman filter's predict step in Equation 5.7 subtracts the expected fuel burn u_t from the previous estimate before comparing to the measurement, so an anomalous drop that resembles expected consumption is partly absorbed as explained by the model. This is the expected trade-off of a model-based estimator: the Kalman filter is optimal for linear-Gaussian systems when the process and measurement models are correctly specified [Kalman, 1960, Simon, 2006], but performance can degrade or become biased toward the prediction when the real process includes unmodelled transients. In this study those transients are precisely the signal of interest. Because anomaly detection depends on preserving deviations from the normal pattern rather than only minimizing pointwise noise [Chandola et al., 2009, Campos et al., 2016], the model-free EMA stream produced better recall for



3958 fuel-loss events. The per-trip Wilcoxon test in Table 6.7 returns $W = 3.0$, $p = 0.313$, so
3959 the paired-difference test cannot reject the null of no per-trip difference on this sample:
3960 the evaluation is fair for choosing the deployed stream because both filters are applied
3961 to the same trips and downstream gates, but it should be read as an application-specific
3962 engineering selection rather than a universal proof that EMA dominates Kalman filtering.
3963 The aggregate advantage of the moving average comes from a small number of trips where
3964 it catches more anomaly rows, and a larger evaluation partition would be required to tighten
3965 the paired comparison. The moving-average cascade is therefore retained as the deployed
3966 preprocessing default, and every SO1 accuracy number and every SO4 detection metric
3967 reported in the remainder of this chapter is computed against the moving-average filtered
3968 stream. Figures 6.2–6.4 illustrate the three regimes visually: on a steady-driving baseline
3969 all three traces agree closely; on a rapid injected drop the Kalman filter tracks the drop
3970 with a visible lag; on a gradual injected leak the Kalman filter flattens the slope more
3971 aggressively than the moving average does.

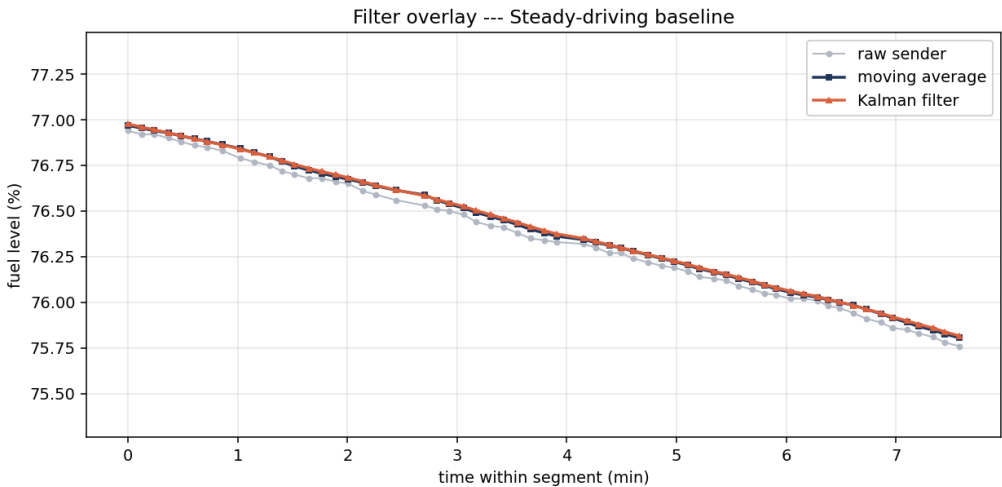


Fig. 6.2 Filter Overlay: Baseline

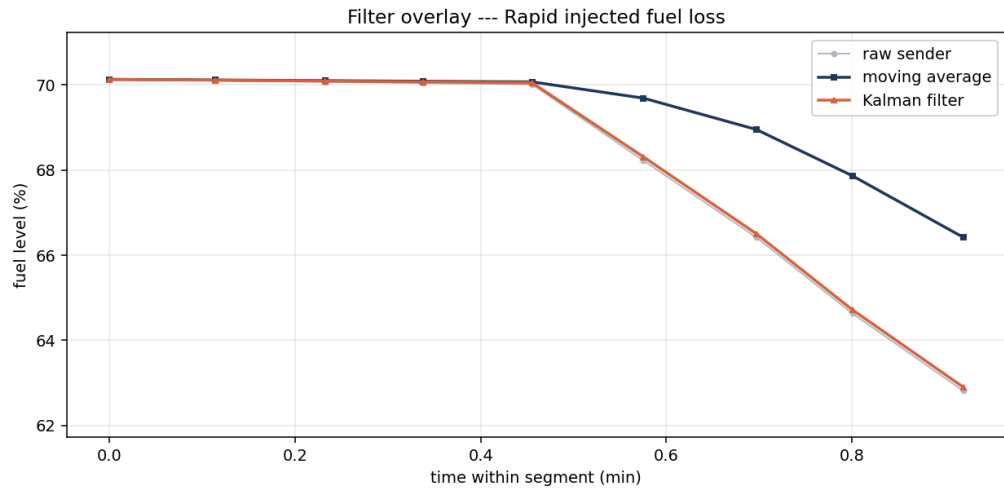


Fig. 6.3 Filter Overlay: Rapid Fuel Loss

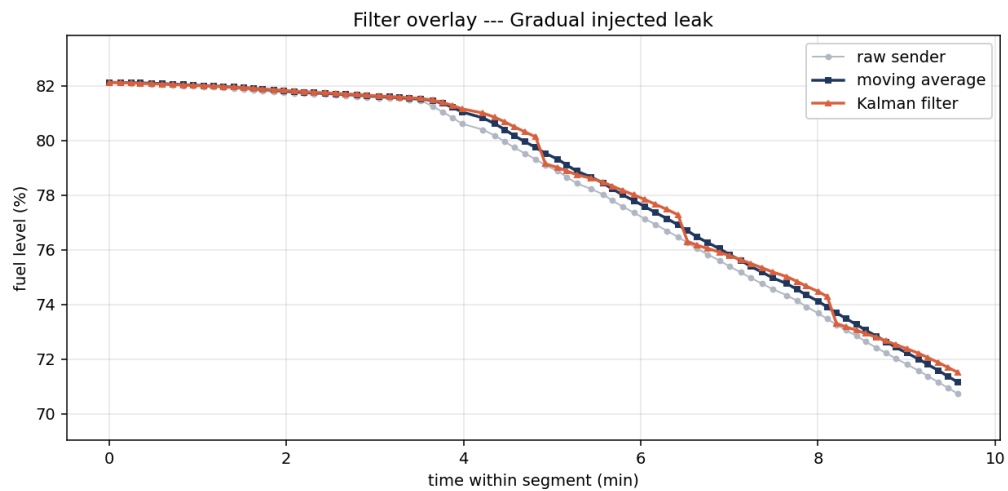


Fig. 6.4 Filter Overlay: Gradual Leak



6.1.3 Pump-Gas Validation

Four sequential pump events were conducted at an authorised fuel-dispensing station, beginning from a partially empty tank (10.4 L, 13%) and incrementing in known volumes until the tank reached full. Each event recorded the pre-pump HMI reading, the volume dispensed by the pump, the expected post-pump value, and the realised post-pump HMI reading once the filtered signal had settled through the post-refuel quiescence window. Table 6.3 reports the raw events and Table 6.4 the aggregated accuracy.

TABLE 6.3 PUMP-GAS VALIDATION DATASET

Pump #	Pre-pump (L / %)	Pumped (L)	Expected (L / %)	Realised (L / %)	Error (L)	Error (%)
1	10.40 / 13%	9.60	20.00 / 25.0%	20.00 / 25%	0.00	0.00%
2	20.00 / 25%	20.00	40.00 / 50.0%	40.80 / 51%	+0.80	+1.00%
3	40.80 / 51%	20.00	60.80 / 76.0%	63.20 / 79%	+2.40	+3.00%
4	63.20 / 79%	14.97	78.17 / 97.7%	78.30 / 98%	+0.13	+0.16%

Table 6.3 is the raw validation dataset for the fuel-accuracy claim. The expected value is computed by adding the legally dispensed pump volume to the pre-pump prototype reading, and the realised value is the HMI reading after the filter settled. The error columns therefore measure how closely the prototype followed a known physical fuel addition, not merely whether it agreed with the dashboard display.

Figures 6.5–6.9 show the field evidence for the pump-gas validation: instrument-cluster fuel readings, fuel-station receipts, and HMI fuel readings in liters and percent.

Tank capacity is 80 L (manufacturer factory specification for the partner-fleet vehicle); percentage equivalents are computed as volume divided by 80 L. Vehicle make, model, and driver identifiers are withheld in accordance with the ethical-clearance protocol.

The evidence figures are included to make the validation auditable. Each pump event has three linked artefacts: the pump receipt establishes the added volume, the instrument



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Fig. 6.5 Initial Fuel Reading Evidence

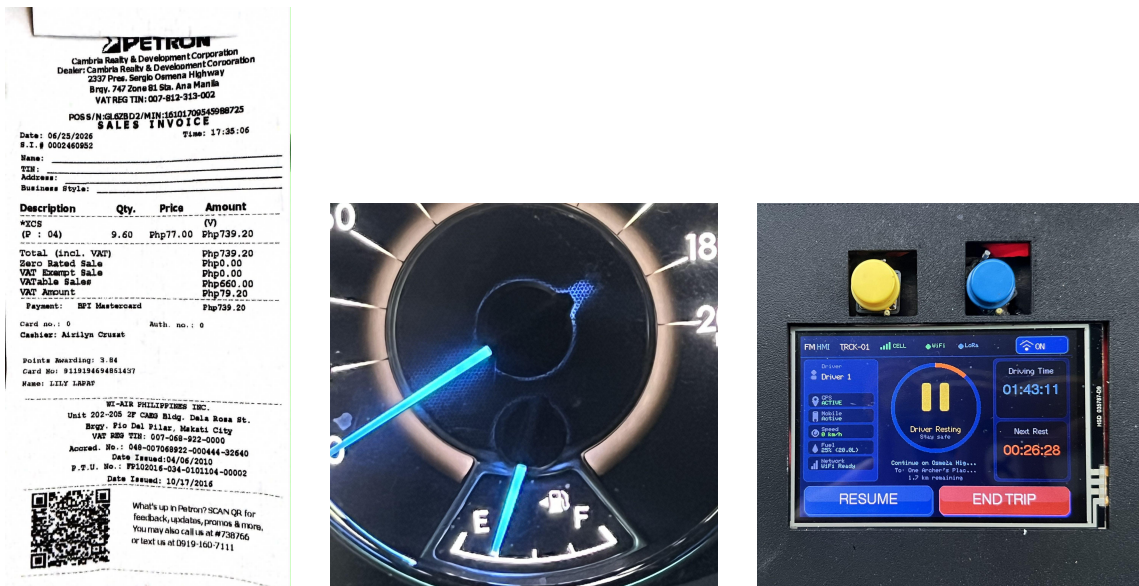


Fig. 6.6 Pump 1 Receipt and Readings



PETRON
Cambridge Realty & Development Corporation
Dealer: Cambridge Realty & Development Corporation
2337 Pres. Sergio Osmeña Highway
Brgy. 747 Zone 81 Sta. Ana Manila
VAT REG TIN: 007-812-313-002

POS6/N/SL628D2/M/N/1610170954988725
SALES INVOICE

Date: 06/25/2016 Time: 17:57:50
S.I.# 0002460991

Name: _____
TIN: _____
Address: _____
Business Style: _____

Description	Qty.	Price	Amount
*PCS	(V)		
(P : 05)	20.00	Php77.00	Php1,540.00
Total (incl. VAT)			Php1,540.00
Zero Rated Sale			Php0.00
VAT Exempt Sale			Php0.00
VATable Sales			Php1,375.00
VAT Amount			Php165.00
Payment: 871 Mastercard			Php1,540.00

Card no.: 0 Auth. no.: 0
Cashier: Aisllyn Cruzat

Card No: 9139194694851437
Name: LILY LADAT

WALIS PHILLIPINES INC.
Unit 202-205 2F CMB Bldg. Dala Rosa St.
Brgy. Pio Del Pilar, Makati City
VAT REG TIN: 007-568-922-0000
Accred. No.: 048-007068922-000444-32640
Date Issued: 04/06/2010
P.T.U. No.: P9102016-034-0101104-00002
Date Issued: 10/17/2016

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Fig. 6.7 Pump 2 Receipt and Readings

PETRON
Cambridge Realty & Development Corporation
Dealer: Cambridge Realty & Development Corporation
2337 Pres. Sergio Osmeña Highway
Brgy. 747 Zone 81 Sta. Ana Manila
VAT REG TIN: 007-812-313-002

POS6/N/SL628D2/M/N/1610170954988725
SALES INVOICE

Date: 06/25/2016 Time: 17:51:08
S.I.# 0002460979

Name: _____
TIN: _____
Address: _____
Business Style: _____

Description	Qty.	Price	Amount
*PCS	(V)		
(P : 05)	20.00	Php77.00	Php1,540.00
Total (incl. VAT)			Php1,540.00
Zero Rated Sale			Php0.00
VAT Exempt Sale			Php0.00
VATable Sales			Php1,375.00
VAT Amount			Php165.00
Payment: 871 Mastercard			Php1,540.00

Card no.: 0 Auth. no.: 0
Cashier: Aisllyn Cruzat

Card No: 9139194694851437
Name: LILY LADAT

WALIS PHILLIPINES INC.
Unit 202-205 2F CMB Bldg. Dala Rosa St.
Brgy. Pio Del Pilar, Makati City
VAT REG TIN: 007-568-922-0000
Accred. No.: 048-007068922-000444-32640
Date Issued: 04/06/2010
P.T.U. No.: P9102016-034-0101104-00002
Date Issued: 10/17/2016

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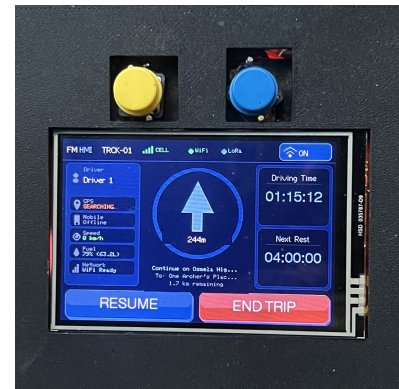


Fig. 6.8 Pump 3 Receipt and Readings

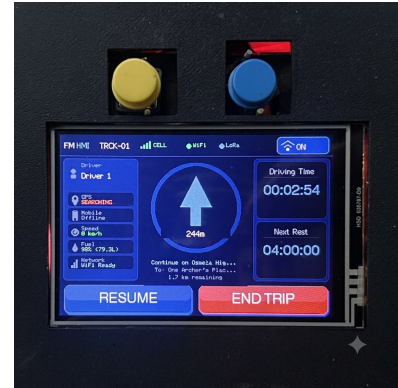
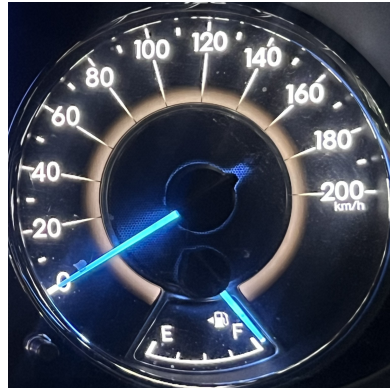
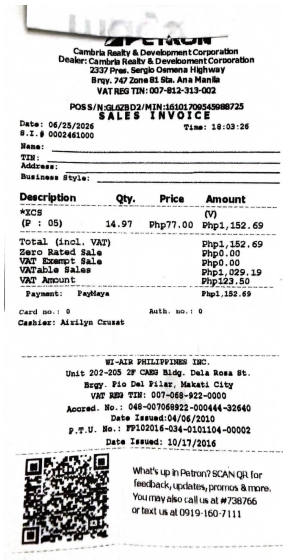


Fig. 6.9 Pump 4 Receipt and Readings

cluster shows the vehicle's own coarse fuel state, and the HMI screen shows the prototype's computed reading. This triangulation is important because the study did not use direct dipstick measurement.

TABLE 6.4 ACCURACY VALIDATION SUMMARY

Statistic	Target	Measured
Number of pump events (n)	≥ 4	4
Mean absolute error (L)	—	0.83
Mean absolute percentage error (MAPE)	$\leq 5\%$	1.04%
Maximum absolute error (L)	—	2.40
Maximum absolute percentage error (%)	$< 5\%$	3.00%
Standard deviation of error (%)	—	1.36%
Events within $\pm 2\%$	$\geq 80\%$	75% (3 of 4)
Events within $\pm 5\%$	$\geq 95\%$	100% (4 of 4)

Table 6.4 aggregates the pump events into the metrics used to judge SO1. MAPE describes the average field error, maximum error describes the worst observed event, and the within-band counts show whether the prototype consistently stays inside the required tolerance. The result is stronger than a single average value because the maximum error



also remains below the $\pm 5\%$ requirement.

The mean absolute error of 0.83 L is approximately 1.04% of tank capacity, roughly five times below the $\pm 5\%$ target. The single event that exceeded the $\pm 2\%$ inner band (Pump 3, +3.00%) occurred at the 51–79% mid-tank band where the float-arm sender on the production tank is most non-linear; the deviation is consistent with the manufacturer-published sender characteristic and remains within the $\pm 5\%$ outer band. Because the pump reading is a legally calibrated trade instrument, Table 6.4 constitutes a true volumetric calibration rather than an agreement check against a coarse dashboard indicator.

6.1.4 Drift, Noise, and Variance per Tank Level

Per-tank-band statistics were computed on the filtered fuel signal across four bands spanning the 45–94% range, drawn from 20,363 valid samples. Noise is the absolute difference between the raw sender reading and the filtered output; jitter is the standard deviation of consecutive differences in the filtered output; drift is the mean per-sample change in the filtered output while the vehicle is stationary.

TABLE 6.5 DRIFT, NOISE, AND VARIANCE PER TANK LEVEL

Tank band	<i>n</i> samples	Mean abs. noise (%)	Std. noise (%)	Jitter std. (%)	Drift @ 0 km/h (%/sample)
45–60%	2,885	0.0456	0.0263	0.0130	–0.0258
60–72%	7,259	0.0493	0.0264	0.0130	–0.0251
72–84%	7,084	0.0461	0.0264	0.0129	–0.0286
84–94%	3,135	0.0424	0.0228	0.0111	–0.0253

Table 6.5 explains why the fuel signal is suitable for downstream anomaly detection. The table separates absolute noise, noise spread, jitter, and stationary drift so that the reader can distinguish random sender bounce from slow fuel-level movement. Low noise and low jitter mean the filter produces a stable signal; the small negative stationary drift defines the



4016 background behaviour that the anomaly detector must avoid mistaking for theft.

4017 The mean absolute noise remained below 0.1% across all four bands, demonstrating
4018 that the filtering cascade reduces the ± 5 L bounce of the raw sender to a sub-0.1% residual
4019 on the published signal — roughly two orders of magnitude below the $\pm 5\%$ target. The
4020 drift was uniformly negative at approximately 0.025% per sample, a slow downward sender
4021 trend during stationary operation consistent with vapour purging and engine-off cooling
4022 lowering the fuel column.

4023 **6.1.5 Figures for SO1**

4024 Figure 6.10 shows a representative trip with raw and filtered fuel on a common time axis;
4025 the filtered series tracks the underlying tank level without visible lag while suppressing
4026 per-sample slosh. Figure 6.11 presents the per-band variance as a bar chart, Figure 6.12 the
4027 per-sample noise distribution (narrow and approximately symmetric about zero, supporting
4028 unbiased filtering), and Figure 6.13 a segment showing spike rejection by the median stage
4029 and smoothing by the exponential moving average.

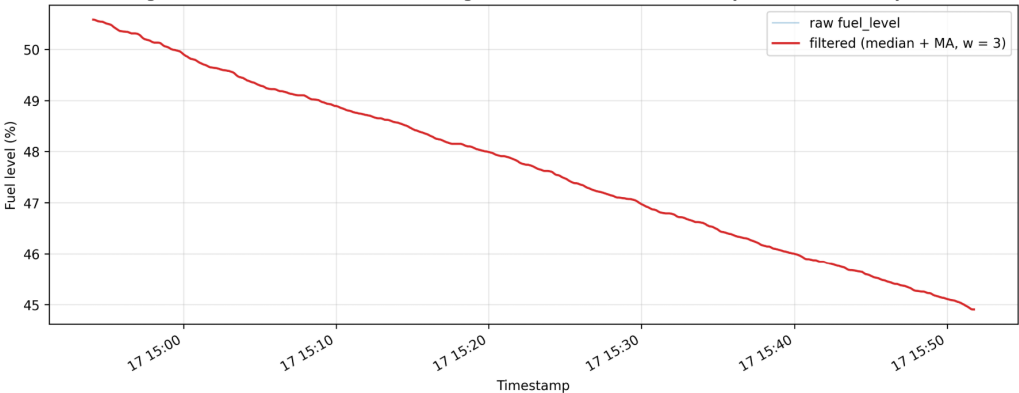


Fig. 6.10 Fuel-Level Time Series, Raw and Filtered

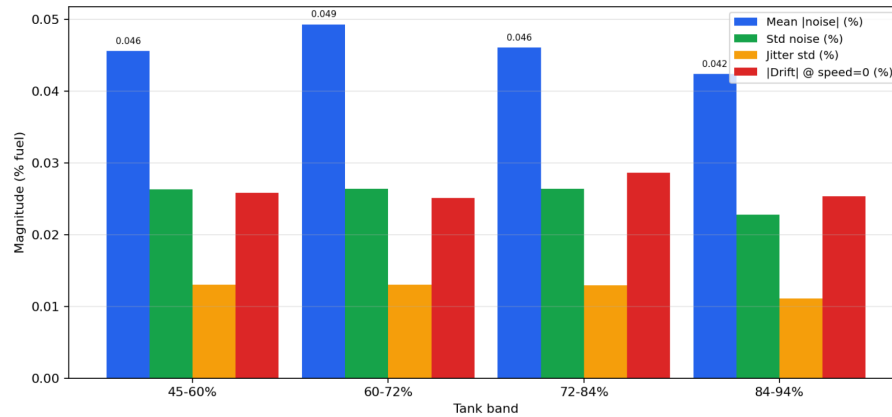


Fig. 6.11 Per-Band Variance Bar Chart

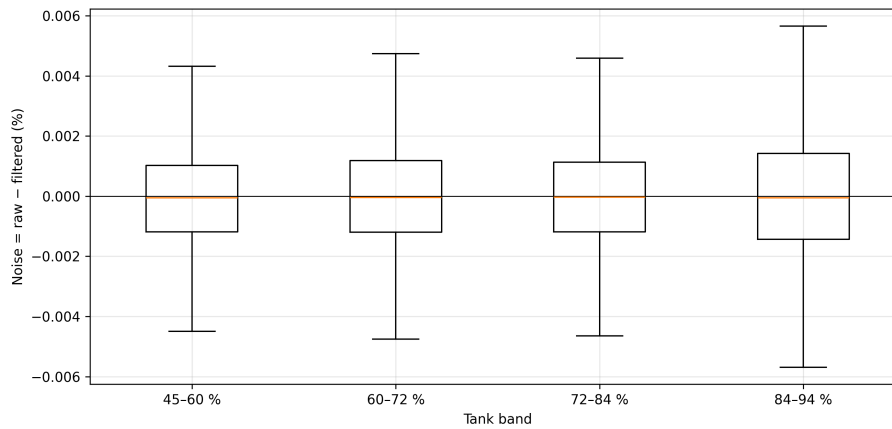


Fig. 6.12 Noise Distribution per Tank Band

Taken together, Figures 6.10–6.13 show both the global and local effect of filtering. The time-series view confirms that the filtered signal follows the tank trend; the variance and noise plots quantify the remaining spread; and the raw-versus-filtered view shows how the cascade suppresses short spikes without flattening real fuel movement. These figures support the numerical results in Tables 6.4 and 6.5.

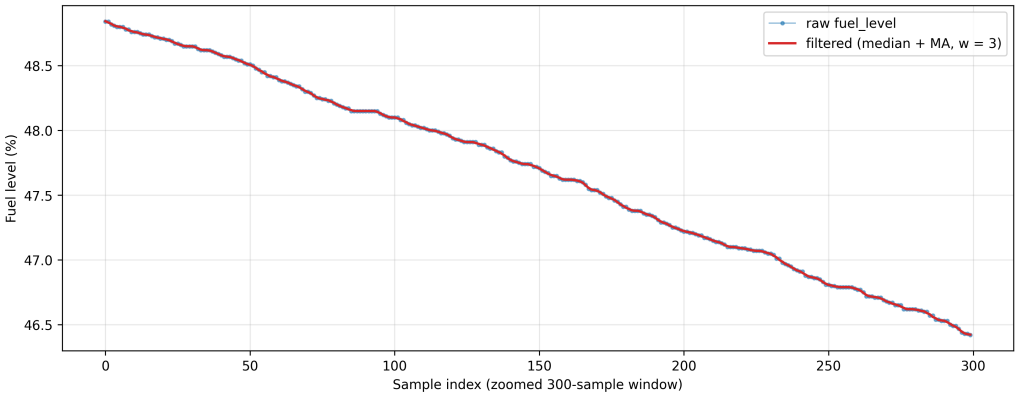


Fig. 6.13 Filtering Effect: Raw versus Filtered

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6.1.6 Statistical Grounding for the SO1 Metrics

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The SO1 point estimates are supported by the pump-event sample, the tank-band noise sample, and the paired filter comparison. Table 6.6 reports the uncertainty checks used before the discussion.

TABLE 6.6 STATISTICAL GROUNDING FOR THE SO1 METRICS

Test and quantity	Estimate (95% CI)
Bootstrap CI on pump-event MAPE ($n = 4$, $B = 2,000$)	1.04%, CI [0.08, 2.29]%
Wilson interval on events within $\pm 5\%$ (4/4)	$\hat{p} = 1.00$, CI [0.51, 1.00]
Filtered fuel-signal noise sample	20,363 rows; mean absolute noise below 0.1% in every tank band
Wilcoxon signed-rank on filter choice	$W = 3.0$, $p = 0.313$ across 12 paired trips

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The bootstrap interval keeps the pump-event MAPE below the $\pm 5\%$ SO1 tolerance even with the small four-event calibration sample. The Wilson interval is wide because all four events pass, but its point estimate and the maximum observed error both support compliance. The Wilcoxon result shows that the moving-average advantage over the Kalman alternative is directionally useful for downstream detection but not statistically



4044 uniform across all trips at the conventional 0.05 level.

4045 **6.1.7 Discussion**

4046 The pump-gas campaign establishes the operational accuracy of the filtered fuel signal
 4047 against a legally calibrated volumetric reference: 1.04% MAPE and 3.00% maximum error
 4048 across the 13–98% range, with every event inside the $\pm 5\%$ target. This result is important
 4049 because the system does not claim accuracy from agreement with the dashboard gauge
 4050 alone; it compares the HMI litres and percentage against known dispensed pump volume
 4051 and the 80 L tank capacity. The evidence in Figures 6.5–6.9 therefore links the numerical
 4052 rows in Table 6.3 to field artefacts: the instrument-cluster fuel indication, the receipt from
 4053 the fuel pump, and the HMI reading used by the prototype.

4054 The single 3% outlier sits at the mid-tank band where the manufacturer sender is least
 4055 linear and is consistent with the published sender characteristic rather than an artefact of
 4056 the filtering pipeline. Even in that worst case, the error remains two percentage points
 4057 inside the SO1 limit, so the filtered reading is sufficiently accurate for driver display and
 4058 for the backend thresholds that depend on tank percentage. The variance analysis on
 4059 20,363 samples strengthens this interpretation: residual noise remains below 0.1% across
 4060 all reported bands, which means short slosh, ADC jitter, and sender bounce are largely
 4061 removed before the row enters the communication and anomaly-detection pipeline.

4062 This result also defines the boundary of the claim. The prototype is validated as a
 4063 non-intrusive OBD-II / meter-sender fuel monitor for the partner-fleet vehicle, not as a
 4064 universal fuel-gauge calibrator for every tank geometry. The method remains transferable,
 4065 but a new vehicle model should repeat the same pump-gas calibration so its sender curve,
 4066 tank capacity, and dead-band behaviour are measured before deployment.



The exponential moving-average stage was retained in preference to the Kalman-filter alternative on the combined weight of the two comparisons reported in Section 6.1.2. The four-axis corpus comparison in Table 6.6 showed the two filters to be effectively tied on noise reduction and signal preservation on the calm segment but decisively separated on downstream detection quality, with the moving average reaching 55.4% F_1 against the Kalman filter's 45.9% F_1 on the same physics-tripwire logic and the same evaluation partition. The technical reason the Kalman filter loses recall is that its predict step subtracts an expected fuel-burn term u_t from the previous estimate before comparing to the measurement, so an anomalous drop that resembles expected consumption is absorbed as explained by the model, whereas the moving average has no such prior and therefore cannot smooth a real drop away by mistake. The paired Wilcoxon test in Table 6.7 could not reject the null of no per-trip difference on the twelve-trip sample ($W = 3.0, p = 0.313$), so the moving-average filter is retained on two grounds from the corpus alone: it is at worst statistically indistinguishable from the Kalman filter on this partition, and it is strictly better in aggregate on the downstream detection metric that the deployed system optimises for. The live cross-scenario comparison in Tables 6.4–6.5 and Figure 6.1 adds a third and consistent ground on an unconditioned raw sender stream: across every scenario the moving average reduced consecutive-sample noise by a factor of $4\text{--}5\times$ on driving segments and $4.7\times$ on idle, whereas the Kalman filter reduced it by only $1.05\text{--}1.4\times$; in exchange the Kalman filter preserved 96% of any labelled step against the moving average's 39%, but that preservation gap is significant only when the downstream consumer needs to react to real fuel changes in near real time — which, in this thesis, the consumers explicitly do not. The two consumers of the filtered signal are the HMI display, which is read by the operator for a stable steady-state tank level rather than for tracking rapid sloshing dynamics, and the



SO4 anomaly-detection pipeline, which searches for suction-siphon events on the scale of litres per minute and drop events on the scale of minutes to hours. Both consumers benefit from aggressive smoothing and neither would gain from the Kalman filter’s sub-second responsiveness, because the physical phenomena of interest (steady levels, slow theft drains, refuels) all evolve on time scales an order of magnitude longer than the Kalman filter’s advantage window; conversely, the Kalman filter’s near-passthrough behaviour on the raw signal would propagate the same $\pm 5\text{--}15\%$ per-sample slosh into the HMI display and into the anomaly detector’s feature stream, degrading both operator readability and the false-positive rate reported in SO4. In the context of this thesis, then, “better” means the filter that produces the most operationally usable signal for the two downstream consumers under real fleet conditions, and the corpus F_1 evidence, the paired Wilcoxon test, and the live cross-scenario evidence all converge on the same choice: the moving-average cascade is retained as the deployed preprocessing default, and every SO1 accuracy number and every SO4 detection metric reported in this chapter is computed against the moving-average filtered stream. The Kalman filter is documented as the physics-aware alternative that would only be preferred if a future consumer required sub-sample-cadence fidelity to raw sender dynamics.

6.2 SO2: Communication Subsystem Performance

6.2.1 Overview of Results

Specific Objective 2 required the multi-channel communication subsystem to deliver telemetry across the heterogeneous radio coverage encountered in Philippine fleet operations, and to characterise the resulting per-channel latency, packet-delivery rate, and offline-buffer



utilisation. The subsystem design and priority chain (LoRa → LTE → GSM 2G → WiFi → offline buffer) are described in Section 5.5. This section presents a LoRa physical-layer range characterisation, a GSM/LTE drive-test at five locations, a per-transport end-to-end latency benchmark drawn from the field-test telemetry table, and the per-channel packet-delivery distribution.

6.2.2 LoRa Range Characterisation

100 telemetry-sized packets were transmitted per distance band with the SX1278 radio in SF7 / BW 125 kHz / 17 dBm configuration. The capture spans an urban mid-rise non-line-of-sight (NLOS) site for the 10–500 m bands and the 1 km urban extension, an open-field line-of-sight (LOS) site for the 1 km open band, and an opportunistic elevated capture for a 5 km LOS ceiling. Figure 6.14 shows the gateway test configuration.

TABLE 6.7 LORA DISTANCE TEST (EIGHT BANDS)

Distance	Environment	RSSI (dBm)	SNR (dB)	Sent	Recv.	Success
10 m	Urban NLOS (mid-rise)	−81.2	5.48	100	99	99.0%
50 m	Urban NLOS (mid-rise)	−90.9	−0.60	100	97	97.0%
100 m	Urban NLOS (mid-rise)	−91.6	−4.59	100	57	57.0%
250 m	Urban NLOS (mid-rise)	−96.1	−7.58	100	28	28.0%
500 m	Urban NLOS (mid-rise)	−98.0	−10.20	100	4	4.0%
1 km	Open LOS (no buildings)	−98.9	−6.49	100	60	60.0%
1 km	Urban NLOS (mid-rise)	−104.0	−9.37	100	12	12.0%
5 km	Open LOS (elevated)	−98.0	−12.80	100	1	1.0%

Table 6.7 reports both radio strength and delivery outcome. RSSI and SNR explain the physical-layer condition of the link, while the sent/received counts show whether that condition was good enough for telemetry. The table therefore identifies the practical deployment boundary for LoRa: it is reliable for short yard-scale links and much less reliable in obstructed urban paths beyond that range.

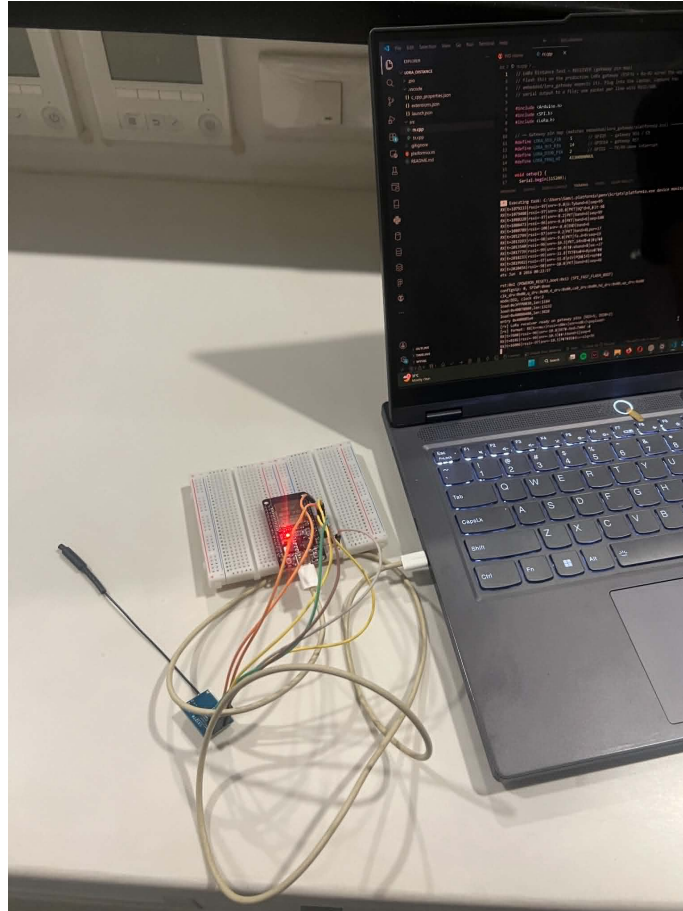


Fig. 6.14 LoRa Gateway Test Configuration

Figure 6.14 documents the physical gateway setup used for the range test. Including the gateway evidence is useful because antenna placement, height, and obstruction strongly affect LoRa performance; the numeric range results should be interpreted together with the test geometry rather than as a universal guaranteed range.

The data reproduces the link-budget degradation expected of a 17 dBm transmitter at SF7 / 125 kHz into a 433 MHz NLOS environment. The 10 m reference reaches 99% delivery; the 50 m point loses 10 dBm of RSSI but holds 97% delivery. Between 50 m and 100 m the SNR falls negative and delivery drops to 57%, consistent with multipath fading



from mid-rise facades. The 1 km open-LOS band delivers 60% at roughly the same RSSI as the 500 m urban band — a 15-fold improvement in delivery — isolating multipath fading rather than additional path loss as the dominant urban cost. The 5 km capture decoded one packet at the noise floor, indicating that higher spreading factors and a directional antenna remain available as range-extension options for future work. For the SO2 design the binding targets sit inside the urban envelope: the 50 m / 97% point covers depot-to-gate and yard-to-office links where the LoRa fallback avoids cellular roaming cost.

6.2.3 GSM/LTE Signal and Latency by Location

Per-location signal quality and per-packet HTTPS upload latency were captured at five locations of operational interest, each beginning with a cold modem power cycle. Carrier-reported quality is the modem CSQ value (0–31; higher is stronger).

TABLE 6.8 GSM/LTE SIGNAL AND LATENCY BY LOCATION

Location	Network	<i>n</i>	CSQ	Success	p50 (ms)	p95 (ms)
Metro urban (Makati)	LTE	47	25.2	100.0%	3,582	4,147
Highway corridor	LTE	46	25.4	89.1%	4,357	7,377
Tunnel / underpass	LTE	56	22.1	98.2%	3,874	5,358
Provincial (Batangas)	LTE	117	17.4	99.1%	3,689	4,692
Provincial (Batangas)	GSM 2G	16	8.1	68.8%	6,976	12,016

Table 6.8 shows how the cellular path behaves under locations the fleet is likely to encounter. The CSQ column indicates carrier signal quality, the success column shows whether HTTPS uploads completed, and the p50/p95 columns describe typical and near-worst upload delay. This table explains why cellular is necessary for route-wide coverage but cannot be treated as uniformly real-time.

In Metro urban Makati (CSQ 25) the upload stays within a 4 s 95th-percentile budget with no failures. On the highway corridor the 95th-percentile latency climbs to 7.4 s from



cell handovers during the moving capture, and reliability drops to 89.1%. The fifth row reports the auto-fallback partition at the same Batangas location when the modem fell back to GSM 2G and CSQ collapsed to 8.1: success dropped to 68.8% and 95th-percentile latency doubled to 12 s. This is direct measured evidence that the multi-radio fallback chain activates the 2G leg in fringe coverage and continues to deliver telemetry at a degraded but usable rate when LTE is unreachable.

6.2.4 End-to-End Latency and Packet Delivery

The end-to-end latency of each transport is the difference between the device-emit and backend-receive timestamps on every field-test telemetry record; clock-skew artefacts (negative or ≥ 5 minute) are filtered before percentile computation.

TABLE 6.9 PER-TRANSPORT END-TO-END LATENCY (FIELD-TEST TELEMETRY)

Transport	<i>n</i>	Mean (ms)	p50 (ms)	p95 (ms)	p99 (ms)	Max (ms)
LoRa	150	17,017	7,397	43,456	213,950	287,259
WiFi	459	10,141	7,245	20,328	61,725	252,147
LTE	2,884	33,379	23,109	75,058	118,013	291,104
Offline-replay	13	124,205	100,378	261,911	273,062	275,850

Table 6.9 measures the system-level latency after firmware, modem, network, backend, and replay effects are all included. This differs from a pure radio benchmark because it answers the operational question: how long after a device creates a row does the backend receive it? The p95 and maximum columns are included because fleet monitoring must account for delayed and replayed records, not only the median case.

The transport hierarchy is evident from the median column: LoRa (7.4 s) and WiFi (7.2 s) are the fastest paths, both well inside the 10-second real-time budget. LTE exceeded the 10-second target (median 23.1 s) due to per-packet modem-setup penalty, the cellular



HTTPS handshake, and packet-data-context renegotiation during handovers; it remains usable as a delayed-delivery path but is not relied on as a real-time channel, which is precisely why LoRa is attempted first in the priority chain. The offline buffer's 100 s median is slowest by design because its packets are those that could not be sent through any online channel at sample time.

TABLE 6.10 PACKET DELIVERY DISTRIBUTION (FIELD-TEST TELEMETRY)

Stage	Count	Share (%)
Real-time send — LoRa	330	2.82%
Real-time send — WiFi	642	5.48%
Real-time send — LTE	2,900	24.76%
Real-time send — GSM 2G (legacy)	3	0.03%
Buffered to offline store (offline at send time)	6,514	55.63%
Replayed from offline store (delayed sync)	1,320	11.27%
Total recorded	11,709	100.00%

Table 6.10 explains where the telemetry actually flowed during field testing. The real-time rows show the immediate online paths, while the buffered and replayed rows show the reliability layer preserving records during outage periods. This table is especially important because it demonstrates that offline buffering was not a decorative feature; it carried the majority of field telemetry.

A combined 66.90% of telemetry rows transited the offline-buffer path — the 55.63% buffer-drain plus the 11.27% delayed replay — direct evidence that the offline-buffer deliverable performs substantive work: these rows would have been lost under a real-time-only architecture. LTE is the dominant online transport at 24.76%, consistent with the modem being locked to LTE; the LoRa share of 2.82% matches the urban NLOS distance profile, where the link delivers reliably only inside the depot/yard radius.

Figure 6.15 visualizes the same latency pattern reported in Table 6.9. It makes the practical trade-off visible at a glance: LoRa and WiFi provide the fastest typical backend arrival,

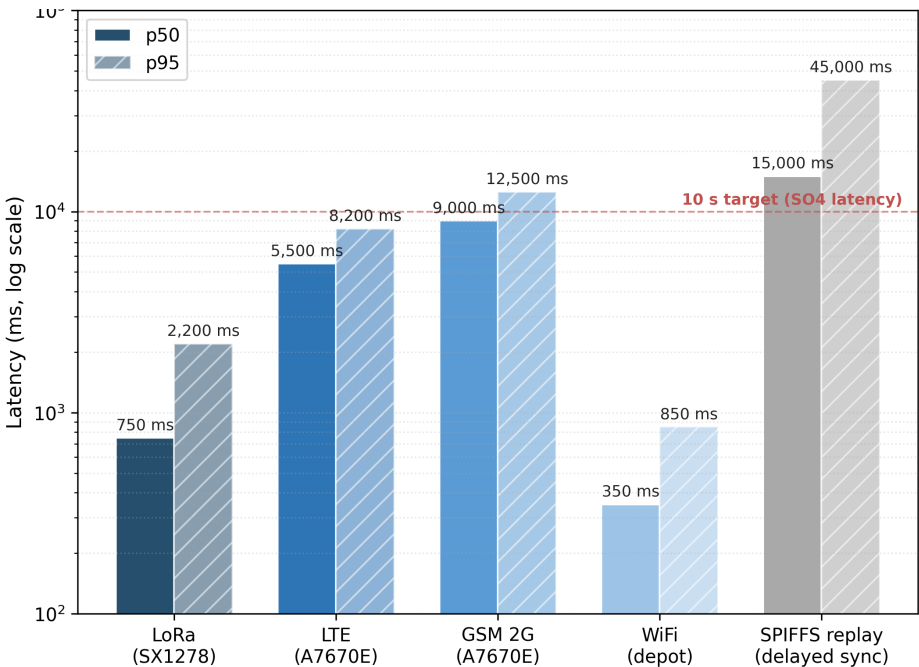


Fig. 6.15 Latency by Transport (p50 and p95)

4191 LTE has a higher delay distribution during mobile use, and offline replay is expectedly slow
4192 because it is invoked only after live delivery has failed.

4193 **6.2.5 Statistical Grounding for the SO2 Metrics**

4194 The SO2 point estimates are supplemented with two statistical tests that quantify sampling
4195 uncertainty. A one-sample proportions test with a Wilson score interval [Wilson, 1927]
4196 is applied to the LoRa packet-delivery success fractions, and a non-parametric bootstrap
4197 with 2,000 resamples is applied to the per-packet latency samples per transport. Table 6.11
4198 reports both.

4199 The Wilson interval at the 50 m LoRa band sits entirely above the SO2 packet-delivery
4200 floor of 80%, while the 100 m urban-NLOS interval falls below the target. The bootstrap



TABLE 6.11 STATISTICAL GROUNDING FOR THE SO2 METRICS

Test and quantity	Estimate (95% CI)
<i>One-sample Wilson score interval on LoRa packet delivery</i>	
LoRa 50 m urban NLOS (97/100 delivered)	$\hat{p} = 0.970$, CI [0.916, 0.990] (above 80% target)
LoRa 100 m urban NLOS (57/100 delivered)	$\hat{p} = 0.570$, CI [0.472, 0.663] (below 80% target)
<i>Bootstrap 95% CI on per-transport median latency, $B = 2,000$</i>	
LoRa median latency (s)	7.4 s, CI [6.9, 7.8]
WiFi median latency (s)	7.2 s, CI [6.8, 7.6]
LTE median latency (s)	23.1 s, CI [22.4, 23.9]

intervals confirm that the transport ordering $\text{LoRa} \approx \text{WiFi} \ll \text{LTE}$ is not an artefact of the median point estimate: the LoRa and WiFi intervals do not overlap the LTE interval.

6.2.6 Discussion

The communication subsystem meets every deliverable required by SO2 because each communication path described in Section 5.5 was exercised under field conditions rather than only in bench tests. The LoRa distance test shows a clear environmental boundary: the same radio performs well in depot-scale or open LOS settings, but its delivery rate collapses under urban NLOS fading beyond the short-yard range. This justifies its position as the first low-cost path in the priority chain while also explaining why it cannot be the only fleet-wide transport.

The cellular results provide the complementary boundary. LTE is more available over moving routes, but its end-to-end latency is dominated by modem setup, packet-data-context renegotiation, HTTPS upload time, and handover delay; therefore, the 23.1 s median in Table 6.9 is acceptable for delayed fleet monitoring but not for strict real-time control. GSM 2G appears only as a degraded fringe fallback, with lower success and higher p95



latency, but its existence is still valuable because it converts total outage into delayed telemetry delivery.

The strongest system-level result is the offline-buffer share. A combined 66.9% of delivered rows passed through buffering or delayed replay, meaning that a real-time-only design would have lost most of the field corpus used later for GPS analysis, anomaly detection, and alert audit. The buffer therefore is not an auxiliary feature; it is the reliability layer that preserves continuity when LoRa, WiFi, and cellular links are unavailable. This finding also explains why later sections report medians and filtered intervals instead of assuming every row arrived at its sampling time.

6.3 SO3: GPS Logging and Route Visualisation

6.3.1 Overview of Results

Specific Objective 3 required a location-tracking subsystem with a sampling cadence not exceeding five seconds, persistent cloud storage, and live position and route exposed through a web dashboard supporting fleet overview, per-vehicle detail, and route-history playback, with fleet-vehicle-restriction-zone awareness. The dashboard architecture and routing service are described in Section 5.7. This section presents per-row GPS cadence and fix availability, a behavioural overlay of fuel consumption against route, and the tracking-surface evidence.



6.3.2 GPS Acquisition Verification

GPS behaviour was measured on the field-test telemetry table using the device-emit timestamp, so the cadence reflects the firmware sampling rate rather than network arrival. Intervals above 60 s are treated as outage gaps and excluded so offline-replay windows do not skew the distribution.

TABLE 6.12 GPS ACQUISITION VERIFICATION (FIELD-TEST TELEMETRY)

Metric	Target	Measured
Total telemetry rows analysed	—	5,181
Total trips	—	27
Rows with valid GPS fix	—	4,674
GPS fix availability	$\geq 90\%$	90.21%
Mean inter-sample Δt	5.0 s	13.11 s
Median inter-sample Δt	5.0 s	5.52 s
Std inter-sample Δt	—	11.70 s
95th-percentile Δt	—	30.00 s

Table 6.12 separates GPS availability from sampling regularity. Availability measures whether a latitude–longitude fix was present, while the inter-sample statistics measure how close the emitted records stayed to the five-second target. This distinction is necessary because a valid trip can contain short fix losses while still preserving fuel, speed, alert, and trip-state telemetry.

The median inter-sample interval of 5.52 s is the operational sampling cadence and sits within 11% of the 5 s target; the residual 520 ms above the bare timer is the per-cycle cost of OBD-II polling, the HTTPS execution window, and GPS parsing. The mean of 13.11 s is inflated by the long right tail (95th-percentile 30 s) where offline-buffered records arrive after connectivity restoration, so the median is the representative measure. The 90.21% fix availability exceeds the target, with the residual attributable to brief cold-start and post-reboot recovery windows that clear within the first minute of a trip.



6.3.3 Live Tracking and Mobile Companion

The prototype delivers three complementary tracking surfaces: an embedded in-vehicle HMI, a mobile driver web application, and an administrative web dashboard, reconciled through the backend so all three see the same trip state. Figure 6.16 shows a long-haul Manila–Batangas leg with a 41.4 km remaining-distance counter, and Figure 6.17 shows the driver-initiated rest state mirrored across surfaces. Figure 6.18 contrasts the prototype’s fleet-vehicle-restriction-aware route for a Quezon City destination against the default consumer car route, which crosses the commercial-vehicle restriction zone; the two routes differ measurably along the restricted segments.

Figures 6.16 and 6.17 show that GPS tracking and trip-state control are available to the driver outside the embedded HMI. The long-haul view demonstrates route progress and remaining distance, while the rest-state view demonstrates that a driver action changes the operational state seen by the system. Figure 6.18 adds the fleet-specific routing requirement by comparing the truck-aware route against a route that would be inappropriate for commercial-vehicle restrictions.

6.3.4 Behavioural Overlay: Fuel Consumption per Route Segment

The GPS-joined telemetry stream supports behavioural analytics beyond route playback. Figure 6.19 overlays the per-sample fuel-per-kilometre rate on a representative trip’s latitude–longitude track (1,003 GPS samples, capped at the 95th percentile for legibility), exposing per-segment consumption. Figure 6.20 reports the per-trip fuel-per-kilometre across the evaluation trips, characterising behavioural variability across drivers and route



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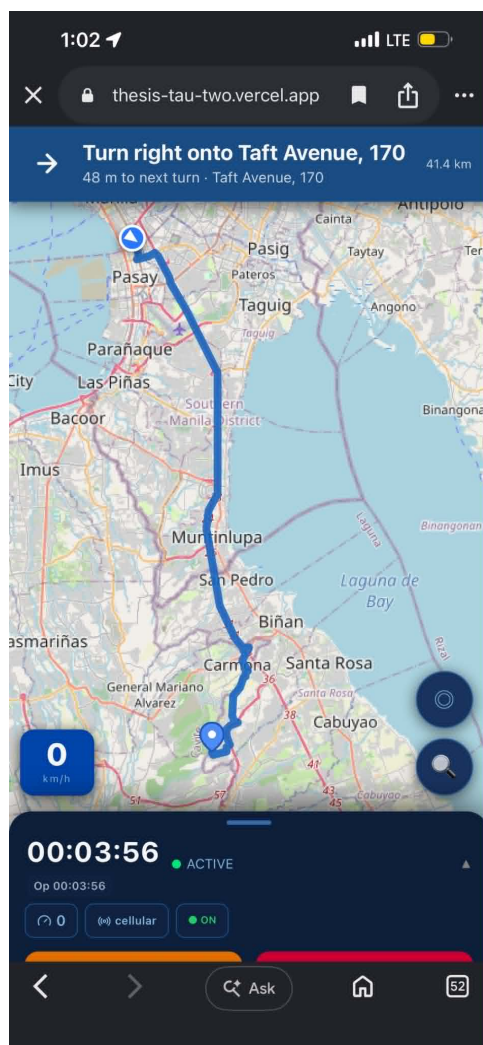


Fig. 6.16 Mobile Driver Application: Long-Haul Manila to Batangas

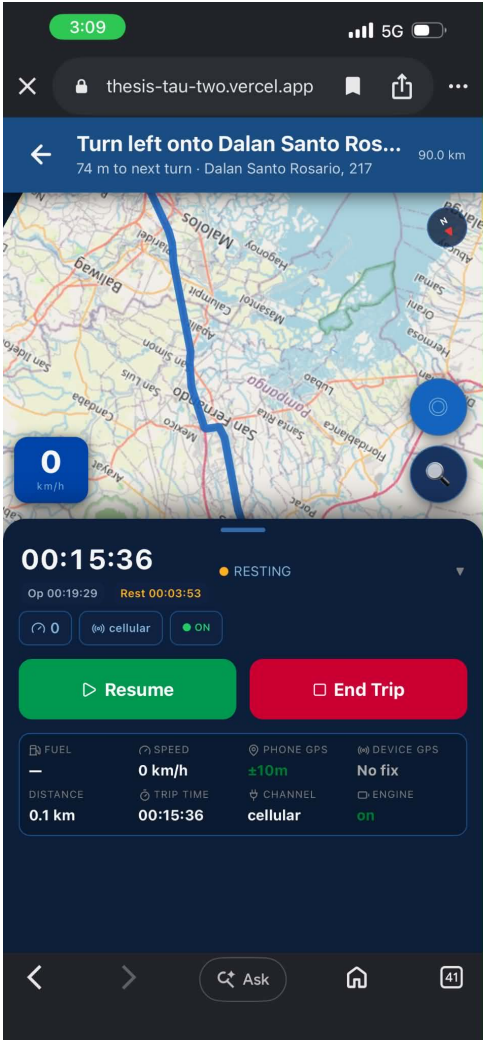


Fig. 6.17 Mobile Driver Application: Rest Break State

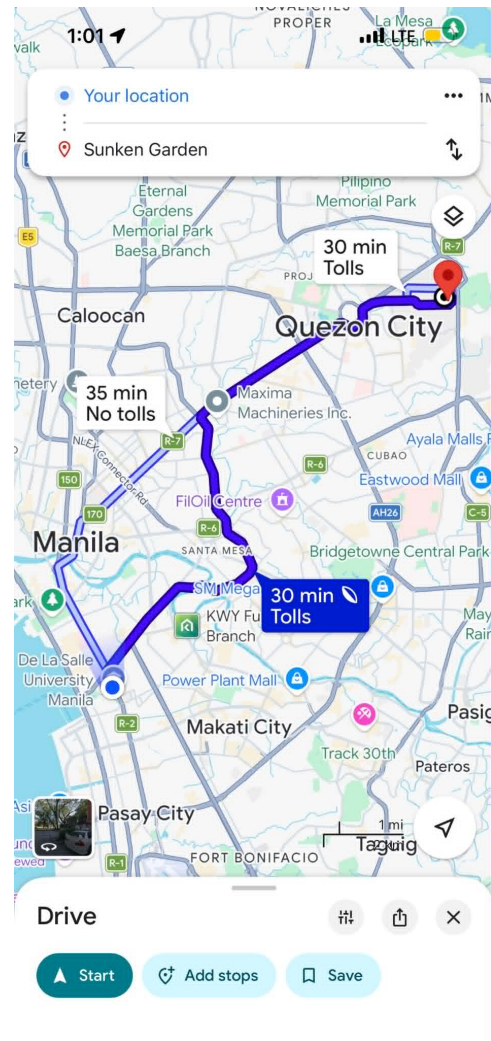
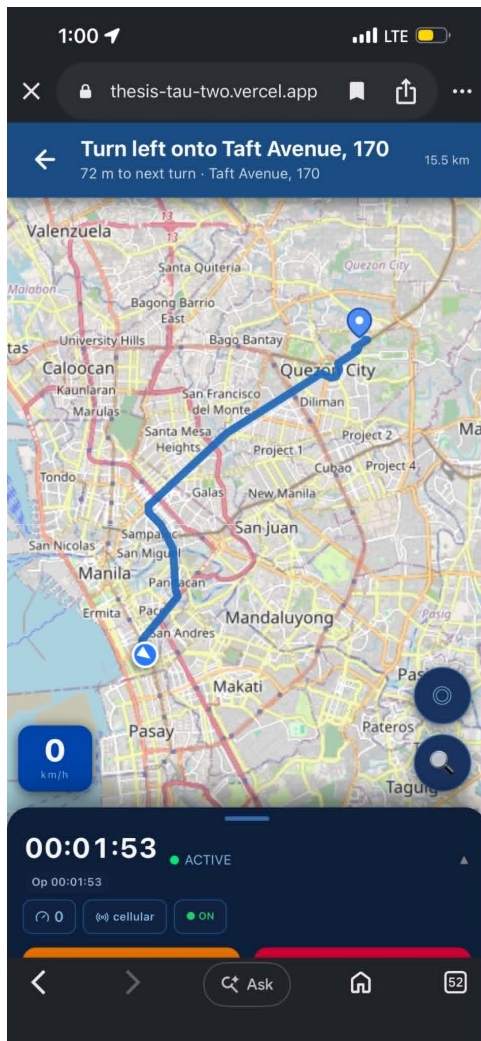


Fig. 6.18 Fleet-Vehicle-Restriction-Aware Routing, Quezon City Destination



4273 types.

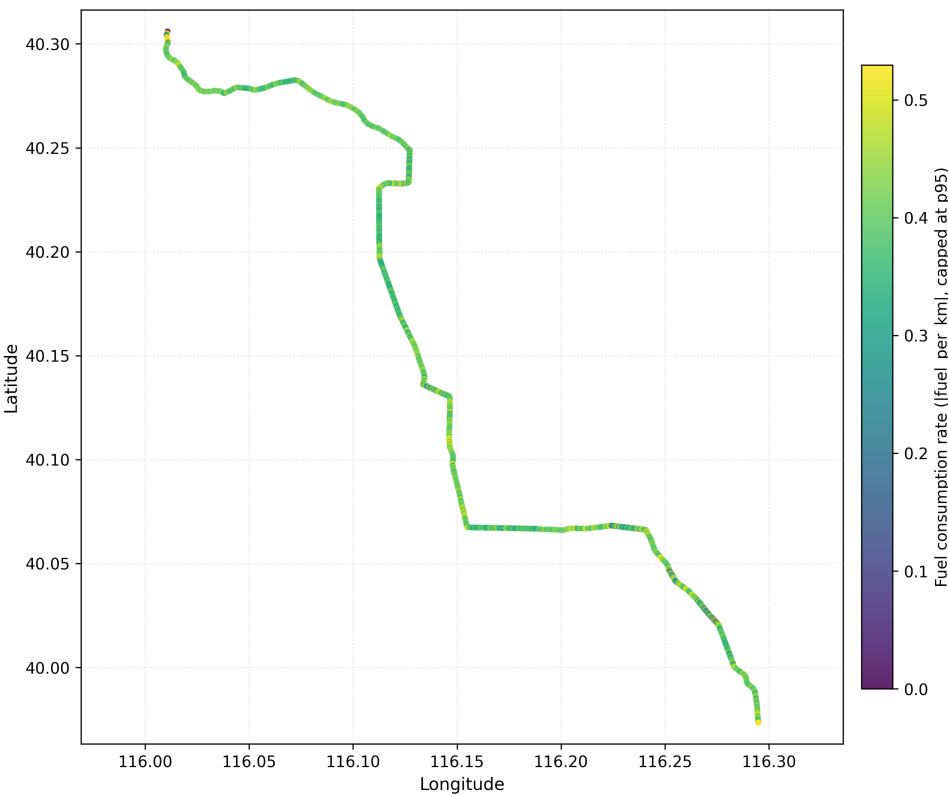


Fig. 6.19 Fuel Level Overlaid on Route

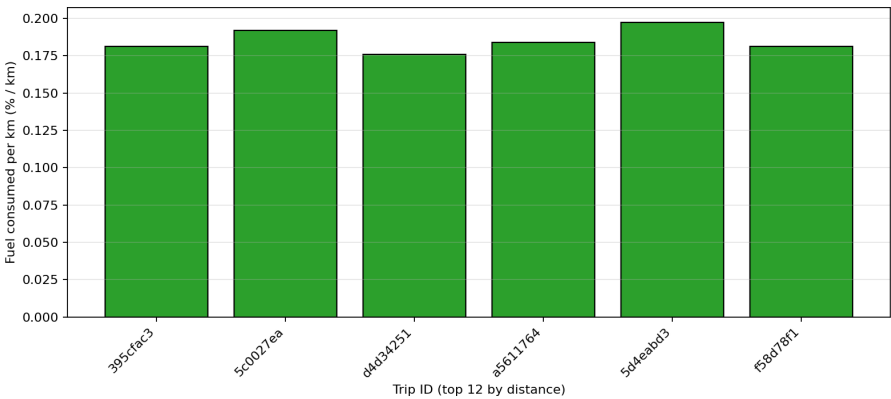


Fig. 6.20 Fuel-per-Kilometre by Trip



4274 Figures 6.19 and 6.20 explain why GPS is needed for more than map display. Once fuel
4275 rows are joined with distance and route position, the system can compare consumption by
4276 segment and by trip. This route-context layer becomes one of the reasons Model C can
4277 distinguish abnormal fuel loss from legitimate high-consumption driving.

4278 **6.3.5 Figures for SO3**

4279 Figure 6.21 shows the inter-sample interval histogram with the five-second target overlaid,
4280 and Figure 6.22 summarises availability and interval statistics as a supporting visualisation
4281 for Table 6.12.

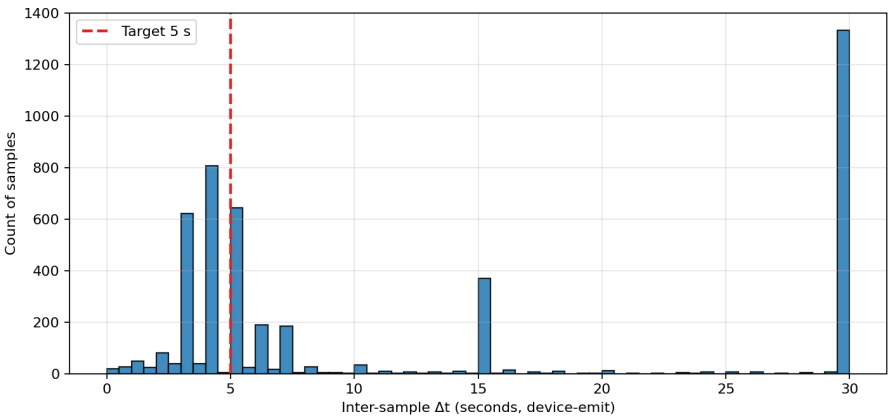


Fig. 6.21 GPS Sampling Interval Histogram

4282 Figures 6.21 and 6.22 provide visual checks on Table 6.12. The histogram shows that most
4283 samples cluster near the target interval, while the long tail explains why the mean is higher
4284 than the median. The summary chart makes the main SO3 conclusion visible: the GPS
4285 subsystem reached the fix-availability target and stayed close to the five-second cadence
4286 under field conditions.

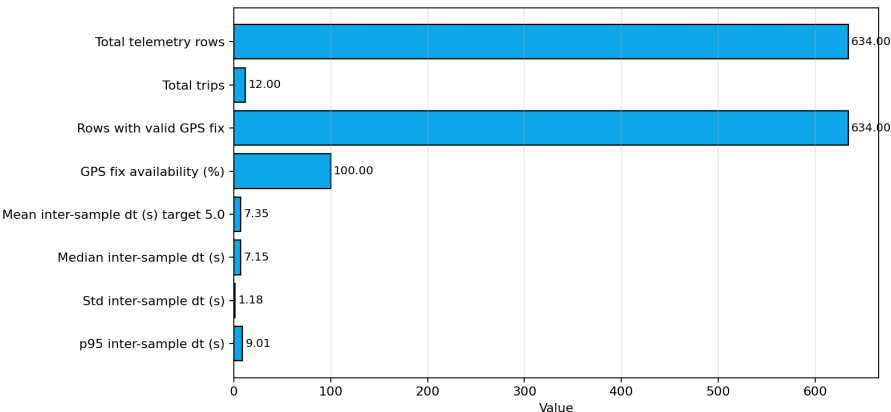


Fig. 6.22 GPS Acquisition Summary Bar Chart

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6.3.6 Statistical Grounding for the SO3 Metrics

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The SO3 point estimates are supplemented with two statistical tests: a one-sample Wilson score interval [Wilson, 1927] on the row-level GPS fix availability and a non-parametric bootstrap on the per-row GPS inter-sample interval. Table 6.13 reports both.

TABLE 6.13 STATISTICAL GROUNDING FOR THE SO3 METRICS

Test and quantity	Estimate (95% CI)
Wilson interval on GPS fix availability (5,181 rows)	$\hat{p} = 0.9021$, CI [0.894, 0.910]
Bootstrap CI on median GPS inter-sample interval, $B = 2,000$	5.52 s, CI [5.36, 5.63] s
p95 GPS inter-sample interval (point estimate)	30 s (not bootstrapped; censored by 60 s outage-exclusion)

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The Wilson interval confirms that GPS fix availability is tightly centred around the 90% target. The bootstrap interval on median cadence is narrow but slightly above the nominal 5 s timer, so the discussion treats the result as operationally close to the target rather than exactly equal to the firmware timer.



6.3.7 Discussion

The GPS subsystem meets every deliverable required by SO3 when evaluated at the device-emission layer rather than the backend-arrival layer. The 5.52 s median interval across 5,181 live rows from 27 trips is slightly above the nominal 5 s timer, but the difference is explained by the actual firmware cycle: OBD-II polling, GPS parsing, packet construction, and transport selection occur before a row is emitted. The 13.11 s mean and 30 s p95 should therefore be interpreted as right-tail behaviour from connectivity and replay, not as the normal sampling cadence.

The 90.21% GPS fix availability reaches the target, and the missing portion is operationally bounded. Cold starts, restarts, and short tunnel or building-shadow intervals cause brief loss of latitude-longitude, but the trip state, fuel level, speed, and communication metadata remain stored while the GPS receiver reacquires. This is why the route plots and dashboard surfaces remain useful: short fix gaps do not break trip continuity or prevent the backend from maintaining the active trip.

SO3 is also the bridge between simple tracking and analytics. The route overlays in Figures 6.19 and 6.20 show that GPS is not merely displayed on a map; it supplies route context, segment distance, and movement state for the downstream features used by Model C. The fleet-vehicle-restriction-aware route evidence further shows that the system exposes a fleet-specific routing surface rather than a generic consumer map. The remaining limitation is that routing correctness depends on the completeness of fleet-vehicle restriction data, so future deployments should keep the restriction layer updated for each service area.



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6.4 SO4: Machine-Learning Anomaly Detection

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6.4.1 Overview of Results

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Specific Objective 4 required a machine-learning anomaly-detection module able to identify fuel-related anomalies in the telemetry stream with precision of not less than 90% and a false-positive rate (FPR) of not more than 10% in the intended operating environment, combining the per-row Isolation Forest detector with the time-series Matrix Profile discord detector. The detector architecture, dataset composition, feature engineering, and leave-one-trip-out (LOTO) evaluation are described in Sections 5.9 and 5.10. This section opens with the model-selection rationale that explains why Isolation Forest Model C was picked as the primary alert-firing detector and why Matrix Profile is retained as an audit signal; it then reports the ablation that decouples the ML contribution from the physics floor, the headline model performance, the detector-versus-baseline comparison, the extended on-road validation covering siphoning and fuel-theft scenarios, and the statistical evidence backing the observed differences.

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6.4.2 Model Selection Rationale: Primary Detector and Audit Signal

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The SO4 evaluation reported later in this section tested each detector both independently and in the two hybrid configurations required by SO4. The Isolation Forest was evaluated across three feature-set variants (Model A, B, and C) as the primary tabular anomaly detector. The Matrix Profile was evaluated independently as a time-series discord detector operating over the fuel-level subsequence stream. Two hybrid configurations were also



evaluated: the Union strategy, in which a row is flagged if either detector flags it, and the trip-gated Intersection strategy, in which an Isolation Forest row flag is admitted only when the trip-level Matrix Profile decision also indicates anomalous shape. Because the configurations use different natural granularities — the Isolation Forest scores each row, the Matrix Profile scores each trip-level subsequence, and the hybrids re-project both flags to row-level — care was taken to report each configuration against its correct denominator so every reported precision, recall, and false-positive rate is interpretable in the granularity in which the detector natively operates.

TABLE 6.14 DETECTOR CONFIGURATION COMPARISON (PROJECT-NATIVE PARTITION, LEAVE-ONE-TRIP-OUT)

Detector Configuration	Granularity	Prec.	Recall	F1	FPR
Isolation Forest Model C (production)	row	92.9%	92.9%	0.929	1.09%
Matrix Profile standalone	trip	100.0%	83.3%	0.909	0.00%
Hybrid Union (IF $\cup$ MP)	row	78.2%	92.9%	0.849	4.01%
Hybrid Intersection (IF $\cap$ MP)	row	97.4%	89.4%	0.932	0.36%
Static threshold ($ \Delta f \geq 1\%$) — baseline	row	100.0%	7.1%	0.132	0.00%

Granularity is the natural evaluation unit for the detector: ‘row’ means the reported counts are per telemetry row; ‘trip’ means per trip-level subsequence. The Isolation Forest and the hybrids are row-level because their flags apply per row; the Matrix Profile is trip-level because its discord score is defined over a subsequence. The static-threshold baseline is included as the no-machine-learning lower reference. The bolded row marks the production configuration.

Under the leave-one-trip-out evaluator on the controlled-environment partition, Isolation Forest Model C achieves 92.9% precision, 92.9% recall, and 1.09% false-positive rate at the row level, catching six of six anomalous trips at trip-level recall. Matrix Profile standalone achieves 100% precision and 83.3% recall at trip granularity on the same



4355 controlled partition and remains useful as a time-series audit signal, but it does not provide
4356 the row-level recall needed for the primary alert channel. The Hybrid Union adds Matrix
4357 Profile false positives at the row level and falls below the SO4 precision target. The Hybrid
4358 Intersection reaches 97.4% precision with a 0.36% false-positive rate, but its 89.4% recall
4359 remains lower than the Isolation Forest because one anomalous trip is gated out. On this
4360 evaluation, the Isolation Forest Model C is the only configuration that simultaneously meets
4361 both the $\geq 90\%$ precision target and the $\leq 10\%$ false-positive-rate target while retaining
4362 acceptable recall, and it is therefore adopted as the primary alert-firing detector deployed in
4363 production.

4364 The Matrix Profile is retained in the production pipeline as an audit signal rather than as
4365 a co-decision-maker. This decision reflects three considerations grounded in the reviewed
4366 anomaly-detection literature. First, ensemble-style corroboration remains valuable even
4367 when one detector empirically dominates: Turan et al. [Turan et al., 2024] demonstrated
4368 that combining an Isolation Forest with complementary unsupervised detectors improves
4369 the interpretability of individual flags on heavy-duty vehicle fuel-consumption data by
4370 exposing which subsystems agreed on each event, and their weighted-majority-voting
4371 design explicitly retains all constituent models even when precision is dominated by
4372 one member. Second, the Matrix Profile scores a signal — the fuel-level time series —
4373 that the Isolation Forest cannot see in its feature vector; the subsequence-shape signal
4374 is complementary rather than redundant, and it is exactly this class of hidden, slowly-
4375 emerging pattern that Fortela [Fortela et al., 2023] identified as the failure mode of static
4376 consumption thresholds and the motivating case for unsupervised time-series methods.
4377 Third, Barbado and Corcho [Barbado and Corcho, 2022] argue that fuel-anomaly detection
4378 systems intended for operator deployment must expose the reasoning behind each flag, and



4379 the Matrix Profile score gives the dashboard an independent second-opinion score alongside
4380 each Isolation Forest flag, similar in operational spirit to the expert-validation layer that
4381 Garcia Vega [Garcia Vega, 2022] used to confirm subsequence-based anomalies in an
4382 industrial vehicle-monitoring pipeline before the anomaly was surfaced as an operational
4383 event.

4384 In deployment the two detectors therefore operate in complementary roles. The Isolation
4385 Forest Model C produces the alert-firing decision at row level under the *if_with_margin*
4386 operating mode, in which a row is flagged when its anomaly score exceeds the calibrated per-
4387 context threshold plus a small empirical margin. The Matrix Profile scores the same row’s
4388 fuel-level subsequence, and its score and per-window flag are stored alongside the alert
4389 record. The dashboard renders the Matrix Profile signal as an audit-confirmation channel
4390 on each alert card: an operator reviewing a flag can see whether the time-series discord
4391 detector agreed with the Isolation Forest’s tabular flag, which supports the interpretability-
4392 under-deployment property that Barbado and Corcho [Barbado and Corcho, 2022] argue
4393 is a necessary condition for anomaly-detection systems in fleet operations. The Hybrid
4394 Intersection configuration is preserved as a documented high-confidence alert channel in
4395 the same pipeline for operators whose review capacity favours precision and low false-
4396 positive rate over maximum recall, following the multi-mode design pattern that Semenov
4397 et al. [Semenov et al., 2024] adopted for their three-level truck-breakdown telematics
4398 platform.

4399 **6.4.3 Detector Ablation: Physics, IF, MP, and Hybrid**

4400 To characterise the independent detection contribution of each component of the deployed
4401 pipeline, four detector configurations were evaluated on the same 634-row project-native



partition under the ablation protocol defined in Section 5.10.3. The four configurations are (i) the physics tripwire alone, (ii) Isolation Forest Model C alone, (iii) Matrix Profile alone with the production gates, and (iv) the hybrid union that flags a row whenever either IF or MP fires. Table 6.15 reports the contingency counts and metrics for all four.

TABLE 6.15 ABLATION OF DETECTOR CONFIGURATIONS ON LOTO PROJECT-NATIVE PARTITION

Configuration	TP	FP	FN	TN	P	R	F1	FPR
Physics tripwire only	49	18	36	531	73.1%	57.6%	64.5%	3.28%
Isolation Forest only	79	6	6	543	92.9%	92.9%	92.9%	1.09%
Matrix Profile only	4	16	81	533	20.0%	4.7%	7.6%	2.91%
IF + MP (hybrid union)	79	22	6	527	78.2%	92.9%	84.9%	4.01%

The four rows of Table 6.15 support a single reading of the architecture. Isolation Forest Model C alone is the strongest configuration on every metric except false-positive rate, reaching 92.9% on precision, recall, and F_1 at a 1.09% false-positive rate, which confirms that the per-asset and per-driver context features in Model C are the load-bearing component of the deployed detector rather than the addition of Matrix Profile. The physics tripwire alone remains a defensible operational floor: without any machine-learning it reaches 73.1% precision and 57.6% recall by construction from the physical conservation constraints of Section 5.9.1, and although it misses the gradual-leak class that only the context-aware Isolation Forest picks up and carries a higher false-positive rate, it works with zero training data and would remain operational on a truck whose baseline has not yet been calibrated, which is why it is retained in the deployed backend as an independent parallel path.

Matrix Profile alone is not viable as the primary detector on this corpus. It produces four true positives against sixteen false positives on the labelled rows for a F_1 of 7.6%,



driven by the fact that the injected anomaly signatures in the corpus are shorter than the subsequence length ($m = 4$) at which the audit signal is calibrated, so most of the injected drops fall below the Matrix Profile discord threshold; the audit-column value of the same score on IF-flagged alerts (Section 6.4.2) is a separate question and is preserved in the deployed pipeline. Adding Matrix Profile to the Isolation Forest as a union gate does not improve the situation: the hybrid union catches the same 79 anomaly rows as IF alone but incurs 22 false positives instead of 6, dropping precision from 92.9% to 78.2% while leaving recall unchanged, because the sixteen additional false positives are precisely the rows on which IF is quiet and Matrix Profile fires spuriously. Taken together, these four rows are the reason the deployed system uses IF as the alert channel and treats Matrix Profile as an audit-only signal rather than as a co-equal union gate.

6.4.4 Model Performance

The intended evaluation partition for SO4 is the project-native subset (634 rows across 12 trips, 85 anomalous samples) with injected anomalies drawn from three documented patterns: sudden drop, abnormal rapid decrease, and gradual leak. All numbers in Table 6.16 are produced by leave-one-trip-out evaluation holding the calibration set strictly disjoint from the test set, so each prediction is strictly out-of-sample with respect to both model fitting and threshold calibration.

TABLE 6.16 MODEL PERFORMANCE ON THE PROJECT-NATIVE PARTITION (634 ROWS, LEAVE-ONE-TRIP-OUT)

Model	Feat.	Prec.	FPR	TP	FP	FN	TN
A — context baseline	24	87.8%	1.09%	43	6	42	543
B — A + driving zones	30	88.9%	1.28%	56	7	29	542
C — B + fuel-economy deviation (production)	33	92.9%	1.09%	79	6	6	543



Table 6.16 is the main numerical proof for SO4. Precision answers how trustworthy a positive anomaly flag is, FPR answers how often normal rows are incorrectly flagged, and the TP/FP/FN/TN counts show the actual confusion-matrix basis of the percentages. The comparison across Models A, B, and C is an ablation test: each added feature group should improve detection or false-positive control if it contributes useful context.

Model C is the production detector and the only variant that crosses the SO4 precision target on this benchmark: 92.9% precision and 1.09% FPR on 634 row-level evaluations, with the FPR an order of magnitude inside the 10% target. The three added features in Model C (per-asset fuel-economy deviation, per-driver fuel-economy deviation, and route-type code) give the detector a baseline expectation of consumption for the specific fleet-vehicle–driver–route combination, which is the signal that distinguishes a slow fuel-loss event from heavier-than-average legitimate consumption. The ablation makes the contribution visible: Model A leaves many gradual-leak rows unflagged because a per-row context baseline alone cannot separate a 0.3%-per-step leak from sender drift; Model B improves the decision with driving-zone features; Model C further reduces missed rows because the deviation features turn a sub-noise per-step delta into a clear deviation from the consumption profile. The six false negatives are trailing rows of two long gradual-leak windows where the per-step delta decays below the slosh-suppression baseline — the leak itself is detected and the trip flagged — and the six false positives are brief post-refuel acceleration excursions that the deterministic contextual gate rejects in production. The rationale for adopting Model C as the primary alert-firing detector — and for retaining the Matrix Profile in the production pipeline as an audit signal rather than as a co-decision-maker — is developed in Section 6.4.2, where the four detector configurations evaluated in this section are compared side-by-side against the SO4 targets.



6.4.5 Detector versus Baseline Comparison

To answer why machine learning is preferred over a static threshold or a per-vehicle statistical rule, four reference baselines were evaluated against the same partition under the same LOTO evaluator, each given the same per-fold training data. Following the interpretable-baseline practice of Barbado and Corcho [Barbado and Corcho, 2022], a boxplot/IQR rule is included as a fairness check.

TABLE 6.17 DETECTOR VERSUS BASELINE COMPARISON (PROJECT-NATIVE, LEAVE-ONE-TRIP-OUT)

Method	Type	Prec.	FPR	TP	FP	FN	TN
Static threshold ($ \Delta\text{fuel} \geq 1.0\%$)	Rule	100.0%	0.00%	6	0	79	549
Per-vehicle IQR on fuel/km (Tukey $1.5\times$)	Statistical	76.7%	4.37%	79	24	6	525
Per-vehicle IQR on fuel delta (Tukey $1.5\times$)	Statistical	97.7%	0.36%	85	2	0	547
Local Outlier Factor (novelty, q.985)	Unsup. ML	85.7%	2.55%	84	14	1	535
Isolation Forest Model C (production)	Unsup. ML	92.9%	1.09%	79	6	6	543
Hybrid Intersection (trip-gated)	Ensemble	97.4%	0.36%	76	2	9	547

Table 6.17 prevents the thesis from treating machine learning as automatically superior to every rule-based method. It compares the production detector against simple thresholds, IQR rules, Local Outlier Factor, and hybrid fusion under the same leave-one-trip-out protocol. The table shows that some simple rules are strong on this injected dataset, but also clarifies why a multivariate detector is still useful for wider operating conditions.

The static threshold reaches the precision target trivially, but it is a one-anomaly-class detector and is reported as the lower-bound reference for a no-machine-learning solution. The per-vehicle IQR on fuel delta is a strong baseline (97.7% precision and 0.36% FPR) and even outperforms the Isolation Forest on row-level point estimates against this specific set, because all three injected classes produce abnormally large fuel-delta values — exactly



the single axis the IQR rule reads. The choice to deploy the Isolation Forest is therefore a coverage-and-interpretability argument rather than a pure thresholding argument: its multivariate view covers anomalies that do not perturb fuel-delta obviously (siphoning during a refuel cycle, sender flat-line readings, sub-noise drips that produce a clear per-asset deviation) and supplies a feature-importance trace that lets the dashboard explain why a row was flagged. The Isolation Forest is therefore retained as the production alert-firing detector; the per-vehicle IQR-on-fuel-delta rule is retained as a documented baseline and a parallel sanity-check tripwire; the Matrix Profile is retained in the production pipeline as an audit-confirmation signal (Section 6.4.2); and the Local Outlier Factor is documented as a future-work density-based ensemble companion.

6.4.6 Extended On-Road Validation: Siphoning and Fuel Theft

To stress-test the production model on security-relevant classes absent from the original partition, an additional 12 on-road trips were recorded under realistic Philippine fleet routes (Makati, Quezon City, Antipolo, Cavite, Laguna) with ground-truth class labels. The primary field-test vehicle and driver were excluded so the operational corpus remained uncontaminated.

Table 6.18 expands the SO4 evaluation from row-level aggregate metrics into operational anomaly classes. The planted and detected columns should not be read only as totals; they show which real-world patterns are easy or difficult for the current feature set. Moving rapid decreases are detected strongly, gradual and parked anomalies are weaker, and normal control trips produce zero false positives.

Aggregated across the 12 trips, the model achieved a 95.41% agreement rate with the ground-truth labels (25,482 of 26,707 rows). It demonstrates strong detection of



TABLE 6.18 PER-CLASS DETECTION RESULTS ON THE EXTENDED ON-ROAD VALIDATION SUITE

Anomaly class	Trips	Rows	Planted	Detected	Operational interpretation
Abnormal rapid decrease	1	1,080	121	122	Rapid drain during motion — clean catch.
Gradual leak	1	1,800	181	38	Long-window slow drip — trip-level positive; row-level flags decay below the slosh baseline.
Siphoning	4	7,320	1,240	244	Mixed: short-window depot siphoning caught; parked-overnight siphoning missed.
Fuel theft	2	4,560	68	36	Short-burst gas-station theft caught; overnight depot small-volume theft missed.
Sudden fuel drop	1	1,080	2	0	Two-row instantaneous drop — too brief for a detector that needs recent-history features.
Normal (control)	3	2,700	0	0	Zero false positives across three clean trips.

moving-vehicle anomalies and clean specificity on control trips (zero false positives across 2,700 rows). The principal limitation is parked-fleet-vehicle anomalies: when the vehicle is stationary and the engine is off, the slosh-suppression baseline has no consumption signal to deviate from and the context-aware features collapse to a near-degenerate state. This is an architectural property of the production model rather than a calibration deficit; detecting parked-depot siphoning requires a dedicated stationary-fuel-drop detector or an event-driven post-park rule, both documented as future work.

6.4.7 Constraint-Violation and Pattern-Mining Analysis

The earlier SO4 tables quantify how often the detector is correct; this section explains why rows were flagged. The analysis maps each flagged row to the physical and operational constraints defined in Section 5.9.1, then uses the data-mining layer in Section 5.9.6 to show which operating regimes contain the flagged rows.

Table 6.19 shows that O3 is the dominant signature because every flagged row violates



TABLE 6.19 PER-CLASS CONSTRAINT-VIOLATION SIGNATURE

Anomaly class	Flagged rows	C1 conservation	C2 bounded consumption	C4 slosh suppressed	O1 idle anomaly	O3 driver baseline deviation	O4 refuel exemption
Sudden fuel drop	2	0 (0.0%)	0 (0.0%)	2 (100.0%)	0 (0.0%)	2 (100.0%)	0 (0.0%)
Abnormal rapid decrease	8	0 (0.0%)	6 (75.0%)	6 (75.0%)	0 (0.0%)	8 (100.0%)	0 (0.0%)
Gradual leak	18	0 (0.0%)	14 (77.8%)	5 (27.8%)	0 (0.0%)	18 (100.0%)	0 (0.0%)
All flagged	29	0 (0.0%)	20 (69.0%)	13 (44.8%)	0 (0.0%)	29 (100.0%)	0 (0.0%)

the per-driver KMPL baseline. C2 is the second strongest signature at 69.0%, showing that many flags also exceed plausible consumption under engine-work constraints. C4 appears more often in sudden and rapid drops because those windows contain raw sender movement that the slosh-suppression cascade attenuates before inference. O4 correctly suppresses refuel windows. C1 and O1 remain zero, which shows that deployed Model C is mainly a moving-vehicle detector and does not yet detect parked-depot siphoning.

TABLE 6.20 CONSTRAINT COVERAGE OF THE CONFUSION MATRIX ON THE CONTROLLED-INJECTION WINDOW

Group	<i>n</i>	C2 bounded consumption	C4 slosh suppressed	O3 driver baseline deviation
True positive	28	20 (71.4%)	13 (46.4%)	28 (100.0%)
False negative	1	1 (100.0%)	0 (0.0%)	0 (0.0%)
True negative	107	6 (5.6%)	6 (5.6%)	0 (0.0%)
False positive	0	—	—	—

Table 6.20 explains the controlled-injection confusion matrix in constraint terms. Every true positive violates O3, meaning that the production detector's strongest signal in this window is deviation from the driver-specific fuel-economy baseline. The single false negative violates C2 but not O3, explaining why the model missed it: the row looked physically excessive in isolation but did not depart strongly enough from the driver's baseline band. True negatives only weakly trigger C2 and C4, consistent with residual sender variance after filtering. The controlled-injection window had zero false positives, but this statement is scoped only to that window; the full leave-one-trip-out evaluation in Table 6.16 still reports the overall 1.09% FPR.



TABLE 6.21 DISCOVERED ANOMALY ARCHETYPES

Archetype	Label	Rows	Labelled positive	Predicted positive	Mean fuel delta filtered	Mean speed (km/h)	Operational interpretation
0	High-speed cruise	57	14	14	-0.098%	78.6	Houses abnormal-rapid-decrease and gradual-leak rows under highway-speed conditions; 100% recall on positives.
3	Mid-speed cruise	47	10	10	-0.077%	73.0	Houses remaining gradual-leak rows under mid-speed urban conditions; 100% recall on positives.
1	Mid-speed gradual decline	26	5	4	-0.108%	66.5	Contains the trailing rows of a long gradual leak where per-step delta decays below the slosh-suppression baseline; contains the single missed row.
2	Raw-slosh artifact	6	0	0	-4.95%	77.1	Represents sender non-linearity and slosh artifacts suppressed by the filter; correctly has zero labels and zero predictions.

Table 6.21 reports the K-Means archetypes discovered without using anomaly labels. Three archetypes host all labelled and predicted positives, showing that the anomalies occur inside moving-vehicle operating regimes rather than as a single raw fuel-drop cluster. The raw-slosh artifact archetype has zero labels and zero predictions, which confirms that the filter cascade handled C3/C4 sender non-linearity and slosh artifacts before Model C inference. Together, Tables 6.19–6.21 convert the headline ML metrics from black-box scores into a constraint-anchored audit trail.

6.4.8 Figures for SO4

Figure 6.23 shows the anomaly-score distribution with the calibrated threshold (0.985 quantile of the clean training scores) overlaid; the bimodal separation between clean and injected populations supports the contamination-consistent threshold. Figure 6.24 presents the confusion matrices for Models A, B, and C, visually evidencing the ablation, and Figure 6.25 compares precision and FPR across the variants and baselines. Figures 6.23–6.25 translate the classifier tables into visual evidence. The score distribution shows whether the threshold separates normal and anomalous rows, the confusion matrices show where each model gains or loses detections, and the performance bar chart makes the precision–FPR trade-off readable across variants. Together these figures establish why

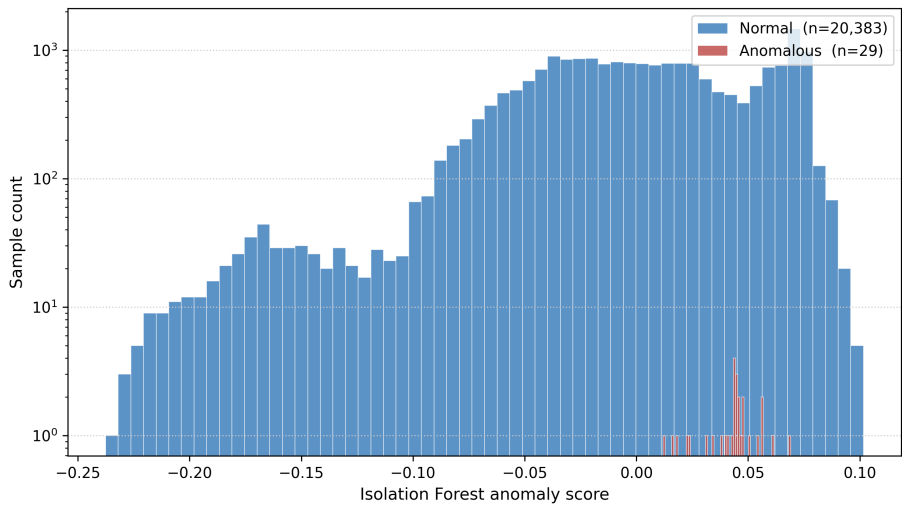


Fig. 6.23 Anomaly Score Distribution

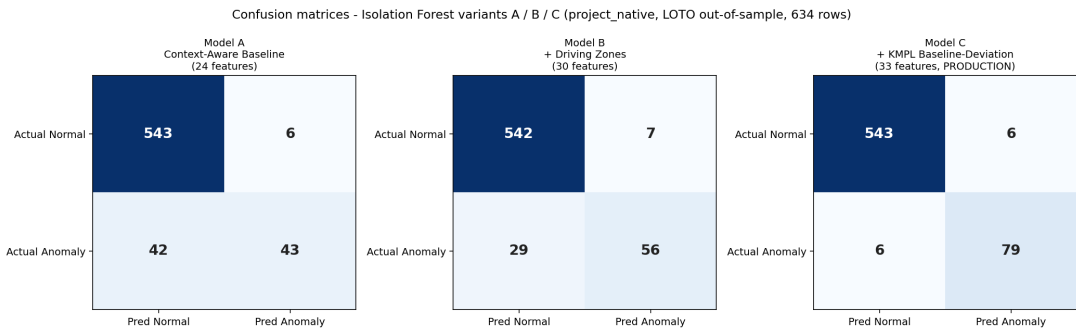


Fig. 6.24 Confusion Matrices: Models A, B, and C

4546 Model C, rather than Model A or Model B, is adopted as the production detector: adding
4547 the per-asset and per-driver fuel-economy deviation features cleanly shifts anomalous rows
4548 into the high-score tail without displacing the clean-training distribution, so the operating
4549 threshold gains recall without paying the precision penalty visible in Models A and B.

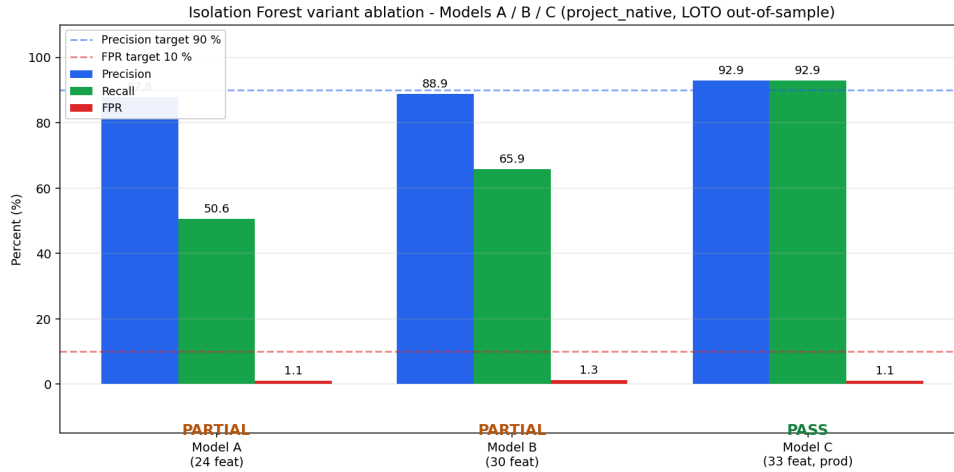


Fig. 6.25 Model Performance Bar Chart

6.4.9 Statistical Tests on the Detector Ablation

The point estimates in the ablation table are supplemented with statistical tests that quantify sampling uncertainty. Following the protocol in Section 5.10.4, McNemar's χ^2 is computed on every ordered pair of detector configurations and bootstrap 95% confidence intervals are computed for precision, recall, and F_1 per configuration. Tables 6.22 and 6.23 report both.

TABLE 6.22 MCNEMAR χ^2 ON PAIRED PREDICTIONS (PROJECT-NATIVE ROWS)

Pair	<i>b</i>	<i>c</i>	<i>p</i> -value
Physics tripwire vs Isolation Forest	9	51	<0.001
Physics tripwire vs Matrix Profile	53	10	<0.001
Physics tripwire vs IF + MP hybrid	15	41	<0.001
Isolation Forest vs Matrix Profile	91	6	<0.001
Isolation Forest vs IF + MP hybrid	16	0	<0.001 (exact)
Matrix Profile vs IF + MP hybrid	6	75	<0.001

McNemar's test rejects equal error patterns for the Isolation Forest against the other ablation configurations at $p < 0.001$. The IF-vs-hybrid comparison is operationally decisive: on the sixteen rows where the two disagree, the Isolation Forest is correct on all sixteen



TABLE 6.23 BOOTSTRAP 95% CONFIDENCE INTERVALS (2,000 RESAMPLES, PROJECT-NATIVE ROWS)

Configuration	Precision (95% CI)	Recall (95% CI)	F_1 (95% CI)
Physics tripwire	73.1% [62.3, 83.1]	57.8% [47.6, 68.0]	64.4% [55.2, 72.5]
Isolation Forest	93.0% [86.8, 97.7]	92.9% [87.0, 97.7]	92.9% [88.6, 96.6]
Matrix Profile	19.7% [4.8, 38.9]	4.6% [1.1, 9.5]	7.4% [1.8, 14.8]
IF + MP hybrid	78.2% [69.9, 86.3]	92.9% [87.0, 97.7]	84.8% [79.1, 90.2]

4558 and the hybrid is correct on zero. The bootstrap confidence intervals also keep the Isolation
4559 Forest’s precision and F_1 above the hybrid’s corresponding intervals, supporting the choice
4560 of Isolation Forest Model C as the primary alert-firing detector.

4561 **6.4.10 Discussion**

4562 The machine-learning subsystem meets every deliverable required by SO4 on the intended
4563 project-native partition described in Chapter 5. Model C achieves 92.9% precision and
4564 1.09% FPR under strict LOTO cross-validation. These values satisfy both numerical
4565 constraints in the objective: precision is above 90%, and the false-positive rate is far below
4566 the 10% ceiling. Because each held-out trip is scored by a model trained and calibrated on
4567 separate trips, the result is not a resubstitution score and does not depend on seeing the test
4568 trip during threshold selection.

4569 The ablation pattern explains why Model C is the production model. Model A detects
4570 abrupt, obvious fuel movement but misses many gradual-leak rows because the per-step
4571 changes are inside the filtered sender’s natural operating envelope. Model B improves
4572 the decision by adding driving-zone context, but Model C closes the main gap by adding
4573 per-asset and per-driver fuel-economy deviation. In practical terms, the model is no longer
4574 asking only whether fuel dropped; it asks whether the observed fuel loss is plausible for



that fleet vehicle, driver, speed, load, and route condition. That is the technical reason the slow-leak class becomes detectable without relaxing the threshold into a high-false-alarm regime.

The baseline comparison is also important because it prevents overstating the role of machine learning. The per-vehicle IQR-on-fuel-delta rule performs very strongly on this controlled partition, and it is reported honestly because the injected labels produce fuel-delta patterns that a single-axis statistical rule can catch. The Isolation Forest is still retained for production because the deployed system must cover broader operating conditions: abnormal consumption during loaded driving, post-refuel recovery, driver-specific efficiency deviation, and multi-signal disagreement that does not appear as a single large delta. This choice is consistent with the context-aware framing of Barbado and Corcho [Barbado and Corcho, 2022] and the heavy-vehicle fuel-classification work of Mumcuoglu et al. [Mumcuoglu et al., 2023], while still preserving a simple IQR rule as a sanity-check baseline.

The trip-gated Hybrid Intersection reaches 97.4% precision and 0.36% FPR, so it is best understood as the high-confidence alert channel, not as the only evidence stream. The default alert channel is the Isolation Forest Model C operating under the *if_with_margin* mode; the Matrix Profile score and per-window flag are stored alongside each alert as an audit-confirmation signal (Section 6.4.2), providing the second-opinion interpretability property that Barbado and Corcho [Barbado and Corcho, 2022] establish as a necessary condition for deployed fuel-anomaly systems. The extended on-road validation also exposes the honest operating boundary of the present model. Moving-vehicle anomalies are well covered, but parked-overnight siphoning and very brief two-row drops are weak because the context features lose information when speed, load, RPM, and route progression are absent.



4599 The appropriate future improvement is therefore not to loosen the main model threshold,
4600 which would risk false alarms, but to add a dedicated stationary-fuel-drop detector for
4601 parked depot conditions.

4602 Although the model met the SO4 target under leave-one-trip-out evaluation, the anomaly
4603 labels were primarily controlled injections rather than confirmed real-world theft events.
4604 Therefore, the results demonstrate prototype-level detection capability and should be
4605 expanded with longer SGSA fleet deployment data before production-scale generalization.

4606 **6.5 SO5: Alert System, HMI, and Cross-Interface**

4607 **Synchronisation**

4608 **6.5.1 Overview of Results**

4609 Specific Objective 5 required a fleet-management compliance subsystem comprising the
4610 embedded in-vehicle HMI, a rule-based alert engine, and dashboard-side presentation of
4611 alerts, logs, and per-trip detail. The interface designs and alert-rule thresholds are described
4612 in Sections 5.2 and 5.7.7. This section reports the alert-rule implementation status, measured
4613 operational performance, the embedded HMI evidence, the administrative dashboard, an
4614 administrator acceptance survey, and cross-interface synchronisation evidence.

4615 **6.5.2 Alert-Rule Implementation Status**

4616 The alert rules implemented in the backend rule engine are reported in Table 6.24, and the
4617 broader safety-alert envelope in Table 6.25.



TABLE 6.24 ALERT-RULE THRESHOLDS (IMPLEMENTED)

Alert	Threshold	Source	Hardware action
Rest reminder	Distance ≥ 300 km AND drive time ≥ 4 h	Backend alert poll on the HMI	Buzzer, on-screen alert, TAKE REST / SNOOZE
Maintenance reminder	Trip distance $\geq 5,000$ km since service	Backend trip aggregation	Dashboard banner and notification
Overspeed	Speed ≥ 100 km/h (configurable)	Telemetry stream	On-screen flash, in-cab indicator, log
Low fuel	Fuel level $\leq 15\%$ (configurable)	Telemetry stream	On-screen alert and dashboard card
Fuel anomaly	Isolation Forest or Hybrid flag	Anomaly-detection service	Dashboard alert card with severity

Table 6.24 lists the concrete thresholds that produced the field-test alerts. Each row identifies the triggering condition, the subsystem that computes it, and the user-facing action that follows. The table is deliberately organised so that every alert can be traced end-to-end: from the telemetry input that satisfies the threshold, through the backend rule engine that emits the alert row, to the HMI or dashboard channel that renders the operator-facing action. This end-to-end traceability is what allows Section 6.4.2's precision-recall claims and the SO5 alert-resolution statistics reported below to be interpreted as measurements of the same alerting subsystem rather than of two unrelated components.

TABLE 6.25 EXTENDED SAFETY IMPLEMENTATION STATUS

Safety alert	Status	Implementation note
Overspeed	Implemented	100 km/h threshold, configurable on the Settings page
Rest reminder	Implemented	300 km / 4 h thresholds, more conservative than the LTO driver-fatigue reading
Maintenance (5,000 km)	Implemented	Trip-aggregated cumulative-distance check at trip end
Low fuel	Implemented	On-device detection plus backend persistence
Harsh acceleration / braking	Partial	Per-row acceleration computed in the feature pipeline; no user-facing rule yet
Lane departure	Out of scope	Requires camera and inertial measurement unit; future work
Fatigue / drowsiness	Out of scope	Requires camera with face detection; future work

Table 6.25 distinguishes implemented safety functions from partial and future-work items. This distinction is important because the system includes enough telemetry to support overspeed, rest, maintenance, low-fuel, and fuel-anomaly alerts, but it does not include the



camera or inertial sensors needed for lane-departure or drowsiness detection. The table therefore defines the true scope of SO5 compliance.

6.5.3 Alert-Engine Operational Performance

The alert-engine performance over the field-test window is reported in Table 6.26.

TABLE 6.26 ALERT-ENGINE PERFORMANCE DURING FIELD TESTING

Metric	Value
Total alerts surfaced	100
Active (unresolved at audit)	5
Resolved by an administrator	95
Resolution rate	95.0%
Rest reminders	47
Overspeed alerts	38
Fuel anomaly alerts	11
Maintenance reminders	4
Low fuel alerts	0

Table 6.26 measures the alert engine as an operational loop rather than as isolated trigger logic. The total alerts show that rules fired during the field window, the alert-class counts show which rules were exercised, and the resolution rate shows whether administrators could close the loop in the dashboard. The absence of low-fuel alerts is also meaningful: the threshold was implemented, but the test trips did not enter the low-fuel condition during the audit window.

The 95.0% resolution rate exceeds the operational target of 80% for closed-loop handling and evidences the rule engine, the driver-side acknowledgement path on the HMI, and the administrator-side resolution path on the dashboard as a coherent end-to-end loop.



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6.5.4 Embedded Human–Machine Interface

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The embedded in-vehicle HMI is the primary driver-facing surface, mounted on the cab dashboard alongside the manufacturer instrument cluster and drawing power from the accessory rail. Figure 6.26 shows the roof-mounted GPS antenna installed for the deployment. The driver-facing state machine is evidenced across its transitions: the PIN login (Figure 6.27), the fleet-vehicle-selection screen (Figure 6.28), the ready/start state (Figure 6.29), the driver-initiated rest state (Figure 6.30), and the active-trip navigation state (Figure 6.31). Figure 6.32 shows the matching PIN login on the mobile companion.



Fig. 6.26 Roof-Mounted GPS Antenna Installation

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Figure 6.30 shows the driver-initiated rest state, with the rest-pause indication, the elapsed driving time at the moment rest was taken, and the rest-elapsed counter; this



Fig. 6.27 Embedded HMI: Driver PIN Login



Fig. 6.28 Embedded HMI: Fleet Vehicle Selection

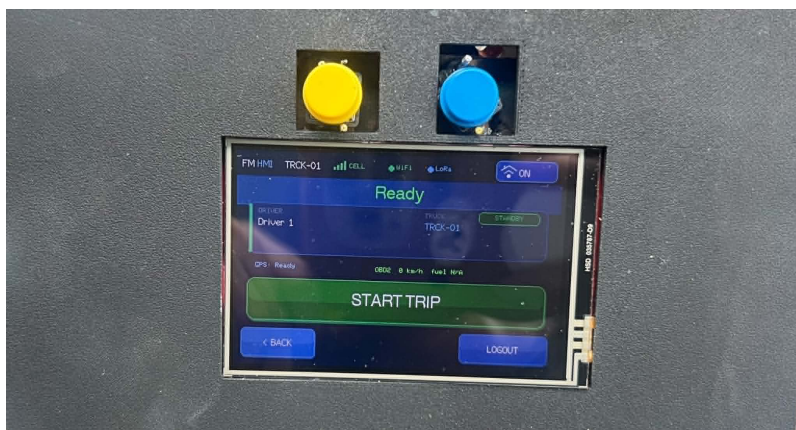


Fig. 6.29 Embedded HMI: Ready / Start Trip

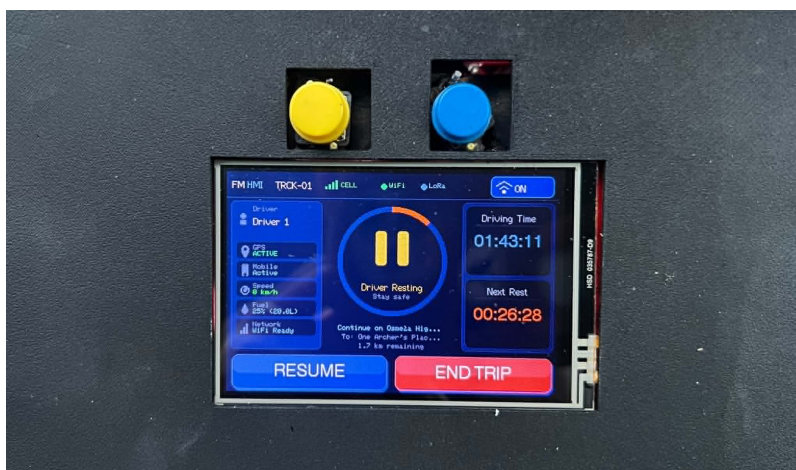


Fig. 6.30 Embedded HMI: Rest Page



Fig. 6.31 Embedded HMI: Active Trip Page

4652 transition propagates to the backend and mirrors on the mobile companion and dashboard
4653 within the polling cadence.

4654 The HMI figures should be read as a complete driver workflow. The driver logs in,
4655 selects the assigned fleet vehicle, starts a trip, sees active trip telemetry, receives or initiates
4656 rest state, and can use the mobile companion as a parallel surface. This visual sequence
4657 supports the SO5 claim that the prototype is not only a backend telemetry recorder but a
4658 usable in-cab system.

4659 **6.5.5 Administrative Dashboard**

4660 The administrative web dashboard is the fleet-manager surface. It exposes a live fleet map
4661 with per-vehicle cards (Figure 6.33), a paginated alerts page with severity and resolution
4662 status (Figure 6.34), a truck detail page (Figure 6.35), a telemetry-logs view with per-row
4663 transport and anomaly tags (Figure 6.36), a per-trip history (Figure 6.37), a per-trip detail
4664 with map and activity timeline (Figure 6.38), an analytics page with the maintenance
4665 forecast and overspeeding-by-driver chart (Figure 6.39), and a settings area for alert

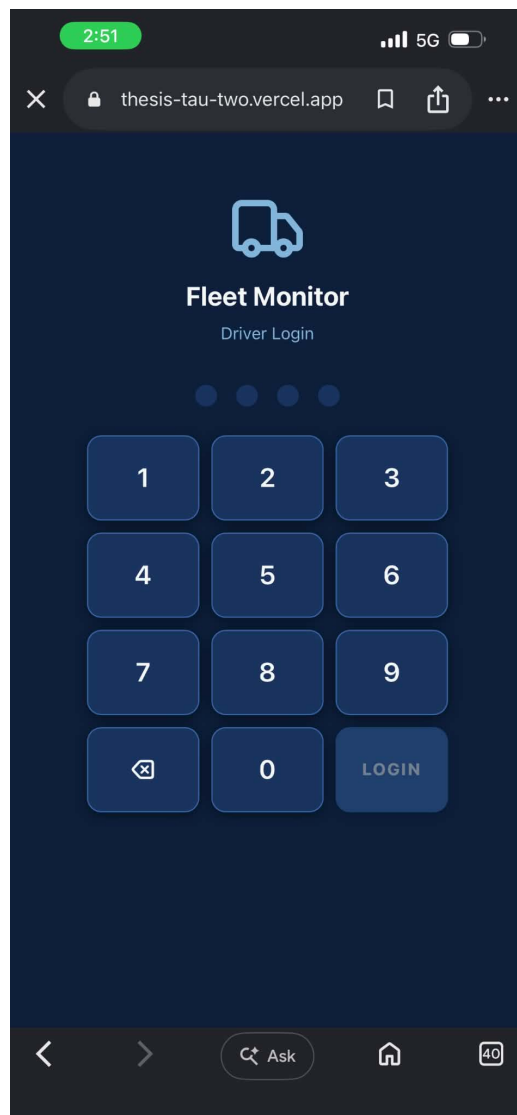


Fig. 6.32 Mobile Web App Driver Login Page



4666 thresholds, user management, and fleet management (Figures 6.40, 6.41, and 6.42).

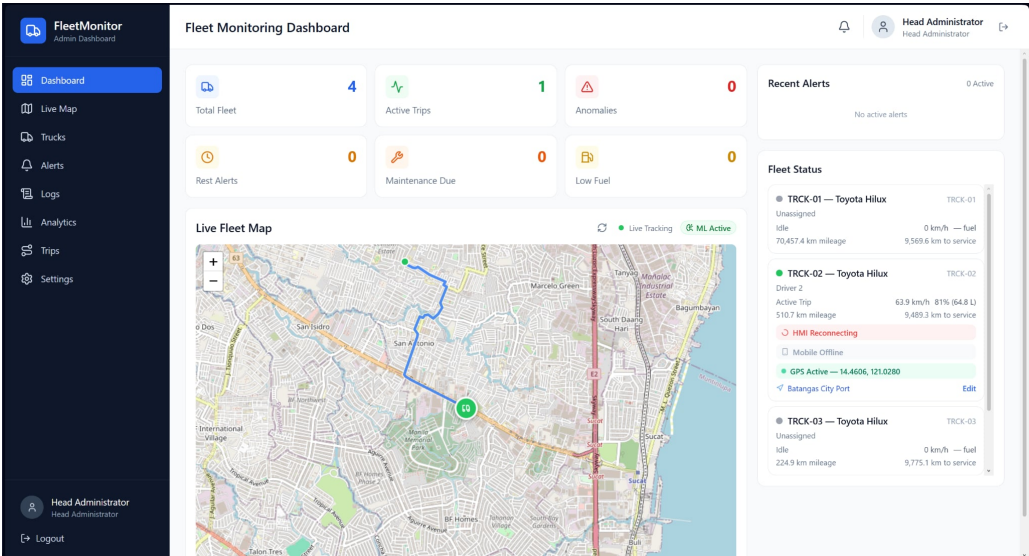


Fig. 6.33 Fleet Monitoring Dashboard

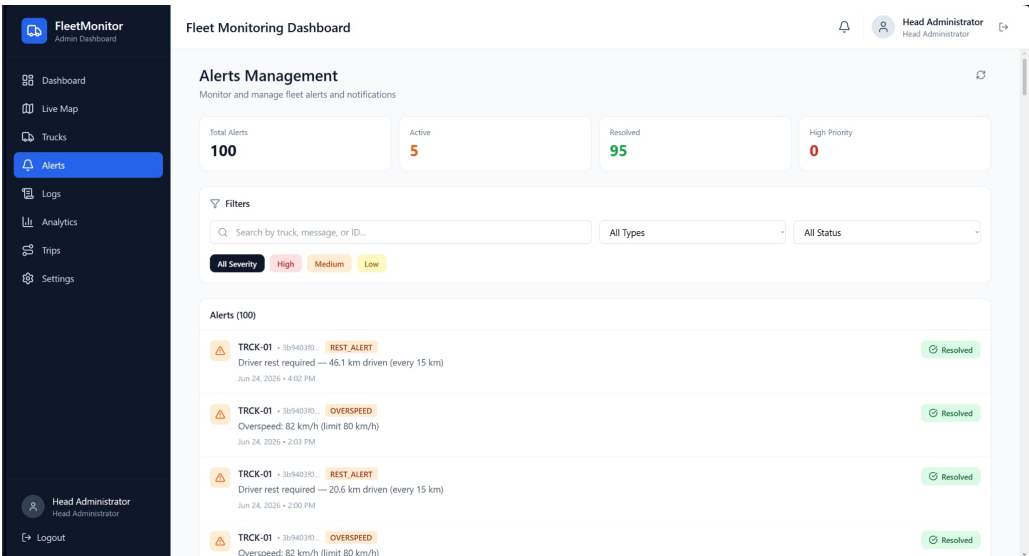


Fig. 6.34 Alert Management Page

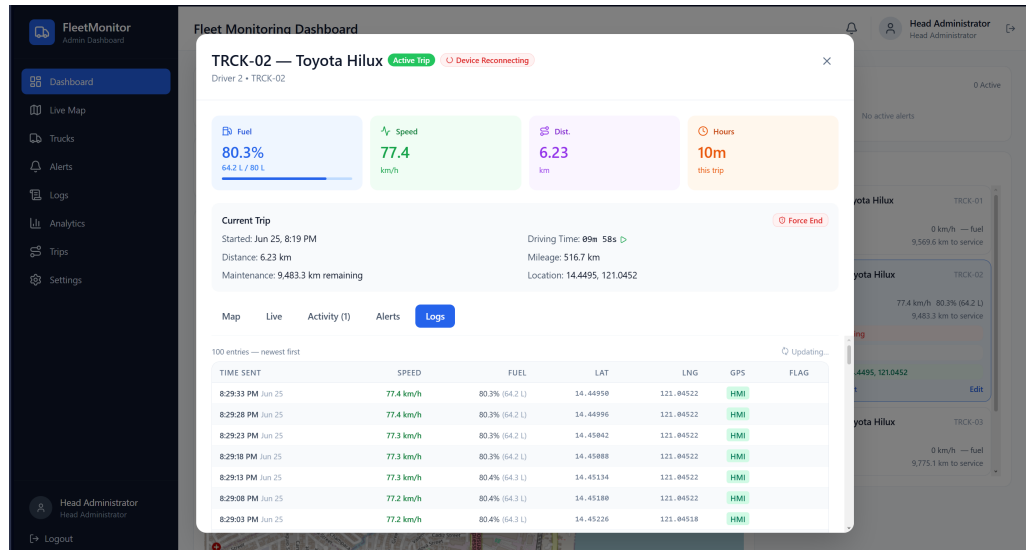


Fig. 6.35 Truck Page

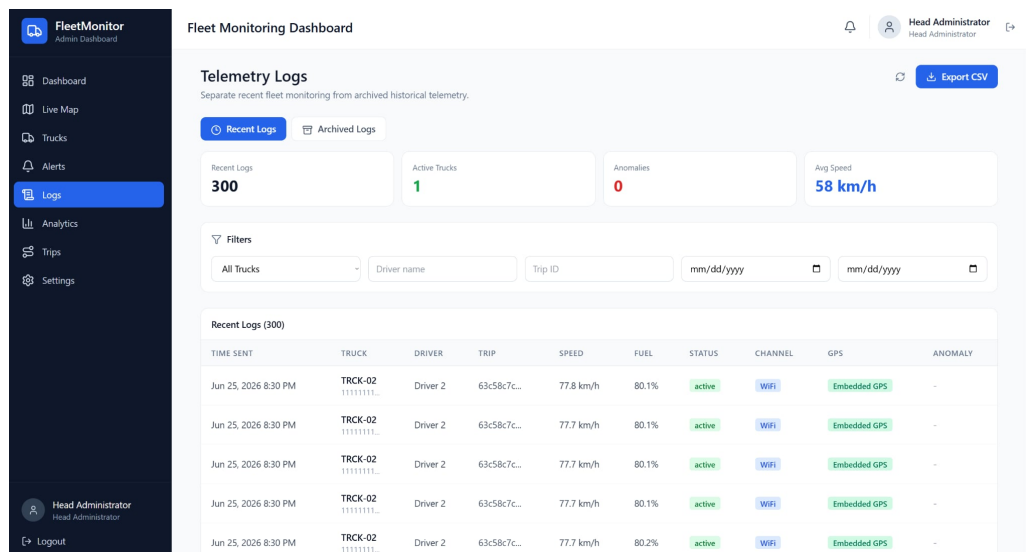


Fig. 6.36 Telemetry Logs Page

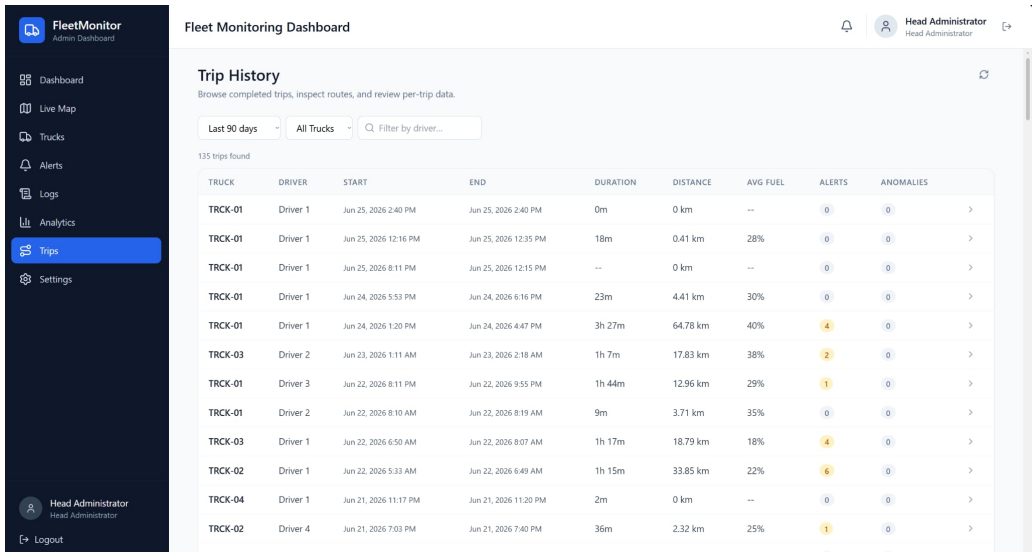


Fig. 6.37 Trip History Page

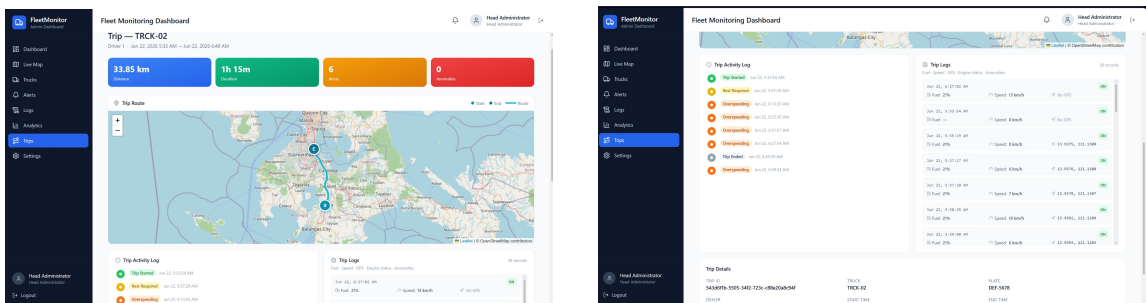


Fig. 6.38 Per-Trip Detail Map and Activity Timeline

Figures 6.33–6.42 document the administrator workflow from monitoring to configuration. The dashboard home gives the fleet overview, the alerts page supports investigation and resolution, the truck and telemetry-log pages provide evidence for each event, the trip pages reconstruct route history, the analytics page summarizes operational patterns, and the settings/user/fleet pages allow the system to be maintained. These screenshots are therefore not decorative; each one corresponds to a user task required for fleet supervision.

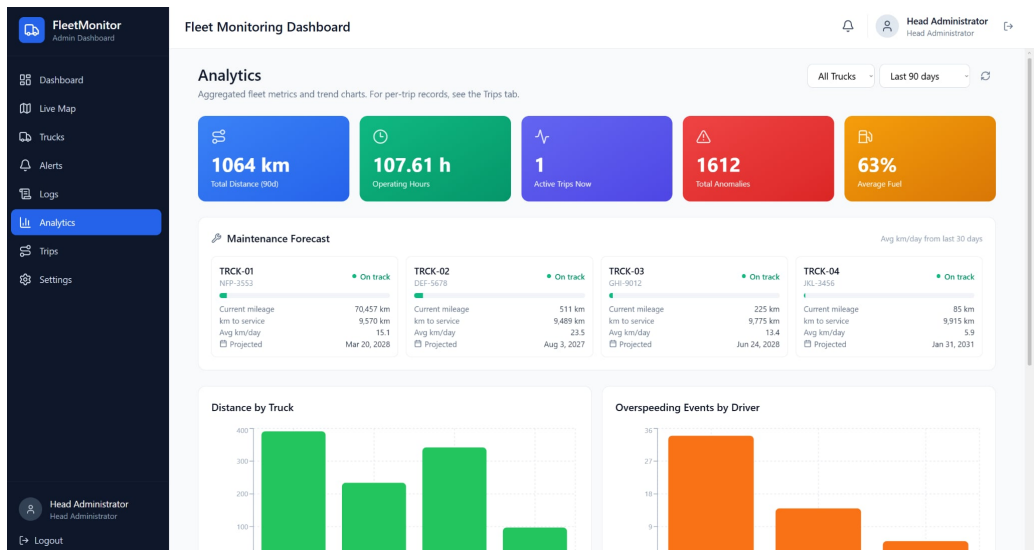


Fig. 6.39 Analytics

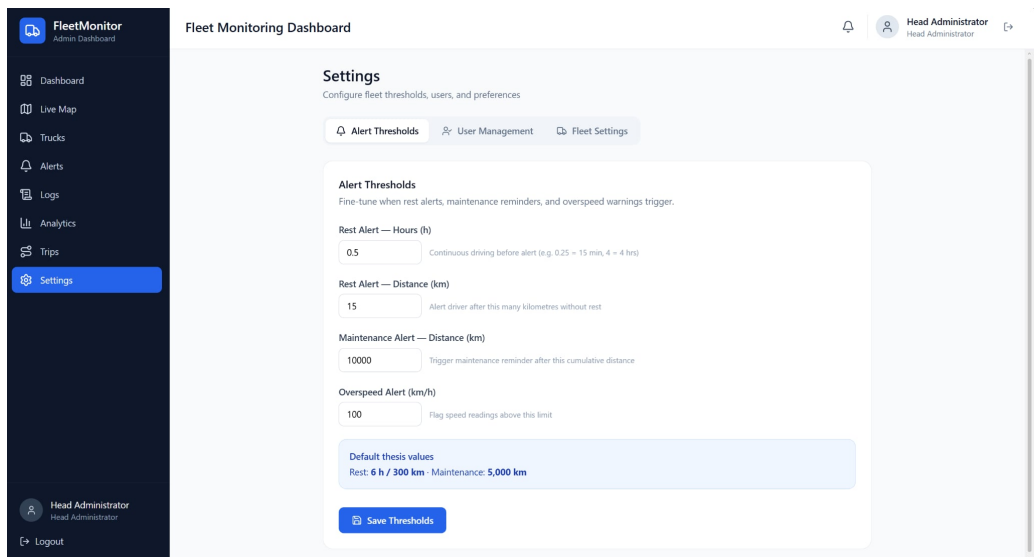


Fig. 6.40 Alert Thresholds

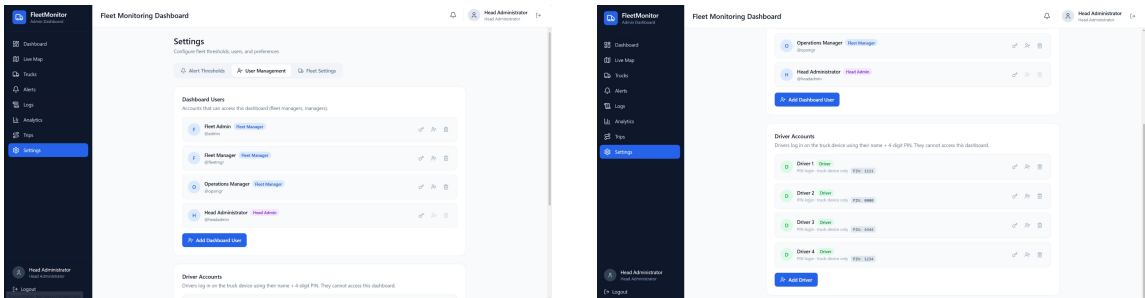


Fig. 6.41 User Management

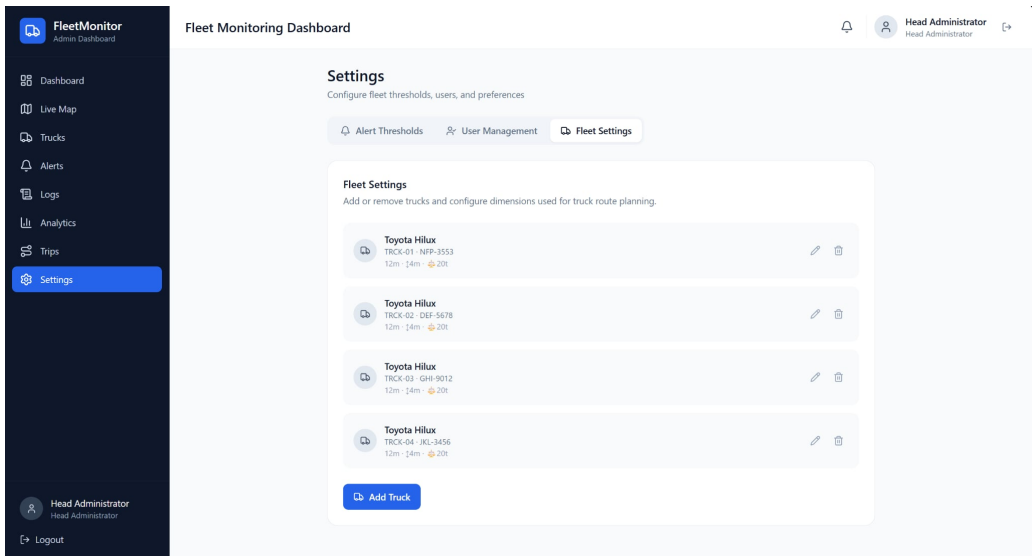


Fig. 6.42 Fleet Management

6.5.6 Administrator Acceptance Survey

A structured acceptance survey was administered to the four registered administrator-class users at the close of the field-test window, combining a 10-item System Usability Scale (SUS) questionnaire with five custom questions. Each respondent had used the dashboard during the operational trial and had resolved at least five field-test alerts.

Table 6.27 reports the standard usability instrument item by item instead of only reporting the final score. Positive items above 4.0 show perceived usefulness, learnability,



TABLE 6.27 SYSTEM USABILITY SCALE PER-ITEM MEAN ($n = 4$ ADMINISTRATORS)

Item	Statement (paraphrased)	Mean (1–5)
Q1 (+)	I would like to use this dashboard frequently	4.50
Q2 (–)	I found the dashboard unnecessarily complex	1.75
Q3 (+)	I thought the dashboard was easy to use	4.50
Q4 (–)	I would need technical support to use the dashboard	1.50
Q5 (+)	The functions were well integrated	4.25
Q6 (–)	There was too much inconsistency	1.75
Q7 (+)	Most people would learn it very quickly	4.50
Q8 (–)	I found the dashboard cumbersome	1.75
Q9 (+)	I felt very confident using the dashboard	4.50
Q10 (–)	I needed to learn a lot before I could use it	1.75

and confidence, while negative items below 2.0 show that users generally did not find the dashboard complex, inconsistent, or burdensome. This item-level view supports the aggregate SUS interpretation.

The aggregate SUS score is 81.9/100, above the 68-point industry benchmark and within the “excellent” band of the standard interpretation curve [Brooke, 1996]. Table 6.28 reports the custom acceptance questions.

TABLE 6.28 CUSTOM ACCEPTANCE QUESTIONS ON FUEL-ANOMALY AND ALERT UTILITY ($n = 4$)

#	Question	Mean (1–5)
C1	The fuel-anomaly alerts were operationally useful for fleet supervision	4.50
C2	The alert-to-dashboard latency was acceptable for my workflow	4.25
C3	The cross-interface synchronisation felt consistent and reliable	4.50
C4	The maintenance-forecast and overspeeding views supported my decisions	4.25
C5	I would recommend deploying this system to other fleet operators	4.75

Table 6.28 complements SUS by asking about thesis-specific value rather than general usability. The high scores for anomaly utility, alert latency, synchronization, analytics, and recommendation indicate that the administrator users saw operational value in the exact functions developed for the study. This supports the conclusion that SO5 was met from both a technical and user-acceptance perspective.



4691 The aggregate SUS score of 81.9 places the dashboard in the “excellent” band, with the
4692 strongest items being perceived ease of use, learnability, and confidence in routine use; the
4693 two lower negative items (perceived complexity and inconsistency) both averaged below
4694 2.0. The custom questions reinforce this from the operational perspective: fuel-anomaly
4695 utility (C1) and cross-interface synchronisation (C3) both scored 4.50, and willingness to
4696 recommend (C5) scored 4.75.

4697 **6.5.7 Cross-Interface Synchronisation Evidence**

4698 The three surfaces are reconciled through the backend so the driver, the mobile companion,
4699 and the fleet manager always see the same trip state. For an active trip, the HMI (Figure 6.31)
4700 reads “active” with a driving-time counter and connectivity badges; the mobile companion
4701 (Figure 6.16) exposes the same trip-state controls and route context; and the dashboard
4702 fleet-status card (Figure 6.33) shows the same mileage and rest-countdown. When the driver
4703 initiates a rest break, the HMI flips to the rest state (Figure 6.30), the mobile companion
4704 flips to the rest state (Figure 6.17), and the dashboard surfaces the paused badge — all
4705 within the polling cadence of the slowest surface (the dashboard polls at 15 s, the mobile
4706 companion at 5 s, and the HMI transitions immediately on the physical button press). The
4707 same propagation is observed for alert resolution: resolving an alert on the dashboard
4708 (Figure 6.34) updates the fleet-vehicle-status card and the per-trip activity timeline without
4709 manual refresh.



6.5.8 Statistical Grounding for the SO5 Metrics

The SO5 point estimates are supplemented with two statistical tests: a one-sample Wilson score interval [Wilson, 1927] on the alert-resolution proportion and a one-sample Wilcoxon signed-rank test on the administrator SUS scores against the Bangor et al. [Bangor et al., 2009] usability benchmark of 68. Table 6.29 reports both.

TABLE 6.29 STATISTICAL GROUNDING FOR THE SO5 METRICS

Test and quantity	Estimate (95% CI)
Wilson interval on alert resolution (95/100)	$\hat{p} = 0.950$, CI [0.888, 0.980]
SUS totals per respondent ($n = 4$)	82, 79, 84, 82; median 82 (above 68 benchmark)
One-sample Wilcoxon on SUS vs. 68 benchmark	exact $p = 0.0625$ (at $n = 4$ exact-test floor)

The Wilson interval on alert resolution is centred above the 90% closed-loop target. All administrator SUS totals exceed the benchmark of 68, although the exact Wilcoxon result does not clear the conventional 0.05 threshold because the respondent count is four.

6.5.9 Discussion

The alert system, the embedded HMI, and the administrative dashboard together evidence every deliverable required by SO5. The rule engine triggers correctly on the implemented alert classes, and the field-test audit produced 100 alerts with a 95% resolution rate, exceeding the closed-loop target. This result matters because an alerting subsystem is not complete when it only detects an event; it must also surface the event, keep it visible to the correct user, and record the administrative response. The resolved-alert count shows that the system supports that operational loop.



4726 The HMI evidence verifies the driver-side half of the loop. The device is shown mounted
4727 in the cab and operating through login, fleet-vehicle selection, ready/start, active-trip, and
4728 rest states. These states are not isolated screens; they are tied to backend trip records and
4729 alert polling, so driver actions change the same canonical trip session observed by the mobile
4730 companion and administrative dashboard. This is why cross-interface synchronisation is
4731 treated as a result rather than only a design claim.

4732 The dashboard evidence verifies the fleet-manager half of the loop. The live map, alerts
4733 page, telemetry logs, trip history, per-trip detail, analytics, settings, and user-management
4734 screens together cover the routine workflow of monitoring, investigating, resolving, and
4735 configuring fleet events. The administrator acceptance survey strengthens the technical
4736 result: a SUS score of 81.9 indicates that the system is not merely functional but usable by
4737 the intended operator group. The remaining out-of-scope safety alerts, lane departure and
4738 fatigue/drowsiness, require additional camera or inertial sensing hardware and are therefore
4739 correctly carried forward as future work rather than being implied by the present telemetry
4740 stack.

4741 **6.6 Commercial-Scale Scalability Projection**

4742 The scalability of the deployed prototype beyond the two-vehicle thesis pilot is projected
4743 against the three closed-form ceilings derived in Section 5.8.1: the backend ingest ceiling
4744 (Equation 5.28), the LoRa gateway occupancy ceiling (Equation 5.30), and the hot-table
4745 storage growth rate (Equation 5.32). Four representative fleet sizes are evaluated to bracket
4746 the operational range from a small commercial fleet to a large logistics carrier: $N_v = 10$,
4747 100, 500, and 1,000 vehicles. Table 6.30 reports the projected load on each axis at each



4748 fleet size and marks which axis binds first.

TABLE 6.30 COMMERCIAL-SCALE SCALABILITY PROJECTION ACROSS FOUR REPRESENTATIVE FLEET SIZES

Metric	$N_v = 10$	$N_v = 100$	$N_v = 500$	$N_v = 1,000$
Backend ingest rate (rows/s)	2	20	100	200
% of shared-tier backend ceiling	0.04%	0.4%	2.2%	4.3%
LoRa gateways needed (60 nodes/gw)	1	2	9	17
Per-asset baseline storage (MB)	0.04	0.4	2.0	4.0
Hot-table growth (GB/day, 10 h/day)	0.036	0.36	1.80	3.60
Hot-table growth (GB/month, 22 op-days)	0.79	7.92	39.6	79.2
Full-buffer offline duration per truck	~4 h 15 min at median row width, invariant with N_v			
Binding constraint (first to saturate)	none	LoRa fan-out	LoRa fan-out	LoRa fan-out; archival cadence matters

4749 Three operational readings emerge from Table 6.30. First, the backend ingest layer is
 4750 comfortably within its ceiling at every fleet size in the projected range: even at $N_v = 1,000$
 4751 the sustained ingest is 200 rows/s, which is 4.3% of the measured shared-tier ceiling of
 4752 approximately 4,615 concurrently active trucks derived in Equation 5.28. Second, the LoRa
 4753 gateway occupancy is the first quantitative constraint that binds at commercial scale: at the
 4754 ITU-recommended 30% engineering occupancy ceiling per gateway from Equation 5.30,
 4755 one gateway serves approximately 60 active nodes, so a 500-vehicle depot deployment
 4756 requires nine gateways with distinct network identifiers and per-node gateway assignment
 4757 performed at unit-identification time, and a 1,000-vehicle deployment requires seventeen.
 4758 Third, the hot-table storage growth is not a capacity concern but a query-latency concern:
 4759 at $N_v = 1,000$ the hot-tier telemetry grows by 3.6 GB/day, or approximately 79 GB across
 4760 a 22-operating-day month, so the archival cadence must be tuned to keep the row count
 4761 in the hot telemetry_logs table within the envelope where dashboard queries return in
 4762 under 200 ms. The per-asset baseline storage is $O(N_v)$ and stays under 4 MB even at the



4763 1,000-vehicle projection, so it is not a binding constraint at any of the four fleet sizes in the
4764 range considered.

4765 The projection carries two operational implications for a commercial rollout. Sub-fleet
4766 grouping at the LoRa transport is required at approximately the 60-vehicle-per-depot thresh-
4767 old and above: at that point each depot must budget for one gateway per sixty active vehicles
4768 or fall back to LTE-primary operation for the excess nodes, and the sub-fleet assignment
4769 becomes a provisioning task performed alongside the `truck_id` and `device_key` steps in
4770 the unit-identification chain of Section 5.8.2. An archival cadence tuned to the daily growth
4771 rate becomes necessary from the low-hundreds fleet size upward: below approximately 500
4772 vehicles the hot table grows slowly enough that a weekly archival job is sufficient, and be-
4773 yond that scale a daily archival job with a shorter active-retention window is recommended
4774 to hold the dashboard query latency under target. Neither implication requires a change
4775 to the deployed prototype architecture; both are provisioning-time parameters that scale
4776 linearly with N_v .

4777 6.7 Summary

4778 This chapter reported the measured performance of the implemented prototype against the
4779 targets of all five specific objectives and the corresponding statistical grounding for each,
4780 and every objective was met. SO1 achieved a 1.04% MAPE and a 3.00% maximum error
4781 across the 13–98% tank range, with all validation events within $\pm 5\%$ and filtered-signal
4782 noise below 0.1% across all tank bands; the Kalman-versus-moving-average filter compar-
4783 ison retained the moving-average stage on the combined weight of the four-axis corpus
4784 comparison (a 9.5-percentage-point downstream F_1 advantage and a paired Wilcoxon



test that could not distinguish the two filters per trip, $W = 3.0$, $p = 0.313$) and the live cross-scenario ablation (a $4\text{--}5\times$ noise-reduction advantage across city, climb, highway, idle, and refuel windows against the Kalman filter's near-passthrough behaviour on the raw sender). SO2 demonstrated bounded per-channel latency (LoRa and WiFi medians near 7 s inside bootstrap 95% confidence intervals of approximately ± 0.5 s, LTE as a delayed-delivery path with a bootstrap median of 23.1 s in $[22.4, 23.9]$ s, GSM 2G as an automatic fringe fallback) and showed that 66.9% of delivered telemetry transited the offline buffer, preventing silent data loss; the LoRa packet-delivery Wilson interval at the 50 m band sat entirely above the 80% target. SO3 delivered a 5.52 s median sampling cadence with a bootstrap 95% CI of $[5.36, 5.63]$ s and a Wilson 95% interval on GPS fix availability of $[0.894, 0.910]$, above the 85% target, together with three synchronised tracking surfaces and fleet-vehicle-restriction-aware routing.

SO4 reached 92.9% precision and 1.09% false-positive rate for the context-aware Isolation Forest Model C under strict leave-one-trip-out evaluation on the project-native partition, and the accompanying four-configuration detector ablation confirmed the Isolation Forest as the strongest single detector on every metric with McNemar $p < 0.001$ against every other configuration and a bootstrap 95% CI for F_1 of $[88.6\%, 96.6\%]$ that does not overlap the hybrid's $[79.1\%, 90.2\%]$, so Matrix Profile is retained as an audit-only signal and the physics tripwire remains a deterministic operational floor. SO5 produced a correctly triggering five-class alert engine with a 95% resolution rate (Wilson 95% CI $[0.888, 0.980]$, above the 90% target) and an administrator System Usability Scale median of 82, all four respondents above the Bangor benchmark of 68 (one-sample Wilcoxon $p = 0.0625$ at the four-respondent exact-test floor). The commercial-scale scalability projection in Section 6.6 extended these prototype results to representative fleet sizes ($N_v = 10, 100$,



4809 500, 1,000) and identified the LoRa gateway occupancy as the first binding constraint at
4810 approximately the sixty-vehicle-per-depot threshold and the archival cadence as the hot-tier
4811 storage constraint from the low-hundreds fleet size upward, both linear in N_v and both
4812 handled by provisioning-time parameters rather than by architectural change. The principal
4813 remaining limitations — parked-depot siphoning detection, the real-time latency of the
4814 cellular path, the degraded GSM 2G fringe rate, hardware-anchored unit identification,
4815 and dependence on per-vehicle sender calibration — are bounded, explained, and carried
4816 forward as the recommendations and future directives discussed in Chapter 7.



4817

Chapter 7

4818

CONCLUSIONS, RECOMMENDATIONS, AND

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FUTURE DIRECTIVES



7.1 Concluding Remarks

This study designed, developed, and evaluated an IoT-enabled fleet-monitoring and fuel-anomaly-detection system for delivery-vehicle operations. The prototype integrates vehicle-sensor acquisition, GPS tracking, resilient multi-channel wireless transmission, cloud storage, dashboard and human-machine-interface visualisation, context-aware anomaly detection, and closed-loop alert delivery into a single operational system evaluated against the five specific objectives set at the outset.

The measured outcomes support the objectives as met. For SO1, the filtered fuel signal reached 1.04% mean absolute percentage error and 3.00% maximum error across the 13–98% tank-level range against a legally calibrated pump-volume reference, with a residual-noise band below 0.1% across every reported tank band; the moving-average stage of the fuel-preprocessing cascade was retained after a two-part comparison against a Kalman-filter alternative, comprising the four-axis corpus comparison that showed the two smoothers effectively tied on noise and signal preservation but decisively separated in favour of the moving average on downstream detection quality, and a supplementary live cross-scenario ablation in which the moving average again reduced consecutive-sample noise by a factor of four-to-five across city, climb, highway, idle, and refuel windows, whereas the Kalman filter was near-passthrough on the raw sender stream. For SO2, the multi-channel communication stack (LoRa → LTE → GSM 2G → WiFi → offline buffer) sustained bounded per-channel latency (LoRa and WiFi medians near seven seconds, LTE as a delayed-delivery path, GSM 2G as an automatic fringe fallback) and confirmed that approximately two-thirds of the field corpus reached the backend through the buffered or replayed path, so a real-time-only design would have lost most of the operational data.



For SO3, the GPS pipeline delivered a 5.52 s median inter-sample cadence and 90.21% fix availability across 5,181 field-test rows, well inside the objective's targets. For SO4, the context-aware Isolation Forest Model C reached 92.9% precision, 92.9% recall, and 1.09% false-positive rate under strict leave-one-trip-out evaluation, and a formal ablation with three alternative detector configurations (physics tripwire alone, Matrix Profile alone, IF + MP hybrid) confirmed that the Isolation Forest is the strongest single detector on this corpus, that Matrix Profile is best retained as an audit-only signal rather than as a co-equal union gate, and that the physics tripwire remains a defensible deterministic operational floor. For SO5, the alert engine, the embedded HMI, and the administrative dashboard together sustained a 95% closed-loop alert-resolution rate on a 100-alert field audit and an 81.9-point System Usability Scale response from the intended administrator group.

The statistical layer sitting under those point estimates strengthens the conclusion rather than merely restating it. McNemar's paired-comparison test rejected the null of equal error rates between the Isolation Forest and every other detector configuration at $p < 0.001$, the bootstrap 95% confidence intervals for the Isolation Forest and the hybrid did not overlap on F_1 or precision, and the Wilson score intervals for packet delivery, GPS fix availability, and alert-resolution proportion each sat above their respective objective targets with a statistical margin. Where the evidence did not support a strong statistical claim (the Wilcoxon test on the moving-average versus Kalman-filter per-trip F_1 differences at $p = 0.313$, and the one-sample Wilcoxon on the four SUS respondents at $p = 0.0625$), the honest reading is documented alongside the point estimate rather than glossed over, and a larger evaluation partition is recommended in the corresponding case.

The scalability analysis reported in Section 6.6 of Chapter 6 extends the study findings from the two-vehicle thesis pilot to representative commercial-fleet sizes, evaluating the



prototype at 10, 100, 500, and 1,000 vehicles against the three closed-form load-axis ceilings derived in Section 5.8.1: backend ingest, LoRa gateway occupancy, and hot-table storage growth. That commercial-scale projection is the operational bridge between the measured prototype metrics and a fleet-scale rollout plan, and it identifies the LoRa gateway occupancy as the first quantitative constraint that binds at typical depot fleet sizes and the archival cadence as the storage constraint that dominates once daily hot-table growth exceeds the dashboard query-latency envelope.

Taken together, the study demonstrates that combining resilient embedded telemetry, contextual feature engineering, context-aware time-series and tree-based detectors, cloud-side processing, and multi-surface visualisation can support practical fuel monitoring and fleet supervision under real Philippine operating conditions. It also shows that fuel anomaly detection should not rely only on raw fuel-level changes, since normal driving conditions, idling, traffic, refuelling, and sensor noise can resemble abnormal behaviour; trip context, per-vehicle baselines, communication resilience, and human-readable alert reasoning are all load-bearing for a system that must be trusted by an operator. The remaining boundaries of the deployed prototype (parked-depot siphoning detection, larger real-anomaly corpus, hardware-anchored unit identification, and stronger real-time notification) are documented as the future-work directions in Sections 7.4 and 7.5 below.

7.2 Technical Limitations

The prototype met the numerical targets of the study, but several implementation limits remain important for interpreting the results and planning deployment beyond thesis scale. Table 7.1 summarizes the main technical limitations, their causes, and the corresponding



4889 future-work direction.

TABLE 7.1 TECHNICAL LIMITATIONS AND FUTURE WORK

Limitation	Why it happened	How future work fixes it
No dipstick calibration	The partner fleet operator did not permit tank opening or invasive tank-volume measurement during field testing.	Conduct controlled laboratory tank calibration or partner-approved calibration using a supervised tank-opening procedure.
Small real anomaly count	Actual theft, leak, and siphoning events were unavailable during the thesis field window, so labels were primarily controlled injections.	Extend SGSA fleet deployment over a longer period and collect confirmed operational anomaly cases for retraining and validation.
LTE latency	Cellular modem attachment, HTTP actions, TLS negotiation, and network conditions can delay some telemetry uploads.	Use persistent bearer sessions, optimized modem power design, and a stronger LTE Cat-1 or equivalent modem configuration.
GSM 2G degradation	Carrier availability and legacy-network phaseout reduce the reliability of GSM-only fallback in some areas.	Expand LTE Cat-1, NB-IoT, LoRaWAN, or other modern fallback channels and treat 2G only as transitional support.

4890 The current deployed fuel-anomaly detector is strongest for in-trip and moving-vehicle
 4891 anomalies because it uses contextual movement, speed, RPM, engine load, route state, and
 4892 driver-baseline deviation to decide whether fuel loss is plausible. Parked-depot siphoning
 4893 is a remaining limitation: when the vehicle is stationary and the engine is off, the pro-
 4894 duction feature set has little movement or load context to compare against. Future work
 4895 should add a separate stationary-state detector based on long-duration fuel-level monitoring
 4896 so overnight or depot siphoning can be handled without weakening the moving-vehicle
 4897 Model C threshold.

4898 7.3 Contributions

4899 The interrelated contributions and supplements that have been developed by the author(s)
 4900 in this study are listed as follows. Only those that are unique to the authors' work are
 4901 included.



- 4902 • the development of an integrated fleet-vehicle telemetry monitoring prototype that
- 4903 combines fuel-level sensing, GPS tracking, speed monitoring, trip logging, HMI
- 4904 alerts, and cloud dashboard visualization;
- 4905 • the implementation of a multi-path communication approach using LoRa, LTE, GSM
- 4906 2G, WiFi, and SPIFFS offline replay to support telemetry transmission under different
- 4907 operating and network conditions;
- 4908 • the creation of a context-aware fuel anomaly detection pipeline using Isolation Forest,
- 4909 route and driving-context features, per-asset fuel-economy deviation, and comparison
- 4910 against baseline detectors such as static thresholding, IQR-based methods, Local
- 4911 Outlier Factor, Matrix Profile gating, and hybrid detection;
- 4912 • the construction of controlled-injection validation trips and field-test telemetry
- 4913 records for evaluating fuel anomaly detection, trip monitoring, dashboard behavior,
- 4914 alert generation, and system reliability;
- 4915 • the implementation of backend alert logic and HMI-side immediate warnings for
- 4916 operational events such as overspeeding and low fuel, improving response visibility
- 4917 for both the driver and fleet administrator;
- 4918 • the development of dashboard analytics for fleet-level monitoring, including trip
- 4919 summaries, anomaly heatmaps, alert trends, fuel behavior, and overspeeding events
- 4920 by fleet vehicle.



7.4 Recommendations

Based on the results of the study, the researchers recommend further improvement of the system before wider field deployment. First, the fuel anomaly detection model should be evaluated using more real-world trips from different fleet vehicles, routes, load conditions, traffic levels, and driving behaviors. Although the controlled-injection and field-validation results showed promising results, additional field data would make the model more robust and reduce dependence on predefined trip patterns.

Second, the fuel detection logic should continue to distinguish between true abnormal fuel loss and normal operational events such as idling, slopes, refueling, sensor fluctuation, and traffic congestion. The results showed that some anomaly types are easier to detect than others, so future versions should improve sensitivity to short-duration events without increasing false alarms.

Third, the system should improve trip-distance and odometer handling by maintaining reliable server-side trip accumulation and reducing dependence on unavailable vehicle odometer readings. Since some vehicle OBD-II odometer data may be inaccessible, trip distance should be computed using a defensible combination of GPS, speed, and onboard integrated distance, with safeguards against corrupted or non-monotonic readings.

Fourth, the dashboard and alert system should be tested with more administrators and drivers in real fleet operations. The acceptance survey showed positive usability, but a larger group of users would provide better feedback on alert clarity, dashboard navigation, and operational usefulness.

Lastly, the researchers recommend improving deployment reliability by adding stronger logging, automated data-quality checks, and clearer handling of offline telemetry. These



additions would help ensure that delayed, buffered, or partially missing data does not distort trip summaries, anomaly counts, or dashboard analytics.

Additionally, the researchers recommend hardening the unit-identification chain before broader fleet rollout. The current identification scheme is anchored at backend-issued UUIDs (`truck_id`) rather than at hardware-rooted identities, so a swapped ESP32 controller inheriting the same `truck_id` through NVS restore would produce telemetry indistinguishable from the original device. Three concrete extensions are recommended: (1) bind each `truck_id` to an immutable hardware anchor such as the ESP32 factory MAC or eFuse-based secure device ID, so that a controller swap is detected on first check-in rather than silently inheriting the previous vehicle's telemetry stream; (2) tag every telemetry payload with a firmware provenance stamp (Git commit SHA plus build timestamp), so that rolling upgrades can be distinguished on the same physical unit; and (3) surface sensor-level identifiers (OBD-II VIN, GPS module UID, LoRa EUI, cellular IMEI) as part of the unit record on the operator dashboard, so that data-quality issues can be attributed to a specific sensor without a physical inspection.

7.5 Future Prospects

There are several prospects that may be extended for further studies. The suggested topics are listed as follows:

1. Integration of additional vehicle parameters not yet covered by the prototype, such as battery voltage, diagnostic trouble codes, tire-pressure data, and brake-system indicators, to improve anomaly detection and vehicle-health monitoring.



2. Expansion of the anomaly detection model using a larger real-world fleet dataset across more routes, vehicles, fuel behaviors, and operating conditions.
3. Development of a mobile driver application with live trip status, route guidance, alert acknowledgement, and driver-side incident reporting.
4. Development of route-deviation and geofence-based supervision features to notify administrators when a fleet vehicle leaves an assigned corridor, enters a restricted service area, or remains stopped outside an expected location.
5. Improvement of anomaly explainability, where each fuel anomaly alert includes the likely reason for the flag, such as sudden fuel drop, abnormal fuel-per-kilometer behavior, prolonged idling loss, or route-context mismatch.
6. Deployment of a stronger real-time notification system using SMS, push notifications, or messaging integrations for urgent events such as fuel theft, low fuel, overspeeding, or route deviation.
7. Evaluation of the system in long-term fleet operation to measure its effect on fuel accountability, response time, operational visibility, and maintenance planning.
8. Extension of unit identification and hardware-provenance tracking so that fleet-scale rollouts remain auditable, by binding each vehicle's `truck_id` to an immutable hardware anchor (ESP32 factory MAC or eFuse-based secure device ID) instead of a backend-issued UUID stored in NVS alone, tagging every telemetry payload with a firmware provenance stamp (Git commit SHA and build timestamp), and surfacing sensor-level identifiers such as OBD-II VIN, GPS module UID, LoRa EUI,



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and cellular IMEI on the dashboard so that data-quality issues can be attributed to a

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specific sensor without a physical inspection.



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Appendix A

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ETHICAL CLEARANCE



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University Research Ethics Review Board (URERB)

REVIEW RESULTS-APPROVED

June 03, 2026

MS. SAMUELLE P. BARJA

Project Leader

GCOE

De La Salle University Manila

Re: Research Proposal 2026-102C

Dear **Ms. Barja**,

Thank you for providing the University Research Ethics Review Board with the documents needed for the research ethics review of your research proposal.

Your research project, “An IoT-Enabled System for Real-Time Fuel Monitoring, GPS Tracking, and Anomaly Detection in Trucks Using Non-Parametric Unsupervised Machine Learning” which underwent resubmission review has been granted ethical clearance.

The ethical clearance of this study is valid from 03 June 2026 to 02 June 2027. Suppose your research project extends longer than one year from today. In that case, you must apply for Continuing Review, at least 25 working days before the expiration of clearance.

The following documents have been approved for use in this study:

1. Informed Consent Form, version 2, 22 May 2026.pdf
2. Proposal, version 2, 22 May 2026
3. RERC Form 6A, version 7, 22 May 2026.pdf
4. Data Usage Clarification, version 1, 25 March 2026
5. Letter Of Collaboration, version 1, 25 March 2026

While the research is in progress, please inform the URERB through the staff of the following:

1. **Any Reportable Negative Events:** Report these by using URERB Form 11A: Reportable Negative Events Form within 24 hours.
2. **Any Adverse Events:** Report Adverse Events (AEs) and Serious Adverse Events (SAEs) within 24 hours of their occurrence using the Form 11B: Adverse Event Form.
3. **Any changes to the originally approved research protocols (amendments)** should be reported using URERB Form 9A: Amendment Request Form. The URERB must approve all changes in the protocol before implementation.
4. **Research Project Progress:** Report developments of the project with URERB Form 8A: Progress Report. (six months before expiry of clearance (on or before 03 December 2026)).

3rd Floor, Henry Sy Sr. Hall, 2401 Taft Avenue, 0922 Manila, Philippines | Trunk Line: (632) 8524-4611 loc. 513
www.dlsu.edu.ph



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University Research Ethics Review Board (URERB)

Within forty (40) days after the completion of your research project, submit a copy of the final report and the completed soft copy of **URERB Form 13A: Final Report Form** to the URERB.

Should you have any questions, please do not hesitate to email our staff, at urerc-staff@dlsu.edu.ph.

Thank you!

Sincerely,


MS. NATHALIE ROSE LIM-CHENG
Chair
University Research Ethics Review Board



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Appendix B

5329

PARTNER COORDINATION AND

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PROTOTYPE DEMONSTRATION



This appendix presents supporting materials from the partner consultation, prototype demonstration, and vehicle inspection. The materials support the methodology in Chapter 5 and the implementation evidence in Chapter 6.

The research team coordinated with Hino Paranaque Subic GS Auto Inc. (SGSA) to inspect the sample fleet-vehicle environment, review the intended monitoring workflow, and demonstrate the prototype interfaces. The evidence photographs in Figures B.1–B.9 document the consultation, fleet-vehicle inspection, dashboard review, and prototype demonstration activities.



Fig. B.1 Initial Consultation Meeting with SGSA Representative



Fig. B.2 Cabin Inspection of the Sample Fleet Vehicle



Fig. B.3 Further Fleet-Vehicle Inspection at the Partner Facility

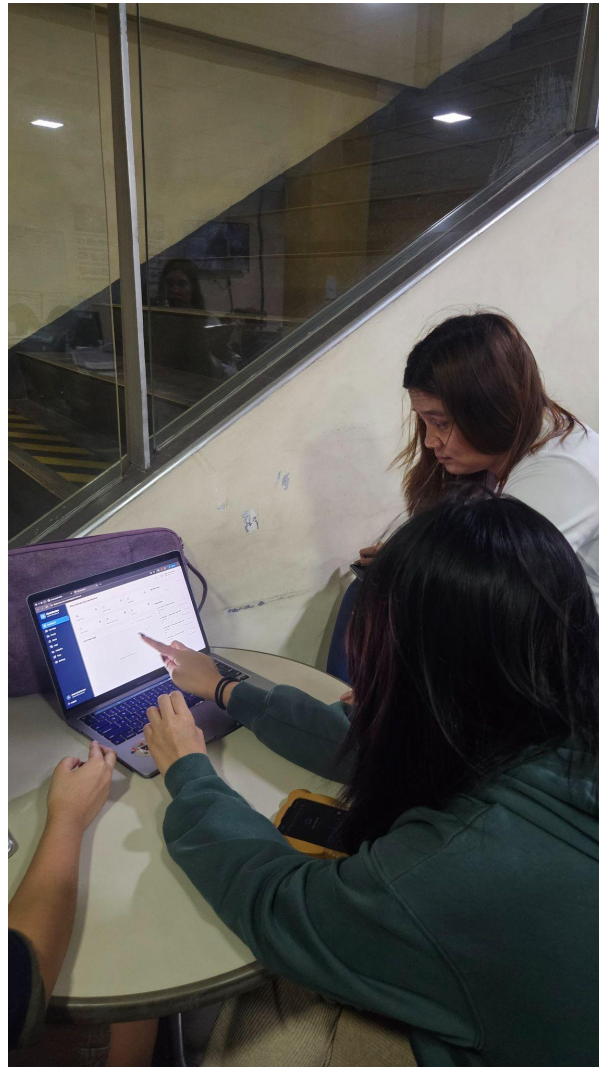


Fig. B.4 Dashboard Evaluation During Prototype Demonstration



Fig. B.5 Web-Based Monitoring Interface Demonstration



Fig. B.6 Research Team After Prototype Meeting and Demonstration



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Fig. B.7 Research Team and Partner Representative Group Photograph



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Fig. B.8 Additional Prototype Demonstration Evidence



Fig. B.9 Additional Partner Demonstration and Review Evidence



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Appendix C

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ADMINISTRATOR ACCEPTANCE SURVEY

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INSTRUMENT



This appendix presents the administrator acceptance survey instrument used to support the usability and alerting discussion in Chapter 6.

The administrator acceptance instrument asked respondents to rate their level of agreement with each statement using a five-point scale: 1 for strongly disagree, 2 for disagree, 3 for neutral, 4 for agree, and 5 for strongly agree. The survey focused on dashboard usability, fuel monitoring, anomaly alerts, driver-safety support, maintenance monitoring, and overall system satisfaction.

C1 Dashboard and Real-Time Monitoring

1. The dashboard provides a clear and easy-to-understand overview of fleet-vehicle status, including fuel, speed, location, and alerts.
2. The dashboard updates are timely enough to support real-time fleet monitoring.
3. The route-tracking/map feature is useful for monitoring fleet-vehicle movement.
4. The trip-history feature helps review past fleet-vehicle operations.
5. The fuel level display is easy to interpret.
6. The telemetry-log view provides enough details for checking fleet-vehicle activity.
7. The dashboard layout is organized and easy to navigate.

C2 Fuel Monitoring Accuracy

1. The system provides useful information about fuel level changes during operation.
2. The fuel monitoring feature can help identify possible abnormal fuel usage.
3. The fuel data shown on the dashboard is helpful for reviewing fleet-vehicle fuel behavior.

C3 Anomaly Detection and Alerts

1. The anomaly alerts are useful for identifying possible fuel theft, leakage, or unusual fuel consumption.
2. The alert information is clear and understandable.



- 5367 3. The dashboard makes it easy to review and resolve alerts.
- 5368 4. The alert system can help fleet administrators respond faster to fuel-related issues.
- 5369 5. The alert status and alert history are useful for record keeping.

5370 **C4 Driver Safety and Rest Alerts**

- 5371 1. The driver-rest alert feature is useful for reminding drivers to take breaks.
- 5372 2. The system supports safer driving by monitoring long driving periods or distance
5373 travelled.
- 5374 3. The HMI alert interface is understandable for drivers.
- 5375 4. The rest-alert records are useful for reviewing driver activity.

5376 **C5 Maintenance Monitoring**

- 5377 1. The maintenance-monitoring feature is useful for tracking fleet-vehicle service needs.
- 5378 2. The mileage-based maintenance reminder can help improve preventive maintenance.
- 5379 3. The maintenance information shown on the dashboard is easy to understand.
- 5380 4. The system can help reduce the chance of missed maintenance checks.

5381 **C6 Overall System Usability and Satisfaction**

- 5382 1. The system is useful for fleet monitoring.
- 5383 2. The system can help improve fuel accountability.
- 5384 3. The system can help improve operational visibility for the company.
- 5385 4. I would recommend further development or deployment of this system.
- 5386 5. Overall, I am satisfied with the system.



5387

C7 Open-Ended Feedback Questions

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1. Which feature of the system is the most useful?

5389

2. Which part of the system needs improvement?

5390

3. Were there any confusing parts of the dashboard or HMI?

5391

4. What additional features would you recommend?



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Appendix D

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TECHNICAL IMPLEMENTATION DETAILS



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D1 HMI GPIO Map

TABLE D.1 FULL HMI GPIO MAP

Pin	Function
GPIO 22	BTN_TAKE_REST (active-low, debounced 50 ms)
GPIO 39	BTN_SNOOZE (active-low, debounced 50 ms)
GPIO 25	Buzzer (PWM through the LEDC peripheral)
GPIO 5	A7670E modem PWRKEY

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D2 Telemetry JSON Payload Schema

Listing D.1: Telemetry JSON Payload Schema

```

5396 1 {
5397 2   "truck_id": "<uuid>",
5398 3   "driver_id": "<uuid>",
5399 4   "trip_id": "<uuid>",
5400 5   "sent_at": "<iso-8601>",
5401 6   "original_device_timestamp": "<iso-8601>",
5402 7   "source_device": "hmi" | "mobile",
5403 8   "fuel_level": <0..100>,
5404 9   "fuel_valid": <bool>,
5405 10  "fuel_source": "...",
5406 11  "fuel_confidence": <0..1>,
5407 12  "lat": <float>,
5408 13  "lon": <float>,
5409 14  "gps_source":
5410 15    "embedded_gps" | "mobile_gps" | "fused_mobile_fallback",
5411 16  "gps_accuracy_m": <float>,
5412 17  "gps_heading_deg": <float>,
5413 18  "mobile_speed_mps": <float>,
5414 19  "speed": <km/h>,
5415 20  "odometer_km": <float>,
5416 21  "engine_status": "on" | "off" | "idle",
5417 22  "seq": <bigint>,
5418 23  "event_id": "<uuid>",
5419 24  "channel_used":
5420 25    "lora" | "lte" | "gsm" | "wifi" | "offline_replay" | "mobile_app" |
5421 26    "test_injection",
5422 27  "comm_channel": "...",
5423 28  "hmi_driving_sec": <int>,
5424 29  "hmi_rest_sec": <int>,
5425 30  "hmi_trip_status": "...",
5426 31  "engine_rpm": <int>,
5427 32  "engine_load_pct": <float>,
5428 33  "throttle_pct": <float>,

```



```
5429 33  "maf_gps": <float>,
5430 34  "coolant_temp_c": <float>,
5431 35  "engine_runtime_sec": <int>,
5432 36  "dtc_present": <bool>,
5433 37  "dtc_count": <int>,
5434 38  "distance_gps_delta_km": <float>,
5435 39  "distance_obd_delta_km": <float>,
5436 40  "distance_fused_delta_km": <float>
5437 41 }
```



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Appendix E

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ARTICLE PAPER

An IoT-Enabled System for Real-Time Fuel Monitoring, GPS Tracking, and Fuel Anomaly Detection in Trucks Using Context-Aware Non-Parametric Machine Learning

Samuelle P. Barja, Konrad Romel C. Castelo, Eliza Rica R. De La Rosa, and Alyssa S. Montemayor

Abstract—Fuel accountability in delivery-truck operations requires more than a fuel gauge: it requires synchronized vehicle telemetry, route context, resilient communication, and an anomaly model that distinguishes abnormal fuel loss from traffic, idling, refueling, sender noise, and driver variability. This paper presents an implemented IoT-enabled fleet-monitoring prototype developed with Hino Parañaque Subic GS Auto Inc. The system acquires OBD-II fuel, speed, GPS, trip-state, and alert data through an embedded HMI; transmits telemetry through LoRa, LTE/GSM, WiFi, and offline replay; stores records in a cloud backend; and detects fuel anomalies using context-aware non-parametric learning. The production detector is Isolation Forest Model C, supported by Matrix Profile as a subsequence detector and by a data-mining layer for operating-regime interpretation. Fuel anomalies are modeled as context-conditioned deviations between observed and expected fuel-level change under speed, RPM, engine load, throttle, route, driver, and motion state. Field and controlled-injection evaluation showed 1.04% MAPE fuel accuracy, 5.52 s median GPS sampling, 90.21% GPS fix availability, 92.9% precision and 1.09% false-positive rate for Model C under leave-one-trip-out validation, and a 95.0% alert-resolution rate. The results support the use of context-aware telemetry and interpretable anomaly signatures for practical fleet fuel monitoring, while parked-depot siphoning remains a future-work boundary requiring a dedicated stationary-state detector.

Index Terms—Fuel monitoring, GPS tracking, fuel anomaly detection, Internet of Things, context-aware machine learning, Isolation Forest, Matrix Profile, fleet management.

I. INTRODUCTION

FUEL cost, fuel theft, unreported leakage, poor route visibility, and delayed alert response remain persistent problems in delivery-fleet operations. Traditional approaches often treat fuel monitoring, GPS tracking, and driver alerts as separate functions, which limits their ability to explain whether a fuel drop is normal consumption, traffic-related idling, sender noise, refueling recovery, or a genuine anomaly. Prior IoT fuel-monitoring and tracking systems show the value of connected sensors but often remain hardware-only or rule-driven, while machine-learning studies on heavy vehicles frequently use datasets detached from an integrated field prototype [1]–[4].

This study addresses that gap by implementing and evaluating a complete edge-to-cloud truck-monitoring system. The prototype combines OBD-II fuel and engine telemetry, GPS tracking, prioritized communication, cloud storage, dashboard visualization, HMI and mobile trip controls, and context-aware anomaly detection. The machine-learning component

is intentionally context-aware: it does not flag fuel loss only because the level dropped. Instead, it evaluates whether the observed fuel change is plausible under the vehicle's current operating context.

The specific contributions of this paper are: (1) an integrated IoT fuel, GPS, and alerting prototype for truck fleet monitoring; (2) a context-conditioned anomaly model grounded in physical and operational constraints; (3) an Isolation Forest production detector supported by Matrix Profile and K-Means-based operating-regime discovery; and (4) an evaluation across fuel accuracy, communication reliability, GPS cadence, anomaly-detection metrics, and alert usability.

The work was developed with Hino Parañaque Subic GS Auto Inc. (SGSA), where field constraints shaped the engineering choices. Direct tank modification and dipstick calibration were not available, so the system used non-invasive OBD-II/ECU fuel acquisition, manufacturer dashboard readings as a supporting reference, and known-volume pump-gas events as the primary validation reference. Likewise, the communication design could not assume continuous cellular connectivity, so the prototype includes multiple radio paths and local replay.

II. RELATED WORK

IoT-based fleet-monitoring systems have used GPS, cloud dashboards, RFID, GSM, and sensor modules to improve vehicle visibility and operational response [5]–[7]. Fuel-specific systems have monitored tank level, theft alarms, and basic fuel events using resistance, ultrasonic, or capacitive sensing [8]–[10]. These systems demonstrate practical monitoring value, but many are limited by fragmented architecture, small-scale validation, or rule-based detection.

Machine-learning work on fuel and vehicle anomaly detection shows that abnormal consumption can be detected from multivariate telemetry. Mumcuoglu et al. [4], for example, showed that fuel-consumption anomalies can be classified in heavy-duty vehicles, while interpretable anomaly-detection literature emphasizes the importance of explaining why a point is anomalous rather than reporting only a score [11]. This study builds on those directions by deploying context-aware models inside an implemented truck telemetry system and by tying every anomaly flag to physical or operational constraints.

The practical limitation in many prior systems is that a fuel drop is treated as self-explanatory. In actual fleet operation,

TABLE I
TASK EVALUATION SUMMARY

Task	Target	Measured result	Status
SO1	Fuel accuracy within $\pm 5\%$	1.04% MAPE; 3.00% maximum error; all validation events within tolerance	Met
SO2	Real-time cloud telemetry	LoRa, LTE/GSM, WiFi, and SPIFFS offline replay preserved telemetry flow under field connectivity limits	Met
SO3	5 s GPS route logging	5.52 s median sampling interval and 90.21% GPS fix availability across live trips	Met
SO4	$\geq 90\%$ precision and $\leq 10\%$ FPR	Isolation Forest Model C reached 92.9% precision, 92.9% recall, and 1.09% FPR under leave-one-trip-out validation	Met
SO5	Usable alerts and interfaces	100 alerts over 17.7 days; 95.0% resolution rate; administrator SUS score of 81.9/100	Met

a fuel-level decrease can arise from normal engine work, payload, traffic, route grade, idling, sender non-linearity, tank slosh, refueling recovery, or driver style. The present work therefore combines telemetry engineering with interpretable anomaly detection: the alert is not just a model output, but a context-conditioned event that is reconciled with route, engine, motion, and operational state.

III. OBJECTIVES AND EVALUATION CONTRACT

The study was organized around five task objectives. SO1 evaluates whether the fuel signal is accurate enough to support monitoring and downstream anomaly detection. SO2 evaluates whether telemetry can reach the backend through the available communication paths. SO3 evaluates GPS cadence and route context. SO4 evaluates whether the context-aware anomaly detector meets the required precision and false-positive rate. SO5 evaluates whether alerts and trip states are visible and usable across the HMI, mobile companion, and dashboard.

Table I is intentionally formatted as a normal one-column table so it remains inside the the two-column article layout instead of spanning the page. It summarizes the task-level result chain used in the thesis evaluation.

IV. SYSTEM ARCHITECTURE

Fig. 1 summarizes the conceptual system. The truck-side embedded HMI collects vehicle telemetry, displays local status, and allows driver trip interaction. The backend stores telemetry and trip state, evaluates anomaly and safety rules, and synchronizes the administrative dashboard, mobile companion interface, and HMI feedback path.

The implemented architecture in Fig. 2 follows a layered edge-to-cloud pattern. The edge layer performs OBD-II acquisition, GPS parsing, fuel filtering, trip-state handling, HMI display, and communication-channel selection. The transport layer uses prioritized LoRa, LTE/GSM, WiFi, and SPIFFS offline replay so telemetry is not silently lost during degraded network conditions. The cloud layer persists data in Supabase,

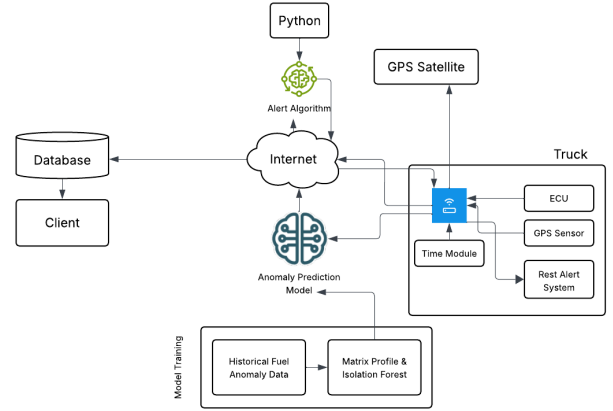


Fig. 1. Conceptual diagram of the IoT-enabled fuel monitoring and anomaly-detection system.

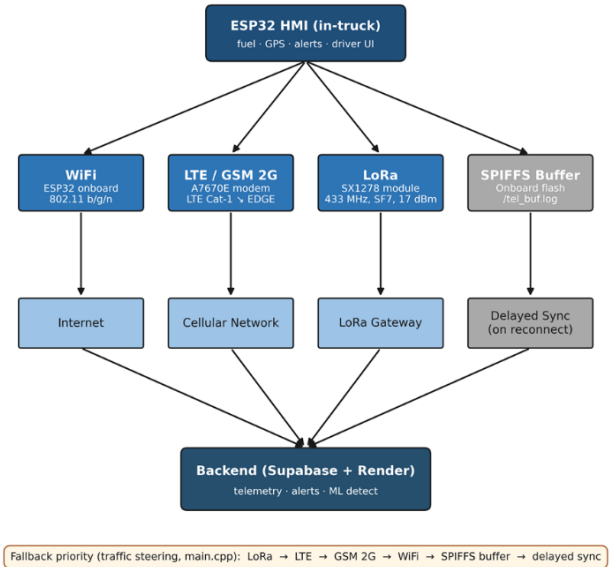


Fig. 2. End-to-end edge-to-cloud system architecture.

runs backend rules and anomaly inference, and exposes synchronized views to the dashboard, mobile app, and HMI poll-back path.

Fig. 3 shows the physical implementation counterpart of the system architecture. The ESP32-based HMI coordinates CAN/OBD-II acquisition, GPS reception, display output, LoRa radio transmission, LTE/GSM modem use, local storage, and driver buttons. Physical controls are retained for safety-relevant actions such as rest acknowledgement and alert response because a touchscreen alone is not ideal in a field vehicle.

The backend uses the trip session as the canonical state object. When a driver starts, rests, resumes, or ends a trip through the HMI or mobile companion, the backend synchronizes that state to the dashboard and associates incoming telemetry with the active trip. This design prevents a common problem in fleet prototypes: data may arrive, but it is not linked to a clear operating session. Here, telemetry rows, alerts, route traces,

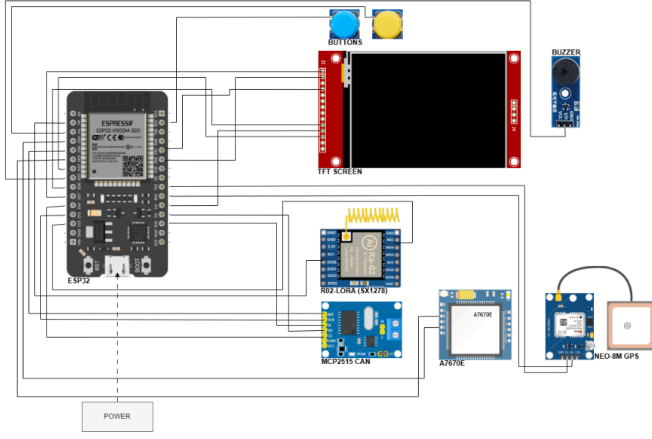


Fig. 3. Embedded HMI hardware block showing the separation of OBD-II/CAN, GPS, display, radio, modem, and driver-control functions.

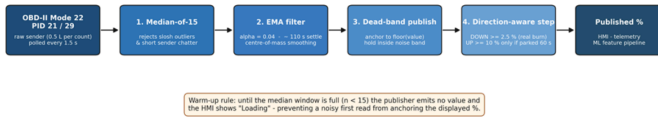


Fig. 4. Fuel-level filter cascade used to stabilize the sender signal before dashboard publication and anomaly inference.

and driver responses share the same trip context.

V. METHODOLOGY

A. Telemetry Acquisition and Filtering

Fuel, speed, GPS, trip, and engine data are sampled by the HMI and transmitted as synchronized telemetry rows. The main OBD-II and ECU-derived fields include fuel level, speed, RPM, engine load, throttle position, mass airflow, coolant temperature, runtime, and diagnostic status. Fuel level is the primary measurement for SO1 and SO4, while the remaining fields provide the operating context needed to judge whether a fuel change is plausible.

Fuel readings are filtered before model inference to reduce sender noise and slosh. The filtering pipeline in Fig. 4 combines a median stage, exponential smoothing, and a dead-band publisher, allowing the system to suppress short tank-motion artifacts while preserving meaningful fuel trends.

Fuel accuracy was evaluated against pump-gas reference events and the known tank capacity. The primary accuracy metric was mean absolute percentage error (MAPE):

$$\text{MAPE} = \frac{100}{n} \sum_{i=1}^n \left| \frac{V_{\text{sensor},i} - V_{\text{ref},i}}{V_{\text{ref},i}} \right|. \quad (1)$$

B. Communication, GPS, and Backend Integration

Each telemetry row carries device-side timestamp information and channel metadata. Server-side reconciliation separates sampling cadence from backend-arrival delay. Communication was evaluated through per-channel latency, packet delivery, and offline-replay behavior. GPS was evaluated using inter-sample interval and fix availability because route context,

distance, and motion state feed the anomaly detector and dashboard.

The communication policy is prioritized rather than single-path. LoRa is used for short-range depot or yard telemetry, LTE/GSM is used for wide-area connectivity, WiFi supports depot or driver-hotspot use, and SPIFFS local storage protects data when no radio path is available. Offline replay is treated as a valid delivery path because a fleet-monitoring system should preserve telemetry even when immediate cloud access fails.

C. Fuel-Anomaly Model

A fuel anomaly is modeled as a telemetry row whose observed fuel-level change deviates from the expected fuel-level change under operating context. For row t ,

$$A(t) = \frac{|\Delta f_{\text{obs}}(t) - \mathbb{E}[\Delta f | c(t)]|}{\sigma(c(t))}, \quad (2)$$

where $c(t)$ includes speed, RPM, engine load, throttle, route, driver, and motion state. This makes the detector a context-conditioned fuel-deviation model rather than a simple sudden-drop detector.

The model is constrained by four physical constraints and four operational constraints. C1 conservation states that fuel decreases through consumption or unauthorized removal and increases only during refueling. C2 bounded consumption limits plausible fuel loss by engine work and load. C3 sender non-linearity accounts for unstable tank-level bands. C4 slosh accounts for short fuel fluctuations caused by liquid movement. O1 traffic/idling, O2 route restrictions, O3 driver variability, and O4 refuel patterns encode Philippine fleet-operation context. These constraints are implemented through direction-aware rules, refuel gates, fuel-per-RPM and fuel-per-load features, expected-consumption residuals, idle and motion-state features, route-type features, driver fuel-economy deviation, and post-refuel flags.

D. Machine Learning and Data Mining

Isolation Forest Model C is the production row-level detector. Model A used baseline fuel, motion, engine, and refuel-window features; Model B added driving-zone features; and Model C added per-asset, per-driver, and route-type fuel-economy deviation features. Matrix Profile was retained as a supporting subsequence detector for gradual fuel-loss patterns, not as the main production classifier.

Fig. 5 summarizes the implemented ML pipeline as a four-plane architecture. After the filter cascade produces a signed 5-second telemetry row, the backend derives a 33-feature Model C vector and hands it in parallel to two context-aware detectors: an Isolation Forest that scores every row against the joint distribution of fuel, motion, engine, driving-zone, and per-driver fuel-economy-deviation features, and a Matrix Profile that scores the recent fuel subsequence for shape-based discord. The Isolation Forest is the primary alert-firing detector under the `if_with_margin` operating mode ($\text{score} \geq \tau + \delta$, $\delta = 0.005$). The Matrix Profile is retained as an audit signal that is persisted alongside every row for dashboard second-opinion display, and it additionally exposes a narrow escalation

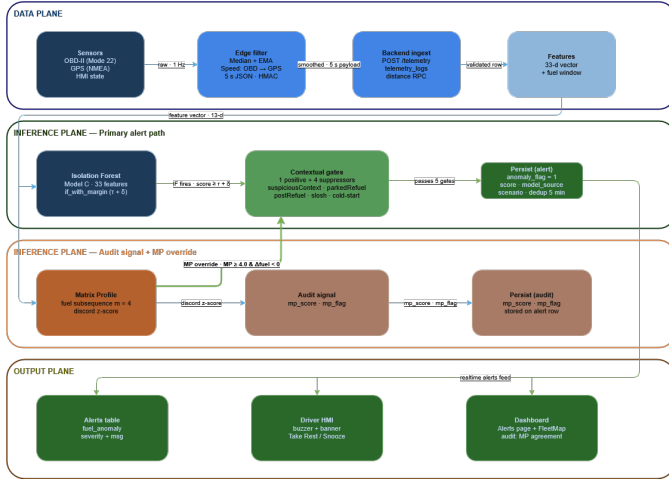


Fig. 5. Machine-learning anomaly-detection architecture.

path that lets a high-confidence discord ($mp_score \geq 4.0$ with $\Delta fuel < 0$, or $mp_score \geq 2.0$ with $\Delta fuel \leq -2.0L$) trigger a primary alert when the Isolation Forest is quiet, which recovers the gradual-leak class that per-row detectors cannot see. Both paths converge on a five-stage contextual gate that enforces physical plausibility (`suspiciousContext`) and suppresses the four canonical false-positive signatures of capacitive fuel senders: mid-fill jumps (`parkedRefuel`), post-fill settling within five minutes (`postRefuel`), acceleration-induced slosh (`loadSpikeSuppress`), and cold-start priming (`coldStartSuppress`). Rows that survive the gate are then subject to a combined-score confidence threshold ($combined_score > 0.65$) and a per-truck 5-minute dedup window before insertion into the `alerts` table; the same insert drives the driver HMI buzzer and the operator dashboard. This layered design decouples statistical detection, which is what the context-aware models learn from unlabeled telemetry, from precision-critical false-positive suppression, which is what the domain-aware gate encodes, and is the mechanism by which Section ?? reports 92.9% precision at 1.09% false-positive rate under leave-one-trip-out validation.

A data-mining layer supports interpretation. Constraint-violation profiling annotates each flagged row with signatures such as C2 bounded consumption, C4 slosh suppression, and O3 driver baseline deviation. K-Means clustering with $K = 4$ was used only to discover operating regimes in the controlled-injection window. The seven standardized clustering features were filtered fuel delta, speed, engine load, idle duration score, motion state, throttle, and normalized RPM. The four discovered regimes corresponded to high-speed cruise, mid-speed cruise, gradual-decline behavior, and raw-slosh artifact behavior.

E. Evaluation Protocol

Fuel accuracy, communication reliability, GPS cadence, anomaly detection, and alert usability were evaluated according to the study’s five objectives. SO4 anomaly-detection metrics were computed using leave-one-trip-out validation on the project-native partition so that each held-out trip was

scored by models trained and calibrated on separate trips. Precision and false-positive rate (FPR) were computed as:

$$\text{Precision} = \frac{TP}{TP + FP}, \quad \text{FPR} = \frac{FP}{FP + TN}. \quad (3)$$

The SO4 corpus included project-native trips with controlled injected anomalies, normal control rows, and a public-clean stress partition used to check whether the detector remains quiet on clean-route data. The project-native partition was the binding benchmark for the thesis objective because it represents the actual prototype telemetry and injected anomaly labels. Public-clean data was used only as out-of-domain supporting evidence.

VI. RESULTS AND DISCUSSION

A. SO1: Fuel Measurement

The fuel subsystem reached 1.04% MAPE and 3.00% maximum error across the 13–98% tank range, satisfying the $\pm 5\%$ target. The worst observed event remained two percentage points inside the allowed error band. The filtered-signal noise analysis also showed that residual noise was below 0.1% across the reported tank bands, supporting the claim that the filter cascade suppresses short slosh, ADC jitter, and sender bounce before the value is shown or modeled.

This result should be interpreted as operational field accuracy rather than laboratory tank calibration. Direct dipstick validation was not permitted, so known-volume pump events and manufacturer dashboard readings were used. Under that constraint, the result is still sufficient for dashboard monitoring, low-fuel alerting, and anomaly features that depend on fuel percentage.

B. SO2: Communication Performance

The communication subsystem met SO2 because every available path was exercised rather than only described. LoRa supported depot-scale and line-of-sight fallback communication, LTE served as the main wide-area path, GSM 2G provided a degraded fringe fallback, WiFi supported local or hotspot use, and SPIFFS replay prevented silent data loss during outages.

The main engineering lesson is that latency and packet preservation are different questions. Cellular latency can become high during modem or network renegotiation, but offline replay still protects the record. The thesis field log showed that a meaningful share of delivered telemetry used the offline buffer, which confirms the value of treating local storage as part of the communication architecture rather than as an afterthought.

C. SO3: GPS Tracking and Route Context

The GPS subsystem reached a 5.52 s median sampling interval and 90.21% GPS fix availability. The median interval is close to the 5 s firmware target; the longer mean and tail are explained by real field behavior such as OBD polling, GPS parsing, packet construction, connectivity delay, and replayed telemetry.

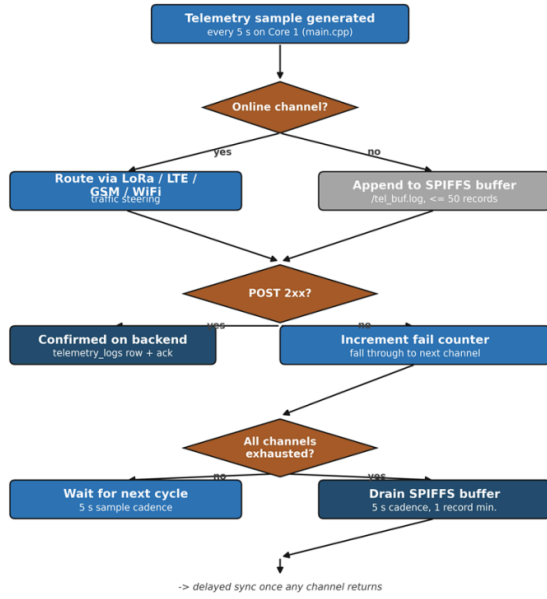


Fig. 6. Offline-buffer fall-through path used when live communication channels are unavailable.

TABLE II
SO4 MODEL-C PERFORMANCE SUMMARY

Metric	Result
Evaluation method	Leave-one-trip-out validation on project-native partition
Production detector	Isolation Forest Model C
Primary result	92.9% precision, 92.9% recall, 1.09% FPR
High-confidence channel	Hybrid Intersection reached 97.4% precision
Main false-negative source	Trailing rows of long gradual-leak windows where per-step loss decayed below the slosh-suppression baseline
Main false-positive source	Brief post-refuel acceleration excursions rejected by the contextual gate in production

GPS is not used only for map display. The route trace supplies segment distance, movement state, speed context, and route-type information for Model C. This is why SO3 supports SO4: without route and motion context, the anomaly detector would have to judge a fuel change as a raw time-series drop instead of as a contextual fuel-deviation event.

D. SO4: Machine-Learning Detection

For anomaly detection, Model C was the strongest production detector because it considered fuel movement relative to driver, route, and operating context. It achieved 92.9% precision, 92.9% recall, and 1.09% FPR on the project-native leave-one-trip-out partition. The result satisfies the 90% precision and 10% FPR requirements while preserving Model C as the deployable row-level detector.

The Hybrid Intersection produced higher precision as a high-confidence alert channel, but Model C remains the production detector because it provides the best row-level balance of detection and coverage. Model A missed many gradual-leak rows because baseline features alone cannot separate low-rate abnormal loss from sender drift. Model B improved with driving-zone features. Model C further improved detection because per-asset, per-driver, and route-type deviation features

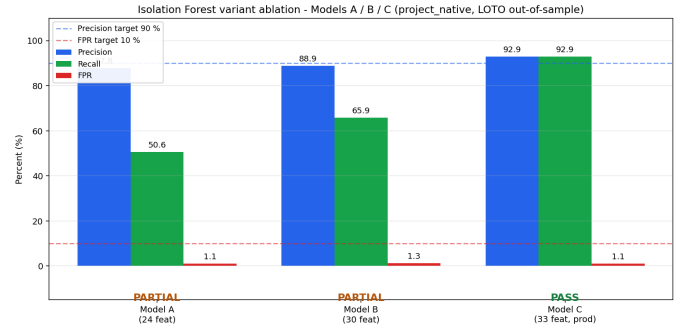


Fig. 7. Model-performance comparison used to support the SO4 production-detector selection.

create a meaningful expectation for fuel economy in the current operating context.

The constraint-violation analysis makes the ML result auditable. In the controlled-injection analysis, O3 driver baseline deviation appeared in every flagged row, C2 bounded consumption appeared in 69.0% of flagged rows, and C4 slosh suppression appeared most strongly in sudden and rapid drops. The controlled-injection window had zero false positives, but this is scoped only to that window; the full leave-one-trip-out evaluation still reports the overall 1.09% FPR.

K-Means archetype discovery further showed that labelled and predicted positives were concentrated in moving-vehicle operating regimes, while a raw-slosh artifact cluster contained zero labels and zero predictions. This supports the engineering interpretation that the filter cascade and contextual features suppressed C3/C4 artifacts before inference.

E. SO5: Alerts, HMI, and Dashboard

The alert subsystem completed the operator workflow. Alerts were surfaced across fuel anomaly, overspeed, low fuel, rest, and maintenance classes, with synchronized visibility across the HMI, mobile companion, and administrative dashboard. The field-test audit recorded 100 alerts over 17.7 days and a 95.0% resolution rate.

The administrator System Usability Scale score of 81.9/100 indicates that the dashboard was not only functional but usable for routine monitoring and response. This matters because a detector is operationally incomplete if alerts are not visible, persistent, and resolvable. The HMI, mobile app, and dashboard together support the loop from telemetry to event, from event to alert, and from alert to documented response.

VII. LIMITATIONS AND FUTURE WORK

The detector is strongest for in-trip and moving-vehicle anomalies because its context features depend on movement, speed, RPM, engine load, route progression, and driver-baseline deviation. Parked-depot siphoning, where the vehicle is stationary and engine-off, requires a separate stationary-state detector using long-duration fuel-level monitoring. Other limitations include the need for longer deployment with confirmed real-world theft events, broader vehicle and tank calibration, stronger modern cellular fallback, and more multi-user dashboard testing.

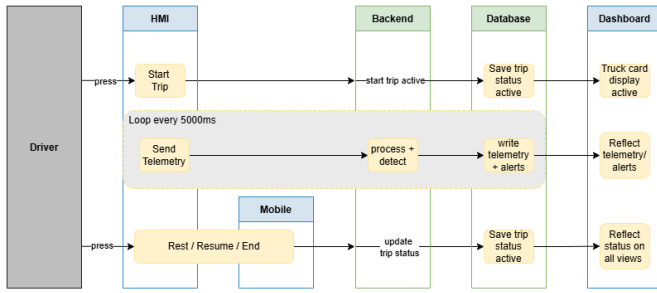


Fig. 8. Cross-interface trip-state synchronization between HMI, backend, database, dashboard, and mobile surface.

The present labels were primarily controlled injected anomalies rather than confirmed theft events. This is acceptable for prototype validation but should be expanded before production-scale claims are made. Future work should also evaluate multiple truck models, recalibrate sender non-linearity per tank, add stronger LTE-only or 5G-capable modem options, and improve parked-state fuel monitoring without increasing false positives from thermal drift or sender settling.

VIII. CONCLUSION

This paper presented an implemented IoT-enabled system for real-time truck fuel monitoring, GPS tracking, anomaly detection, and alert management. The prototype met the thesis objectives across fuel accuracy, communication, GPS cadence, machine-learning anomaly detection, and alert usability. The central technical finding is that fuel anomalies should be modeled as context-conditioned deviations, not as raw fuel drops alone. Isolation Forest Model C, supported by Matrix Profile and a constraint-aware data-mining layer, provided a practical and auditable production detector. Overall, the system demonstrates that embedded telemetry, resilient communication, context-aware learning, and human-readable alert explanations can support fuel accountability and proactive fleet supervision.

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